MEAT CONSUMPTION AND POTENTIAL REDUCTION FOR ENVIRONMENTAL AND PUBLIC HEALTH BENEFITS

by

GARRETT LENTZ

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Abstract

The focus of this thesis was to better understand meat consumption and investigate how a shift to more plant-based diets may best be promoted. The various environmental impacts linked to animal agriculture were explored and a movement towards more plant-based diets was found as a solution that could alleviate environmental impacts, along with the added benefit of improving public health and helping to safeguard future food security. Shifting a behaviour that is as prevalent as meat consumption is no easy task however, as high rates of meat intake have become normalised in many developed nations, being influenced not only by the desires of individual actors', but also structures within society that encourage continued production and consumption. Potential economic, regulatory, and informational measures to encourage meat reduction were explored and after weighing multiple factors, the potential for information provision to shift consumer meat intake held promise. However, before further inquiry into potential information provision measures, it was advisable to first obtain a more thorough understanding of consumers' meat consumption within the relatively understudied nation of New Zealand.

Thus, the first study of the thesis sought to better understand New Zealand consumers' meat intake through the distribution of a nationwide questionnaire. Awareness of meat's environmental impacts was determined to be low and the most common motivations for reducing meat were considerations of cost and health. Attitudes and attachment to meat were found to be strong predictors of willingness and intentions to reduce personal meat intake, as well as agreement with structural measures aiming to reduce meat consumption at a broader scale (e.g. environmental meat tax).

With a more thorough understanding of New Zealand consumers, the second study of the thesis sought to test different forms of information provision (in the form of documentary films) through the use of questionnaires previously validated in the first study in order to determine the effectiveness of different informational framings. Exposure to a film about meat's impacts on personal health, environmental sustainability, or animal welfare all increased intentions to curb meat intake, while reducing self-reported meat intake frequencies and positive attitudes and attachment towards meat. Animal welfare information was most effective at reducing positive attitudes and attachment towards meat, health information was particularly effective at reducing perceived dependence on meat, and environmental information showed the greatest declines in self-reported meat

intake frequencies. All three informational frames were found to increase agreement with proposed meat-reduction structural measures.

Findings from this thesis have implications for future studies and/or interventions seeking to understand or promote meat reduction, as it provides a clearer understanding of New Zealand consumers' meat intake, adds validation to tools that can be useful in the study of meat consumption, and confirms that information provision can be an effective method for promoting reduced-meat diets. This thesis adds to the growing body of literature on meat consumption and its potential reduction in efforts to promote environmental sustainability and public health, offering insightful and practical information to both researchers and practitioners.

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1. Lentz, G., Connelly, S., Mirosa, M., & Jowett, T. (2018). Gauging attitudes and behaviours: Meat consumption and potential reduction. *Appetite*, *127*, 230-241.

CHAPTER 1

INTRODUCTION AND THESIS OUTLINE

Throughout history, humans have utilised resources from the environment in order to provide food, water, shelter, heat and other amenities in efforts to not only survive, but live in a more comfortable manner. Over time, and especially following the agricultural and industrial revolutions, humans became more efficient at gathering and converting these raw resources into more 'usable materials' (e.g. easily consumed foods, tools, technologies, etc.). This has led to a rapid growth in human population over the past century, and this trend is expected to continue into the near future (although the rate of growth *is* slowing), with the current world population of 7.3 billion being predicted to reach 9.7 billion by 2050 (UN DESA, 2015). With continued population growth and improbable stabilization within the century (Gerland et al., 2014), feeding the world while avoiding environmental degradation will be a challenge.

Food demand in 2050 is expected to be 50% higher than it was in 2012 (FAO, 2017), and although food production has currently kept pace with rising populations, the consequences of continued resource overexploitation coupled with resulting environmental degradation has undermined the integrity of earth's natural ecosystems. Agriculture occupies more than a third (37%) of the earth's land surface (Smith et al., 2008), has heavily contributed to the deforestation of biodiverse tropical rainforests (Gibbs et al., 2010), is responsible for 92% of the planet's water footprint (Hoekstra and Mekonnen, 2012), and is a major contributor to global climate change, accounting for 17-32% of global anthropogenic greenhouse gas (GHG) emissions (Bellarby et al., 2008; See figure 1.1 below for breakdown of global GHG emissions by sector). Under current methods of food production, these figures along with fertilizer and pesticide use are likely to increase with population growth and rising food demands, all of which will have a significant impact on the diversity and function of our terrestrial and aquatic habitats, and their ability to support livelihoods through their ecosystem services (Tilman et al., 2001). Ecosystem services, as defined by Daily (1997, pg. 3), are, "the conditions and processes through which natural ecosystems, and the species that make them up, sustain and fulfill human life".

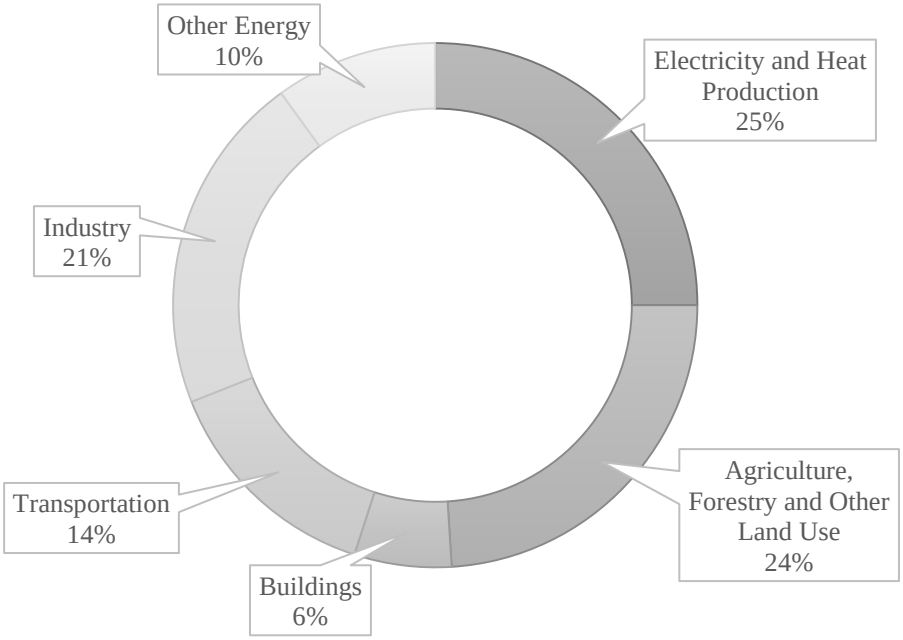

Figure 1.1 Global GHG emissions by sector. Adapted[reprinted] from ”Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change”, by the Intergovernmental Panel on Climate Change, 2014.

The agricultural sector is performing an extremely difficult balancing act, with productivity and sustainability on opposite sides of the scale. Food security strives to ensure that all people have access to safe, sufficient, affordable and nutritious food to support a healthy life (FAO, 2015), while ecological sustainability involves the protection of natural capital that acts as both a source of environmental goods and a sink for wastes (Goodland, 1995). These concepts are intimately linked, as ensuring food security through unchecked agricultural intensification can threaten ecological sustainability, while ignoring sustainability threatens food security through the loss of essential ecosystem services that support food production and human life in general. This has led researchers such as Tilman et al. (2002, pg. 671) to endorse a dual perspective which values both agricultural efficiency and ecosystem services, as both will be, “crucial if we are to meet the demands of improving yields without compromising environmental integrity or public health”. Of course, the improvement of yields may not be necessary if other inefficiencies in the food system can be addressed (e.g. roughly a third of global food production is either lost or wasted; Gustavsson et al. 2011) and it is certainly not the only (or even main) issue in regards to food security, as resources are not distributed evenly across the globe (both naturally and by people).

However, the effects of agricultural intensification are still very important as they are already being felt, as deforestation and its accompanying soil degradation continue to rise (Lepers et al., 2005; Barona et al., 2010) and water sources quickly diminish in numerous parts of the world (McKinsey & Co., 2009), threatening the integrity of ecosystems and their ability to support humans and their livelihoods. These bleak outlooks have sparked researchers to investigate many facets of the food industry to determine where sustainability measures such as GHG emissions, land use, and water use can be improved. This is summarised perfectly in a recent *Lancet* report, which states,

> "Providing a growing global population with healthy diets from sustainable food systems is an immediate challenge…Strong evidence indicates that food production is among the largest drivers of global environmental change by contributing to climate change, biodiversity loss, freshwater use, interference with the global nitrogen and phosphorus cycles, and land-system change…Food production depends on continued functioning of biophysical systems and processes to regulate and maintain a stable Earth system; therefore, these systems and processes provide a set of globally systemic indicators of sustainable food production…Because food systems are a major driver of poor health and environmental degradation, global efforts are urgently needed to collectively transform diets and food production" (Willett et al., 2019; pg. 447-448).

By observing how and where agricultural products are being produced, distributed, and consumed, scientists have narrowed down key 'leverage points', that if addressed, could lead to significant improvements for both the natural environment and food security (West et al., 2014). One 'leverage point' that has been under recent scrutiny is the livestock sector. In a United Nations Environment Programme (UNEP) report, animal products are listed besides fossil fuels as one of the largest contributors to worldwide environmental problems (Hertwich, 2010). Globally, the livestock industry accounts for 18% to 51% of GHG emissions (Steinfeld et al., 2006; Goodland and Anhang, 2009), 80% of anthropogenic land use (Stehfast, 2009) and almost a third (29%) of the agricultural water footprint (Mekonnen and Hoekstra, 2012). Moreover, livestock are a major source of pollution (Sutton et al., 2011; Bouwman, 2013; Liu et al, 2016) and are inefficient converters of feed crops (Aiking, 2011; Gonzalez et al., 2011; Westhoek et al., 2014). Thus, if increases in livestock production mirror population growth in the coming

decades, the negative impacts of the food system on already fragile and degraded natural ecosystems will likely be exacerbated.

Thus, the aim of this thesis is to establish the importance of critically examining and scrutinising our current food systems both in terms of production and consumption, and to provide a foundation for why the raising of animals for food, in particular, is harmful for both the environment and public health. Chapter 2 will explore the environmental impacts of livestock production including GHG emissions, land and water footprints, pollution, etc. and will outline how a shift to more plant-based diets could not only alleviate such impacts, but also improve public health and help protect future food security. Chapter 3 will explore how a societal shift to more plant based diets might occur by outlining what creates and shapes behavioural patterns within societies (e.g. individual agents or societal structures) and how behaviour change models based upon such discussions have been developed. Consumption's current place and form in society, including meat consumption specifically, will also be discussed in order to understand and identify barriers that might impede efforts to shift the status quo. Chapter 4 builds upon Chapter 3 by exploring potential pathways for promoting meat reduction throughout society, including through economic, regulatory, and/or information provision measures. After exploring these potential pathways and their strengths and weaknesses, Chapter 5 offers a summary of consumers and their current awareness of meat's environmental impacts; consumer motivations for meat reduction; specific behavioural models that could be useful for the study of meat-related beliefs, opinions, and behaviours; and why New Zealand is an appropriate location to further study meat consumption and its potential reduction. Chapter 6 outlines the methodologies and design of the two studies that comprise the thesis, along with how they individually add to the current literature and synergistically contribute to the overarching goal of the thesis. Chapter 7 and 8 cover the methods, results, discussions, limitations and future directions, and conclusions of Study 1 and Study 2, respectively. The final chapter, Chapter 9, summarises the findings of both Study 1 and Study 2, discusses their implications for the future study of meat consumption, and offers advice and insight to individuals, governments, and/or organisations seeking to promote societal shifts away from meat-based diets to those that are more plant-based.

CHAPTER 2

IMPACTS OF LIVESTOCK PRODUCTION AND MEAT CONSUMPTION

2.1 Chapter outline

This chapter will begin with a brief introduction and overview of current and projected food production and the various environmental impacts linked to the current food system (2.2), and how animal agriculture, specifically, contributes to these impacts. Current levels of meat production along with its associated impacts will be explored in depth, both at a global scale and within New Zealand, in order to clearly establish its contributions to a variety of environmental issues, including climate change, land use, soil degradation, deforestation, water use, eutrophication, and pollution (2.3 and 2.4). Next, the potential for these various impacts to undermine future food security, especially in poorer regions of the world, will be discussed along with the potential for the increased uptake of plant-based diets to alleviate some of the pervasive public health issues currently being witnessed in many developed nations (2.5). Finally, a chapter summary will be provided and a case will be made for the widespread adoption of more plant-based diets, on the basis that they can promote global environmental sustainability, future food security, and improved public health (2.6).

2.2 GHG contribution of the livestock Industry

With severe and potentially irreversible negative impacts from climate change (e.g. more severe weather phenomena, ocean acidification and coral bleaching, shifting habitats and potential species declines, etc.) expected to occur with a global temperature rise of 2 °C (Schellnhuber et al., 2006), the Intergovernmental Panel on Climate Change (IPCC, 2018) has advised that by the year 2030, global GHG emissions be reduced by 45% of 2010 levels and that net zero emissions be reached around 2050 in order to meet a target of only 1.5 °C global temperature increase. Reaching this target is not expected to fully eliminate climate-related damages to human societies and the environment, but is instead predicted to limit damages to more ‘manageable’ levels than expected with a 2 °C

increase. To attain this level of reduction, major contributors of GHGs in all sectors of industry must be identified and addressed accordingly.

The production of animal products for human consumption plays an important role in regards to the GHG emissions produced by the food industry (Steinfeld et al., 2006; Smith et al., 2007; Opio et al., 2011), with estimates on global emissions from the livestock sector ranging from 18% to 51%, depending on whether respiration, land use, emissions from fish farming, unequivocal efficiencies between regions, and potential corrections for undercounting livestock numbers are taken into account (Steinfeld et al., 2006; Bellarby et al., 2008; Goodland and Anhang, 2009). Although exact figures are still debated, there is a consensus that the livestock sector plays a significant role in global climate change (Steinfeld et al., 2006; Smith et al., 2007; Carlsson-Kanyama and Gonzalez 2009; Goodland and Anhang, 2009; Stehfest et al., 2009; Stuart, 2009; Gill et al., 2010; Thøgersen 2010; Garnett, 2011; Hedenus et al., 2014; Larios et al., 2016). New Zealand is a relatively small and sparsely populated nation, and therefore only contributes about 0.17 % to global gross GHG emissions (Ministry for the Environment "MfE", 2017); however, the nation has the fifth highest level of per capita emissions among the 35 nations in the Organization for Economic Cooperation and Development (OECD, 2017) and this is in large part due to its agricultural industry. Total cattle within New Zealand have increased from 8.9 million in 1994 to 10.1 million in 2017, of which 6.5 million are dairy cattle (which have increased in number in recent years) while 3.6 million are beef cattle (which have slightly decreased from 5 million since 1994). Sheep numbers have also decreased since 1994 from almost 50 million to 27.5 million (Stats New Zealand, 2019). According to a study by Zonderland-Thomassen et al. (2014), which utilised survey data from Beef and Lamb New Zealand, there are 426 beef and sheep farms in New Zealand which can be classified into 7 different farm categories. Thus, not all farms will contribute equally to discussed environmental impacts like greenhouse gas emissions, and thus these differences (if significant) will be mentioned where appropriate. Agricultural emissions are the largest contributors to New Zealand's gross GHG emissions (49%, see Figure 2.1 below), with a majority of these emissions being comprised of methane (about 72%) resulting from enteric fermentation of livestock and nitrous oxide emissions (about 22%), which are partially driven by deposits of animal waste onto soils (MfE, 2018). Thus, for these reasons, taking a critical look at animal agriculture's contribution to GHG emissions is important for addressing the threats of

climate change and reaching emissions targets both globally and within New Zealand, specifically.

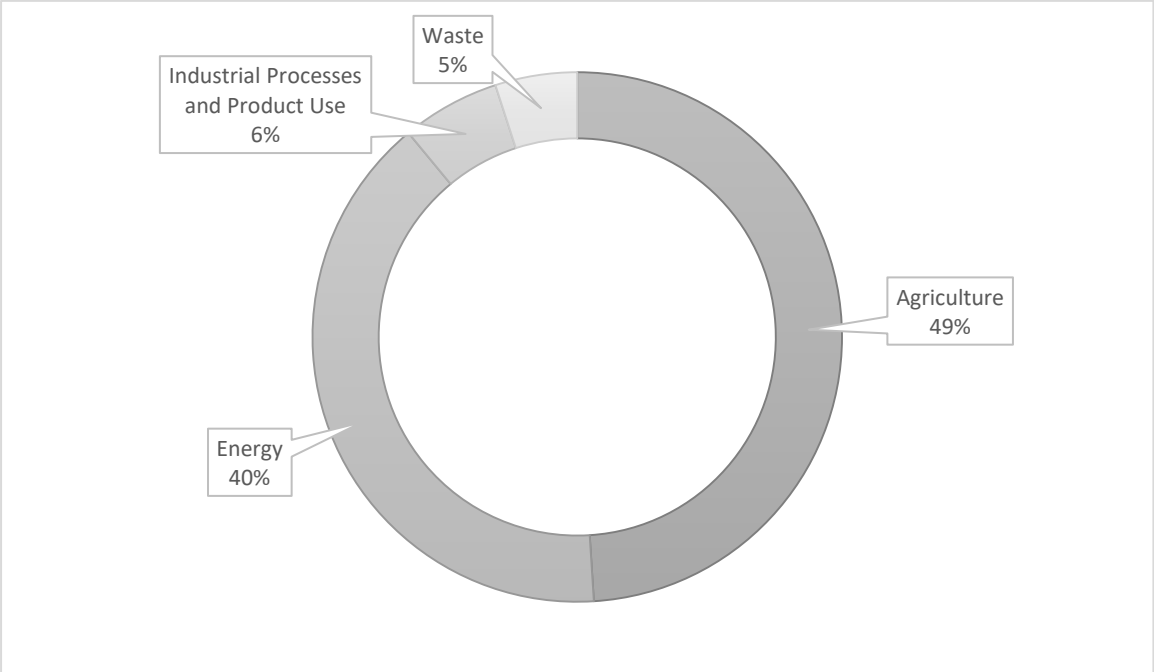

Figure 2.1 New Zealand's 2016 emissions profile. Percentage of gross GHG emissions are outlined by sector, measured in megatonnes (Mt) of carbon dioxide equivalent (CO_2-e). Adapted [reprinted] from "New Zealand Greenhouse Gas Inventory 1990-2016", by the Ministry for the Environment, 2018.

Livestock only contribute about 9 percent of total anthropogenic carbon dioxide emissions (Steinfeld et al., 2006); however, carbon dioxide is only one of many greenhouse gases which include methane and nitrous oxide. In terms of global methane and nitrous oxide emissions, the livestock sector contributes 37 and 65 percent, respectively (Steinfeld et al., 2006). Although methane and nitrous oxide emissions only account for small percentages of total anthropogenic GHG emissions when compared to carbon dioxide (16%, 6% and 76% respectively; IPCC, 2014), they can still have a monumental impact on climate change. This is because greenhouse gases are not created equal, as global warming potentials (GWP) and lifespans in the atmosphere can differ greatly. GWP is the capacity for a greenhouse gas to trap heat in the atmosphere; carbon dioxide acts as a baseline to which all other gases are compared, and is thus given a GWP of 1. According to the IPCC (2013), methane's lifespan in the atmosphere is 12.4 years and its GWP is 86 over a 20 year metric, while nitrous oxide's lifespan is 121 years with a GWP of 268 over a 20 year metric. Carbon dioxide, in comparison, has a lifespan of at least 100 years (Goodland and Anhang, 2009) and being a reference figure, has a GWP of

1. Put simply, gram for gram methane and nitrous oxide have greenhouse effects that surpass carbon dioxide by a factor of 86 and 268, respectively. This explains their relatively large contribution to radiative forcing (28%; EPA, 2012), the capture and retention of radiant energy by Earth's atmosphere, despite their small fraction of global emissions. Thus, in addition to carbon dioxide, methane and nitrous oxide emissions from the livestock sector also require proper attention when discussing climate change mitigation strategies.

Specifically, methane has been targeted as a priority greenhouse gas for short-term climate change mitigation (Moore and MacCracken, 2009; Ramanathan and Xu, 2010), as its lifespan in the atmosphere is only 12 years. Thus, managing methane emissions will mitigate climate change faster than targeting carbon dioxide, a GHG which has a much longer atmospheric lifespan. By no means does this suggest that carbon dioxide emissions can be ignored if methane emissions are addressed, but rather that carbon dioxide should be considered more of a *stock* that lingers in the atmosphere, while methane is more of a *flow* which is constantly being introduced and removed from the atmosphere (IPCC, 2014). However, although methane is a short-lived gas, it is important to note that if its flow is never reduced, or instead increases in the future, its contribution to climate change can become a long-term issue, as its entrance rate into the atmosphere would be equivalent to (or even outpace) the rate of its dissipation, resulting in no (or positive) net change in its atmospheric concentration over time. Therefore, in addition to carbon dioxide, methane will likely need to be addressed at some point in time if long-term climate goals are to be reached.

Methane emissions mainly result from enteric fermentation by ruminants, with smaller amounts resulting from the management of animal manure and wastes. There has been an 11% increase in methane emissions from enteric fermentation between 2001 and 2011, accounting for 39% of the agricultural sector's total global GHG emissions (FAO, 2014), and is now the largest source of anthropogenic methane emissions on the planet (Ripple et al., 2014b). With such a large share of the GHG budget, methane emissions have become one of the prime targets for reduction recommendations. Remember, because methane's atmospheric lifespan is about a tenth of carbon dioxide's, mitigation in this sector would result in relatively quick reductions in global GHGs over the coming decades in comparison to strategies that target carbon dioxide emissions alone (e.g. renewable energies and alternative fuels), which will take a century or more to see reductions in

global GHGs. Also, with a GWP 86 times that of carbon dioxide, even small reductions can result in relatively powerful strides towards slowing the progression of climate change.

Besides direct emissions of methane, the livestock industry also contributes heavily to deforestation which has a substantial impact on the carbon cycle and climate change. Biomass burning releases GHGs (Andreae et al., 2001), while replacing rainforest with grasslands also results in a concurrent drop in carbon storage from 200 to just 8 tons per hectare (Goodland and Anhang, 2009). So not only does clearing of rainforest directly contribute to GHG emissions, but its negative effects are compounded by the indefinite removal of significant carbon sinks. This can be seen in the Amazon, with pastures for grazing now covering 70% of previously forested lands (with feed crops using a large share of the remainder) (Aiking, 2011), and animal agriculture accounting for 80% of the total GHG emissions resulting from land use changes (Bustamante et al., 2012). Based on the average deforestation rate of 19,400 km^2 per year, the emissions from Brazilian Amazonian deforestation alone in the year 2007 represented more than 2% of global GHG emissions (Garnett, 2009). This estimate is conservative, seeing as it does not account for additional gases released from disturbed soils, nor does it include the increased fertilizer and manure input from livestock production.

Of course, emissions from livestock will vary depending on the type of animal being raised, the environment the animals are raised in, and the practices that are implemented, as some might argue that grass-fed cattle mitigate or avoid many of the impacts associated with more concentrated operations. For example, through the act of grazing, cattle can actually reduce a small portion of GHG emissions by killing microbes that produce large amounts of nitrous oxide in soils (Wolf et al., 2010). Reducing nitrous oxide emissions might therefore be a potential benefit for raising grass-fed cattle, but the study only looked at specific areas where there are relatively frigid winters accompanied by warm summers, and so areas that do not fit this criteria may not have this added benefit from grazing. In addition, a U.S. study found that cattle who are grass-fed produce about one and a half times the amount of methane as cattle who are raised conventionally (i.e. held in confined spaces and consuming mostly feed crops such as soy, corn, grain, etc.), stressing the importance of considering relative methane emission outputs depending on the nature of production (Capper, 2012). So yes, variations exist between livestock operations, and they each have their own positives and negatives in terms of

climate change impacts, but the fact remains that when taken on a global scale, it is clear that the raising of livestock for food is a major contributor to GHG emissions. Therefore, whether grass or grain fed, raised conventionally or in more 'natural' settings, the amount of cattle we are raising for food must be addressed if we hope to mitigate climate change and its impacts in an effective manner.

2.3 Land and water footprints of livestock

The ways in which humans shape landscapes are important for a number of reasons. Replacing natural habitats with grasslands or crops not only threatens plant and animal biodiversity (Tscharntke et al., 2005; Phalan et al., 2011), but also limits the capacity for land to perform natural ecosystem services such as carbon sequestration, water filtration and nutrient cycling (Dale and Polasky, 2007). For example, heavy grazing can lead to soil erosion as well as loss of flora and fauna (both terrestrial and aquatic) in riparian regions (Batchelor et al., 2015; Beschta et al., 2013). These land use changes as a result of livestock production continues to be a serious problem, as wildlife are deprived of needed habitats (Gibbs et al. 2009; Flynn et al. 2009) which provide valuable ecosystem services. In fact, excess use of land through the raising of livestock is one of the major contributors to population declines in both large carnivores and herbivores (Ripple et al., 2014; Ripple et al., 2015), and if feed crop expansion is to keep pace with estimated increases in demand for meat (e.g. 80% increase in soy production by 2050), many habitats could face further encroachment from agriculture, threatening numerous species in many regions around the world (WWF, 2017). Biodiversity loss, especially in traditionally biodiverse regions like the tropics (Machovia et al., 2015), becomes even more worrisome as the global rate of species extinction is currently 1000 times that of the estimated background (i.e. baseline) rate (Pimm et al., 2014). If continued habitat loss and species extinctions are to be abated, extensive land use by the livestock industry will likely need addressing.

With such extensive areas of land being dedicated to raising animals and their feed, the livestock industry is the largest user of land in the food industry. This also holds true in New Zealand, where about 45% of all land is used for agricultural purposes, with farming of beef and sheep accounting for almost 32%, followed by dairying at almost 10% of all land (Stats NZ, 2018). This places the livestock sector in direct competition with other uses of land, including crop production for human consumption, bioenergy crop

production, and natural habitats for wild species, as a few examples. Therefore, if lower livestock numbers were actualised, the freed lands could be utilised to provide more crops destined for direct human consumption, provide space for the rising demand for biofuel crops (Dicks et al., 2009) or provide spaces for rewilding, which would aid both conservation efforts and enhance carbon sequestration as natural habitats are restored (Smith et al., 2010; Smith et al., 2013).

So it is clear that a portion of land is utilised as the physical space in which livestock live and grow, but why is the land dedicated to feed crops an issue? In terms of sustainability, does it matter whether crops are fed to humans or to livestock? The simple answer, is yes. Feed efficiencies for livestock are very low when the input of feed crops to the output of meat produced are compared. For example, for every six kilograms of plant protein from grains and forage, only one kilogram of animal protein is produced (Pimentel and Pimentel, 2003), meaning feeding plant foods directly to humans would be six times more efficient. In addition, extraneous GHG emissions associated with the feeding, raising, and processing of livestock, as well as the packaging and storage of the final product (i.e. meat) would also be eliminated. Yes, it is true that grazing animals such as cattle can produce high quality food from relatively low quality grasslands; however, the production volumes from grass fed stocks are quite low when compared with more intensive production systems that supplement with feed crops (Nijdam et al., 2012), and the amount of land that would be required to produce enough grass-fed beef for the world is not feasible, especially considering the growing need for habitat restoration and conservation-dedicated lands in order to mitigate environmental impacts such as ecosystem service and biodiversity loss, as discussed previously. In addition to these nuances, the increased methane emissions of grass-fed cattle in comparison to crop-fed must be considered as this would likely accelerate global climate change and its resulting negative impacts. A study conducted by Gerbens-Leenes and Nonhebel (2002), found a hypothetical diet based on wheat uses six times less land than current affluent European diets that include meat, and as a result, the authors suggested reduced meat consumption as a powerful option to moderate the current over-utilisation of land and other natural resources. As an example, in 2009, 41% of the world's feed went to the poultry industry due to its threefold increase in consumer demand as compared to the mid-1960s (WWF, 2017). With around 40 percent of cereals and 94 percent of soy being used for global livestock feed (D'Silva and Webster, 2010), the amount of land and food allocated to produce meat

becomes a serious issue, especially when confronted with future projections of population growth and associated increases in food demand. Therefore, reducing meat consumption and switching to a diet with a greater emphasis on plant-based sources like wheat will not only save land for other uses (e.g. rewilding for conservation purposes), but also help to ensure global food security by partly eliminating the inefficient process of feeding crops to livestock rather than directly to humans.

In addition to excessive land use, livestock also contribute heavily to the degradation and pollution of soils and waterways. In the United States (U.S.), 85 percent of topsoil loss can be attributed to livestock ranching (Millstone and Lang, 2003), making the sector a significant contributor to soil erosion and desertification (Asner et al., 2004). Moreover, livestock are responsible for 37 percent of pesticide use and contribute a third of the nitrogen and phosphorous loads that pollute U.S. freshwater resources (Steinfeld et al., 2006). Beef and sheep farming, which is a significant contributor to the New Zealand economy, at around 10 billion in revenues in 2019 (dairy cattle also high at around 18 billion in 2019; Ministry for Primary Industries 2019) generally have less water scarcity footprints in comparison to non-New Zealand pastoral farms (Zonderland-Thomassen et al., 2014). In addition, water footprints were found to differ greatly between farming classes, dependent upon whether dairy bull calves were purchased and placed on the property. However, in New Zealand the majority of dairy farms have already shifted to more intensive farming practices (e.g. more heads of cattle in same land area, heavier application of nitrates to promote greater growth of grasses, etc.) and some beef farms are shifting as well (White, 2010). It is this shift to greater intensification within New Zealand (and in many other countries) that has been found to contribute to more numerous and severe environmental impacts like deterioration to soil quality and pollution of waterways with pathogens and excess nutrient leaching (White, 2010). In terms of greenhouse gas emissions and excess nutrient inputs, the dairy industry is actually a more major contributor within New Zealand, but the raising of beef cattle and sheep are still major contributors, along with being tied to more clear negative public health impacts after consumption. Thus, intensive livestock practices (both for meat and dairy) lead to large amounts of nutrients from animal wastes being deposited into freshwater systems, and although nitrogen and phosphorous act as important fertilizers for soil health and plant growth, overabundance can cause eutrophication and acidification of waterways leading to algal blooms and dead zones, resulting in serious repercussions for

aquatic biodiversity (Erisman et al., 2008 and Townsend and Howarth, 2010). One example is the infamous Gulf of Mexico dead zone, an area of hypoxic waters 22,000 square kilometres in size created by nutrient runoffs from crops and livestock that enter and flow down the Mississippi and Atchafalaya Rivers (Rabalais et al., 2002). Dead zones have been identified as a key stressor for marine ecosystems and have been estimated to affect more than 400 systems, covering a total area of more than 245,000 square kilometres (Diaz and Rosenberg, 2008). Although nutrient inputs from crop fertilizers also play a major role in creating algal blooms and dead zones, livestock contributions are an important factor to address considering their nutrient inputs from manure runoff and their share of crop consumption coupled with inefficient feed-to-meat conversions. This has led Steinfeld and colleagues (2006) to claim the livestock sector is, "…probably the largest sectoral source of water pollution, contributing to eutrophication, "dead" zones in coastal areas, degradation of coral reefs…", which leads to the assumption that a decrease in meat consumption and therefore, production will potentially alleviate the strains currently placed on our freshwater resources and saltwater ecosystems. This is especially relevant in a nation like New Zealand, where animal agriculture is a major sector of the economy, as the intensification of livestock has contributed significantly to not only degraded lands, but also waterways that are now too polluted for drinking and/or recreation (Foote, 2015).

Adding to livestock's role in water pollution, the sector is also a major consumer of global water supplies. The production of animal products accounts for nearly one-third of the water footprint for agriculture as a whole, with the footprint from the feed consumed by animals accounting for 98% of that one-third (Mekonnen and Hoekstra, 2012). With most of the water footprint coming from feed, one may make the assumption that the animals themselves have very little to do with the overall amount of water used, but this thinking is flawed. As seen with land use, feed inefficiencies are to blame once again, as animals consume much greater quantities of food than they themselves can produce. This can easily be shown with the example of beef, which has a water footprint 20 times larger than for cereals and starchy roots on a per calorie comparison (Mekonnen and Hoekstra, 2012). Feed efficiencies are not equivalent for all types of meat however, with beef requiring ~8 and ~11 times more feed per kilogram of meat produced when compared with pork and poultry, respectively (see Figure 2.2 below).

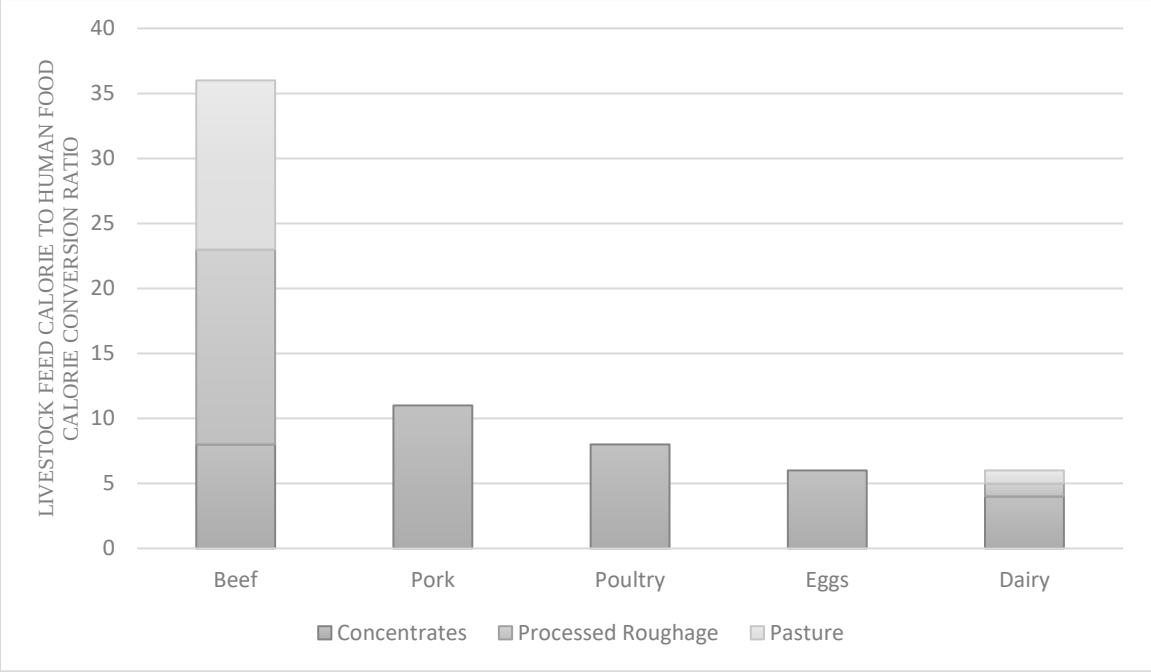

Figure 2.2 Feed to food calorie conversion factors for different livestock types in the United States. Bar heights are determined by the calories used to feed the animal divided by the resulting calories of the final product. For beef and dairy, the contribution of different feeds are divided by category running from the bottom to the top: concentrates (blue), processed roughage (orange), and pasture (grey). Adapted [reprinted] from "Land, irrigation water, greenhouse gas, and reactive nitrogen burdens of meat, eggs, and dairy production in the United States", by G. Eshel et al., 2014, *Proceedings of the National Academy of Sciences, 111,* 33.

Although there are significant differences in water footprints between different meat products, Mekonnen and Hoekstra predict that replacing all meat with an equivalent amount of plant crops would result in a 30% reduction in the water footprint of an average American citizen's diet (2012). Henning (2011, pg. 71) asserts that the use of freshwater for livestock is not only an inefficient and wasteful use of an increasingly scarce resource, but is immoral when the desire for the taste of meat threatens something so vital for human survival. Virtual water flows, the water embedded in products like meat when traded to other nations, are also an important factor to consider, as countries with plentiful water resources may be "virtually" taking water from other countries whose water sources are scarce when purchasing food products like meat (Hoekstra and Hung, 2005). Therefore, with limited freshwater supplies and the need for increased irrigation for food production to keep pace with population growth, it may be necessary to address the production and/or consumption of meat if we hope to conserve water resources, especially in nations where water is already scarce.

2.4 Future food security and recommendation of plant-based diet

Despite steady improvements over the past two decades, food security is still a major issue with around 821 million people malnourished (FAO, 2018). Food insecurity is the result of a combination of contributing factors, including overconsumption in certain nations, inequitable dispersal of resources, food waste, and inadequate distribution systems (WWF, 2017). The problem of feeding the world's hungry becomes more worrisome as global meat consumption is projected to be 72% higher in 2030 when compared to 2000 levels (Fiala, 2008), with increases in both developed and developing countries (See Figure 2.3, Garnett, 2009), as increased production is expected to exacerbate environmental impacts which could undermine food production and human safety, especially in poorer regions. A report from the Global Panel on Agriculture and Food Systems for Nutrition Experts warns that pressures on the current food system and those expected in the near future are likely to have serious consequences for global food security and that a novel transition to a less meat-based diet is necessary if issues like malnutrition, health, and environmental damages are to be alleviated effectively (Haddad et al., 2016). Thus, homogenization of diets in developing nations to match those commonly found in more developed 'western' countries (i.e. high rates of meat consumption) is not only an issue of potentially losing traditional and culturally appropriate foods, but also a major threat to environmental sustainability and future food security.

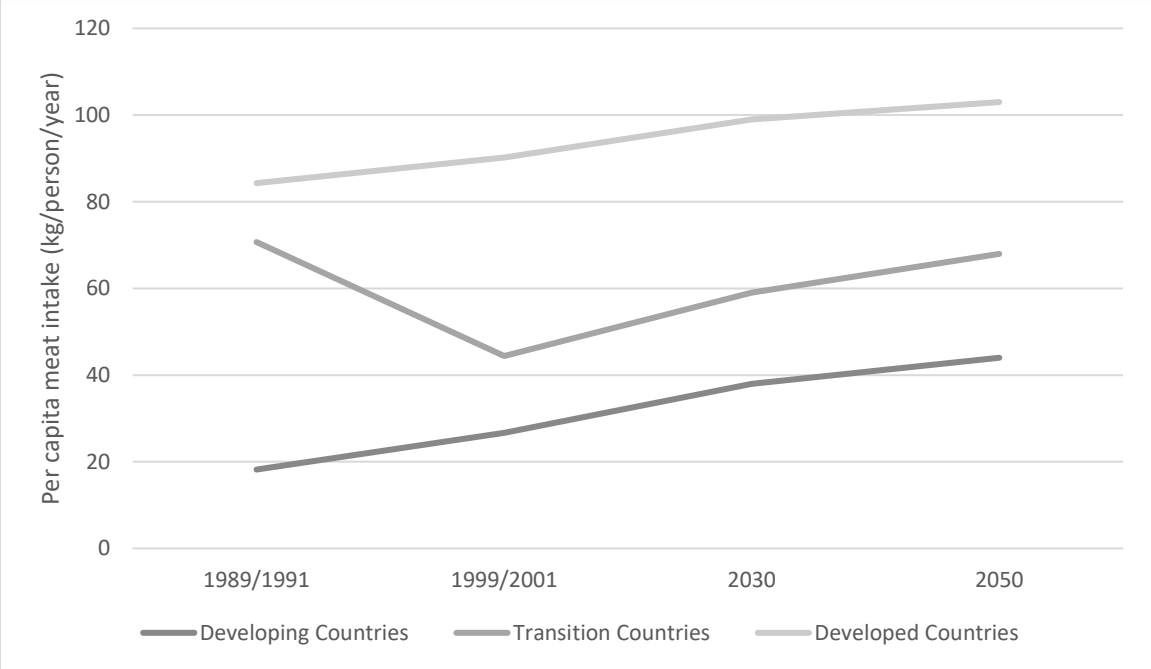

Figure 2.3 Projected per capita meat consumption (kg/person/year) to 2050 in developing, transitioning, and developed countries. Adapted [reprinted] from "Livestock-related greenhouse gas emissions: impacts and options for policy makers.", by T. Garnett, 2009, *Environmental Science and Policy*, *12*, 498.

Livestock already require a significant share of finite resources (land, water, crops) and contribute a great deal to environmental impacts (climate change, deforestation, pollution), so sharp rises in global meat demand will strain these resources and threaten fragile ecosystems even further. If resources become too scarce and ecosystems are degraded to the point where their services can no longer function, the very foundations that support food production and therefore, human life may be threatened. According to Pelletier and Tyedmers (2010), livestock's contribution to humanity's safe operating space in regards to proposed "sustainability boundaries" including GHGs (52%), biomass appropriation (72%), and reactive nitrogen (117%), all of which could compromise the integrity of natural environments and the attainment of food security, are expected to increase to 70%, 88% and 294% respectively by the year 2050. This is assuming the livestock sector grows as forecasted, while maintaining current shares of the safe operating space for each sustainability boundary. If the impacts from the livestock sector are to be minimized to respect the proposed sustainability boundaries, the GHG emissions, biomass appropriation, and reactive nitrogen mobilization per unit livestock protein would have to be reduced by 87%, 75% and 86% of year 2000 levels (Pelletier and Tyedmers, 2010).

Implementation of the latest technologies and mitigation strategies on a global scale will take large sums of time, money and human effort and has been projected to reduce the environmental impact of the livestock sector by only 20% (Weidema et al., 2008); therefore, addressing production concerns alone will not be enough to mitigate negative impacts. This has led many authors to strongly suggest a reduction in meat consumption, with some claiming it a necessity if goals for environmental health and future food security are to be reached (Carlsson-Kanyama and Gonzalez, 2009; Garnett, 2009; Goodland and Anhang, 2009; McMichael et al., 2007; Pelletier and Tyedmers, 2010; Raphaely and Marinova, 2014; Stehfast et al., 2009; Tilman and Clark, 2014; Weber and Mathews, 2007; Weidema et al., 2008). For instance, if crops that are currently fed to livestock are redirected to direct human consumption, the amount of food calories available to the human population would increase by an estimated 70%, which could feed an additional 4 billion people (Machovina et al., 2015). This could help tremendously with efforts to maintain or increase food production while simultaneously saving resources such as land and water in the process. Of course, whether increased food production would occur and whether these additional foods would make it to those most in need is another issue, but a dietary shift to primarily plant-based foods would open up possibilities for feeding more people with less resources. The question then becomes: If meat consumption is to be reduced in efforts to promote environmental sustainability and future food security, where should efforts be directed? In the following paragraphs, the assertion that developed nations should take primary responsibility to reduce meat consumption is explained.

Developed nations have consistently eaten the greatest amount of animal products, including meat (Bellarby et al., 2013), and are still projected to eat more than double the amount of meat per capita in 2050 when compared with developing nations (Garnett, 2009). Now, although total meat consumption in a country like China outpaces that in the U.S. by a factor of 2, the U.S. per capita meat consumption outweighs China by a factor of 2 (Larsen, 2012). That is, although China has more than four times as many people as the U.S., their total meat consumption is only two times as large, since the average Chinese citizen eats only half the meat of an average American. With citizens in developed countries like the U.S. eating at least double the meat of citizens in developing nations like China (i.e. overconsumption), efforts should be primarily focused on lowering per capita meat consumption levels in developed nations, where a greater

magnitude of change in personal meat consumption is possible and already being recommended in some nations (See Canada's 2019 food guide)[1]. In addition to high per capita meat consumption rates, most consumers in developed nations have the luxury of greater food choice, with less of a reliance on particular food products. Meanwhile, in parts of developing nations, livestock may be necessary not only for meeting nutritional requirements (Charles et al., 2010; Neumann et al., 2002; Walker et al., 2005) but also for supporting livelihoods, as one billion of the world's poor rely on animal rearing for their livelihood (FAO, 2019). For example, the rearing of chickens within rural African communities can have positive impacts on both local food security and economic opportunity (Kitalyi, 1998). Asking communities that rely on animal agriculture as their primary source of food or income to reduce meat intake is therefore an impractical and unjust request, especially when communities in more developed nations eat more meat (per capita) while having less reliance on the raising of animals for survival.

Livestock may be indispensable in areas of developed nations as well, but these are exceptions as a majority of consumers in developed countries have greater financial freedom and relative flexibility in food choices when compared to average consumers in most developing nations. In addition, the growing desire for more meat and biofuels can compete for the same crops eaten by people in developing nations (Rosegrant, 2008; Rosegrant, 2010), driving supplies down and prices up, sometimes to the point where food security is threatened in poorer areas due to affordability issues. Without the resources or necessary funds to be autonomous in regards to food choices, focusing on consumers in developing nations would not only be difficult, but unfair considering the highest rates of per capita meat consumption occur in developed nations. For instance, the food consumption patterns in developed nations exceed sustainability levels by a factor of 4 on average (Carlsson-Kanyama, 1998), so the room for improvement in these nations is quite high and if we are to promote global environmental justice, it is only fair to ask those who have contributed the most to environmental and food insecurity issues (i.e. developed nations) to take the lead in changing their environmentally harmful practices and lifestyles (e.g. high meat intake).

If the consumption of meat from livestock is to be reduced in efforts to aid the environment and those who depend upon its stability in order to meet their basic needs,

[1] The new Canadian food guide recommends choosing protein rich plant-based foods over those of animal origin. See here: https://food-guide.canada.ca/en/healthy-food-choices/

something must be substituted that can safely and efficiently replace the missing calories, proteins, fats and nutrients in consumer's diets, while minimizing environmental impacts and resource use. Some may think that increased reliance on seafood is a potential dietary alternative to land-based meats, but evidence points to the contrary. First, overfishing is a major issue, with more than half of assessed species being fished at the maximum rate within sustainable boundaries and just under a third being fished at unsustainable rates (i.e. above sustainable boundaries; FAO, 2016). In addition, non-target species such as turtles, birds, marine mammals, and sharks are also threatened by fishing practices due to accidental capture, also known as by-catch, which can have significant impacts on their populations (Dulvy et al., 2008; Lewison et al., 2012; Read et al., 2006; Wallace et al., 2010). These issues could not only put pressure on the populations of affected species, but also progress into more widespread negative impacts if these species play pivotal roles in maintaining the established food webs and/or functionality within their respective ecosystems (McCauley et al., 2015). Some fishing practices, such as trawling (drag netting the seafloor with fishing vessels), have been linked to habitat modifications through the loss or reduction of both structural and functional biodiversity within marine ecosystems (Thrush and Dayton, 2002). Additional environmental impacts of fishing practices include GHG emissions from fuel combustion of fishing vessels, release of antifouling substances into the marine environment (from protective paint used to prevent growth of subaquatic organisms on hull of vessel), waste discharge, and release of cleaning agents and refrigerants (Avadí and Fréon, 2013). Although some of these issues may be avoided through the farming of fish species, this practice brings other challenges such as waste management, contaminant pollution, ecological impacts of fish feed, and possible interactions that could negatively affect wild fish populations (Cole et al., 2009; Holmer, 2009). Yearly per capita fish consumption has risen from 9.9kg to 19.7kg over the past 50 years, and this ever increasing demand has led to more industrialized aquaculture practices (WWF, 2017). Intensive farming of fish requires crop-based feeds, and in 2010 these crops amounted to 26.4 million hectares, or a land area about the size of the U.K. (WWF, 2017). Due to fishing's varied and significant impacts on species populations as well as marine ecosystems, it is currently not a viable alternative to land-based meats in terms of sustainability, and taken a step further, the consumption of fish and seafood should also be reduced alongside land-based meats in order to promote improved environmental health in oceanic ecosystems as well.

Plant-based alternatives have been suggested as viable replacements for meat, due to their lower energy input and reduced environmental burden. Replacing animal proteins with vegetable proteins have been shown to reduce environmental impacts (Carlsson, 1998; Finnigan et al. 2010; Håkansson et al., 2005; Head et al. 2011; Nonhebel and Raats 2007; Soret et al., 2014), with meat-based meals having 1.5 to 2 times the environmental load when compared to equivalent meatless meals (Baroni et al., 2007; Reijnders and Soret, 2003). For instance, when comparing carbon footprints on a per kilogram of product basis, ruminant meats have 19 to 48 times and non-ruminant meats 3 to 10 times the carbon footprint of high-protein plant based foods (Ripple et al., 2014b).

Gonzalez and colleagues (2011) have shown that plant based foods can provide protein more efficiently not only in terms of energy input, but also in terms of GHG emissions per unit protein (see Figure 2.4).

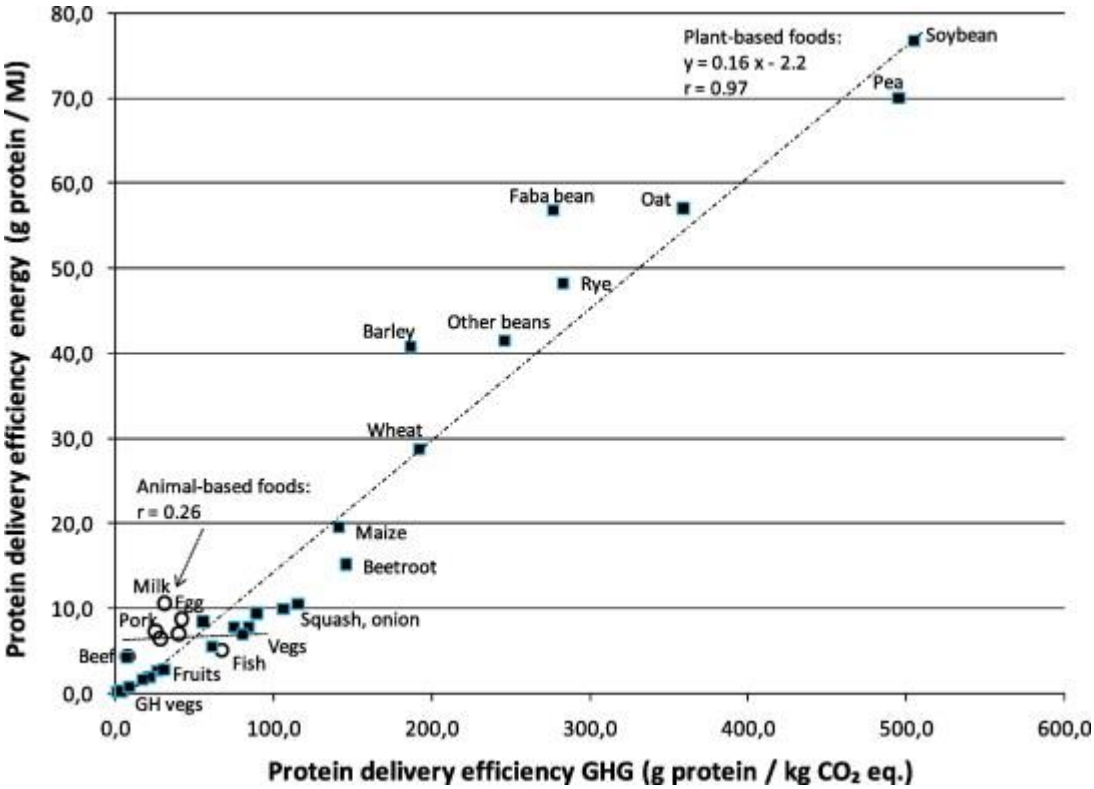

Figure 2.4 Correlation between protein delivery efficiency per unit energy and protein delivery efficiency per unit GHG emission. Circles are animal based foods while squares are plant-based. Reprinted [adapted] from "Protein efficiency per unit energy and per unit greenhouse gas emissions: Potential contribution of diet choices to climate change mitigation", by A. Gonzalez, B. Frostell, and A. Carlsson-Kanyama, 2011, Food Policy, 36, 567. Copyright 2011 by "Elsevier". Reprinted [or adapted] with permission.

This means that not only do plant-based foods provide far more protein for the same amount of energy input, but also contribute less to global climate change when compared to their animal-based counterparts. In fact, Harwatt et al. (2017) used emission figures based on a mixture of life cycle assessments (LCA) (those performed both in the U.S. and those representing global averages) in order to determine the potential climate change impacts of substituting beans for beef in the U.S. diet. They found that projected GHG emission reductions could contribute 46 to 74% (depending on whether U.S. or global LCA were used) towards the 2020 U.S. climate change target (i.e. 17% reduction in net emissions compared to 2005 levels), with minimal difference (1%) when beef was substituted at an equivalent caloric or protein ratio with beans. This is supported even further by Weber and Mathews (2008), who when looking at yearly GHG emissions from U.S. consumers' food habits, determined that a person who replaces red meat and dairy with vegetables for one day a week would have a reduction in emissions equivalent to a 1,160 mile car ride. A recent LCA analysis conducted by University of Michigan's Centre for Sustainable Systems found that the quarter pound "Beyond Burger", a 'meaty' plant-based burger created by the company *Beyond Foods*, produced 90% less GHG emissions, required 46% less non-renewable energy, and was 99% and 93% less impactful on water scarcity and land use than an equivalent beef-based quarter pound burger (Heller and Keoleian, 2018). These findings support the replacement of meat products with plant-based alternatives, showing that they are more efficient both in terms of energy input and GHG emissions when substituted calorie for calorie or protein for protein.

The same pattern holds true for land use as well, according to Audsley et al. (2010) who estimated that a 50% reduction in meat consumption among UK consumers would release 0.6 Mha of arable land (accounting for the offset of increased land use for direct feed crops for humans), and an additional 5 to 10 Mha of permanent grassland that would become available for re-wilding or other uses (Mha = mega hectare (ha x 10^6)). When investigated on a global scale, if consumers in developed countries reduced their protein intake (currently above recommended levels) by one third while simultaneously substituting meat with plant-derived proteins, 87% to 94% of agricultural land used for feed crops could be set free, totalling 400 million hectares (Aiking, 2011). In addition to land use, switching to more plant-based diets could alleviate both food-related water footprints and pollution tied to the raising of livestock. Non-vegetarian diets use almost three times as much water, thirteen times more fertilizer, and one and a half times the

pesticides in comparison to vegetarian diets (Marlow et al., 2009). Therefore, it can be concluded that a shift from the currently high rates of meat consumption in developed nations to more plant-based diets would mitigate multiple environmental impacts, while also freeing resources such as land and water so they can be utilised for other purposes.

Furthermore, medical and public health researchers have called for reduced meat consumption to mitigate the negative health effects it has created in the developed world. To laymen, increased meat consumption is probably seen as necessary and beneficial for health, due to the perception of animal proteins being "complete" since they contain all the essential amino acids, while vegetable proteins are seen as "incomplete" as they are missing one or more. However, it has been shown that all essential amino acids can be acquired from eating a mixture of plant-based proteins (Harvard, 2016), and the American Dietetic Association has supported the claim that vegetarian diets can be appropriate and healthy for people in all stages of life (Melina et al., 2016). Aiking et al. (2006) assert that global meat consumption could easily be cut by one third, without risking the health or nutrition requirements for any individuals, and a recent *Lancet* report advises a 50% reduction in unhealthy foods like red meat and a 100% increase in healthy foods like fruits, vegetables, nuts, and legumes in order to promote environmental sustainability and human health (Willett et al., 2019). Although the consumption of meat has been and can be a valuable source of protein, iron, B12, and other B vitamins in the human diet (Pereira and Vicente, 2013), it seems that meat consumption in developed countries has risen to rates where its association with health risks has become more prominent than its ties to health benefits. Meat consumption, especially that of red and processed meats, has been linked to higher rates of obesity, heart disease, specific cancers, diabetes and other non-communicable diseases (Aston et al., 2012; Aune et al., 2009; Bouvard et al., 2015; Campbell and Campbell, 2006; Ekmekcioglu et al., 2017; Friel et al., 2009; Huang et al., 2012; Kelemen et al., 2005; Larsson and Wolk, 2006; Pan et al., 2012; Sinha et al., 2009; Snowdon and Roland, 1985; Wang and May, 2009). Recent large meta-analyses have related increased consumption of processed meats (50g increase per day) to higher rates of coronary heart disease (42% increase), type 2 diabetes (19% increase) and colorectal cancer (18% increase) (Ashton et al., 2012). The increased prevalence of these diseases as a result of increased meat consumption are not only a public health concern, but also an economic burden as health care costs associated with meat consumption have been estimated at $28.6 to $61.4 billion dollars in the U.S. alone (Barnard et al., 1995). Diets

that are more plant-based offer an alternative, being linked to multiple health benefits including the reduction of risk factors and prevention of diseases, including cardiovascular diseases, obesity, and prostate cancer (Esselstyn et al., 2014; Ferdowsian et al., 2010; Hu, 2003; Knekt et al., 1996; Ornish et al., 1998; Ornish et al., 2005; Tantamango-Bartley et al., 2013; Wright et al., 2017; Zong et al., 2018).

Although there are many health issues linked to excessive meat consumption, the public health impacts related to its production also deserve attention. Over the past decades, a growing number of small farms have been replaced with large industrial scale 'factory farms' in order to meet growing global demand for meat while keeping production costs low (Pluhar, 2010). The issue with factory farms is that massive numbers of animals are often kept in confined conditions, which results in an increased risk of acquiring and spreading diseases among the animals as well as high concentrations of waste being produced. Pathogens that are present within wastes can pollute drinking water when storage pits leak into groundwater, potentially infecting humans resulting in the formation of diseases that are sometimes resistant to antibiotics (as a result of using antibiotics in animal feed), which could lead to serious illness and even death for those who are immune compromised, like infants (Acar et al., 2009). In addition, air pollution in the form of ammonia, sulphur dioxide and dust that is released from these factory farms can be harmful for both workers and residents of the surrounding community (Walker, 2005). Many slaughterhouse workers incur psychological distress as a result of their occupation (Dillard, 2008), about a third suffer from occupational respiratory diseases (Horrigan et al., 2002), and residents that live near these operations report higher incidence of respiratory problems, nausea, and headaches among other ailments (Thu et al., 1997; Wing and Wolf, 2000). A more recent study by Sneeringer (2009) looked at infant mortality in the U.S. based on spatial variations in livestock productions over a span of two decades, and found that every 100,000 animal unit increase within counties (i.e. province within a state) corresponded to 123 more deaths of infants under a year old. Using regression analyses, they found that a doubling (100% increase) in the number of animal units within a county resulted in a 7.4% increase in the infant mortality rate, even when controlling for a variety of potential covariates. It is clear from these studies that the public health impacts of meat production are not to be taken lightly, as serious health issues can occur among workers and surrounding communities, with some of the most vulnerable individuals (i.e. infants) even facing higher probabilities of death. Therefore,

reducing meat consumption in order to promote greater environmental sustainability and food security is not only the responsibility of developed nations, but is also highly recommended from a self-preservation viewpoint, given the associated environmental and public health impacts that result within regions where concentrated livestock farming occurs.

2.5 Chapter summary

It becomes apparent, when faced with widespread and significant environmental impacts, that humans may need to rethink and reshape how society functions in order to prevent environmental disasters, promote current and future food security, and live longer, healthier lives. Our current food production practices, and more specifically our desire to produce and consume meat, has contributed heavily to climate change, land degradation, deforestation, water footprints, pollution, habitat and biodiversity loss, future food insecurity, prevalent non-communicable diseases, and other health complications. When taken together, these impacts are too overwhelming to ignore, as our appetite for meat has not only threatened our own health, but also the health and wellbeing of others with whom we share the planet, potentially jeopardizing the ability for future generations to not just thrive, but simply survive. The adoption of plant-based diets has been suggested as an alternative that could alleviate environmental impacts, help protect future food security, and promote public health. Based upon the discussions in this chapter, it is obvious that more plant-based diets should be adopted in order to reduce rates of meat intake; however, one may question how such a societal transition might occur. This question shall be explored and discussed further in Chapter 3.

CHAPTER 3

MEAT REDUCTION: SHIFTING SOCIETY

3.1 Chapter outline

The imperative for promoting meat reductions throughout society have been established, but with meat consumption being so ubiquitous, where does one begin in addressing the problem? This chapter intends to act as a starting point, building upon the ideas and paradigms about what comprises "society", whether established structures or individuals are more influential on its form and function, and what this may mean for changing consumptive behaviours, like meat intake. Discussions will begin with a short introduction (3.2), a brief history of the structure-agency debate, along with the resulting dialectic approach and how it can be used as a theoretical basis for studying society and potential transformation (3.3). Next, the ways in which behaviour change scientists have dealt with the structure-agency dilemma and how behaviour change has been traditionally addressed by policymakers is explored (3.4); followed by a brief history of consumption, its transformation over time, and how various economic paradigms have shaped responses to associated impacts (3.5). Finally, meat's place and significance in society will be outlined in order to gain a clearer picture of how meat intake has become a prevalent and embedded part of the western diet (3.6), concluding with a chapter summary (3.7). By the end of Chapter 3, the reader will be prepared to begin Chapter 4, which will explore the variety of specific options available for promoting societal meat reduction.

3.2 Introduction

Important questions regarding how societies form, how they transform, and what they are comprised of have been asked throughout human history and are still discussed to this day. One intellectual dilemma, described as the "structure-agency" debate, came as a result of theoretical discourse regarding whether individuals within a society, or the institutions/ideas that have been established by groups of citizens (e.g. corporations) or the collective (e.g. government, culture), are the most significant drivers of social norms. Theorists, such as Emile Durkheim and Max Weber, have stressed the importance of either structure or agency in terms of influencing societal composition and function, and

thus, these seemingly contradictory positions will be outlined further along with what this may mean for potential meat reduction in section 3.3 below. Resulting from this debate, and of considerable interest to the author, is the dialectic approach which strives to take both structure and agency into consideration, providing a framework that bridges the gap between both individual agency and societal structure, by recognizing their importance and interplay in a unified manner. As pointed out by behaviour change theorists (Guagnano et al., 1995; Williams, 2003), the dialectic approach acts as a useful theoretical base from which to explore how agency and structure may independently and synergistically promote and/or discourage certain behaviours, and Beverland (2014) argues this holds true for the behaviour of meat consumption as well.

Establishment of a theoretical base for social norms like meat intake is a logical starting point, but if behaviour change is ultimately sought (i.e. lower meat consumption), it would be useful to understand how agency and structure are interpreted and addressed by those who study behaviour change. Not only can this place agency and structure into a more useful context for shifting behaviour, but the knowledge and findings of past authors can also help identify how behaviour change has traditionally been addressed in society and whether agency and structure have both been given proper attention by policymakers.

If the ultimate goal is to reduce societal meat consumption, it would be useful to understand how general consumption has shifted over time in human societies and whether current levels of meat intake are following trends seen in other forms of consumption. In addition, societal norms that specifically apply to meat must also be understood, including meat's status in society and the drivers and/or barriers behind people's decision to eat meat (both societal and personal), if meat intake is to be properly addressed. For example, besides fulfilling nutritional needs, meat can act as a symbol with special significance in many social and cultural contexts. Also, it has become a standard food item in recent years as a result of increased wealth and productivity, especially in many developed nations. These factors make the goal of reducing meat consumption much more challenging than what it would first appear, as some drivers for the behaviour and barriers to change are embedded in culture, resulting in high meat intake being a default for both society and the individual.

3.3 Structure-agency debate and the dialectic approach

Since the 1970s, two main paradigms have emerged as dominant approaches to addressing issues surrounding consumption, with one focusing primarily on structural change and the other on individual agency (Spaargaren, 2011). Structure refers to the systems created by individuals that determine the possibilities and limitations of their actions within society, and can be both material (e.g. governments, organizations, and businesses) or cultural (norms, customs, traditions). These systems can hinder or promote certain choices, through limiting or providing options to individual agents (Barker, 2003). Durkheim (2014), one of the main figures of the structural paradigm, proposed that there are ever-persistent and external 'social facts' like laws, social norms, and traditions that coerce and act upon individuals. Thus, society and how it functions is not simply comprised of an amalgamation of individual thoughts and behaviours, but is rather more influenced by a combination of 'social facts' that remain as fixed and objective entities, due to their independence from the personal opinions, feelings, and wants of individuals. Also, since these 'social facts' are often a driver and/or result of social solidarity, deviations from societal structure are often repressed or met with resistance in order to maintain said structure (Radcliffe-Brown, 1940). Thus, structure has a tendency to stagnate changes within society through repressing deviant thoughts and actions and resisting systematic transformations.

Agency, in contrast, is described as the capacity with which individuals are able to use free will to make independent choices. The agency paradigm is founded upon the idea that the combination of these choices, based upon each actor's personal opinions and behaviours, determines the structure and function of society. Weber (1978) believed it was imperative to not underestimate the importance of the individual in society, and that 'social actions' would lose their meaning and purpose without individual interpretation and directed behaviour. He further suggests that individuals, rather than being passive entities who are simply acted upon by social forces, are motivated actors that are dynamic in the sense that they can adapt and change their opinions and actions depending upon social context. Barth (1967) takes this idea further by arguing that social norms and societal structures are simply patterns amongst populations, which results from the interaction of individuals who are using available strategies to compete amongst and survive with one another. Thus, the agency paradigm asserts the independence of

individuals, while stressing how their self-directed and self-motivated thoughts and actions actually help to create, sustain, or dismantle social contexts within society.

Each paradigm includes its own theoretical assumptions and favoured strategies to promote social change, with differing opinions on how society, and the way people function within it, is constructed, maintained, and/or transformed. Those arguing from a structural paradigm would claim that for effective societal change to occur, efforts would best be focused on adjusting larger systems in our society rather than trying to influence and change individual actions, because individual action is mostly pre-determined by structure. Thus, if social change is desired, strategies that address issues at a more systematic level (e.g. new international trade policies, country-wide regulations on specific products, etc.) would be advised, based upon this paradigm. The agency paradigm, on the other hand, stresses the importance of observing individuals within society, as their micro-level choices, once conglomerated, determine and give meaning to cultural, social, and economic norms. Through this paradigm, social change is expected to result from influencing individual opinions and actions, as it is the network of people's thoughts and behaviours that ultimately give meaning and structure to society.

Structure-dominant and agency-dominant viewpoints have been discussed in multiple contexts going back decades in the sociological and geographical literature, but more recent debates have focused on teasing apart the interrelationship between agency and structure using dialectic methods (Chouinard, 2008; Tan, 2011). The dialectic approach, as you can imagine, consists of beliefs that both structure and agency have an inextricable and reciprocal effect on one another, with structure influencing agency and agency influencing structure in a simultaneous fashion. The major theories that support each 'school of thought' are outlined below, along with theorists who acted as important contributors to the present understanding of the agency-structure debate (Table 3.1). Structuration (Giddens, 1984) and Practice Theory (Bourdieu and Wacquant, 1992) are two of the main ideologies that have resulted from sociologists taking a dialectic approach to the Structure-agency debate.

Table 3.1 Outline of foundational theorists and theories relevant to the structure-agency debate.

Paradigm	Contributors	Theory
Structure Dominant	Emile Durkheim	Social facts
	Radcliffe-Brown	Structural functionalism
Agency Dominant	Max Weber	Social-theoretical framework
	Frederick Barth	Ethnographic framework
Dialectic approach	Anthony Giddens	Structuration theory
	Pierre Bourdieu	Practice theory

Being dialectic approaches, the theories of Structuration and Practice simultaneously consider the roles that agency and structure take in determining how social systems are created and reproduced. Since their formulation, the theories have been used as tools to understand the intersection of socialization with autonomy in determining how individual actions influence social structure and vice versa. Both theories mend the rift in ideologies between agency and structure by claiming that they act in an intertwined and cyclical fashion, influencing and shaping one another in unison (Giddens, 1984; Bourdieu and Wacquant, 1992). Although slight differences between the theories and their interpretations exist (e.g. to what degree are individuals reflexive and conscientious and therefore, able to transform social norms?; Tan, 2011), these slight differences will not be explored further, as an outline of the structure-agency debate and the resulting dialectic theories is sufficient as a foundation for further discussion on how more specific social changes (e.g. reduced societal meat intake for the sake of sustainability and/or health) might occur.

Referencing consumer culture theory and practice theory, Beverland (2014) proposes that the tensions between the enablers and barriers to changing meat consumption results from interactions between agency and structure. Therefore, understanding the roles of both agency and structure in influencing consumers' choices to eat meat is an important consideration for any discussion aimed at discovering viable options for reducing meat consumption. As explained above by the dialectic paradigm, social structures can influence individual actions, but these same structures can also be changed when agents

decide to replace, ignore or reproduce the actions that form them (Guantlett, 2008). An appropriate example for the case of meat consumption could be in cultural traditions that include the consumption of meat as a central element. For instance, serving specific types of meat for holiday meals, like turkey or ham for Thanksgiving dinners, is a yearly tradition in the United States; however, if a growing number of individuals decided to have a meatless holiday (to aid the environment, promote personal health, etc.) as a result of their own agency, the tradition of serving meat at every holiday may weaken to the point where it is no longer a tradition. If enough people utilised their agency to implement meatless meals during the holidays, despite tradition dictating the exact opposite, the tradition may transform from one centred around meat to one where it is no longer a vital component, completely transforming the 'structure' of the holiday meal. This structural transformation could then, in turn, influence and encourage the agency of many other individuals to also adopt meatless meals during the holidays, creating a cyclical feedback loop where agency to avoid meat and a more 'meatless' holiday meal structure constantly influence and buttress one another.

Recognizing this potential dynamic between agency and structure for traditions, like those based around consuming meat, is a useful initial step, but how do we determine the degree with which individual agency can 'override' (in a sense) meal 'structure' that would otherwise promote meat consumption as a default (and vice versa)? To clarify, when meat is avoided or chosen as a meal, is it more as a result of the consumer's personal views and attitudes (agency) or availability and cultural norms (structure) that influence the final decision? Placing the ideas of agency and structure in a more concrete example, like a supermarket scenario, might allow for better understanding and exploration of their individual and combined potential influences on the decision to (not) purchase and consume meat. When a consumer goes into a store and picks out a package of meat to purchase, one may presume that the person is autonomously choosing the product with no outside influence, making the choice entirely based upon agency. However, certain structural factors may influence this decision, like a certain cut of meat being on sale that week, a store display showing a recipe that includes a specific cut of meat as a base ingredient, or a lack of plant-based alternative products that the consumer would potentially consider purchasing in place of an equivalent meat product. The list of potential structural influences can go on almost indefinitely, with other factors unrelated to the supermarket such as cultural norms, family food preferences and acquired cooking

skills also acting as potential influences in the decision-making process. The same can be said for the influence of agency in the decision to (not) buy meat, as it will likely depend upon the consumer's attitudes towards meat, attachment to the ideas of needing meat for a proper meal, personal knowledge about the potential costs and benefits on one's health when consuming meat, etc. Again, the list of potential influences based upon agency or structure can go on almost ad infinitum; the main point, however, is that when making a choice to (not) purchase and consume meat, an individual can be influenced by both personal factors and the structural systems and contexts within which choices are made. This hypothetical supermarket scenario offers another example of the cyclical feedback loop between agency and structure mentioned above. The consumer's choice to (not) purchase a particular meat product influences the supermarket to supply more/less of that product in efforts to properly meet demand, while in turn the supermarket's decision to (not) supply, advertise and run specials on that meat product encourages (or discourages) the consumer to start, stop or continue purchasing that particular product. Therefore, individuals create, support and influence the systems they work within (i.e. create demand for supermarkets) while the established structure reciprocates by providing and shaping the environments in which individual actions occur (i.e. supermarkets try to stock and sell more of what consumers want).

Therefore, in the case of consumption, and meat intake specifically, there is no clear answer for the 'order' or 'strength' with which structure or agency influence individual decisions to (not) purchase or consume a product. These questions become even more difficult to answer when considering the sheer variation and complexity of (and between) individual actors, in addition to the wide array of contexts that decisions are made. As noted by those who have studied consumption, the proposed strategies for social change resulting from the structure-dominant and agency-dominant paradigms have historically not aligned (Jackson, 2005; Spaargaren, 2003; Spaargaren, 2011; Wangel, 2011), and this is not a surprise given the complexity outlined above. However, exploring the literature on the history and effectiveness of past efforts to shift behaviour could offer more insight into the roles that agency and structure will likely play in the endeavour to reduce meat consumption, specifically.

3.4 Behaviour change: how theory and policy are influenced by the structure-agency debate

The primary goal, as stated in Chapter 2, is to reduce meat consumption levels in order to mitigate food-related negative environmental and public health impacts. With the understanding that addressing production alone will not be enough, addressing high consumption rates through the means of social change will be a necessary component if these goals are to be realised. With a basic understanding of the possible interactions of agency and structure and how they both create and change societal behaviour, it would be useful to discuss behaviour change models that emphasize the inclusion of both factors. As stated by Guagnano et al. (1995), the social sciences have been split into two "think groups" similar to that described above in the agency vs. structure debate. In past studies on behaviour, some researchers focused mainly on internal processes occurring within individuals while others observed mostly external factors, with few integrating the two to better explain behaviour and how to change it. This led Guagnano and colleagues (1995) to create a model for behaviour change that incorporated both internal and external factors, called the ABC Model of Behaviour. This model incorporates internal individual attitudes (A) and external contextual factors (C) in order to explain behaviour (B). Attitudes (i.e. learned tendencies) can range from extremely positive to extremely negative positions on particular actions (e.g. reducing meat consumption is a great idea vs reducing meat consumption is a terrible idea), while conditions (i.e. context) can encourage or discourage specific actions by providing social, legal, physical or financial support or opposition (e.g. reduced meat consumption is affordable and convenient, or expensive and challenging). These external conditions can work in several ways and to various degrees, with some creating strong or weak motivations and others creating strong or weak barriers towards a proposed action. At various scales, these attitudes, behaviours, and contexts can change greatly over time (e.g. current century, decade, time of day) and space (e.g. country, community, immediate environment, etc.), while also contributing to the current time-space structures that individuals behave within (Dyck and Kearns, 2006). Put simply, not only do agents and structures merely exist and act within space and time, but they actually shape and create space-time as well. Thus, according to the authors behind the ABC model, the most important aspect to consider for behaviour change is the relative *relationship* of attitudes and context within specified space-time frameworks.

Obviously, actions that are supported by positive attitudes and context are more likely to be prominent in the population, while those facing negative attitudes and unsupportive contexts are less likely to occur. The challenge is to tease out the relationships between attitudes and context, and to what degree they support or impede a given action. For example, when thinking of reducing meat consumption, an individual can have positive attitudes towards the idea, but still not incorporate those attitudes into actualized behaviour change due to unconducive conditions. To add to this complexity, an individual can have multiple attitudes and contexts that are both positive and negative towards a contemplated action like reducing meat consumption, dependent upon the specific action under question and the environment in which this action is to take place. Although this complexity can become disorientating, the ABC model simplifies the confusion by asserting that behaviour change relies upon the magnitude of the absolute value of the sum of both attitudes and context. This means that for any intervention aimed at curtailing or encouraging certain behavioural changes, the results will depend on the direction and strength of both attitudes and context, as well as their related interaction (Figure 3.1).

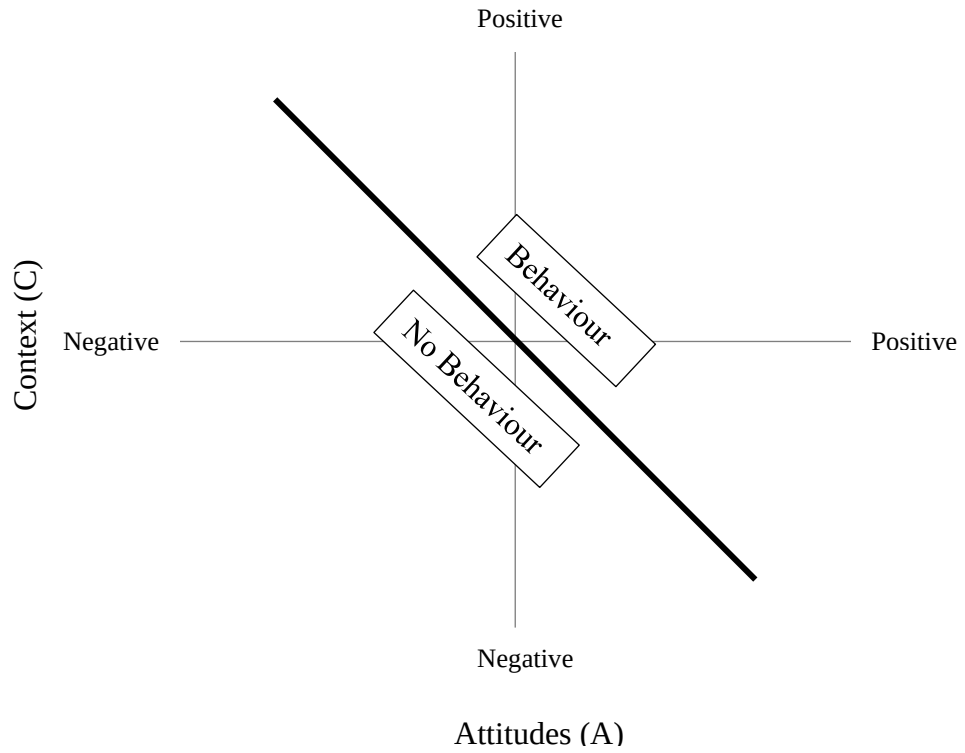

Figure 3.1 A-B-C Model of Behaviour. Adapted [reprinted] from "Influences on attitude-behavior relationships a natural experiment with curbside recycling", by Guagnano et al., 1995, *Environment and Behaviour, 27,* 699 – 718.

For instance, if attitude towards a behaviour is relatively positive and external conditions are neutral or positive, there is a strong chance for the individual to partake in the behaviour. However, even if an individual has a positive attitude towards a specific action, uptake of a behaviour may not occur if external conditions are too unconducive (i.e. negative). This is true for the reverse instance as well, as positive external conditions promote the uptake of a behaviour when attitude is neutral or positive, but potentially fails to encourage the behaviour if the attitude towards that behaviour is strongly negative. The main point of the model is to not only realize the importance of both personal attitudes and external conditions, but to understand their relative influence over individuals when choosing to utilize or avoid certain behaviours. For example, individuals may have a range of positive and negative attitudes and conditions when deciding whether or not to consume meat. One may think that eating meat is unhealthy but that it is also delicious; two attitudes that have an opposite influence on the behaviour. Also, one may believe that eating meat is cheap and convenient in certain circumstances (e.g. fast food burgers) while being expensive in others (e.g. premium price on fresh beef in supermarkets); two external conditions that would have contradictory effects. The myriad of positive and negative attitudes and conditions, along with their independent and synergistic effects in influencing the individual is what determines whether a behaviour is adopted or not. Therefore, the best route to minimize meat consumption and increase plant-based food uptake is to decrease the current positive attitudes and conditions surrounding the consumption of meat while simultaneously increasing the positive attitudes and conditions for adopting a more plant-based diet.

It is imperative to look at how this information has been used in recent sustainable consumption policy decisions. Recent policy reports on shifting towards sustainable consumption have focused on creating social change through emphasizing personal responsibility and individual behaviour change over potential structural transformations (Shove, 2010). These policies have been based upon the theories of the social psychologist Paul Stern (2000), who proposed that environmentally significant behaviour (B) is based upon the interaction of personal attitudes (A) and contextual factors (C), mirroring the ABC model of Guagnano and colleagues (1995). Researchers argue that ABC theory is under-utilised by many policy makers because focus tends to reside on changing attitudes and behaviours (agency) while simply citing contextual factors (structure) as barriers. Some academics claim that common rhetoric focuses on the power

of agency and individual choice, while ironically realizing that consumers can become 'locked-in' to unsustainable behaviours through structural norms and incentives (Shove, 2010). In addition to the ABC model, alternative models that strive to include factors of both agency and structure have been developed as well. For example, after studying energy-related behaviours Stephenson et al. (2010) created the 'energy culture' framework, which includes four main components: material culture (e.g. technologies, home design), energy practices (e.g. electricity use), cognitive norms (e.g. beliefs, understanding), and wider context (e.g. household demographics, building regulations, social marketing, etc.). Material culture, energy practices, and cognitive norms interact and influence one another, and all three components are in one way or another impacted by wider contextual factors. These various interactions occur at all scales (i.e. individuals, businesses, society, etc.), constantly shaping both individual behaviour and behavioural environments. Thus, if one intends to promote effective and long-lasting energy-related behaviour change, or any behaviour change for that matter, understanding how material culture, practices, cognitive norms, and wider context interact to form 'behavioural systems' is an appropriate first step according to the 'culture' framework.

Although it is important to realize the interplay between structure and agency, as well as the mutual effects they have on one another, one can argue that the "overemphasis" on agency in policy decisions may be due to its relative feasibility. That is, changing individual consumer behaviour is often an easier endeavour to undertake when compared to large scale structural changes, which are sometimes in direct opposition to special interests and can therefore be hard to achieve (Beverland, 2014; Spencer, 2002). Of course, the same interests of businesses and organizations can also reside in individuals (e.g. producers want to keep producing meat and consumers want to keep consuming meat), but it is important to make a distinction as individual interests are usually less concentrated and therefore less powerful than when compared to large organizations, industries, and governments who have relatively concentrated money, power, and influence. In addition, implementation of large scale structural changes will likely take large amounts of time and be a hard fought process where somewhat 'revolutionary' transformations will take place in a step-wise fashion over lengthy periods of time. This is due to the fact that current systems and policies would need to be analysed, replacement systems and/or policies would have to be devised, and these devised solutions would have to combat resistance from opposing individuals, political parties, organizations,

businesses, etc. that have an interest in or desire for maintaining the status quo. Encouraging behaviour change through individual agency, however, makes use of established systems and structures, rather than trying to combat them directly, in order to influence consumers to voluntarily change their consumption patterns, impacting the supply and demand dynamic to push society in a more desirable direction. Focusing on agency, therefore, is likely a more attractive option as it requires less analysis of current systems/policies, lacks the need to devise replacement systems/policies, and often faces less opposition from both large organizations and individuals. Thus, agency-focused measures may be a useful way to promote quicker and easier behaviour change to wide audiences, while structure-focused strategies work to enact deeper systematic and societal changes over the long-term.

Behaviour change policies tend to focus more on agency than structure, as practicality and viability are prime considerations, even though accompanying changes in structure would be ideal based upon the ABC model of behaviour change. Unfortunately, proposed policy and systematic changes do not occur in a vacuum, and even though regulations to force structural changes would likely contribute greatly to shifting societal consumption patterns, the truth is that invested parties would most likely not support or even condone such 'drastic' changes. Thus, policymakers have mostly focused on trying to nudge and convince individual agents to voluntarily make modifications to their lifestyles, and although the addition of more structural changes would likely be the most effective and fastest route to societal change, the current political and societal climate in many developed nations makes proposing and implementing transformational policies incredibly challenging. Of course, advantages, drawbacks, and opportunities in targeting agency and/or structure will depend upon the behaviour under question and its accompanying context.

3.5 Understanding consumption: economic paradigms

The influences of structure and agency in forming and shaping society have been discussed, along with how behaviour change theorists and policymakers have wrestled with the two concepts in efforts to promote positive behaviour change across society and/or among individuals. These broad ideas, theories, and outlines are important to establish when contemplating societal transformation through behaviour change, but how

do they factor into changing a *consumptive* behaviour, like meat intake? Establishing a definition for consumption, along with a brief overview of its role throughout history and its current position and influence in modern society is a proper point of entry. Consumption, as defined for clarity, is the process in which humans consciously purchase and/or use products or resources, whether directly or as part of a process to create or obtain another desired product, good, or service. In early human history, consumption was mainly driven by the desire to survive, and so natural resources were obtained and sometimes transformed in order to provide food, shelter, tools, weapons, etc., which all helped individuals and/or communities survive in their environment. As humans began to conquer the natural environment, expand in population, and make technological advancements, consumption became increasingly driven by the desire to satisfy wants, in addition to the satisfaction of needs for survival.

This form of consumption has been referred to as hedonic consumption (Hirschman and Holbrook, 1982), and is loosely defined as any form of consumption that seeks to satisfy wants that lie outside of direct needs, like consuming large quantities of meat because it is pleasurable, not because it is required for nutrition or energy. Hedonic wants can be driven by multiple facets in regards to the consumed good, including the pleasure an individual receives from the sensory, emotional, intellectual, and/or symbolic information that the good coveys or provides (Hirschman and Holbrook, 1982), and it has been claimed that within developed societies today, hedonic consumption has actually overtaken and outpaced that of need-based consumption (Migone, 2004). It would seem that meat intake, at its current levels in many parts of the world, is one behaviour that has shifted from one traditionally based on need to one driven mainly by wants. Whether truly hedonic or not, current levels of meat intake along with many other forms of consumption are severely damaging natural environments and resources, and so it is advisable that patterns of consumption are addressed. If we intend to remedy or at least minimize these associated impacts, we must either completely re-think and/or redefine consumption and its central place within modern society (Firat, 1999) or adjust/substitute our current consumption patterns to be more sustainable, healthier, etc.

The issue of how to address modern consumption in order to alleviate environmental impacts has been thoroughly explored and discussed in the economic literature. Neoclassical (welfare) economics (i.e. mainstream economics) has been criticised for its contribution to many of our current systems which create and perpetuate environmental

degradation and societal inequalities (Goodland and Ledec, 1987; Spaargaren and Van Vliet, 2000). Environmental, ecological and green economics are all paradigms that arose in response to the flaws and critiques of Neoclassical economics, with slight differences in underpinning assertions and approaches to addressing issues resulting from current consumption practices[2].

Environmental economics developed as a result of market failures seen in the mainstream economy, specifically the overextraction and unequal distribution of environmental goods and services (Perman et al., 2003). The environmental economic paradigm is based upon neoclassical concepts and methodologies, and so often places environmental issues within economic contexts (i.e. environment is a component of the economy, not vice versa; La Notte et al., 2015). One core component of environmental economics is the assertion that externalities, which are positive and/or negative impacts on third party entities that result from an economic activity (e.g. pollution from a business that makes water undrinkable in cities downstream), are important and must be taken into account through price changes in the market (e.g. environmental tax). The inclusion of externalities are expected to communicate the true and full price of the good or service to the consumer, making products with high impact have higher prices and vice versa. The higher prices are expected to lower demand for environmentally damaging products, and funds from the taxes are to be used to alleviate environmental damages and issues created by economic activity. Thus, Environmental economics strives to place ecological concerns within the mainstream economy, in order to utilise current systems and markets that are already in place in order to shift production and/or consumption patterns to be more environmentally-friendly.

Ecological economics, in contrast to environmental economics, is based more in natural science than classic economics, and thus places the economy within the global natural environment (La Notte et al., 2015). This paradigm is founded upon the idea that mainstream economic processes transform valuable and limited natural resources into waste products, and that human-made capital is not equivalent to natural capital (Georgescu-Roegen, 1993). Put simply, when natural resources are used in economic activity, the worth of the resulting product is less than the original natural resource, as

[2] Differences between the economic fields and economists within them are not necessarily as distinct as presented and fall within a spectrum rather than rigid categories. They are outlined as such for the sake of simplicity and succinctness.

much (if not all) of its potential energy is consumed and mainly waste products remain. As a result of these foundational beliefs, ecological economists often take a long-term view on resource use and environmental stewardship, and are often very cautious in giving support to mainstream forms of resource extraction and use. Given that many in this field are ecologically trained, metrics are often non-monetary (e.g. ecological footprint, sustainability boundaries) and are therefore sometimes difficult to communicate to others not trained in the field and can be challenging to incorporate into mainstream economic paradigms and systems that place values almost completely on monetary gains or losses (Gowdy and Erickson, 2005).

Green economics offers the strongest critique of Neoclassical economics, proposing complete systematic reforms, where needed, to change how current economies are structured and how they function. The main aim of the paradigm is to promote an economy that works for the equal benefit of all people, protects the natural environment and all life within it, and reduces (or prevents) incidences of inequity and poverty that have occurred and continue to occur in the current economic system (Kennet and Heinemann, 2006). Local production and consumption, the reuse and recycling of goods, the deconstruction of current power relationships, and an emphasis on smaller and more local groups or businesses instead of large multi-national corporations to satisfy human needs and wants are just a few suggestions resulting from the philosophical foundations of Green economics.

The goal of outlining these paradigms is not to give a full synopsis of mainstream economics and proposed alternatives, but rather to help put into context the varying theories and positions put forth and how they differ in terms of how to respond to unsustainable consumption patterns (i.e. adjust these patterns within the current system or re-invent the system entirely). On one end, Environmental economics strives to work within the mainstream economic system and utilise its current structures and systems to promote environmental stewardship, while at the other end, Green Economics argues for a complete overhaul of the mainstream economy, encouraging systematic changes that would not only alleviate environmental impacts, but also serve humanity by more fair and equitable means. It is not certain whether a complete overhaul of our current economic system is required or if enough adjustments to production, consumption, and markets can be created and implemented in order to avoid environmental and ecological catastrophes. In addition to this uncertainty, it is unknowable if and/or when complete economic and

societal transformations would or could occur, especially given the tendency for established structures and systems to perpetuate, rather than dismantle themselves. Thus, given the scope of the current work, the remainder of the thesis will focus mainly on discussions and potential solutions for addressing meat consumption through changes in structure and/or agency that could reasonably occur within the current economic and societal frameworks.

3.6 Meat consumption and meat's place in society

With a clear direction for how this thesis aims to proceed in terms of addressing consumption and its impacts, the next question becomes, "What is the current role and importance of consumption within society, and how does meat intake, specifically, compare?" In modern society there are a myriad of roles that consumption plays, from satisfying needs for survival (e.g. food, shelter, medicine), fulfilling hedonistic wants, and serving social and culture purposes (e.g. identity formation, social distinction, meaning creation) (Firat, 1999; Jackson, 2005). Material goods are consumed not only out of necessity for life but also as symbols related to identity, status, social cohesion and personal and cultural meaning. This can easily be understood when thinking about the purchase of an automobile, as a potential buyer not only considers functional factors such as gas mileage and durability, but also symbolic ones such as wealth and status that can be displayed through the purchase of a specific brand/style of vehicle. These functional and symbolic factors are present throughout the different forms that consumption takes in current society, with the purchase of food being no exception.

Food is more than just a means to an end, as the provision of sustenance plays only a part in decisions regarding food preferences (McDonagh and Prothero, 2005). What we choose to eat and how much we eat of it can be a powerful signifier of who we are and what we believe (Cronin et al., 2014; Fischler, 1988). This same tenet holds true in regards to meat consumption, as individuals along the spectrum from complete avoidance to enthusiastic engagement find identity and meaning in their choices to either avoid or embrace the consumption of meat (Fiddes, 1994; Fox and Ward, 2008; Jabs, Sobal, and Devine 2000). For example, meat consumption has commonly been associated with psychosocial values and beliefs like that of power (Gaard, 2002), masculinity (Kildal and Syse, 2016; Stevens, Kearney, and Maclaran 2013; Stuart, 2006; Tobler, Visschers, and

Siegrist 2011), wealth (Cronin, McCarthy, and Collins, 2014; Rozin et al., 2012b; Ruby, 2012; Scho¨ sler, Boer, and Boersema 2012), and even political affiliation (Dhont and Hodson, 2014; Pfeiler and Egloff, 2018). This signifies that consuming meat is (at least partially) an act of fulfilling psychosocial values and beliefs as mentioned above (described as *carnism* by Joy, 2011), in addition to its role in providing energy and nutrients to fuel the human body.

Furthermore, higher rates of meat consumption have been tied to economic growth and wealth creation, with a diet high in meat being referred to as a "diet of affluence" (Campbell and Campbell, 2004). This can be understood when looking at countries that have experienced a relatively recent spike in growth, like China, as their levels of meat consumption continue to rise with concurrent increases in wealth. This process is referred to as the substitution stage of the nutrition transition, where diets consisting mostly of grains, legumes, fruits and vegetables are replaced with ones containing more processed foods, products of animal origin, sugars, fats and alcohol as a result of development and industrialization (Vranken et al., 2014). Since many developed nations have been in or have completed the substitution stage for a relatively long period of time when compared to developing nations, it is likely the case that high levels of meat intake are perceived as more of a standard when compared to nations where the substitution stage is still relatively novel.

In addition to meat's status and psychosocial importance, it has also become deeply integrated into the West's ideas of proper meal structure, nutrition, and cooking to a degree where it has become expected, even though the prevalence of such high rates of meat eating is a relatively recent phenomenon (Cross et al., 2011). This can easily be realized when looking at a standard restaurant menu or when thinking of a good recipe to make for dinner, as most of the menu's entrée options will likely include meat and the recipe ideas that would first come to mind would likely have meat as a central component of the dish. Meat's place as a 'default' in the diet, in addition to the positive values attached to its consumption has had more time to become ingrained into Western society, permeating through psychosocial beliefs, making efforts to reduce meat consumption a complex and difficult task. Lusk and Norwood's (2009) findings support this assertion by showing that despite understanding the economic benefits of a plant-based diet, consumers still place a value on the status of meat that overrules concerns about cost. This demonstrates that meat consumption is indeed a structure-agency dialectic, as the

decision is not only a rational choice whereby individuals weigh costs and benefits such as financial (dis)incentives, but is also a partially ingrained habit that results from the association of meat with positive traits and attributes like status, wealth, and tradition.

Within the psychology literature, this dynamic of contemplative choice and ingrained habit has been referred to as ‘slow’ and ‘fast’ thinking, where slow thinking involves more deliberative attention and analysis in regards to a behaviour, while fast thinking is a quicker and more intuitive process whereby individuals reduce cognitive strain by performing behaviours with minimal contemplative thought (Kahneman and Egan, 2011). An example of slow thinking would be an individual weighing out costs and benefits of whether to consume meat or not based upon relevant information (e.g. health impacts, environmental burden, animal welfare, etc.), while fast thinking would in contrast involve choosing what to buy at a supermarket based upon already ingrained beliefs and/or past experiences and traditions (e.g. buy and eat a steak because that’s what I usually do on Fridays). Fast thinking allows individuals to go about their day-to-day lives without the constant need for potentially stressful and time consuming thoughts, while slow thinking occurs less frequently due to a tendency for people to avoid cognitive strain. Thus, the symbolic and functional roles of meat that tie into its current position as an almost ‘automatic’ habit for many people likely place the behaviour in the realm of fast thinking (at least when at the specific point of making a decision of whether to purchase or consume meat). Of course, each individual relies on fast or slow thinking to varying degrees, with some people living more by intuition and others by more contemplative thought, and so any efforts to change behaviours will likely depend upon the target audience (Kahneman and Egan, 2011). Structural changes that shift default options in society are therefore more likely to be effective for those who rely more on fast thinking when making decisions, while interventions seeking to build capacity and agency among individuals will likely be more effective for those who rely more on slow thinking. This psychological basis ties together structure and agency with attitudes and context in order to give a clearer picture of how the behaviour of high meat intake might be addressed in an effective manner, which will be important when outlining the specific options available for promoting societal meat reduction.

3.7 Chapter summary

Overall, there are many aspects to consider when seeking to change any societal behaviour, including meat consumption. Not only are there multiple structures and systems that encourage the continuation of high meat intake, but there are also individual agents who make choices within these given structures. Behaviour change scientists have incorporated such debates into their models, which stress the importance of considering both structure and agency when seeking to promote change in a societal behaviour like meat intake, as realised behaviour is a balancing act between the attitudes one holds and the relative contexts one acts within. Past behaviour change policies, however, have mainly focused on addressing individual agency while largely ignoring structural influences, but this is not surprising given the increased organization and time needed to enact such transformative and widespread endeavours, in addition to the resistance put forth by organizations and individuals invested in maintaining the status quo. With meat intake being a consumptive behaviour, it is important to outline the criticisms of consumption and its form and place in the current neoclassical economic paradigm. Although it is important to consider whether consumption can be adapted and changed for the better, or whether a complete overhaul of our global economic system is mandatory in order to properly address associated impacts, it is beyond the scope of the current work to speculate further. With this understanding, future discussions on how to reduce meat intake will focus on potential changes to structure and agency that are reasonable and possible within the current economic system. Being a consumptive behaviour, one must take care not to simplify meat intake to a matter of only nutrition and taste, as neglecting the social and cultural importance of meat in people's daily lives would ignore a potentially vital factor when seeking to shift the behaviour. Despite structural barriers, variation between individual agents, and meat's central role in the western diet, opportunity exists to promote more plant-based diets if proper care is taken to understand the multitude of factors that contribute to its consumption. Although determining an objective 'best course of action' for reducing meat intake seems unlikely given the diversity of structural influences and the variation between individual agents, options that address structural barriers and/or individual agency do exist. In Chapter 4, these options will be explored, along with evidence and support from past behaviour change endeavours that can provide relevant background and guidance for proposing and implementing potential meat reduction measures.

CHAPTER 4

PATHWAYS TO SOCIETAL MEAT REDUCTION

4.1 Chapter outline

In Chapter 4, upstream and downstream strategies that may address the current meat consumption dilemma will be introduced, reviewed and placed within a market governance mechanism (MGM) framework (4.2), which is broadly defined as formal or informal guidelines to promote behaviour change among individuals, businesses, organizations and/or governments in efforts to encourage sustainable development in markets (Blackmore, 2011). Within the meat reduction literature, suggested MGMs have mainly fallen into three categories: economic, regulatory, and informational. Therefore, suggested measures will be organised and reviewed by category, and their strengths and weaknesses in terms of potential implementation and effectiveness will be discussed (sections 4.3, 4.4, and 4.5). In addition, this chapter will explore how these measures may be implemented (both independently and synergistically), what obstacles impede their enactment, and what their respective trade-offs are when used with the intended goal of making a societal shift towards a reduced meat future. By the end of Chapter 4, I will provide a summary and argue for the importance and practicality of including information provision initiatives in the MGM toolkit for reducing meat consumption, and how downstream measures can be used as initial stepping stones to begin the process of shifting society away from current ‘meat heavy’ diets (4.6).

4.2 Introduction

There are a myriad of motivations for organisations to continue selling and individuals to continue consuming large quantities of meat products. This, along with meat intake increasing in popularity and pervasiveness across the globe, makes a complete societal transition difficult, but also necessary if we hope to mitigate associated environmental impacts and prevent a rise in their prevalence and severity in the coming decades (Schösler et al., 2012). Tackling such a pervasive issue like meat consumption will therefore be an extremely complex endeavour, given that the chain of consumption can be influenced by multiple entities (e.g. government, industry, consumers) on various levels

(e.g. regulation, innovation, consumer behaviour) and is currently perpetuated by cultural and social norms. This complexity is present in New Zealand as well, and is a unique case because it is a country of about 4.8 million people that produces enough food to feed 40 million, and thus about 90% of its production is exported to other markets, including dairy and meat which are some of the country's main exports (Siegel, 2016; Tucker, 2018). In addition, unlike the U.S., there are no government subsidies for farmers in New Zealand, as they were removed in 1984 and government financial support for agriculture went from 34% of revenues to 0% by 1995 (Le Heron and Roche, 1999).

This complexity makes changing the current meat consumption paradigm an enormous task, but it also allows for a wide array of potential solutions that can work from multiple angles to encourage changes in consumption patterns. For example, the "carrots and sticks model" proposed by Rothshild (1999) is structured to address societal behaviours that result in negative impacts (e.g. environmental damage, public health impacts, etc.) through three primary avenues: education, marketing, and law. Education and marketing are both similar tactics, in that they promote changes in an uncoercive manner (although marketing typically promotes specific options), while law threatens punishment for non-compliance. These three avenues have been incorporated into the more recent MGM framework, although the MGM framework utilizes economic, regulatory, and informational MGMs (Blackmore, 2011). Economic MGMs can combine both marketing and law and includes any measures that seek to make pro-social behaviours more financially appealing (either for consumers or for industry). Regulatory MGMs typically incorporate both law and marketing, as new regulations can force state-run organizations to adopt more pro-social behaviours while simultaneously using marketing to encourage consumers (or state employees) to choose more pro-social behaviours once available. Informational MGMs can include both education and marketing and seek to either promote pro-social behavioural choices (amongst those already available) through awareness raising or through marketing of new products. The carrots and sticks model and MGM framework are both viable frameworks when discussing the various measures for promoting societal meat reduction, but the author believes the MGM framework categorizes the proposed meat reduction measures in the literature (e.g. meat taxes, plant-based public procurement, awareness raising) quite well, and thus this model was chosen to explore these measures in more detail.

Within this chapter, the MGM framework is used in order to outline potential behaviour change initiatives, which can be categorised into two broad groups, commonly labelled in the literature as “upstream” or “downstream” initiatives (Verplanken and Woods, 2006). Upstream approaches focus on changing macro-level structures, usually through choice editing or price manipulations so that the context within which individual behaviours occur is more conducive for desirable choices (e.g. plant-based meat alternatives priced lower than their meat-based counterparts). Obviously, enacting these types of large-scale measures requires a great deal of control and power and are therefore normally utilised by governments, institutions and other large organizations that have the resources and authority necessary to steer consumers towards preferred products. Downstream approaches, on the other hand, are those that directly target individuals, usually through education, informational campaigns or capacity building programs in order to shift current attitudes and behaviours towards those that are deemed to be more beneficial for society.

With similar goals in mind, upstream and downstream approaches can be combined or transitioned between in order to promote an effective transformation of consumer behaviour. For example, downstream meat reduction initiatives like awareness raising and vegetarian cooking campaigns could be introduced and implemented first in order to gain wider support from individuals, which could eventually help promote and expand the possibilities for more upstream initiatives such as taxes that account for the environmental externalities of meat production, or the increased provision of meatless meals in public institutions. For the sake of organisation and clarity, a MGM framework will be utilised in order to place such initiatives into one of three categories: economic, regulatory, and informational.

4.3 Economic MGMs

Economic MGMs are market-based instruments that balance costs and benefits through negative (e.g. taxes) or positive (e.g. subsidies) financial (dis)incentives to promote progressive social and environmental outcomes (Blackmore, 2011). Basically, these MGMs are a form of full-cost accounting, a concept which stresses the importance of including indirect environmental, social, and economic impacts into the price of goods (i.e. the triple-bottom line; Schaltegger and Burritt, 2017). In other words, full-cost accounting and the triple-bottom line are designed in order to create markets where the

"right" or "true" cost of an item is reflected in its price, based proportionally on the variety and severity of negative impacts associated with its production, distribution, and consumption. These concepts align with the field of environmental economics (discussed previously in section 3.5), and therefore take an economic perspective in terms of valuing nature. Thus, it is important to note that these concepts still rely on neoclassical economic assumptions, like the market being the most efficient and effective way to allocate resources

Economic MGMs or "price-based measures" could be applied in multiple markets to encourage more sustainable and healthy diets through price manipulation, and when used in addressing meat consumption, could include implementing an environmental tax (or cap-and-trade scheme[3]) on meat products and/or applying subsidies to lower-impact alternatives such as fruits, vegetables, grains and meat analogues. The idea behind an 'environmental tax' is that foods with high environmental impacts (e.g. meat) would be taxed at a higher rate than foods with lower impacts (e.g. grains), so that externalised environmental impacts are directly reflected in the personal cost to the consumer at the point of purchase. Such a tax could act alone or be coupled with subsidies for low-impact foods to make sustainable products more affordable and competitive, while making resource intensive and environmentally damaging counterparts more expensive (Garnett, 2012). Thus, consumers are expected to be discouraged from buying unsustainable products due to relative price increases while simultaneously being encouraged to purchase more sustainable products due to increased affordability.

Taxes and subsidies on food or food ingredients (e.g. fat and sugar) to promote public health have been discussed by a large number of scientific papers (e.g. see review by Mytton et al., 2012), however such actions have only been recently discussed as a potential avenue for pushing more environmentally sustainable diets through the taxation of meat products (Chalmers et al., 2016; Edjabou and Smed, 2012; Sall and Gren, 2015; Wirsenius et al., 2011). Since the idea of a 'meat tax' is relatively new, and has therefore not been implemented in any nation in order to address associated environmental or public health impacts, researchers have only been able to speculate and model the effects of proposed meat taxes on demand and consumption rates (4.3.1). These estimates and

[3] Cap-and-trade is an extension of a tax, in a sense, where the impacts of a good (e.g. meat) are included in the price of production. Producers usually pay this price through the purchase of credits, which allow them to produce goods that have negative impacts, but the total credits available for purchase on the market for are limited or 'capped', so that total environmental impacts are restricted to a specific goal or target.

predictions will be discussed along with supplementary studies looking at the effects of previously enacted taxes on other food ingredients/products such as sugary beverages and foods containing high levels of saturated fats (4.3.2). Although these additional studies may not be directly related to meat consumption, an analysis of their implementation and effectiveness at promoting public health through changing consumption patterns is a worthwhile exploration given the current lack of real-world meat tax scenarios. Lastly, potential roadblocks to implementing proposed economic measures will be discussed (4.3.3).

4.3.1 Environmental tax on meat consumption

An environmental tax to discourage meat consumption could be applied in two ways, as input taxes which are based upon the direct emissions measured at production of the livestock product, or as output taxes that are applied at the point of consumption. Wirsenius and colleagues (2011) have argued in favour of implementing output taxes at the consumption level over input taxes (e.g. taxes proportional to methane and nitrous oxide emissions), because although emission taxes based upon measured data at site of production may be better at directly linking private and environmental cost, they are not as cost-effective to implement and monitor and may be undermined through the increased purchase of cheaper imported meats. Other researchers agree, choosing to use consumption taxes when modelling the potential effects of price change schemes on the consumption of different food products (Chalmers et al., 2016; Edjabou and Smed, 2012; Sall and Gren, 2015).

Wirsenius et al. (2011) estimated the effects of consumption taxes on the production and consumption of animal products in the European Union (EU) and found that the implementation of a tax scheme with differentiated GHG emission taxes for separate food items could result in a mitigation effect of approximately 32 million tons of carbon dioxide equivalent (CO_2e), or about 7% of the total GHG emissions from EU agriculture, even when considering substitution effects. These findings are further supported by a more recent study by Chalmers et al. (2016) in Scotland, who found that a carbon consumption tax has the potential to reduce household meat demand, thereby leading to a 10.5% reduction in total meat-related emissions. These findings show that the use of taxes on high impact animal products could potentially reduce consumption and therefore be an

effective strategy in mitigating GHG emissions related to this dietary choice. However, it is important to note that differentiation between animal products through individually tailored taxes is highly advised as the GHG emissions from each form of meat production varies widely, with projected reductions in ruminant meat alone accounting for 80% of the total emission reductions calculated by Wirsenius et al. (2011). In addition, substitution effects must be considered to accurately predict emissions savings as decreases in consumption for one meat category (e.g. beef) may lead to increases in another (e.g. pork and chicken).

Edjabou and Smed (2013) created a models that considered a GHG emission tax on 23 different foods and projected that a household's carbon footprint would be reduced by 2.3-8.8% or 10.4-19.4%, depending upon whether the added tax per CO_2e was low (0.15-1.73 Danish krone [DKK]) or high (3.53-6.90 DKK). Unsurprisingly, higher tax rates that result in higher prices would be more effective at limiting the consumption of high-impact foods, with low tax rates having the potential to create only minimal changes in consumption patterns. However, the application of a meat tax could jeopardize public health by creating unfavourable substitution effects (e.g. higher consumption of 'junk' foods), especially among lower income consumers who could be disproportionately impacted by such a regressive tax (Garnett, 2012; Jensen et al., 2013; Leicester et al., 2004; Wirsenius et al., 2011). To explain further, if a meat tax was implemented, consumers of low socio-economic status (in comparison to those of high socio-economic status) would have to spend a higher proportion of their disposable incomes in order to purchase their desired meat products. If these poorer consumers are compensated for the tax in efforts to maintain equality between socio-economic classes, they may substitute more expensive meat products with cheaper unhealthy options and consume more total kJ than if not compensated at all, ultimately threatening their health and wellbeing. However, if they are not compensated the meat tax may seem 'unjust' as richer consumers could continue to afford eating high impact meat products while options for poorer consumers would more than likely be perceived as restricted. Given that a meat tax would have to raise prices by a significant margin in order to deter consumption, this point must not be ignored. This is why great care and consideration must be taken when suggesting any tax on food items that consumers deem to be 'essential', as it has the potential to create negative public health repercussions, while also being viewed as an unfair and divisive measure that makes food choice socio-economically determined.

In a recent study, Sall and Gren (2015) estimated that demand for beef, pork, and chicken in Sweden could be reduced by 19%, 8% and 5%, if applied taxes created a paralleled total price increase of 33%, 11% and 9% respectively. The increase in prices would result from a weighted tax that corresponds to marginal damage costs from four environmental pollutants (GHG, nitrogen, ammonia, phosphorus) associated with livestock raising. Similar to the models created by Wirsenius et al. (2011) and Edjabou and Smed (2013), this study used the econometric approach of constructing an almost ideal demand system (AIDS) model (Deaton and Muellbauer, 1980), which estimates the income, demand and price elasticities for different meat products. The price elasticity of demand is a figure determined by a multitude of factors (e.g. household income, consumer preferences, availability of alternatives, etc.), and it predicts the percentage change of demand for a product with a 1% change in that product's price (Andreyeva et al., 2010). Put simply, products with higher elasticities (i.e. elastic) are predicted to have purchase rates that are tightly linked to corresponding price changes while products with low elasticities (i.e. inelastic) are predicted to have weaker links, making price changes not as effective at altering demand. This study supports previous findings that food products, including meats, are relatively inelastic (Andreyeva et al., 2010; Chouinard et al., 2007; Kuchler et al., 2005; Tiffin and Arnoult, 2010), but adds to these findings by showing that the elasticities between different meat products can vary. This is an important consideration because beef products carry a much higher GHG burden than other types of meat, meaning that reducing beef intake would provide the greatest environmental benefit; however, if beef is *hypothetically* more inelastic than another type of meat, equivalent price increases as a result of taxes will be less effective at changing consumption rates for beef products in comparison to alternative meat products. Therefore, in this hypothetical scenario, beef prices would have to be taxed at a higher rate than alternative meat products in order to reach an equivalent decrease in consumption. So when thinking about the design and implementation of a meat tax to influence consumption, price elasticities of the different meat products should be used as guidelines to ensure that individualised tax rates effectively lower intakes of meat products with the greatest environmental impacts.

These studies about how meat consumption can be influenced by taxes are performed in a somewhat speculative manner, given that these predictions are based upon models which can only be as accurate as current data will allow. When taxes are actually implemented

there may be overlooked factors, misunderstood interactions or unintended consequences, especially when applied to a necessity such as food where public health and/or social equity could be jeopardized. These inherent complexities could impact the effectiveness and lasting appeal of a meat tax once fully implemented; therefore, it would seem helpful to reference past food taxes that have been employed in countries for reasons of public health to determine their effectiveness in shaping consumption patterns.

4.3.2 Implementation and effectiveness of food taxes in promoting public health

One study conducted in Ireland found that a soft drink tax implemented in the 1980s reduced consumption by 11% for every 10% increase in price (Bahl et al., 2003), however it is important to note that soft drinks were determined to be slightly more elastic than meat in another study (Andreyeva et al., 2010), meaning that a similar tax on meat would most likely result in smaller reductions. This is not surprising, as soft drinks are likely viewed by the majority of consumers as an "optional" choice, while meat is seen as a "need", and so consumers are more likely to cut back on "optional" soda consumption while "necessary" meat may not see reductions at all. More recently in 2011, Denmark, France and Hungary introduced taxes on 'unhealthy' foods and beverages that are believed to be linked to high rates of obesity (Villanueva et al., 2011). A study analysing the effects of one such tax, specifically the 'fat tax' implemented in Denmark, found that products taxed at the highest rate (30% price increase on butters, margarines, blends and oils) resulted in a 10-15% drop in intake levels when compared to those before the tax (Jensen and Smed, 2013). The authors also cite that the results obtained from the analysis support the findings of previous modelling studies, meaning that the simulations were relatively accurate at predicting consumer response to price changes that resulted from food taxes that were actually implemented in a real-world scenario. This is an important finding as it offers support for the potential validity and accuracy of the meat tax modelling studies discussed previously. In particular, Andeyeva et al. (2010) estimated meat to be more elastic than fats (butters, oils, etc.), meaning that a proportional price increase on meats should result in an even greater decrease in consumption than those seen in the above 'fat tax' example.

The support for implementing a meat tax is further buttressed by Charlebois et al. (2016), who found that in a survey of Canadian respondents, 43% agreed that price increases

have had an impact on their beef consumption habits, showing price fluctuations to already have a significant influence over consumers' choices in purchasing meat products. Knowing this, the careful implementation of a meat tax as a means to lower meat consumption through price manipulation seems to hold promise. In addition, subsidies on healthier, low-impact foods such as fruits, vegetables, grains and legumes can also be helpful if used alongside proposed meat tax measures. Studies have determined that price reductions can increase the purchases of healthier foods in community-based settings and among low socio-economic populations (French, 2003; Powell et al., 2009). An Australian study found that 39% of consumers said that the costs of fruit and vegetables were not affordable and 23% said that these affordability issues caused them to buy less fruits and vegetables than desired (Kathryn et al., 2017). These sentiments were more prevalent among young and low-income consumers, meaning that a plant-based food subsidy could help these consumer groups tremendously. Therefore, if taxes imposed on meat are implemented alongside subsidies for plant-based foods, meat products could be more readily substituted with plant-based alternatives, especially among those who are financially disadvantaged. Combining these measures may make promoting shifts to sustainable diets more affordable and appealing, especially among more financially restricted consumers, with the added potential benefit of improving public health. The funding needed for such subsidies could come from the tax revenue generated by taxing meat alone or from a combination of taxes on multiple food categories deemed 'unhealthy' and/or 'unsustainable', and/or from healthcare savings resulting from lower meat consumption. This type of revenue recycling (i.e. where tax funds are redistributed to the public) has already been proposed and implemented in some carbon tax schemes, in efforts to increase public acceptance of such measures (Klenert et al., 2018; Murray and Rivers, 2015).

4.3.3 Potential roadblocks to effective implementation

Potential challenges and roadblocks must also be discussed to get a full understanding of the feasibility of a proposed meat tax and its effectiveness in promoting more sustainable diets. Multiple authors have cited the lack of political will (Edjabou and Smed, 2012; Sall and Gren, 2015) and potential opposition faced from meat producers and consumers (Nordgren, 2012) in implementing a meat tax. This is likely to be the same in New Zealand, as the government has had a relatively "hands-off" approach when it comes to

farmers and food since the removal of government subsidies and safety nets in the 1980s, and since meat and dairy are two of the country's biggest and most profitable exports. Due to the power and influence of the meat and dairy industries in New Zealand, resulting from their economic significance, something like a meat tax would likely face serious opposition (or disregard) from the industry, government, and people (Radio New Zealand, 2019). Meat and dairy are still seen as part of the national identity for the majority of citizens, and so any suggestion of a tax on meat would likely face strong backlash from consumers, not even considering the money that the meat industry would put behind ad campaigns and outreach against such a tax.

Taxes on real foods (e.g. sugar and fat) for health reasons are relatively recent, and there seems to be a general resistance among policy makers to impose taxes on food products whether it be to improve public health or environmental sustainability (Sall and Gren, 2015). This lack of willingness is, at least partially, due to producers of animal products being strong influencers in the political realm. For instance, over the past few decades the U.S. government has tried to promote meat reduction in order to align with growing research on the health benefits of low-meat, high-plant diets, but has been met with strong resistance from the livestock industry, sometimes resulting in reversed or watered-down dietary guidelines and policies (Nestle, 1999). Therefore, any proposed environmental tax on meat will not only be relatively novel in the policy space, but will more than likely not garner the support necessary to be seriously considered, let alone enacted. This point is made well by MacRae and Winfield (2016) when discussing the potential implementation of an environmental tax on meat and dairy consumption:

> "Dominant institutional and societal actors who feel that their interests are well served by the existing arrangements are likely to resist the entry of new actors or new ideas into the policy process. As a result, it is generally accepted that most policy change will be incremental in nature. Major changes in direction are not the norm, rather policy is likely to be modified slowly over time in response to evident weaknesses, problems and opportunities. (pg. 154)"

Step-wise policy shifts will likely occur at a slow and incremental pace, meaning that the proposition of a meat tax, if ever supported, may take decades to implement across multiple regions, as seen with the slow uptake of carbon taxes around the globe over the

past decade as a result of the Paris Agreement (Funke and Mattauch, 2018). In addition, some countries may support and implement a meat tax while others invest more in the industry, hampering the net global impact of such measures, as seen in regards to carbon taxes where some nations like India and China have reduced investments in coal while other countries like Egypt and Indonesia have increased investments (Edenhofer et al., 2017). Despite these global challenges and setbacks, carbon tax schemes have already been implemented effectively in some nations (e.g. Sweden, Norway, Finland) and with tax frameworks for carbon already being established with relative success, implementation of a meat tax within these countries may be more streamlined due to prior governmental experience, in addition to increased industry and public familiarity with such a tax.

If we assume that a meat tax scheme could overcome currently embedded resistances and be implemented across multiple regions over the next decades, there is still no guarantee that the costs associated with a tax on meat would be fully passed onto the consumer. As noted by Jensen and Smed (2013) when observing the effects of a fat tax in Denmark, retailers may adjust prices or offer discounts when such a tax is implemented in order to keep prices down and sales up, blunting the impact of the tax on final prices where it would ultimately have little to no effect on consumption rates. Also, consumer segments may shift from purchasing products at high-end stores to making purchases at lower priced discount stores. These potential adjustments and/or shifts would thereby undermine the effectiveness of a meat tax, as consumers could bypass increased prices by simply switching to cheaper shopping locations, resulting in distributional changes in meat sales but not necessarily reductions in overall consumption. Also, using price manipulations to incentivise individuals to choose more plant-based foods over meat products is unlikely to lead to continued behaviour change if and when the incentives are removed. As stated by Schultz and Kaiser (2012), behaviour change that results from incentivising individuals usually results in a return back to the original behaviour once the incentives are no longer present. In addition, overjustification effects, where the provision of external incentives decreases a person's intrinsic motivation to perform an act, can sometimes cause the individual to revert back to their original behaviours and perform them at a rate that was even higher than before incentives were first offered (Deci et al., 1999). Thus, if price manipulation policies are to be utilised, the (dis)incentives must be consistent; otherwise, any changes consumers make towards reducing their meat

consumption may revert back, with the added risk of actually increasing their meat intake levels beyond what was initially observed before the incentives were enacted.

Furthermore, as pointed out by Edjabou and Smed (2012) differentiated environmental taxes on meat products would fail to promote mitigation advances and technological improvements in livestock production due to their static nature. These consumption taxes would not automatically adjust when technological improvements or new mitigation strategies are implemented, limiting incentives for producers to reduce production-related environmental impacts. However, the revenues from a meat tax could be recycled and applied towards subsidising producers that seek to lower their environmental impacts (e.g. through technological innovations, uptake of more sustainable production methods, etc.). It is also important to consider that in some nations, producers may be especially resistant to government intervention in food production. For instance, in 1984 agricultural subsidies were removed in New Zealand and now the majority of farmers strongly oppose ever having them implemented again, as they feel government intervention in any form is unnecessary and unwarranted (Sayre, 2019).

4.4 Regulatory MGMs

Regulatory MGMs are created and enforced by governments, are legally binding and can include international agreements (Blackmore, 2011). They are commonly referred to as "hard" MGMs, in comparison to "soft" MGMs which are voluntary or cooperative initiatives, and are intended to promote behaviour change through legal consequences as a result of non-compliance. Regulatory measures to decrease meat consumption could come in many forms, including rations or bans, regulation of meat advertisements and changes to public procurement policies.

With current free market economies it is extremely difficult to imagine any bans or rationing on all or even specific meat products being supported or implemented, as the rationing of meats have only been utilised in the past during extreme crises such as World War II (Wansink, 2002). Although rationing or banning certain meat products would have obvious significant impacts on meat consumption, the acceptance and implementation of such measures is not foreseeable. The same can be said for the implementation of regulations on meat advertisements. Bans and/or regulations on junk food advertisements directed at children have been implemented in only a handful of European countries in

efforts to combat rising rates of obesity (Brambila-Macias et al., 2011), with debates on the idea of implementing them in other nations currently ongoing; making the idea of a full or even partial ban on meat advertisements directed at the general public for the sake of environmental sustainability or public health unlikely in any nation, particularly in the near future. Therefore further discussions will focus on regulatory measures that are more feasible and likely, specifically the potential for utilizing public food procurement (PFP) as a tool to promote healthy and more environmentally sustainable diets through reduced meat consumption (4.4.1 and 4.4.2). In addition, strengths and barriers of employing PFP in efforts to reduce meat consumption will be summarised (4.4.3 and 4.4.4).

4.4.1 Public food procurement to promote sustainable healthy diets

Governments often provide public goods, such as infrastructure (e.g. laws, roads, parks), which are usually 'free' in terms of accessibility and use for all citizens, in addition to merit goods, such as healthcare and educational facilities, that are provided based on merit and/or need (Stiglitz and Rosengard, 2015). Governments often rely on private industry to supply the inputs that are necessary to create and provide such goods (e.g. lights for a park, asphalt for roads, books for libraries, etc.) and therefore use tax revenue to pay private industries for their materials and/or services. This process is public procurement, and although some may argue that having such close relations between government and private industry through the exchange of money and services may encourage corruption, which may be the case in some instances, there is also opportunity to create positive change. In terms of promoting societal change, public procurement could help create new markets and/or shift existing ones through governments utilising their large and influential purchasing power to provide public and merit goods that are produced in a way that is less impactful (e.g. more environmentally-friendly).

Governments and public institutions are large consumer bodies, with industrialized countries spending over 10% of their gross domestic product on public procurement (Zhu et al., 2013). This means that public institutions can act as influential bodies in shaping markets through increased demand for certain products (Testa et al., 2016), with the food sector being no exception. PFP involves the act of government purchasing, preparing and serving food in public institutions like schools, hospitals, community centres, prisons and state companies (Freudenberg, 2016) and can sometimes be used to promote certain goals such as better food quality, improved public health, reduced environmental impacts,

and/or increased support of social 'goods' (e.g. social justice through improving access to quality food for low-income consumers). Green public procurement (GPP) and sustainable public procurement (SPP) are two forms of public procurement that include an environmental component in decision-making processes, encouraging public institutions to purchase and deliver products and services that meet typical price and quality standards while also minimizing negative impacts on the environment. This is usually accomplished through general criteria and/or specific standards defining environmental requirements (e.g. ecolabel certification, carbon footprint, water footprint, seasonality, locality) when calling for tenders (i.e. supply of goods) (Cerutti et al., 2015). Outlining GPP/SPP goals with intentions to reduce the purchase and provision of high environmental impact meat products in public institutions has a strong potential to not only mitigate environmental impacts related to PFP, but additionally offer public health benefits. Not only would suppliers be compelled to provide more sustainable plant-based food products in response to such a large shift in their consumer base (i.e. public institutions), but individuals that rely on these institutions for the majority of their meals would also see health benefits through reduced disease incidence. Therefore, PFP can be used as a tool to promote both environmental sustainability and improved public health through the reduced purchasing and provisioning of meat products.

Most PFP initiatives to date have been employed to provide healthier and more local foods to consumers, with governments focusing mainly on young children in schools and in promoting local economies (Bloomfield, 2015). There is potential for the values and goals of GPP/SPP to play a strong role in PFP policies, but some have argued that governments around the world have largely overlooked opportunities to reduce environmental impacts through the purchase of more environmentally-friendly food products (Garnett, 2012). Some PFP programs have focused on improving health outcomes, while others have placed environmental impact reduction as a primary or secondary emphasis, usually in the form of prioritising reduced carbon footprints through the acquisition of more locally sourced and organic foods (Stefani et al., 2015). Whether for the promotion of health and/or sustainability, PFP programs and studies measuring their successes and shortcomings can provide vital information when considering whether PFP initiatives can act as effective tool for promoting meat reduction in society.

4.4.2 Overview of health and environmentally focused PFP programs

A comprehensive review of healthy PFP policies was conducted by Niebylski et al. (2014) to observe and analyse their impact on eating behaviours and health outcomes when implemented in schools, worksites, hospitals and other institutions. One such policy included England's healthy food procurement standard that was implemented in 2008, affecting 136 primary schools. The purchases of fruits, vegetables and salads rose by 15% while those for processed foods high in fats, sugar and sodium fell by 12%. Also, 74% of students showed a greater desire for healthier food options, accompanied by a 15% increase in healthy food purchases from 2006 to 2009. These findings were supported by another study done on worksites in Denmark, which found that education paired with healthy food procurement strategies led to an increase in healthy food consumption by 70 grams per day in the workplace. Similar positive findings were observed when looking at the effectiveness of policies implemented in other government institutions like hospitals, care homes and correctional facilities. Overall, the review found PFP policies to be an effective tool at promoting healthy dietary change within public sector institutions; however, the authors cite the potential role played by prior and/or simultaneous ancillary education and supportive pricing in helping to successfully implement the initiatives and create positive health outcomes.

In addition to prioritizing health, some public institutions have also placed environmental sustainability as a major tenet when procuring food products. In fact, in the U.K., the procurement of more environmentally sustainable foods by the central government had already received attention in the early 2000s, leading to the launch of the Public Sector Food Procurement Initiative (PSFPI), which included the reduction of food-related environmental impacts as one of its main objectives (Morgan, 2008). Similar efforts have been seen in other nations as well, and a few examples have been mentioned and discussed in a collaborative study conducted by Foodlinks, a project funded by European Commission (Barling et al., 2013). The study looked at instances where PFP programs were utilised to fully or at least partially promote more environmentally-friendly diets in five different nations (Sweden, Italy, Scotland, Denmark and Austria). All five case studies promoted more healthy and sustainable diets through the increased purchase and provision of organic, local and fresh foods, with three of the case studies specifically aiming to reduce meat consumption as well. For example, the school canteens in Malmo (Sweden), offer vegetarian options every day and provide only vegetarian options one day

out of the week, while in Vienna (Austria) the dishes that public institutions serve are comprised of a minimum two-thirds vegetarian ingredients in efforts to reduce the environmental impacts associated with meat consumption. Overall, the initiatives were found to be a success, as positive movements towards reduced obesity rates, increased organic and local food consumption and mitigated environmental impacts were all observed. A common theme arising from the organisers of the PFP programs was that the allocation of funds dedicated to information provision and awareness raising among public workers and the larger community were observed to be important components in the adoption, support and success of these programs.

4.4.3 Strengths of implementing meat reduction through PFP programs

In comparison to meat taxes or other regulations that would be novel and therefore difficult and laborious to implement, PFP is a practice that is already well-established among both institutions and their respective communities. With infrastructure already in place, it will be much easier to re-align current practices to match the goals outlined for a sustainable healthy diet rather than creating new programmes and policies that would be built from the ground up. For example, avenues already exist for institutions to purchase plant-based foods from suppliers and provide them to their constituents, and there may even be a benefit of added savings if low-price plant foods can be obtained and substituted for meat. Yes, adjustments will have to be made such as increasing the quantity and variety of plant-based foods purchased, as well as building the capacity of chefs to prepare and cook satisfying and tasty vegetarian meals; but these are relatively small changes in comparison to the creation and implementation of entirely new meat-based regulations and/or tax codes.

Also, there is a potential to strengthen the positive outcomes of PFP initiatives by pairing them with “nudges” that voluntarily shift consumer behaviour towards “better” options by adjusting current infrastructure, choice defaults and access to alternatives (Thaler et al., 2014). Nudges, in the form of new meal structures for example, brought about by revised procurement guidelines have the potential to break old habits and form new ones for those being served, shifting norms about what a “meal” is comprised of, including expectations of the presence and quantity of meat. These nudges do not force options on individuals, but instead use incentive structures to steer consumers towards better options while still leaving alternative options available. For example, school cafeterias could make fruit and

vegetable options more attractive to students by making them free or low-cost in comparison to processed foods that are high in fats and sugars, thereby "nudging" them to choose healthier options. In an intervention study involving nudging, Campbell-Arvai and colleagues (2012) found that selection of meat-free menu items in a school cafeteria rose from 40% to 90% when placing meat-free options as the default menu items rather than as alternatives on a separate menu. So, the consumption of more plant-based meals can be greatly increased by simply changing the default options on the menu due to consumers' tendency to choose those defaults. In a more recent development, as of 2017 Portugal has made it mandatory by law for all public canteens to have at least one fully plant-based meal option (Nagesh, 2017) Not only are strategies like offering plant-based 'defaults' effective in shaping food choice, but they are also expected to garner greater participation among consumers in comparison to the outright banning of certain foods (Taber et al., 2012). Thus, nudging techniques can be easily utilised in conjunction with reduced-meat PFP policies to not only procure and provide healthier and more sustainable plant-based food options, but to also to encourage their uptake by those being served.

Besides ease of implementation and the potential for 'nudging' consumers, PFP also has the potential added benefit of bringing together public representatives, small-scale farmers, local businesses and educators to create a large force for the promotion of more sustainable healthy diets. This could create a growing support among multiple stakeholders for policies and programmes that take a stronger stance on reducing meat consumption, but that are currently controversial or unsupported (e.g. meat taxes). Healthy food procurement has already been shown to have both short-term and long-term benefits (Campbell et al., 2014; Niebylski et al., 2014; Freudenberg, 2016), meaning that it might be easier for public officials to justify the further implementation of similar food procurement policies (e.g. meat reduction initiatives) in the future. If utilised, the wide adoption of reduced-meat PFP policies could act to garner both public and political support for further steps to reduce meat consumption (e.g. meat taxes and/or plant-based food subsidies) that would otherwise face heavy resistance.

Furthermore, although there is variation depending upon the nation and sector under discussion, public institutions generally serve large population groups and therefore have a major impact on public health and food-related environmental impacts (Freudenberg, 2016). Children in schools, refugees, hospital patients, incarcerated individuals and other citizens who mostly rely upon food programmes in public institutions to meet their daily

caloric intake constitute a significant portion of the population and are all groups that could benefit from reduced-meat PFP policies. Also, exposure to more plant-based foods and recipes has the potential to increase acceptability and preference for a reduced meat diet among these population groups (Reisch et al., 2013). The pervasive effects of changes in public procurement policies have even been shown to influence the activities and practices of private institutions, as Ho et al. (2010) found when reviewing the effects of GPP activities on industry in five Asian countries. The study showed that governments which demonstrate a strong commitment to GPP in efforts to reduce their environmental footprint increase green procurement practices in the private sector as well. This means that GPP can act as a stable source of demand for greener products, while also sparking the adoption of similar green practices among private sector businesses as well. Therefore, PFP aimed at reducing meat consumption can potentially lead to a greater movement among consumers and private businesses to adopt and promote reduced-meat diets. Although PFP programs will likely take various forms depending upon the nation or region, the potential for promoting healthier and more sustainable diets through service of food to the public can be a powerful option (Morgan, 2008).

4.4.4 Barriers to implementing meat reduction PFP programmes

Although reduced-meat PFP initiatives hold promise, the implementation of such measures in public institutions will likely face political and public resistance. According to Brammer and Walker (2011), there are many advocates of ‘limited government” who would look at such programmes as a primary target for budget cuts, while individuals who believe strongly in personal responsibility and choice, would only see such measures as contributing to the so-called “nanny state”. Both of these factors make the likelihood of an extensive overhaul in PFP policies to reduce meat consumption less likely, especially in a country like New Zealand which has a relatively “hands off” approach in regards to food production and consumption, but this does not mean changes are impossible. As stated by Reisch et al. (2013), although public dining facilities have mostly ignored the opportunity to implement any meat reduction initiatives (true in New Zealand as well), there is growing evidence of the health and environmental benefits of a reduced-meat diet, which may put pressure on these institutions. As discussed above, there are many public schools and universities who have already shifted their procurement policies to

promote health and sustainability, and many of these included efforts to reduce meat consumption.

In places where political resistance is strong, a dearth of technical competence will likely also be present, further impeding any shifts towards healthier and/or more sustainable PFP policies. Studies have found procurement managers to have a lack of understanding in incorporating sustainability considerations when calling for tenders, as shown in a survey conducted by Snell (2006), where 83% of purchasing managers considered themselves ill-equipped to deliver such services. As stated by Morgan (2008), this limits a state's capacity to make their procurement practices more sustainable, as knowledgeable, well-trained and competent procurement managers, as well as canteen staff (e.g. chefs) will be necessary if "greening of the realm" is to occur. Therefore, successful implementation of health and sustainability goals into current PFP policies will first require capacity building among procurement managers and staff to increase the knowledge and skills required to properly incorporate such goals into current purchasing guidelines and service of more plant-based meals (Pacheo-Blanco and Bastante-Ceca, 2016). However, as mentioned this necessary training is unlikely to occur if political agendas do not align with public health and environmental sustainability goals.

In cases where political support is present and the training and/or hiring of procurement managers who can successfully incorporate health and sustainability goals into their practices is available, institutions may still have to struggle with acquiring funds and breaking away from long-term contracts with current suppliers. A study of private firms in the U.S. conducted by Min and Galle (2001) found that economic obstacles were sighted as the most frequent barrier for firms when considering the purchase of more "green" products, as they were perceived to be more expensive and therefore incompatible with their allotted budgets. State institutions who intend to implement more "green" PFP policies may face similar challenges, especially those with relatively inflexible price and budgetary constraints like schools (Conner et al., 2012). This was observed first-hand by Kleine and das Gracas Brightwell (2015), who studied Brazil's progressive school meal policy which has placed an emphasis on schools procuring and providing fresh, local, healthy and more environmentally sustainable food for their students. The researchers found that one of the major challenges of implementing such

policies came down to logistical and financial issues, including the need for more storage facilities and school kitchens in order to properly process fresh produce.

Despite such financial challenges, some governments have offered direct aid to public bodies who seek to change their procurement practices. For example, under New York State municipal law, public bodies are permitted to "overpay" by 10% in selecting food tenders that are locally grown in the state of New York versus food tenders that are not local, making the procurement of locally sourced goods more affordable (de Blasio and Camillo, 2015). Another example is the Ontario Local Food Act in Canada, which encourages farmers to donate foods to institutions who distribute foods to the public by offering the farmers who donate a tax credit (Ministry of Agriculture Food and Rural Affairs, 2016). Such measures make acquiring healthier and more sustainable foods more affordable for public institutions, and so although affordability concerns can act as barriers, with political will and initiative these barriers can be overcome as evidenced by these examples. In New Zealand, I believe political will along with citizen support could make similar meat reduction measures feasible, despite resistance from the meat industry like Beef and Lamb New Zealand (2018b), who recognize the negative impacts associated with red meat but still plan to continue producing and marketing meat products until they are no longer viable to domestic and/or international markets. Thus, creating and maintaining political will and citizen support is an uphill battle as it would go against the current structures and culture. Changing consumer opinions may be an initial step that can "get the ball rolling" for such a change, and information provision which will be discussed next may play an important role.

4.5 Informational MGMs

Informational MGMs aim to provide information in the public sphere through altering individual understandings and priorities, ultimately persuading target groups to reassess the significance they attach to the environmental and social issues associated with their current behaviours. These MGMs take a softer, more voluntary approach to behaviour change, usually through highlighting the role of consumer choice and its ability to shape market demand through the products purchased and consumed by individuals. Unlike most measures that fall under the economic or regulatory MGM umbrellas, informational MGMs are usually consumer focused and encourage individual agency as a potential driving force for societal behaviour change. These measures generally materialize in the

form of product labels, marketing campaigns, and/or awareness raising initiatives that all seek to inform and thereby empower consumers to voluntarily change their consumption habits for a desired outcome (e.g. improved health, lower environmental impacts, etc.). Of course, informational MGMs place a strong emphasis on agency and choice, while relying heavily on the assumption that for the most part, consumers are rational agents who weigh out pros and cons before making a purchasing decision, and that multiple options are available for the consumer to choose between. This will obviously vary depending upon the consumer in question, the purchasing scenario they find themselves within, and the behaviour being addressed (e.g. meat consumption). Hence, the potential role of different informational MGMs in promoting meat reduction will be summarized.

To begin, current consumer awareness and knowledge in regards to meat's associated environmental impacts will be reviewed (4.5.1), followed by the potential for labelling and marketing campaigns to be used to promote meat reduction (4.5.2 and 4.5.3). Then, the general effectiveness of awareness raising in changing individual attitudes and behaviours will be discussed, along with awareness raising on meat's impacts, specifically (4.5.5), and how the 'Intention-behaviour' gap may act as a significant barrier to behaviour change (4.5.6). Finally, the strengths (4.5.7) and weaknesses (4.5.8) of informational MGMs will be outlined and summarised.

4.5.1 Low awareness of meat's environmental impact

In order to provide information to consumers in an effective manner, one must understand the target group's beliefs and knowledge on the issue at hand. Studies in multiple nations have shown that awareness of meat-related environmental impacts is low compared to other environmentally damaging behaviours (e.g. plastic use, buying local foods, etc.) among the general public (Lea and Worsley, 2008; Macdiarmid et al., 2016; Siegrest et al., 2015), and this is evident even among university undergraduates in the United States (Campbell-Arvai, 2015). In addition, although switching to a more plant-based diet has been identified as one of the most environmentally-friendly behaviour changes an individual can make, government recommendations and resources in many developed nations (e.g. Canada, United States, Australia, EU countries) have failed to focus on promoting plant-based diets to the public (Wynes and Nicholas, 2017). Therefore, it would seem there is a lack of connection between consumers' food habits, especially meat consumption, and the environmental sustainability impacts related to those choices.

This is important to note, as a lack of awareness among consumers in regards to the environmental impacts of livestock production removes the opportunity for those consumers to consider reducing their meat consumption in order to promote a more sustainable diet. Put simply, if consumers do not see or understand the environmental consequences of their food, their ability to choose alternative options in efforts to promote sustainability will be highly impaired or non-existent. Also, understanding levels of awareness among consumer groups is important to recognize in order to optimize interventions in ways that are most appropriate for changing consumer attitudes and behaviours. For example, if consumers do not understand the impacts of climate change or are sceptical of climate change science, promoting 'climate friendly' products may not shift behaviours due to a lack of connection between food choice and environmental concerns (Jacobsen and Dulsrud, 2007). Knowledge of meat's sustainability impacts is quite low among consumers, so a necessary first step may be to help connect food choices to these issues by means of public education and awareness raising initiatives.

4.5.2 Labelling

Food product labels have grown in popularity over the past few decades, with the introduction and wider use of labels that are both regulated and standardized (e.g nutrition, organic, fair trade) (Grunert and Wills, 2007; Howard and Allen, 2010) and those that are used mainly for marketing purposes (e.g. natural, heart healthy; Skubisz, 2017). Labels communicating nutritional information have become a standard in many developed world countries (Campos et al., 2011); but with rising concerns over the health of the planet, the use of environmental labels (i.e. ecolabels) such as the Carbon Trust Carbon Reduction Label and Rainforest Alliance Label have also begun to rise in popularity (Garnett, 2012). Recently, ecolabelling has even been suggested for meat products (Roos et al., 2013) as a way to effectively communicate the environmental impacts associated with its production to consumers. The act of creating and applying environmental labels to food products is a growing trend, but the most important factor to consider is whether they can act as an effective tool in reducing meat intake levels through changing consumer behaviour. Therefore, the potential for ecolabels to encourage meat reduction among consumers through the communication of environmental impacts is explored.

The UK government commissioned a study to look into the evidence (or lack thereof) of ecolabels' impacts on consumer behaviour and concluded that labelling alone will not be enough to stimulate the changes necessary to significantly reduce food-related environmental impacts (Defra, 2010). This claim is supported by Vandenbergh et al. (2011), who state that although labelling has a place in helping consumers to make product choices that better align with their already held values and beliefs, it is unlikely to radically shift behaviours of large consumer groups. This explains how nutritional labelling has failed to significantly deter the rising rates of diet-related diseases (e.g. heart disease, diabetes) caused by unhealthy consumer food behaviours. Therefore, placing ecolabels on meat products will likely have minimal impact in shifting large numbers of consumers away from the currently pervasive 'meat heavy' diets to more sustainable plant-based ones.

Another issue besides the questionable effectiveness of using ecolabels as a behaviour change tool is whether or not consumers actually trust and utilize the information presented to them. Kikuchi-Uehara et al. (2016) conducted a survey that compared consumer's environmental awareness and trust in environmental information with their preferences for ecolabeled products and found that as consumer awareness and trust decreased, so did the preference for products that claimed environmental benefits through the use of ecolabels. Therefore, when awareness of impacts and trust of information is low, the likelihood of consumers utilizing ecolabels is much lower than when awareness and trust are high. For the case of meat consumption, it has already been established that environmental impact awareness is quite low, meaning that individuals will likely be sceptical of environmental information provided to them through the use of labels. This doubtfulness will impede a consumer's ability to fully trust and thereby incorporate the information provided by an ecolabel into their purchasing decisions, especially if competing labels like "grass-fed environmentally friendly beef" are used by manufacturers. This potential scepticism will likely be enhanced by the fact that ecolabels, especially those designed specifically for meat products, are relatively novel concepts that most consumers have not been exposed to. Adding to this, Sirieix et al. (2013) conducted focus groups with consumers asking about their perceptions and reactions to different sustainable food labels in the UK and found that labels such as 'organic' and 'fair trade' were perceived positively while other, less familiar ecolabels such as 'climate friendly' received scepticism. Therefore, new ecolabels for meat

products will likely face scepticism from consumers, at least until they become more familiar with the environmental impacts of meat production and the use of such labels becomes more prevalent.

Thus, the idea of using labelling as a catalyst for promoting societal reductions in meat consumption appears to hold little promise. Although useful as a tool to help guide consumer choice, labelling will likely not be enough to promote large shifts in meat consumption behaviour. Awareness of meat's impact on the environment is quite low and consequently, consumers are likely to be sceptical and mistrust the information provided by a novel ecolabel for meat products. This will blunt the effectiveness of labels as a tool to convince consumers to reduce the amount of meat they currently consume in their diets. Therefore, it may be necessary to first educate consumers about the impacts of meat consumption and the benefits of plant-based diets before ecolabels can be used effectively. This education could be accomplished through marketing and/or awareness raising campaigns in order to give consumers a basic understanding of meat's impacts, so that they are more likely to both believe and utilize ecolabels on meat products in the future. Increased nutrition knowledge has already been related to higher rates of healthier food choices and as a result, healthier eating behaviours (Kolodinsky et al., 2007), so this is likely to hold true in the case of meat consumption as well. Creating a well-informed consumer base will likely be a necessary first step, with the utilization of labels acting as a supplementary or follow-up measure in order to help consumers more easily translate their newfound knowledge into actualised meat reduction. So, although increased ecolabelling has a potential role to play in shifting meat consumption habits, it is unlikely to be effective until lack of consumer awareness in regards to meat's environmental impacts is addressed.

4.5.3 Marketing Campaigns

Educating consumers through marketing campaigns has been a tool used widely by governments, organizations and individuals wishing to shift consumer behaviour into more 'desirable' directions. Although their effectiveness has been criticized due to examples of informational campaigns failing to make significant changes to behaviour (Schultz and Kaiser, 2012), there is an understanding that individuals who are more knowledgeable about the benefits of a behaviour are more likely to engage in that behaviour (Schultz, 2002). So, even though information campaigns may not always be

effective, there is strong evidence linking understanding of consequences to behavioural tendencies.

In addition, an interesting point to consider is that although increasing knowledge generally does not produce significant changes in behaviour, exceptions to this finding exist where there is evidence that the target population is generally unfamiliar with the issues related to a particular behaviour (Lee and Kotler, 2012; Mckenzie-Mohr, 2013). This seems logical, as providing more information to those who are already informed about an issue will likely not have much of an impact on behaviour, while educating consumers on issues related to a particular behaviour where knowledge is lacking would seem more promising for encouraging change. With low awareness of meat's impacts on environmental sustainability, informative marketing campaigns may prove to be an effective first step towards changing consumer behaviours, especially if they are strategically designed to counteract marketing campaigns that will likely offer contradictory information (e.g. meat industry TV advertisements).

4.5.4 Effectiveness of awareness raising in shifting attitudes and/or behaviours

There are many successful initiatives that have relied upon raising awareness and educating consumers in order to shift attitudes and/or behaviours, including smoking cessation (Jha, 2006), energy conservation (Steg, 2008) and public health campaigns (Moore and Johnson, 2015). One study looking at the effects of anti-smoking media messages on university students found ads that focused on health consequences not only led to a significant increase in knowledge, but also to an increase in negative attitudes and beliefs associated with smoking (Murphy-Hoefer et al., 2010). Although measuring changes in behaviour were not included in the study, it is assumed that as negative attitudes and beliefs towards smoking increases, the likelihood for individuals to perform the behaviour decreases; hence, it would seem that anti-smoking media messages would likely have a negative impact on the behaviour of smoking cigarettes. Besides changing attitudes and beliefs, there are studies that have actually measured behaviour change as a result of providing information to consumers. For example, the effects of an educational intervention on healthy diets performed in public schools in Brazil were measured to see if it had any significant impact on food consumption behaviour (Dos Santos et al., 2015). The study found significant reductions in consumption of soft drinks and processed juices

($p = 0.007$), meat and sausage ($p = 0.072$) and fried foods ($p < 0.001$) as a result of the educational intervention, but it is unknown how long these effects lasted. In addition, a study in Bangladesh found that simply informing consumers about the potential risks associated with arsenic levels in their well water led to 37% of participants changing to a safer well within one year (Madajewicz, 2007). These studies show that providing information to consumers can and does have an effect on not only the attitudes towards a behaviour, but also the action of the behaviour itself.

4.5.5 Effects of information on meat consumption

Evidence suggests that information provision may be a vital component in efforts to reduce meat consumption, but it would be helpful to review studies that have actually observed changes in consumer meat intake in response to information exposure. One study by Kinnucan and colleagues (1997) created a health information index that measured the amount of scientific research papers published on the health impacts of different meat products in a given year. Using this index, they compared its impacts on demand for different meat products and found that changes in health information had a greater effect on demand for meat products in the United States than when compared to price fluctuations. Compared to the price elasticities of chicken and beef (-0.158 and -0.444, respectively), they found that the elasticity of information on meat's negative health impacts to be higher in absolute value (1.54 and -0.583, respectively), meaning that relatively small changes in health information can induce much larger changes in demand when compared to equivalently small price changes. Obviously news and media acted as a mediating factor, as average citizens get their information from such sources and not from scientific articles directly, but nevertheless this shows that an increase in health information communicated to consumers can actually be more effective than price changes in influencing consumer's to alter their purchasing behaviours in regards to meat products. Later studies support these initial findings, asserting that health information does indeed have a significant impact on demand for various meat products (Ben-Kaabia et al., 2001; Tonsor et al., 2010) and a more recent study involving a multicomponent intervention to reduce meat consumption found that participants identified the informational component, specifically, as the greatest contributor to their observed meat reductions (Amiot et al., 2018).

So it would seem that providing information on the health benefits of a reduced meat diet has an effect on meat demand, but what about other issues such as sustainability and animal welfare concerns that are commonly associated with the production and consumption of meat? One study, conducted by Cordts and colleagues (2014) looked at how negative information (in the form of newspaper articles on various issues commonly tied to meat consumption) affected consumer attitudes towards meat consumption in Germany. They found that those consumers who read articles on meat production's impacts on animal welfare, health and climate change showed an increase in concern accompanied by significant increases in intentions to reduce their meat consumption in the future (from 12.8% intentions to reduce without article to 28.0%, 23.5% and 18.8% intentions to reduce after reading animal welfare, health and climate change articles, respectively). This shows that informing consumers about the various impacts of meat consumption can and does have an effect on not only beliefs and attitudes, but also changes in behavioural intentions to consume less meat in the future. Of course, one cannot assume that intentions automatically lead to actual behaviour change and thus, this potential barrier for promoting meat reduction will be explored.

4.5.6 Intention-behaviour gap

Past behaviour change studies have indeed found there to be a gap between intentions to change behaviour and actual behaviour change due to issues of perceived consumer effectiveness, availability of alternative options, barriers due to dominant social norms, etc. (Kollmuss and Ageyman, 2002; Vermeir and Verbeke, 2006). However, this does not necessarily mean that changing attitudes and intentions of consumers will not remain a necessary component in efforts to promote meat reduction, but rather that it will likely take a combination of changing attitudes, intentions, societal norms, and purchasing environments. In current society, most individuals carry a positive affinity for meat that outweighs their knowledge, beliefs and attitudes about the negative consequences of its consumption, which has resulted in a lack of motivation to change the predominant meat-heavy dietary patterns that are now seen in many countries. This is supported by Klockner (2017), who found that in a study of Norwegian consumers, a majority (76.3% and 72.7% for two different samples) are in the predecision stage when asked about reducing their beef consumption. The predecision stage, as defined by Klockner, is where an individual is addressing the question for why an action is necessary and in the case of beef

consumption, a consumer is contemplating, "Why is it necessary to reduce my beef consumption?" The predecision stage, which is the first of four behavioural stages (predecision, preaction, action and postaction) was shown by Klockner to have the lowest levels of intentions to reduce meat consumption in comparison to the other three groups. Even though intentions led to a significant reduction in beef consumption only at the final stage (i.e. postaction), moving from the predecision stage into and through the preaction and action stages are necessary before the postaction stage can be reached. Therefore, it would seem logical to address the issue of getting consumers out of the predecision stage, through measures like awareness raising efforts, as intentions to reduce and actual reductions in beef consumption can be promoted by helping to move individuals out of the predecision phase and into the later action stages. With that said, it is important to still remember that consumers will likely receive constant streams of information and societal feedback that solidify meat consumption as a norm, which will undermine and/or counter any information aimed at encouraging individuals to move out of the predecision stage (i.e. actively seeking to reduce their meat intake).

Current norms in society are biased towards favouring meat consumption, so inherently it will be a challenge for individuals to translate any meat-reduction intentions into behaviours when the conditions they are surrounded by are not encouraging or even conducive towards such a dietary shift. Nevertheless, changes in these external conditions are more likely to occur as consumer awareness and concern over meat's impacts rise. Attitude changes may lead consumers to be more supportive and accepting of alternative foods and diets, leading to incremental changes in conditional factors like availability of plant-based food products (as a result of increased demand) and greater social acceptance of reduced meat diets, thereby creating an environment that is more supportive of individuals trying to translate their new meat-reduction intentions into actual meat-reduction behaviours. Thus, although awareness raising campaigns aimed at altering attitudes and beliefs towards meat consumption are unlikely to lead to large or quick shifts in societal behaviours, they can act as important 'first steps' to engaging consumers in ways that can lead to greater structural changes in the future (i.e. increased social acceptance, greater variety and availability of food alternatives, etc.). Therefore, even though information provision alone may not be a complete or final solution, it does seem to have a place in the behaviour change toolkit, especially for issues that consumers are relatively unaware of, like meat's environmental impacts. With raised awareness,

consumers will have the opportunity to change their attitudes on meat consumption, which might eventually encourage wider adoption of more plant-based diets as society and industries shift to provide products that align with newfound opinions and preferences. Of course, information is not uni-dimensional or uni-directional, as consumers will likely be exposed to competing pro-meat information from advertisements and/or other messages that challenge and refute information that encourages meat reduction, so this is something that must be kept in mind.

4.5.7 Strengths of information provision in reducing meat consumption

Informational MGMs have multiple strengths and advantages that differ from economic and regulatory MGMs. To start, information provision faces less resistance as opposed to economic and regulatory measures, due to its minimally invasive and voluntary nature. It aligns more with neoliberal values (e.g. autonomy, personal responsibility, free markets, etc.) which are currently prevalent in many western nations, making it a more palatable option for citizens in comparison to economic or regulatory interventions which could be seen as intrusive and/or overbearing on consumer sovereignty (Csutora, 2012). The main challenge is that consumers are both unaware of the environmental impacts of meat, in addition with their predominantly negative attitudes and lack of motivation to replace animal-derived products with more plant-based foods (Hoek et al., 2017). Therefore, raising awareness may be a first step in changing these attitudes and increasing motivation to reduce meat in the diet. In addition, specific information provision measures can enhance the benefits and effectiveness of other initiatives through raised consumer awareness. For example, informed consumers can increase the effectiveness of ecolabels, as having a clearer understanding of the information being communicated can make the consumer more trustworthy of such labels, and therefore help guide them in making a more environmentally-friendly purchase. Furthermore, spill-over effects that are more indirect like individuals becoming more interested in other pro-environmental or healthy behaviours (e.g. exercising more, recycling, using less energy, etc.) as a result of switching to more plant-based diets are also possible.

In addition to more streamlined implementation and potential spill-over effects, information provision is a tactic that can be utilised by individuals, organizations, governments and industry to promote reduced meat diets. Unlike changes to tax codes or public procurement policies, raising awareness through providing information to

consumers can be implemented at multiple levels because it does not require as much centralized power or control to implement, making information provision the most flexible and widely accessible tool among all the MGMs. Awareness raising efforts also have a greater potential to create enduring behaviour change in comparison to the other MGMs, as consumers who fully understand the reasons behind such a lifestyle and have the opportunity to fully incorporate such information into their personal values and beliefs, will likely retain newly established behaviours even if contextual factors like societal norms and purchasing environments change. Rephrased for simplicity, informed consumers are more likely to be determined to do what is beneficial for themselves and society and will also be less likely to fall back into traditional behaviour patterns if and when other incentives for meat reduction (e.g. financial benefits, convenience, etc.) are reduced or removed.

4.5.8 Weaknesses of information provision in reducing meat consumption

Information provision may be less effective than more intrusive and strong-handed measures (i.e. economic or regulatory) in terms of how quickly societal shifts in meat consumption can be encouraged. Behaviour change studies within the psychology literature have mainly found that repeated habits are more likely to be broken by changing the environment in which an individual operates, rather awareness raising which is better suited for one-time decisions (e.g. buying an energy efficient appliance). Since meat consumption is a habit that is encouraged by current structures, consumers may find it difficult to translate their newfound knowledge into actual practice. Even if consumers know the consequences of buying meat, they may find it difficult to reduce its consumption if they lack vegetarian cooking skills, if meat substitutes are unavailable and/or if their common routines normally involve eating meat (e.g. eating with friends or family who cook meat, typical restaurants have few or no vegetarian options, etc.). However, as discussed above, structural changes aimed at reducing meat consumption have been scarce and face strong resistance from many industries and consumers, and so information provision may have to be relied upon as an initial step to begin shifting consumer opinions and behaviours. So although not a perfect solution, awareness raising may create more public and political support for more structural measures aimed at reducing meat intake (e.g. price incentives, more plant-based PFP programmes) with the

additional benefit of encouraging consumers to choose plant-based options if and when their availability increases and they become more mainstream (i.e. societal norm).

Consumers do not receive information in isolation, and contradictory pro-meat information presented through advertisements and even more reputable sources like governmental dietary recommendations can make the idea of meat reduction more confusing and/or less appealing. For example, as stated by Tucker (2018) Beef and Lamb NZ already run ad campaigns with popular celebrities on a regular basis, and information on meat-based meals is prevalent while celebrity endorsements and information on plant-based meals are rarer to find. Due to a lack of clarity and consistency about the impacts and consequences of choosing to eat meat, consumer confidence in choosing more plant-based foods can be low or fickle. Due to contradictory information and/or many other factors, consumers will react differently to meat reduction messages depending upon the individual. For example, findings from Onwezen and Weele (2016) show that distinct consumer groups can be identified based upon how they handle and react to information about meat's various impacts. Consumers who are 'indifferent' do not experience conflicting thoughts at all, those who are 'struggling' experience conflicting thoughts and accompanying negative emotions, and those who are either 'coping' or 'strategically ignorant' find ways to deal with conflicting thoughts, either through changing behaviour or by ignoring meat-related issues, respectively. Therefore, for a multitude of reasons, individual consumers will not think or react in the same ways when given a piece of information, and so this inconsistency must be understood and taken into account when trying to change behaviour through the provision of information.

Even if impacts are made clear to consumers, and short-term behaviour change is observed, there is no guarantee that such changes will be long-term, as effects from messaging tend to diminish over time. For example, Joireman et al. (2009) conducted a series of lab-based studies on the impact of messages warning about the depletion of a natural resource. Such warnings are not uncommon in the media, and they are generally intended to spur conservation efforts among the public. Using a resource dilemma framework in which individuals chose to harvest from a shared resource across multiple trials, the researchers presented participants with a warning that the common resource was running out. Such messages produced a short-term reduction in consumption, and the effect was particularly strong under conditions where the participants were told the resources would last for only a short period of time, but unfortunately, across the trials,

the harvesting quickly returned to pre-alarm levels as the warning about a dwindling resource began to wear off. Thus, studies such as these point to the fact that although awareness raising efforts may promote short-term behaviour change, there is no guarantee that individuals won't revert back to their original behaviours over time or once messaging is no longer present.

4.6 Chapter summary

Through the use of a MGM framework, this chapter gave an overview of the specific measures that could be potentially implemented in order to promote societal meat reduction, along with their associated strengths and weaknesses. In an ideal setting, it would be best to synergistically implement economic, regulatory, and information provision measures in order to promote a dietary culture shift towards more plant-based foods. As asserted in the ABC model of behaviour change discussed in Chapter 3, altering both personal attitudes and the external conditions in which people purchase and consume meat products will be the most effective strategy in shifting dietary behaviours to be healthier and more environmentally sustainable. However, even though a review of the literature shows the potential benefits of implementing more structural changes such as price manipulations and/or adjusting food procurement policies to create conditions that are more conducive to changing consumer behaviour, lack of public and political support for such measures acts as a major impediment to their utilization. As stated by Heiskanen et al. (2009), implemented strategies that have promoted sustainable consumption have mostly relied upon less coercive measures, as policy makers prefer to provide information and incentivize rather than restrict or punish consumers in order to promote behaviour change. This has been observed in New Zealand as well, as the Ministry for the Environment has recommended reduced-meat diets, but governmental dietary guidelines still include meat and many institutions like public schools, hospitals, universities, etc. are still procuring meat and dairy in large quantities, maintaining their place as "staple" foods in the New Zealand diet.

Therefore, educating and empowering consumers in order to change their meat-related attitudes and behaviours may be an initial first step that can lead to more structural changes. This is not to say that efforts to change the current structures that promote high levels of meat consumption are futile. These 'upstream' strategies that aim to shift

consumer environments actually hold much promise and they can play an important role in changing dietary behaviour, especially when used alongside 'downstream' efforts that address issues at the individual level. However, the dilemma is that without strong public and political backing, the utilization of economic or regulatory measures to reduce meat consumption is unlikely, especially in Western nations that are dominated by strong neoliberal ideologies which prioritize personal choice, limited state influence and free markets over collective action, high levels of state involvement and market regulation. Therefore, unless transformative cultural and political revolutions occur in these nations, a more realistic strategy may be to target and address individuals as an initial catalyst for change due to the tendency for Western nations to place a high value on consumer freedom and sovereignty.

Although likely a slow (but arguably more sustainable) approach, awareness raising among individual citizens can eventually lead to societal shifts through a multitude of ways. For example, if individual attitudes become more negative towards meat and more positive towards plant-based eating, social pressure to consume meat can begin to diminish as cultural acceptance of more plant-based meals and diets simultaneously increases (i.e. subjective norms gradually shift to encourage plant-based eating, while discouraging meat consumption). As more and more individuals change their ideas about what they (and those around them) should be eating, social and societal incentives for consuming more plant-based diets will also likely become more favourable. According to the ABC model for behaviour change (Guagnano and colleagues, 1995), as attitudes change and contexts likely begin to change alongside them, plant-based eating behaviours are likely to become easier and more prevalent. For example, social support for plant-based diets will increase as more and more individuals change their attitudes and behaviours, along with those around them (e.g. friends, family, co-workers, etc.). In addition, as individuals change their attitudes and behaviours more positively to plant-based foods, companies are likely to capitalize on shifts in demands for more plant foods over meat products, which has already been observed in many countries including New Zealand (1 News, 2019; Caldwell, 2019). As a result, companies are going to compete for this relatively "new" consumer market for plant-based foods by investing in and creating more plant-based foods for consumers. This influx of plant-based products will not only encourage greater consumption among individuals (e.g. through individuals seeing more appealing plant-based products in the store offered by some of their favourite brands) but

will also eventually lead to cheaper and even more attractive plant-based alternatives as more and more companies put money and resources into providing ever-more-attractive plant-based products that consumers want to purchase. Put simply, by changing individual attitudes and behaviours, cultural and social norms that promote meat consumption will begin to weaken as new 'subjective' norms are created that promote more plant-based eating. These changes will not only provide more social support for plant-based diets, but also encourage companies to place more money and resources into high-quality and attractive plant-based products for these shifting consumers, thereby increasing consumer demands for such products even further as the products become more normalized, attractive, and cheaper. Again, although this will likely be a slow process, it will be sustainable in terms of shifting societal meat consumption because it simultaneously addresses many of the underlying cultural and social barriers that currently encourage meat consumption both within New Zealand and abroad (e.g. convenience and cheapness of meat products, social encouragement from friends and family to eat meat, industries still investing in meat products, etc.)

Even if information provision alone is not an ideal solution in addressing consumer behaviour, in the case of reducing meat consumption in New Zealand and other Western nations, it would seem to be the most feasible option currently available. Furthermore, educating and empowering consumers on the various impacts of meat consumption may lead to additional benefits in addition to personal behaviour change. As consumers become informed and meat-related impacts become more widely recognized, public resistance to economic and regulatory measures are likely to decline, freeing governments from fear of public backlash when contemplating policy actions, like an environmental meat tax. Thus, creating a more well-informed citizenry on the issues tied to meat consumption may act as a first step in a pathway towards the acceptance and utilization of more structural measures that could further encourage the uptake of healthier and more sustainable plant-based diets.

If educating consumers seems to be a more feasible option than the suggested structural measures, the next question then becomes, "What are consumers current attitudes and behaviours in regards to meat consumption?" and "How effective are different types of information at altering meat-related attitudes and behaviours?" In Chapter 5, these questions will be explored, not only to elucidate current consumer beliefs in regards to

meat and its potential reduction, but also to provide insight into the potential effectiveness of different informative framings.

CHAPTER 5

UNDERSTANDING AND TARGETING CONSUMERS

5.1 Chapter outline

To begin Chapter 5, a short introduction is provided (5.2), followed by a synopsis of how increasing awareness of meat's environmental impacts among consumers might provide an opportunity to promote reduced-meat diets (5.3). Next, the various motivators that currently encourage consumers to reduce their meat consumption and how awareness raising efforts may enhance these motivators will be explored (5.4). Behavioural models, and more specifically, the Theory of Planned Behaviour (TPB) and Meat Attachment Questionnaire (MAQ), will be discussed in terms of their potential application in promoting meat reduction (5.5). Lastly, with an experimental framework based upon behavioural models established, justification for choosing New Zealand as a desirable location of study will be provided (5.6), along with a concluding chapter summary (5.7).

5.2 Introduction

It is difficult to devise effective meat reduction strategies without understanding the range of beliefs consumers hold in relation to meat consumption. For example, although the environmental impacts of meat consumption are well established in the literature, this does not mean that average consumers are aware of this information when making dietary decisions. If this is true, motivations to reduce meat intake could potentially be bolstered by raising consumer awareness regarding these environmental issues (assuming that enough consumers care about the health of the environment when learning such information, and are willing to make dietary changes). In addition, understanding the variety of motivations that lead (or potentially lead) to meat reduction behaviours could help identify why some consumers have already shifted their consumption patterns and what a typical shift towards meat reduction might look like for others.

Understanding consumer awareness and motivations for reducing meat consumption is an initial step in the right direction; however, information on how meat-eating behaviours

might be shifted through the provision of information could also be useful. Therefore, applying theories and models based in behavioural research may prove useful for the study of meat reduction. These models can be utilised as both explanatory and predictive tools, offering theoretical structures that have already been tested and verified through their use in past behavioural studies.

One theory in particular that has already been utilised in past meat consumption studies is the Theory of Planned Behaviour (TPB; e.g. Berndsen and Van der Pligt, 2004; Povey et al., 2001), and due to reasons that will be outlined below, focus and attention will mainly be paid to this theory/model. In addition to the TPB, the recently developed Meat Attachment Questionnaire (MAQ; Graca et al., 2015a) will also be discussed as a theory/model that may be of great use when seeking to understand, measure, and predict meat consumption and its potential reduction.

Establishment of the TPB and MAQ as appropriate behavioural and psychological models is an important step, but the choice in location is also important. New Zealand is a country which has received little attention within the literature to date, despite it being an appropriate and intriguing location to study potential meat reduction. With specific models, frameworks, and location established for studying meat eating behaviour, the end of this chapter will lead directly into the first study of the thesis, which will be outlined in Chapter 7.

5.3 Awareness of environmental impacts and behaviour change

Although the environmental impacts linked to meat consumption are well understood within the scientific literature, consumer understanding of these impacts may be more tenuous, if not absent. This is important, as consumers are likely to make purchasing and consumption decisions at least partially based upon their knowledge and beliefs regarding such actions. If consumers lack knowledge of meat's environmental impacts, their ability to use that as a motivation to reduce their meat intake becomes a lost opportunity. Studies have shown that in general, awareness of the environmental impacts of meat is low among the general public in multiple nations (see section 4.5.1). So, there seems to be a lack of connection between consumers' meat consumption habits and the environmental sustainability impacts related to these choices. This lack of awareness among consumers can only be detrimental to meat reduction efforts, as this knowledge would likely reduce

meat intake levels, even if only among consumers who are concerned for the environment. In fact, a study conducted by the UK government (DEFRA, 2011) that included over 3000 survey respondents found that 85% of consumers stated they may change their diet for the benefit of the environment, and a more recent study (Bailey et al., 2014) looking at global public opinions found that willingness to reduce meat consumption (in this study defined as 'likely to take action' or 'action taken') increased from about 30% for those unaware of climate impacts to 60% for those aware of climate impacts. Therefore, evidence so far supports a link between awareness regarding environmental impacts and willingness to change dietary habits, and this applies to the consumption of meat as well.

Raising awareness of the environmental impacts of livestock production and consumption may be an important issue to address, especially since consumers seem to be generally resistant and unwilling to reduce their meat consumption (Hoek et al., 2017; MacDiarmid et al., 2016). Although this lack of understanding exists, currently making environmental sustainability an unlikely motivator for reducing meat consumption for many consumers, those that are already aware of these environmental issues seem to be more willing to reduce their meat intake. This shows that environmental concerns could hold promise as a potential motivator for meat reduction alongside other motivators that are likely more prevalent and salient within society (e.g. health and animal welfare concerns). Exploring the literature on how this variety of motivators influence individuals to adopt reduced-meat and/or meatless diets (e.g. vegetarian, vegan) can act as a promising starting point to inform any studies wishing to gain a greater understanding of the current drivers for meat reduction.

5.4 Motivations for meat reduction

One way to better understand the main motivations that lead to meat reduction is to study consumer groups who have already reduced or eliminated meat from their diets. Jabs et al. (1998) did just this, and found the main motivations for becoming vegetarian could be used to break down the 'vegetarian' consumer base into two main groups, labelled 'health' and 'ethical' vegetarians. 'Health' vegetarians eliminated meat from their diet mainly due to perceived risk of consuming meat and belief in health benefits of reducing its consumption. This has been supported in a more recent study by Graca and colleagues

(2019) who found that consumers with food orientations towards health and naturalness were more likely to have already reduced their meat consumption and adopted a more plant-based diet. Contrary to this, 'ethical' vegetarians were those individuals who decided to eliminate meat from their diets primarily due to concerns for the welfare of animals during the livestock production process. Of course, many individuals shared both 'ethical' and 'health' concerns, but many readily identified which motivation was primary in their decision to stop eating meat. More recently, Fox and Ward (2008) added to these findings by communicating through an online message board with vegetarian respondents from the US, Canada and the UK about their reasons for adopting their meat-free diets. Most of the vegetarian respondents cited either health or animal welfare concerns as their primary reason for adopting a vegetarian diet, with environmental concerns only playing a significant role for one respondent, which is not surprising considering low awareness among consumers in regards to meat's environmental impacts. It is important to note that some of these motivators acted as catalysts into the diet while others were supplemented later on to help solidify the decision to continue a vegetarian diet. For example, some respondents stated that their process began with one motivation (e.g. ethical considerations) but was encouraged by other motivators (e.g. health and environmental concerns) later in the dietary change process. These additional motivators may have acted to buttress the initial decision, complementing the motivation that sparked initial interest, transforming a temporary dietary choice into an enduring lifestyle change.

Penny et al. (2015) added further to these findings by conducting an online questionnaire with not only vegetarians and vegans, but also 'meat reducers' (those who do not eliminate but consciously limit the amount of meat in their diet). Their study looked into the cognitive and behavioural factors associated with consumers' decisions to remove all or some of the animal proteins in their diets. They found that health concerns were the most prevalent reason for reducing meat consumption in all groups; however, meat reducers were more concerned with the cost of meat and weight control, while vegetarians and vegans were more likely to endorse environmental and animal welfare concerns. This means that although health concerns tied to meat consumption are more ubiquitous among consumers, environmental and animal welfare concerns regarding meat production and consumption seem to be more concentrated among those who most consistently reduce meat in their diets (i.e. vegetarians and vegans). One possible interpretation of these results is that although the health impacts of meat seem to spark

wider unease among the general public, environmental and animal welfare concerns may play a vital role in making the commitment to reduce meat even stronger. This is further supported by Hoffman et al. (2013), who conducted an online questionnaire in the United States looking at the differences between vegetarians who adopted their diets for health (i.e. 'health' vegetarians) or ethical (i.e. 'ethical' vegetarians) reasons. They found that ethical vegetarians had stronger convictions and exhibited stronger dietary restrictions in comparison to health vegetarians and concluded that ethical vegetarians may experience stronger feelings of conviction, consume fewer animal products and remain vegetarian for longer.

From these studies, one could safely assume that most consumers identify potential health benefits as their main motivation for trying to reduce their meat intake. This could be due to a multitude of reasons. Maybe the awareness of health impacts of high meat consumption are more prevalent among the general populace in comparison to other impacts (i.e. animal welfare and environmental sustainability impacts). Or, maybe awareness about the issues do not play a significant role and this phenomenon can simply be explained by a tendency for individuals to put concerns about their personal health over the concerns of issues that are more abstracted from their everyday lives (e.g. environmental sustainability, animal welfare). Most people likely think about what foods they should consume, how much they should exercise and when they should go to the doctor's office for a check-up much more often than they do for how animals are treated on farms or the status of environmental degradation. Also, health concerns may be most prevalent because individuals feel that they have a greater degree of control over their own health by performing simple actions like eating healthier foods, exercising more and drinking more water. Therefore, increased levels of awareness and apprehension towards health impacts, coupled with a higher perceived control over personal health and wellbeing (in comparison to potentially more detached issues like environmental degradation and/or animal welfare) may explain why concern for health is currently the most prevalent reason among consumers to reduce meat consumption.

Health concerns, costs and weight control seem to be the primary motivators for 'meat reducers' while health and/or animal welfare concerns provide the main motivation for the adoption of a meatless diet among vegetarians. Environmental concerns do not seem play as significant of a role in either of these groups, however knowledge and understanding of the environmental impacts of meat production and consumption are

quite low as already stated, possibly explaining its limited mention among consumers as a main motivation for meat reduction. So, although beliefs about health benefits tied to lower meat intake are currently more prevalent and salient throughout society in comparison to moral and ethical arguments, it may not be optimal to rely upon health arguments exclusively as an agent of dietary change, as shown by Penny et al. (2015) who found that animal welfare and environmental concerns can further strengthen the convictions and resolve of consumers who have decided to limit their meat consumption. This is extremely useful for the study of meat reduction, because convincing individuals to make the initial decision to reduce or eliminate meat in their diets is only half the battle, with other half consisting of making this change stable and long-lasting. So overall, it seems that health arguments are acting as the main motivation for consumers reducing their meat consumption, however; environmental sustainability and animal welfare considerations could make meat reduction behaviours more consistent and persistent over time, if efforts are placed on raising consumer awareness regarding these issues. Raising awareness on the environmental impacts of meat is key, as a systematic review by Sanchez-Sabate and Sabate (2019) found that the majority of consumers are unaware of meat's environmental impacts, that these impacts are often underestimated or ignored, and that there is confusion amongst consumers on how exactly more plant-based diets are more environmentally-friendly.

More studies looking at these motivations (and others that are possibly less prevalent) and how they change depending on current meat consumption behaviours could prove useful for understanding how to promote meat reduction among individuals and across populations. Established behavioural models may offer a framework from which to more systematically study meat-reduction motivations, and so these models and their use in the meat reduction literature will be discussed next.

5.5 Behavioural models and examples of their use in the meat consumption literature

Behavioural models are tools that can be utilised by those wishing to study behaviour change in a more structured and systematic way. An abundance of behavioural models exist, including the Theory of Planned Behaviour, Health Belief Model, Transtheoretical model, Social Practice Theory, and Diffusion of Innovation Theory (to name a few)

(Morris et al., 2012). These models have mainly been used to study health-related behaviour change (e.g. smoking cessation, dieting, etc.; Armitage & Conner, 2000), but have also found applicability in the study of pro-environmental behaviours (Nisbet & Gick, 2008), including sustainable consumption (Verain et al., 2015) and even meat consumption, specifically (Berndsen and Van der Pligt, 2004; Graca et al., 2015a; Povey et al., 2001). A systematic review on behaviour change studies found interventions which utilised behavioural theories and models had larger effect sizes when compared to interventions that did not include established models or theories (P = .049); in addition, those studies that utilised the Theory of Planned Behaviour in their interventions had the largest effects on behaviour when compared to other behavioural theories/models (Webb et al., 2010). Due to the apparent effectiveness of utilizing established behavioural models to promote behaviour change, especially as seen with the Theory of Planned Behaviour, it seems logical to apply such a theory to the study of meat reduction. Not only would this be advisable based upon the results of past behaviour change studies, but utilizing the Theory of Planned Behaviour, specifically, would allow the researcher to build upon past works that have already implemented the theory in the study of meat consumption behaviour. Thus, an overview of the theory and its potential utility in studying meat reduction will be provided.

The Theory of Planned Behaviour (TPB; developed by Ajzen, 1985) is a theory which has been used to help predict actions that are under volitional control, meaning ones that an individual can make using their own free will, but that can be hindered or promoted based on external factors, or ‘conditions’ as referenced in the ABC model discussed in section 3.4. The TPB assumes that human beings are ‘reasonably’ sensible, meaning that they consider the information and knowledge regarding a decision, in order to better understand the consequences and implications of that action. These considerations are then translated into beliefs about an action, and the resulting intention to (not) perform the action, which is the direct antecedent of actual behavioural performance.

So, there is an understanding that intentions to perform a behaviour should increase the likelihood of actual behavioural performance; however, where do intentions originate from and how are they influenced? According to the TPB, there are three main determinants which create and influence intention: personal, social and control factors (See figure 5.1. below). The personal factor is labelled “attitudes” and is comprised of the positive or negative feelings the individual associates with performing a behaviour; the

social factor is labelled "subjective norms" and is created by the individual's perceptions of whether or not a behaviour will be seen by others in a positive or negative light; and lastly the control factor is labelled "perceived behavioural control" (PBC) and is the individual's perceived degree of control in performing a behaviour. As expected, positive and negative attitudes, subjective norms and PBC associated with performing a behaviour combine to create either an overall positive or negative intention to perform said behaviour. The determination of whether the created intention will be positive or negative is dependent upon the relative influence and direction of these factors associated with the behaviour. Once an intention is formed and resulting behaviour performed, interactions and feedback the individual experiences after the performance are then analysed and incorporated into future intentions and behaviours. So, intentions and behaviours are not static over time, but rather flexible and open to change depending on what feedback an individual takes into account.

According to the TPB, attitudes are comprised of salient beliefs, which link behaviours with outcomes and attributes that can be either positive or negative (Ajzen, 1985). For example, a person may believe that eating a more plant-based diet (behaviour) is tied to a myriad of outcomes that incentivise or deter the behaviour. They may believe that the adoption of a plant-based diet would reduce their food-related carbon footprint and risk for cardiovascular diseases (positive outcomes), but also make them feel weak while also minimizing the consumption of some of their favourite foods (negative outcomes). The attitude towards consuming a more plant-based diet will therefore be determined by an individual's evaluation of the probability and relative importance of each outcome. Therefore, an individual who holds beliefs about a behaviour that are comprised of mostly optimistic outcomes will have an overall positive attitude towards performing the behaviour, while an individual who holds beliefs about a behaviour that consist of mainly undesirable outcomes will have an overall negative attitude towards performing the behaviour. These beliefs are termed "behavioural beliefs" and they form the foundations upon which attitudes are built.

Subjective norms, like attitudes, are based upon beliefs as well. However, these do not arise from internal thoughts and feelings alone but rather from an individual's beliefs about how others view and judge performing or not performing a specific behaviour. For instance, if an individual believes that a majority of their peers will judge an action as "good", there will be social pressure to perform the behaviour; however, if an individual

believes that a majority of their peers will judge an action as "bad", there will be social pressure to *not* perform the behaviour. How others view and judge an action is also mediated by the individual's desire to align themselves with the views of their peers. So, if the individual performing the behaviour does not mind the opinions or judgements of others regarding that specific behaviour, the relative influence of subjective norms will be quite weak. Thus, if an individual cares about the opinions of their peers and holds beliefs that they will have strong opinions (either negative or positive) on what behaviours they are performing, those beliefs will have a significant impact on subjective norm. These beliefs that form the subjective norm are called "normative beliefs".

As with attitudes and subjective norms, PBC also has determinants. These manifest in either internal or external control beliefs. Internal control beliefs can include abilities, emotions, skills and knowledge of the individual in relation to the proposed behaviour, while external control beliefs can include barriers and opportunities to perform the given action. Therefore, individuals who feel that they have the capacity to perform a behaviour and are given opportunities (or lack of barriers) to perform it will have a greater sense of PBC. Thus, the performance of a behaviour is not only dependent on an individual's attitudes and subjective norms, but also on the amount of control an individual has in a given scenario, which can fluctuate depending on circumstances. For instance, imagine if someone wished to make a meat-free Christmas dinner for their family. This person knows there are vegetarian substitutes for some of their favourite traditional meat-based dishes at the market, but they have no experience with vegetarian cooking. Even knowing the market offers an opportunity to purchase vegetarian substitutes (i.e. external control belief), the person may not have enough confidence in their cooking abilities (i.e. internal control belief) to continue with their meat-free Christmas dinner plan. As seen with this example, the combination of internal and external control beliefs amalgamate into the concept of PBC, which in turn acts alongside attitudes and subjective norms to determine intentions. Note, however that the 'perceived' part of PBC is quite important as this only includes how much control a person *believes* themselves to have rather than how much *actual* control they possess. This is why actual behavioural control acts as a mediator for both PBC and the transfer of intentions into behaviour (see figure 5.1 below). A person can *believe* in their control to eat healthy, but their *actual* control may waver when the opportunity to eat a sugary, fat-filled cheesecake presents itself.

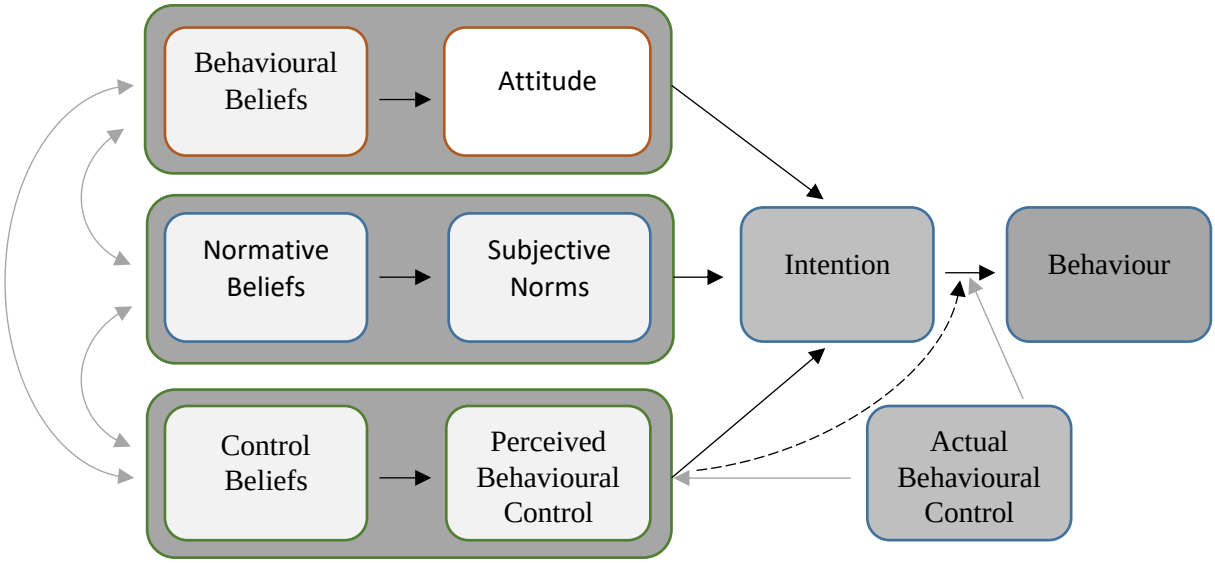

Figure 5.1 Theoretical Model for the TPB. Adapted from Ajzen (2006).

Criticisms of the theory exist, including the overreliance on rationality and the exclusion of more 'unconscious' influences that are not directly linked with the behaviour in the mind of the actor, like underlying emotions or social influences, as found by Conner et al. (2012) when studying blood donation. This criticism has been refuted by Ajzen (2015) however, as he states that the theory does not rely upon the rationality or objectivity of actors, but rather is solely focused on how beliefs influence behavioural intentions, no matter whether they are 'rational' or 'irrational' in nature. Although not the focus of this thesis, if one wished to specifically identify the roles of more underlying factors, other models would likely be more appropriate than the TPB. For instance, the Social Learning Theory (i.e. Social Cognitive Theory) is specifically designed for understanding social influences on behaviour (e.g. Rosenstock et al., 1988) and the Fogg Behaviour Model offers an excellent framework to explore the influences of underlying emotions on behaviour (e.g. pleasure, pain, hope, fear, social rejection, etc.; Fogg, 2009).

In addition to the 'rationality' criticism mentioned and refuted above, the theory has also been critiqued for its static nature (Sniehotta et al., 2014), as intentions and resulting behaviour are a constant process that takes place over time, rather than at one specific point in time. Although questioning this seems worthwhile, this criticism seems to not take issue with the TPB specifically, but rather human behaviour itself. People make

commitments, change their minds, acquire certain behaviours while stopping others, etc. which makes behaviour a variable that is in constant flux over time. Therefore, this is not an issue with the TPB per se, but rather a criticism of not incorporating longitudinal design into experimental behaviour change studies. Of course, all models will have their strengths and weaknesses depending upon the specific goals of the researcher, and if ones wishes to explicitly focus more on changes occurring over time, a model like the Transtheoretical Model of behaviour change may be more appropriate due to its inclusion of "stages of change" (Prochaska and Velicer, 1997).

Furthermore, as stated by Sniehotta et al. (2014) some studies have found the TPB to provide inconsistent results in terms of its predictors and their ability to align with actual behaviour change outcomes. For example, attitude change sometimes did not lead to changes in intentions, or intentions did not lead to changes in behaviour, etc. Ajzen (2015) also responded to this criticism, stating that the TPB is meant to be a framework for studying intentions and behaviour, and that the model is not specifically designed for behaviour change. Rather, the TPB can be used in order to identify the beliefs that are most likely to lead to changes in intentions and behaviours, and does not offer any additional guidance or insight into the strategies or techniques that should be used to actually change said behaviour.

Despite these criticisms, the three TPB components, along with the resulting intentions they form, have been shown to accurately predict both the frequency and quantity of meat consumed; however, attitudes were found to be the strongest predictor, followed by PBC and subjective norm, respectively (Berndsen and Van der Pligt, 2004; Graca et al., 2015a; Povey et al., 2001; Saba and Di Natale, 1998). Findings from these initial studies have been expanded upon with the recent creation of the Meat-Attachment Questionnaire (MAQ), developed by Graca et al. (2015a), which strives to identify how strongly a consumer wishes to continue (or stop) the consumption of meat, and which factors mostly heavily contribute to those wishes. The MAQ is comprised of 16 questions, which are grouped into four first-order dimensions: hedonism, entitlement, affinity and dependence; and one second-order dimension: meat attachment. Hedonism refers to pleasure gained from consuming meat, affinity to positive attributes associated with meat, entitlement to the idea of having a right to consume meat, and dependence to the feeling of meat being a necessity in the diet. These four first-order factors combine to create the second-order factor, meat attachment, which is how devoted a person is to the act of consuming meat.

The MAQ has been shown to predict willingness and intentions to reduce meat consumption above and beyond the predictive power of the TPB components alone (Graca et al., 2015a), making it a useful addition to the core TPB components when studying consumer willingness to reduce meat intake. Therefore, the TPB and MAQ could prove useful in future studies on meat consumption by providing theoretical foundations and practical tools that offer both explanatory and predictive power for both current intake and reduction potential. Given that TPB and MAQ have been selected for studying potential meat reduction, the population that these instruments will be tested on must be established, and I will argue New Zealand is an appropriate and favorable location.

5.6 Reasoning for New Zealand as a location for study

Although the current literature gives some basis and direction for studies involving meat reduction, like the use of the TPB and MAQ, it is clear as stated by Corrin and Papadopoulos (2017) and Ruby (2012) that the study of meat reduction and vegetarianism is still quite young and limited in scope. Most studies on meat consumption and motivations for its reduction have been concentrated in the United States and the United Kingdom (Corrin and Papadopoulos, 2017; Ruby, 2012), with only two studies (to the author's knowledge) conducted in New Zealand (Allen et al., 2000; Potts and White, 2008). Both studies contributed to understanding the social aspects of vegetarianism (e.g. treatment of vegetarians by peers, opposing political views, etc.), but information on motivations for meat consumption and its potential reduction among the broader consumer base in New Zealand is still absent. New Zealand is an important place to study meat reduction for three main reasons, in addition to those outlined more generally for developed nations (refer back to section 2.5):

1. According to the Ministry for the Environment (2018), nearly half of GHG emissions in New Zealand are a result of ruminant fermentation (i.e. methane emissions) and the release of urinary nitrogen (i.e. nitrous oxide emissions).

2. New Zealand has the 6th highest per capita meat consumption (excluding fish) rate in all the world (FAO, 2013), and the majority of this meat is local (raised in New Zealand). This means that reductions in meat intake could result in significant

environmental and public health benefits. New Zealanders now, compared to 10 or 20 years ago, are consuming more chicken and pork while eating slightly less beef and significantly less lamb (Poultry Association of New Zealand, 2018)

3. The raising of livestock is seen as part of the cultural heritage, resulting in meat consumption being associated with ideas of national pride and support (Potts and White, 2008).

As mentioned previously, meat and dairy are two of New Zealand's major exports, where the majority of such products are shipped to other countries for consumption. This is not due to a lack of consumption of meat and dairy among New Zealand consumers, but rather just excesses in production that are traded on global markets. Production of meats (including chicken and pork) and dairy have become more "intensified" on average within New Zealand, but a significant proportion of beef producers are trying to remain "traditional" in the sense that animals are still pasture-raised with minimal confinement. As discussed previously, this increase in animal farming for meat in combination with intensification have resulted in degradation of soil qualities, pollution of waterways, and significant increases in greenhouse gas emissions. Thus, encouraging global meat reduction is likely to lead to the greatest environmental benefits for New Zealand (assuming farmers shift to producing plant foods or rewilding of lands), but enacting changes on the global scale are even more complex and slow in comparison to country-wide changes. Therefore, raising awareness among New Zealand consumers and encouraging them to reduce their own consumption of meats may be a more feasible and readily attainable goal that would not only come with its own environmental benefits, but may also get consumers and the government to support shifting focus of exports to products (or versions of products) that are less environmentally damaging.

Again, major and quick policy changes from the government to actively discourage meat consumption is unlikely to occur due to their "hands off" policy on farmers and agricultural products. Government bodies, like the Ministry for the Environment, can encourage reduced-meat diets for environmental benefits, but these recommendations will likely be overshadowed by dietary guidelines that still recommend meat along with continued procurement of meat products among large institutions like government offices, schools, hospitals, universities, etc. In addition, the vast amounts of money spent

by the economically powerful meat industry to further encourage meat consumption through advertisements and media campaigns is likely to continue. Therefore, shifting to more plant-based diets will be an uphill battle, and is a movement likely to originate and gain initial support from large bodies of consumers becoming more aware of the negative impacts associated with meat production and consumption. Government has an opportunity to take strong steps towards promoting meat reduction, but as seen in many other Western nations, steps taken are likely to be minimal and non-invasive while being largely overshadowed by factors that promote consumption. Also, the meat and dairy industries find ways to distort the findings of prominent food scientists and their reports to encourage maintaining the status quo of meat consumption (Beef and Lamb New Zealand, 2019), like the EAT-Lancet Commission's report on healthy diets from sustainable food systems, which clearly encourages a shift away from meat and towards plant-based diets for environmental and public health benefits.

Consumer groups also differ within New Zealand, as those of Māori and/or Pacific Island descent eat more meat, especially lamb and fish/shellfish, in comparison to those of European descent (Ferguson, 2002). In addition, men consume more meat overall when compared to women, matching trends in other countries. Overall, as mentioned previously, New Zealand consumers have slightly decreased their beef intake, greatly reduced their lab intake, while slightly increasing their chicken and pork intakes. Overall though, New Zealanders still have one of the highest per capita meat countries in the world and thus, reductions in meat consumption, especially beef would greatly improve environmental impacts along with health benefits. This is not to say that reductions in chicken and pork consumption would also have tremendous benefits, but just that reductions in beef would likely have the greatest.

These factors make New Zealand not only an appropriate, but interesting location to study meat consumption and its potential reduction. Therefore, in order to build upon this relatively limited research topic, efforts should first be directed at understanding the thoughts and behaviours of New Zealand consumers in regards to meat consumption and the consideration of its reduction in the diet. Not only will this add more scope to the meat reduction literature, but will also lay the foundation for supplemental studies focusing on how behavioural shifts toward reduced meat intake might occur, both within New Zealand and in the global context.

5.7 Chapter summary

It has been established that awareness of meat's environmental impacts is low, but that higher levels of awareness, if achieved, could result in lower rates of meat intake. Environmental concern is not the only motivation that leads to meat reduction however, as health concerns, animal welfare concerns, financial costs, etc. all can play an influential role in a person's decision to reduce or abstain from meat. It is important to keep in mind that these motivations often do not act in isolation, and that one motivation can be primary while others secondary, and that their relative strengths can vary over time and between contexts. In terms of understanding awareness and motivations, in addition to the general behaviour of meat consumption, it is helpful to utilise behavioural models in order to frame how change may best occur. Despite some criticisms, the TPB has been shown to be an effective model for studying behaviour change, including specifically for the case of studying meat intake and its potential reduction. Due to its success, the TPB along with the recently developed MAQ will be utilised as a framework for studying potential meat reduction in New Zealand; a nation, that for reasons mentioned above, will be the focus of studies contained in this thesis. With the establishment of the TPB and the MAQ as useful models and New Zealand as a proper study location, Chapter 6 will outline proposed research questions and broad methodologies employed in the two studies that comprise this thesis, along with how their findings can complement one another.

CHAPTER 6

METHODOLOGIES FOR STUDIES

6.1 Chapter outline

The purpose of this chapter is to summarise and provide direction based upon the discussions and conclusions made throughout previous chapters. Chapters 7 and 8 that follow are based upon two separate studies that were conducted as part of this thesis work, and the goal of this intermediate chapter is to establish the research questions and methodologies used for each study, and how both studies complement one another and come together to provide additional insights on the thesis as a whole. There are six sections in this chapter: the first is a quick overview of the literature, in order to re-establish the overarching goal of the thesis, in addition to the scope of the two studies and how they fit together (6.2), the second provides an outline of the research questions for Study 1 (6.3), the third discusses the broad methodologies to be employed in order to answer proposed research questions for Study 1 (6.4), followed by a similarly framed section for Study 2 (6.5), and a final section on how findings from both studies can offer insights on meat intake and its potential reduction in New Zealand (and elsewhere), as well as its potential utility for future studies (6.6). The intention of this chapter is to provide the scope, purpose, and methodologies of both studies, in addition to how they fit into the larger thesis. More detailed methodological information for both Study 1 and Study 2 are provided in Chapters 7 and 8 that follow.

6.2 Literature overview and scope for studies

Before discussing the scope of the studies and what information they are designed to provide, a quick overview of what Chapters 1 through 5 have established seems appropriate. Humanity is contributing to a wide array of environmental issues, and the raising of animals for food is one of the most widespread and impactful human activities. Moving society away from meat-heavy diets and towards more plant-based diets could not only mitigate some of these impacts, but also simultaneously promote improved public health. Thus, the overarching goal of this thesis is to better understand how a

transition to a more plant-based society can best be accomplished, to promote environmental sustainability and improved public health.

The issue, however, is that high levels of meat consumption are the norm in many developed nations, and thus changing such a behaviour on a societal scale is not only a matter of individual agency, but also the structures within society that perpetuate and encourage the maintenance of said behaviour. There are a plethora of approaches that can be utilised in efforts to shift society, including those that take an economic, regulatory, and/or informational focus. Due to a multitude of factors, regulatory and economic measures are unlikely to occur quickly enough and at the scale required in order to reduce meat consumption on their own, and thus informational approaches were chosen as the main focus of this thesis. Before exploring informational approaches and designing interventions however, it was advisable to first understand targeted consumers. An overview of the literature found that consumer awareness in regards to meat's environmental impacts is low and that there are some 'typical' motivations for meat reduction (e.g. health, animal welfare concerns, etc.); in addition, behavioural models like the Theory of Planned Behaviour (TPB) were found to be useful when studying meat consumption and its potential reduction. However, these findings have not been replicated in New Zealand, a country that for multiple reasons deserves attention in terms of its current meat intake and potential reduction.

Thus, the first goal of the thesis is to understand New Zealand consumers, their awareness of meat's environmental impacts, their attitudes and attachment towards meat, their willingness and intentions to reduce their meat intake, and their agreement with proposed meat-reduction structural measures (e.g. environmental meat tax). In addition, by understanding the average New Zealand consumer, potential similarities and/or differences between consumers in separate nations can be identified. Understanding New Zealand consumers acts as a foundation for the second goal of the thesis, which is to better understand how (and what types) of informational interventions prove effective at changing people's meat-related attitudes and behaviours. So in basic terms, the purpose of Study 1 is to provide an understanding of New Zealand consumers in terms of potential meat reduction, which provides a solid foundation for Study 2, which focuses on testing different informational approaches/subjects aimed at promoting societal meat reduction.

Due to the difference in aims of Study 1 and Study 2, the methodological approaches for both are also different. Since Study 1 sought to capture meat-related information about the average consumer in addition to the wide variability across the consumer spectrum, a study design that can gain information from large numbers of individuals was needed. Instead of an in-depth or longitudinal exploration of each consumer, Study 1 sought to acquire the opinions of consumers on primary research questions that could easily be compared and analysed as a whole and across sub-groups (e.g. gender, age, meat intake, etc.). Thus, a large sample size was needed to capture enough variation in order to make such comparisons feasible and to determine the attributes of an 'average' consumer. A downside of such a design is that more detailed and thorough information in regards to each consumer's meat-related thoughts, opinions, and behaviours was not captured as 'trends' and 'averages' were being sought.

Study 2 built upon Study 1, and therefore utilised some of the same specific models and/or measures, but with a more narrowed aim of seeking to understand the short and long-term impacts of specific informational interventions. Since Study 2 sought to measure changes over time within consumers in response to different forms/types of information provision, a smaller sample size was sought (in comparison to Study 1) due not only to issues of feasibility, but also in order to gain more detailed and longitudinal data on each individual consumer. In this research design, more detail was desired in terms of what information is impactful, in what ways, with which consumers, and over what time scales. A weakness of this approach was that it did not capture the wide variability between consumers, and thus generalisability of the results was not as powerful as in Study 1. Both the designs and methodologies of Study 1 and Study 2 were pre-approved by Maori consultation and the University of Otago's ethics committee (D17/064**)** before research began (see Appendix A and B, respectively).

6.3 Study 1 research questions

With the scope of Study 1 and Study 2 understood, outlining the research questions and specific methodological approaches utilised in each study is the next issue at hand, beginning with Study 1. Some progress has been made towards understanding consumers' lack of awareness of meat's environmental impacts and how different motivations influence an individual's choice to reduce their meat intake. However, the literature is

still nascent and evolving and requires further study, especially in New Zealand. With only a handful of studies to date having utilised the Theory of Planned behaviour (TPB) to study the phenomenon of meat consumption, its usefulness can be expanded upon with further study. In addition, the Meat-Attachment Questionnaire (MAQ), which has only been used in one study to date (Graca et al., 2015a), could benefit from further analysis as this could potentially add or detract from its validity as an explanatory and predictive tool. Therefore, further study involving the TPB and MAQ could add to the growing literature on meat consumption behaviours.

Not only can both the TPB and MAQ help in understanding individual meat consumption, but they can also be used to better understand consumer agreement with proposed structural meat reduction measures (e.g. meat taxes, plant based food subsidies), which is currently missing in the present literature. This is important, as the implementation of strategies to reduce meat consumption at the structural level is likely to lead to larger and quicker shifts in consumption behaviours. However, current lack of political will and public acceptance for such measures act as impediments to change and thus, understanding what factors influence acceptance and support for such measures could act as a starting point for more impactful and widespread changes in eating behaviours, both at the individual and societal scale. Hence, a study that focuses on consumer acceptance of potential meat reduction structural measures in addition to willingness and intentions to reduce personal meat intake could provide important insights for how individual and public actions might be connected in the case of societal meat reduction. Linking individual and public efforts is seen as vital for making significant social change (Berkes et al., 2000), meaning that substantial alterations in such a prevalent societal behaviour (i.e. meat consumption) will likely require efforts in both the public and private realms.

Based upon these considerations, Study 1 of this thesis aims to add to the literature by studying both consumer awareness of meat-related environmental impacts and consumer motivations for reducing meat intake in the New Zealand context. This will not only provide insight into current trends present in New Zealand, but also provides more evidence for the utility of the TPB and MAQ as explanatory and predictive tools for individual meat consumption and agreement with proposed structural measures. Given the need for more studies on consumer attitudes towards meat reduction and the particular lack of information on New Zealand consumers specifically, the focus of Study 1 includes four principal research questions (RQ):

RQ1.	Do consumers place ‘eating less meat’ below other sustainable food behaviours in terms of environmental benefit?
Hypothesis 1	Consumers will place ‘Eating less meat’ in the bottom half of sustainable behaviours in terms of environmental benefit.
RQ2.	Are there any significant differences in motivations to reduce meat consumption between consumer groups?
Hypothesis 2	There will be significant differences between consumer groups for the various motivations to reduce meat consumption (e.g. weight control, taste, health benefits, etc.)
RQ3.	Can any of the TPB components (attitudes, subjective norms, PBC) accurately predict willingness and intentions to reduce meat consumption? Can they also predict current meat reduction behaviour and agreement with proposed structural measures?
Hypothesis 3	The TPB component ‘attitude’ will accurately and consistently predict willingness and intentions to reduce meat consumption, along with predicting current meat reduction and agreement with proposed structural measures. Subjective norms and PBC will be more inconsistent and will not be a predictor for all variables.
RQ4.	As initially found by Graca and colleagues (2015a), does the MAQ provide further predictive power for these phenomena above and beyond that of the TPB in the New Zealand context?
Hypothesis 4	The MAQ will offer predictive power above and beyond the TPB components for willingness and intentions to reduce meat consumption and agreement with proposed structural measures.

6.4 Study 1 methodological approach

Since thoughts and opinions of the “average” New Zealand consumer were sought, methodologies that lended themselves to studying large populations, like acquiring representative samples and distributing questionnaires (Kelley et al., 2003), was utilised. Acquiring a representative sample of New Zealand consumers allowed for the study of meat reduction motivations across the entire consumer spectrum, from those who consume meat regularly to those who don’t at all, which offered insights into how a typical transition towards meat reduction might occur in New Zealand. Also,

representative information of the broader consumer base, as acquired through the study design, could be vital in guiding strategies and methods of meat reduction campaigns aimed at encouraging the general public to reduce their meat intake, whether it be for sustainability, health, or any other reason. Such a study directly answered calls for more population based studies seeking to explore and understand the motivational aspects involved in shifting consumers toward more sustainable plant-based diets (Hartmann and Siegrist, 2017), while also being only the second study to utilize the MAQ, potentially adding to its cross-cultural validity as both an explanatory and predictive tool.

Since the main purpose of Study 1 was to capture the thoughts, opinions and behaviours of the average New Zealand consumer with regard to meat consumption, and since data was not readily available to properly answer this question, primary sources were used. Since an understanding of the broad consumer base was desired, a methodological approach that provided large sample sizes in order to include the wide variation in consumer thoughts and behaviours was optimal (Kelley et al., 2003). With this in mind, an online questionnaire for Study 1 that could be easily distributed to large groups of consumers seemed appropriate, as it captured the wide array of meat-related thoughts, feelings, ideas, and habits that are present among New Zealand consumers. This cross-sectional approach, as described by Babbie (1990, pg. 89) was, "designed to study some phenomenon by taking a cross-section of it at one time". In other words, utilising a questionnaire allowed for a snapshot in time, capturing the wide variation in meat-related thoughts and behaviours within the studied population. The intention of a cross-sectional study was not to measure effects of a phenomenon over time or in response to an intervention, but rather to provide a general portrait of the phenomenon, within the study population, at time of measurement. Thus, since the aim of Study 1 was to understand New Zealand consumers and their meat-related attitudes, feelings, thoughts, and behaviours, a cross-sectional online questionnaire was distributed to hundreds of consumers is an efficient and proper method of inquiry. Acquiring a representative sample of New Zealand consumers from across the nation would be very difficult and time consuming if sourced alone, thus the acquisition of the sample was facilitated by ResearchNow (now renamed Dynata; www.dynata.com), an online market research company which has connections with thousands of consumers in multiple cities across New Zealand through their online databases.

6.4.1 Strengths of questionnaire

A questionnaire was an appropriate method of data collection in Study 1 for a multitude of reasons. First, expansion of the TPB's and MAQ's predictive and explanatory powers in regards to meat-related thoughts and behaviours was sought. Past studies on meat intake that utilised the TPB and MAQ (discussed previously in section 5.5) utilised quantitative questionnaires, and so a continued use of quantitative questionnaires allowed for more direct comparisons and/or extensions of previous findings. Furthermore, qualitative measures (i.e. interviews, open-ended questions) would have provided depth on some of the studied research questions, but would have failed to provide population-based data on the strengths of different motivations and factors related to meat intake and its potential reduction. Therefore, a quantitative approach was preferred and as a result, a questionnaire comprised mostly of scalar questions was the measurement tool used to answer the proposed research questions for Study 1. Categorical questions were also utilised, but mainly for breaking respondents down by current meat intake habits or demographic variables. The strength of using mainly scalar over categorical questions was that it allowed for attributes to be measured along a spectrum, providing respondents with a format that allowed them to choose from a plethora of points rather than broader categories that may or may not have encompassed their exact feelings and/or thoughts. Not only did this allow for more refined answers, but provided data that was more continuous and normally distributed, which is typically preferred for a variety of statistical techniques (Martz, 2017).

An additional benefit of utilizing a questionnaire in an online format specifically, was the anonymity it provided to participants, which helped to limit bias towards answering questions in a way that was deemed desirable by other participants and/or the researcher (Brace, 2008; pg. 32). This became especially relevant when considering the topics covered (e.g. environmental impacts, health implications, questionable ethics regarding treatment of animals), as some individuals may have felt uncomfortable answering these questions with complete honesty if other individuals were present (e.g. other participants in a focus group setting). Of course, participants could still have held a tendency to answer questions in a way that they believed was favourable to the researcher, and so it was made clear in the questionnaire that their honest opinions were desired and that there were no 'right' or 'wrong' answers for any of the questions.

6.4.2 Considerations for questionnaire

When creating a questionnaire, one needs to ensure that questions are clearly written and that the questionnaire has a natural flow so that participants do not become confused. If a questionnaire is designed in such a way that is confusing or unclear for participants, the quality of data received may be poor due to the responses to the questions failing to accurately represent the respondent's thoughts. The questionnaire designer is not present when participants are answering questions, unlike in structured interviews where questions can be explained or expanded upon to offer more clarity and guidance. Hence, due to this limitation the questionnaire designer must strive to make questions as clear and straightforward as possible to ensure that participants provide accurate answers to the questions posed.

There are many things to consider when designing a questionnaire (Stats NZ, 2015). Simple language should be used throughout to ensure that participants of all educational backgrounds can properly answer each question. Questions should not be ambiguous, meaning that they are broken down to be as specific as possible for the respondent in order to avoid misguided responses. Care should be taken to avoid leading questions, which might influence a respondent to answer in a specific way due to the perception that certain answer choices are "right" or "better" while others are "wrong" or "worse" (Stats NZ, 2015, pg. 43). The questionnaire designer must also strive to minimize the use of language that may pull or influence a respondent to particular answers. Questions that utilize scales as an answer format should be checked for balance, meaning that the number of negative and positive answer options on the scales will be equivalent in both number and perceived 'strength' (Stats NZ, 2015; pg. 43). Also, questions based on presumptions (e.g. thinking back to when you first reduced your meat consumption, what were the most important reasons for doing so?) should only be used when those presumptions are confirmed by previous questions (e.g. have you ever or are you currently reducing your meat consumption?).

In addition, the questionnaire designer must take care to avoid double-barrelled questions that could confuse respondents, leading to invalid data (Blair et al., 2013; pg. 207). For example, a double-barrelled question could be: "Please rate your agreement with implementing a meat tax or plant-based food subsidy in order to reduce food-related

environmental impacts." When contemplating this question, a respondent may disagree with a meat tax but agree with a subsidy on plant-based foods. By having these two concepts together in the same question, the respondent may become confused about how to answer it and the data may be invalid. Not only could the respondent become confused and give an inaccurate answer, but the researcher would not be able to pull any meaningful insights from the data gathered due to not knowing whether the answer selected is relating to one concept or the other. Therefore, caution is required when creating questions to eliminate potential double-barrelled questions.

If the desire is to acquire quantitative data, the questionnaire should be comprised mostly of closed questions, meaning that individuals have to select from pre-determined categories or respond to provided scales to answer a question (e.g. Likert-type scale). When using this design, the researcher must be careful of bias when choosing which answers are provided to respondents, as certain answers could preferentially be chosen even though other answers may be just as valid for respondents. Therefore, the researcher must be sure to carefully consider and include as many relevant answers to questions as possible (within reason), in order to minimize the chances of respondents choosing answers simply due to availability (Stats NZ, 2015; Pg. 44). Despite potential bias, an advantage of closed questions is that the most desirable information is obtained and that this information is provided in a way that is easier to compare and analyse between respondents. Put simply, closed questions offer a structure for participants, thereby limiting their ability to answer tangentially to the question at hand. Therefore, closed questions, with an 'other' answer option included where appropriate, allow the researcher to be clear with the respondent about the type of answers sought, while also providing an outlet for expression of ideas or opinions that are potentially not included in standard answer options.

The ordering of questions is also a vital consideration when designing a questionnaire, as responses to some questions may be influenced by having answered previous ones. In this regard, an online questionnaire has the advantage of breaking questions into pages that make future questions impossible to access until previous ones are answered (Brace, 2008; pg. 33). Also, the questionnaire can be structured in such a way to prevent backtracking, so respondents cannot go back and change a previous answer when looking at and answering future questions. In addition, the placement of questions can be done in such a way to minimize the influence that early questions might have on answers for later

questions, in order to gauge consumer opinions with minimal interference from the questionnaire itself. For example, questions that seek to gauge general attitudes and beliefs of meat's impact on the environment (e.g. rate how much you agree with the following statement: Eating meat is disrespectful towards life and the environment) can be placed before questions that ask to rate agreement with suggested structural solutions for these environmental impacts (e.g. meat taxes). This would ensure that responses on beliefs regarding meat's environmental impacts are solely produced by the participant and not influenced in any way by questions discussing the government's potential use of suggested meat reduction measures.

6.4.3 Study 1 methodology and Study 2 follow-up

From the discussions above, it is apparent that a quantitative online questionnaire distributed to hundreds of consumers to acquire representative, cross-sectional data was a preferable method of choice, which broadly sought to understand the average New Zealand consumer in terms of their awareness of meat's environmental impacts, their attitudes and attachment towards meat, etc. The strengths of using a questionnaire, along with proper design and distribution considerations have been outlined. With the research aims and methodologies roughly described for Study 1 (more specifics provided in Chapter 7), Study 2's aims and methods can be introduced and discussed to allow the reader to understand both Study 1 and Study 2 individually, as well as to discuss how the two studies complement one another contribute to the overarching objective of the thesis (i.e. how best to promote a societal shift to more plant-based diets).

6.5 Study 2 methodological approach and research questions

Results from Study 1 were intended to offer a baseline understanding of New Zealand consumers and their meat-related beliefs and habits. The next logical step, therefore, was to use this information in order to test how New Zealand consumers might be directed away from meat-heavy diets towards more plant-based diets. For instance, the results of Study 1 supported the MAQ as a verified and useful tool for measuring meat attachment, while also being strongly linked to positive attitudes towards meat, willingness and intentions to reduce meat in the diet, agreement with proposed meat reduction structural

measures, etc., so strategies that aimed to reduce meat attachment were hypothesised to be effective at promoting meat reduction. However, the challenge was identifying how such relevant variables (e.g. meat attchment, attitudes, subjective norms, etc.), might be altered. Luckily, the primary motivations of consumers who have already reduced or eliminated meat from their diet were identified in Study 1 and thus this data was utilised in order to create meat-reduction interventions for Study 2. For example, findings from study 1 showed that certain motivations (e.g. health concerns, environmental concerns, animal welfare concerns) were more prevalent among reducers and/or abstainers when compared to standard consumers, meaning that those motivations may play an important role in changing attitudes and meat attachment to a degree that promote actual meat reductions. Utilising similar questionnaire design from Study 1, Study 2 tested changes in meat-related variables over time in a longitudinal format, as short and/or long-term changes may indicate which (and to what degree) interventions are effective. Hence, a design for Study 2 that measured meat-related variables (through previously validated questionnaires from Study 1) immediately before, immediately after, and a longer time period after (e.g. 1 month following) motivation-based interventions was deemed as an appropriate approach to determine how different interventions might promote meat reductions over time.

Providing information to consumers on the potential benefits of reduced meat intake could increase motivations to reduce and also alter attitudes and attachment, possibly to levels that approach that of consumers who are already reducers or abstainers. Understanding the impact (or lack thereof) of these different arguments on consumer attitudes, meat attachment, willingness and intentions to reduce meat intake, and acceptance of hypothetical meat-focused structural measures could be beneficial for any individual, group, or organization that is planning, implementing, or promoting meat reduction initiatives now or in the future. As stated by Joyce et al. (2012, pg. 5), “Further research is required on what messages might resonate to support broader food policy changes and the medium through which these messages are delivered.” Studies have looked into how different messages affect meat-related attitudes, behaviours, and/or policy acceptance (e.g. Bertolotti et al., 2016; Cordts et al., 2014; Graham and Abrahamse, 2017; Stea and Pickering, 2017; Vainio et al., 2018; Whitley et al., 2017), but many of these studies offered limited substance in the information provided to research participants (e.g. short descriptive paragraph, newspaper article, etc.). As suggested by

Whitley et al. (2017), a study that measures consumer response to more thorough information provision would be useful for determining the effectiveness of different arguments / messages, especially in the context of framing potential meat-reduction policies. Furthermore, as stated by Graham and Abrahamse (2017), more longitudinal studies that observe the short and long-term effects of information provision on meat-related variables are needed; in addition, studies that compare the relative effectiveness of different motivational framings (e.g. health, animal welfare, etc.) have been suggested (Stea and Pickering, 2018).

Luckily, information on meat consumption's link to these topics is available in multiple forms of media, one being documentary films. Documentary films are no longer just an outlet for entertainment and/or informing, but also a means by which viewers are encouraged to become active in debates, public policy, and activist work surrounding the film's topic(s) (Nisbet and Aufderheide, 2009). Not only could documentary films provide a more thorough background on the issues discussed (i.e. an hour or two of engagement with the topic versus a few minutes of reading a short descriptive paragraph), but they could also have the potential to be more effective than text alone by offering images, graphs, supporting commentary, etc. that viewers may find more engaging and/or convincing than text alone. In fact, studies have found documentary films to be influential in changing people's attitudes, perceptions, and behaviours (Hans and Kimberly, 2011; Lin, 2013), and this includes topics specifically pertaining to environmental concerns as well (Janpol and Dilts, 2016). Also, as noted by Karlin and Johnson (2011), films have been studied extensively as mediums for telling stories and entertaining audiences, but not enough attention has been given to their potential for sparking social change.

In addition, through years of personal correspondences, interactions and discussions with those in vegetarian/vegan communities, it is clear that documentary films have the potential to act as an influential medium on diet-related thoughts and behaviours, with some individuals even mentioning particular films as primary catalysts behind their meat reduction efforts. Therefore, since the effects of more thorough information on consumer attitudes and behaviours in regards to meat consumption is quite limited, and since documentaries offer more substance while seeming to be particularly impactful on individuals' decisions to reduce meat intake, a study looking at the effects of meat-related documentary films on consumer attitudes and behaviours would contribute significantly

to the current meat reduction literature, while simultaneously building upon findings from Study 1.

It is established that more information is needed on the effectiveness of different "meat reduction" messages, and that documentary films offer a more thorough and promising medium through which this information can be delivered to consumers, but which consumers should be targeted with this messaging? Obviously, the choice of audience for a documentary film is vast, but there may be specific segments of consumers that might be more receptive to take such messaging and apply it to their daily lives in the form of dietary change. Young adults, and specifically young adults attending university are an ideal group to target for multiple reasons. Young consumers, especially those that are educated, are quite receptive to novel and innovative ideas (Ottman et al., 2006) and a large portion are concerned with environmental protection (Connell et al., 1999; Gifford and Nilsson, 2014), so adopting a new diet that can have positive impacts on the environment might be an appealing prospect for this consumer segment. Additionally, young consumers, and more specifically young university students, are at a stage in life where many are no longer under direct supervision of their parents and have more freedom to make their own food choices, especially if they move out of their parent's home and are living independently. These initial stages of independence could be the opportune time to present individuals with meat-reduction messaging, as addressing the behaviour at a relatively early age could influence individuals to reduce or abstain before high levels of meat intake become a heavily ingrained habit.

Furthermore, most students do not have a permanent partner or children and therefore probably have less social pressure from others to cook and consume certain foods. For example, partners and children may sometimes act as barriers to changing dietary habits through direct resistance and/or resulting inconvenience. Especially in the case of meat consumption, conflict may arise if a partner is not amenable to meals without meat and/or if children resist consuming the plant foods necessary for a complete and healthy diet. Even if an individual makes the decision that despite these hindrances, they will still pursue a low meat diet themselves, there is an inconvenience factor that may affect their ability to uphold such a commitment. If partners or children are unwilling to accommodate more plant-based meals, multiple meals may have to be made which results in more complexity and time spent cooking, adding stress and requiring extra effort from the individual trying to reduce their meat intake. Most would find such a task incredibly

challenging, making any ideas of reducing meat consumption in a long-established family unit quite difficult. However, these barriers may be mostly avoided when working with young university students, as most will have more freedom and autonomy over their food choices when compared to those integrated into a family unit. Students will more than likely still have social barriers due to potentiality of living with housemates and going out to eat with friends occasionally, however it is expected that these social barriers would be much easier to overcome in comparison to those found in a tight centralized family unit as described above. Thus, students are at a time in their lives where they have more autonomy over their food choices, but are also not necessarily old enough to be settled into habits and beliefs that are heavily ingrained and routinized and therefore extremely difficult to change; they are at an ideal age and stage in life where they are not too young to where their food choices are strongly tied to their parents (Campbell et al., 2007; Lucas-Thompson et al., 2017), but also not likely old enough to have a completely "established" life situation.

In addition, besides being an optimal target group, changing the beliefs and behaviours of university undergraduates has the potential to spark social change that could endure well into the future. After all, current students will be the workers, leaders and parents of tomorrow, taking their ideas, values and habits with them as they start occupations, take leadership roles, and begin their own families. Therefore, changes in dietary beliefs and behaviours that are adopted during students' years of study could become cemented into values and habits that continue with them into perpetuity, potentially leading to a generational shift that relies less on meat-based meals.

With insights on attitudes, motivations, and behaviours gleaned from Study 1 and the gaps that still remain in the meat reduction literature, Study 2 aims to build upon previous findings through answering specific research questions and their underlying hypotheses:

RQ1	Does the viewing of a meat-related film have any significant impacts on willingness and/or intentions to reduce meat in the diet, reduction frequencies, and/or self- reported meat intake frequencies?
Hypothesis 1	The viewing of a meat-related film will significantly increase willingness and intentions to reduce meat in the diet and reduction frequencies, while decreasing self-reported meat intake frequencies.

RQ2	Do film framings (e.g. human health, environmental sustainability, or animal welfare) have any differing impacts on motivations to reduce meat consumption?
Hypothesis 2	Film framings will have significant, although different, impacts on motivations to reduce meat consumption.
RQ3	Do the meat-related films have an impact on TPB variables, meat attachment (and/or its sub-components) or agreement with proposed meat-reduction structural measures (e.g. meat tax) either within film groups over time or between film groups?
Hypothesis 3	The meat-related films will significantly decrease attitudes and meat attachment, have no effect on subjective norm or PBC, and will significantly increase agreement with proposed meat-reduction structural measures both within and between groups (compared to a control group) over time.
RQ4	Do post-viewing factors (e.g. acquiring additional information, discussing with others, etc.) offer any additional explanatory support for findings related to previous research questions? (No hypothesis as this is purely exploratory to provide more context to results)

6.6 Study 1 and Study 2 overview

The goals of Study 1 were to measure the average New Zealand consumer's awareness of meat-related environmental impacts, the strength of various reduction motivations, and if variables based on behavioural/psychological models (e.g. attitudes, meat attachment, etc.) can accurately predict willingness and intentions to reduce personal meat intake, as well as agreement with proposed structural measures (e.g. meat tax). A quantitative approach, in the form of online questionnaires with scalar questions, was chosen in order to capture variability across hundreds of consumers (who will be acquired from pre-established research panels), while also offering insights on what meat consumption and potential reduction may look like for an 'average' consumer. These methods provided a snapshot of consumers and potential pathways for how societal reductions in meat may be achieved in New Zealand, but offered no further concrete information on how awareness raising efforts, in practice, may actually influence meat-related variables over time.

Thus, Study 2 was designed in order to build upon results obtained by Study 1, as a clearer understanding of consumers would allow for awareness raising interventions to be created in such a way that (hopefully) inspires long-term changes in meat-related variables among consumers (reduction motivations, attitudes towards meat, meat attachment, meat intake frequency, etc.). Films, documentaries specifically (discussed further in Chapter 8), were chosen as the preferred form of information provision due to their increased length and potential for inspiring personal engagement with the provided material. University students were chosen as the target audience due to their positioning in life, where they are (or are becoming) independent of their parents but are not yet 'settled' into families, occupations, etc. Changes in meat-related variables resulting from participation in the educational interventions were measured in a longitudinal manner through the use of multiple questionnaires, so that changes could be analysed over time, something that was lacking in Study 1's design. Hence, Study 2 intended to build upon Study 1 by putting its results into practice, through an informational intervention, in order to see how consumer opinions and behaviours reacted to information on meat's negative impacts and how these reactions diminished, increased, and/or maintained themselves over time.

Taken together, Study 1 and Study 2 contribute to both understanding current meat consumption and potential reduction among New Zealand consumers, while also measuring how an informational intervention can influence such factors over time. Both studies provide information that organizations and/or individuals who wish to promote meat reduction within New Zealand will likely find useful, while also delivering insights into the effectiveness of different forms of information when trying to promote meat reduction through informative campaigns. In addition, future studies that seek to explore meat reduction further can reference both studies in order to both understand meat consumption in New Zealand at a snapshot in time (Study 1) and how different forms of information can influence motivations, attitudes, and behaviours to reduce meat intake (Study 2). Thus, whether a researcher wishes to understand how 'average' consumers have shifted over time or what additional forms/types of information might compare to those previously studied (or both), future studies will benefit from information sought in both Study 1 and Study 2.

CHAPTER 7

STUDY 1 – UNDERSTANDING NEW ZEALAND CONSUMERS

7.1 Chapter outline

The following chapter is dedicated to outlining and explaining the specific methods utilised and the insights gained from Study 1 (published previously by Lentz et al., 2018; see Appendix C), which was designed and executed with the aim to answer four primary research questions:

RQ1 Do consumers place 'eating less meat' below other sustainable food behaviours in terms of environmental benefit?

RQ2 Are there any significant differences in motivations to reduce meat consumption between consumer groups?

RQ3 Can any of the Theory of Planned Behaviour (TPB) components (attitudes, subjective norms, perceived behavioural control [PBC]) accurately predict willingness and intentions to reduce meat consumption? Can they also predict current meat reduction behaviour and agreement with proposed structural measures?

RQ4 As initially found by Graca and colleagues (2015a), does the Meat Attachment Questionnaire (MAQ) provide further predictive power for these phenomena above and beyond that of the TPB in the New Zealand context?

This chapter will begin with an overview of the methods implemented in Study 1, including questionnaire design, testing, and implementation, as well as the statistical analyses used for data interpretation (7.2). Next, sample characteristics, descriptive statistics, and findings relevant to the main research questions will be outlined in the results section (7.3), followed by a discussion section (7.4), which will include a brief results summary as well as a general discussion outlining what these results may mean in the broader context. Lastly, limitations and future research directions (7.5), final conclusions for Study 1 (7.6), and implications for Study 2 (7.7) will be discussed.

7.2 Methods

The following section will outline the methods used for Study 1, beginning with the general design and formatting of the questionnaire that was provided to participants. The questionnaire design section is organised and discussed in relative order of the proposed research questions. After reviewing the questionnaire design, how the questionnaire was tested and checked for errors along with to who and how it was distributed is explained. Finally, the statistical software used and the specific analyses that were performed in order to answer the proposed research questions is outlined.

7.2.1 Questionnaire design

Using Qualtrics, an online questionnaire design service (www.qualtrics.com), an online questionnaire was designed in order to address the research questions proposed for Study 1 (see Appendix E). When devising the scales for the questionnaire, the designer wanted to ensure that enough points were provided in order for respondents to answer with enough precision, but also not overwhelm the participants or provide too many points to where answer options closer to the centre of the scale became less meaningful. According to Finstad (2010), a seven point scale offers a well-balanced and meaningful scale that limits participant interpolation while also being optimal for electronically distributed questionnaires. Thus, a seven point scale was used and kept consistent throughout the questionnaire (where appropriate) in efforts to obtain the most accurate answers from participants.

A short passage was given before the start of the questionnaire introducing the topic of the study, along with the information sheet for participants that outlined the study (Appendix D), confidentiality and security of data, ethics information, and contact details of researchers involved. After reading the information sheet, participants were asked to give consent before being directed to the questionnaire. Before being distributed to participants, the preliminary questionnaire was reviewed by two fellow researchers and was sent to students (16 in total) in multiple departments throughout the University of Otago in order to determine if any adjustments were needed to make the questionnaire clearer. Once changes to questions were complete (mostly grammatical and word choice adjustments) and the preliminary data was checked for compatibility with proposed analyses, the questionnaire was deemed ready for distribution.

Perceived environmental friendliness of sustainable food behaviours. Participants were first asked, "On a scale from 1 to 7, rate how you believe each of the following eating behaviours affects the environmental friendliness of your diet." The sustainable food behaviours participants were asked to rate included 'buying foods with less packaging material', 'eating seasonal fruits and vegetables', 'buying locally sourced foods', 'avoiding foods that are transported by air', 'buying organic foods', and 'eating less meat' (modelled after Tobler et al., 2011). The bottom anchor (i.e. 1) was labelled 'very small environmental benefit', while the top anchor (i.e. 7) was labelled 'very large environmental benefit'. This set of questions aimed to understand where consumers place the eating of less meat, as far as sustainability benefits, in comparison to other well known 'sustainable' food behaviours.

Current meat consumption habits with willingness, intentions and motivations to reduce. Before being asked any questions about meat-related habits and attitudes, participants were given a definition of meat, as defined by the researchers: "Please keep in mind when answering questions, that the word "meat" refers to red and white meats (e.g. beef, lamb, pork, chicken, turkey, fish, seafood etc.) that are either unprocessed (e.g. chicken breast, steak, fish filet) or processed (e.g. sausage, salami, meat mince, chicken nuggets, crab cakes)." After reading this short definition, participants were asked, "On average how often do you consume meat or products that include meat?", and the provided answer options included 'never', 'rarely', 'several times a week', 'daily', and 'several times a day'. Next, participants were asked, "Have you already or are you currently making any efforts to reduce your personal meat consumption?", with the answer options of 'yes' or 'no'. Answers from these two questions were used in order to break the respondents into three consumer groups (abstainers, reducers, and standard). Abstainers were defined as those who answered 'never' to the question of "On average how often do you consume meat or products that include meat?", reducers were defined as those who answered 'yes' to "Have you already or are you currently making any efforts to reduce your personal meat consumption?", and standard consumers were all remaining participants who answered that they consumed meat but have not reduced.

Willingness, intention and motivations to reduce were all measured on seven point scales. For willingness, participants were asked, "On a scale from 1 to 7, how willing would you be to consider reducing your meat consumption sometime in the near future?" and for intentions, "Specifically, in the next six months do you intend to reduce your meat

consumption?". Willingness had the anchors 'not at all willing' and 'extremely willing' while intentions had 'do not intend at all' and 'fully intend'. For motivations to reduce, participants were asked one of two questions depending upon how they answered previous questions. Standard consumers who said they eat meat and have not reduced its consumption were asked, "How important, if at all, would each of the following factors be in strengthening your consideration to reduce your overall meat intake?", while reducers and abstainers were shown an alternate question, "Please think back to when you first decided to reduce your meat consumption. How important were each of the following factors in influencing your initial decision to lower your overall meat intake?" Six potential motivations to reduce meat consumption (with an optional 'other' box) were provided to participants in order to gauge what already has or hypothetically would motivate them to reduce their meat intake. These answer options included 'health benefits', 'more environmentally friendly', 'animal welfare concerns', 'high cost of meat', 'taste preferences', and 'weight control'. The anchors included 'not at all important' and 'extremely important', with an additional middle anchor 'moderately important'.

TPB components. The third part consisted of questions aimed to understand each TPB component (i.e. attitudes, subjective norms, PBC) in regards the participant's meat consumption (as consistent with Berndsen and van der Pligt, 2004; Graca et al., 2015a). Participants were asked, "On the scales provided, please choose what most closely aligns with your thoughts and attitudes towards the act of consuming meat. NOTE: Scores closer to 1 mean you agree more with the attribute on the left and scores closer to 7 mean you agree more with the attribute on the right." Five sematic differential scales followed, each ranging from 1 to 7 for each set of items: 'bad to good', 'unpleasant to pleasant', 'against to for', 'unfavourable to favourable', 'negative to positive'. Each negatively associated term was placed as a bottom anchor (i.e. 1), while each positively associated term was placed as a top anchor (i.e. 7). All five scales were averaged to produce an 'attitude' score for each participant.

Internal consistency of these five items in forming the construct 'attitude' was determined with Chronbach's alpha (α) (calculated in SPSS), consistent with normal procedure when analysing summated Likert-type scales (Gliem and Gliem, 2003). General 'rules of thumb' were used to determine acceptability as suggested by George and Mallery (2016): $\alpha > .9$ – excellent; $\alpha > .8$ – good; $\alpha > .7$ – acceptable; $\alpha > .6$ – questionable; $\alpha > .5$ – poor;

α < .5 – unacceptable. Internal consistency between the five items was calculated in SPSS and was found to be excellent (α = .94).

Subjective norm was determined with two scalar questions: "People who are important to me think that I should eat meat" from 1 (strongly disagree) to 7 (strongly agree), and "In regards to people who are important to you, how much do they influence your actions to either consume or not consume meat?" from 1 (not at all) to 7 (a lot). Internal consistency was poor (α = .55), however, lower alpha values are common in scales with only two or three items (Tavakol and Dennick, 2011), in which case composite reliability can be used as an alternative. The formula provided by Raykov (1997) was used for calculations, and the composite reliability for the three item scale was 0.81 (reliability above 0.7 is considered acceptable; Hair and Lucas, 2014). Answers on these two items were multiplied together to produce a subjective norm score (as consistent with Graca et al., 2015a). The scores on these two scales were expected to have a multiplicative rather than additive effect, and were therefore multiplied together to produce a subjective norm score, as consistent with Graca et al. (2015a). To clarify, the overall subjective norm score is the product of how much others care about an individual eating meat crossed with how much the individual cares about these others' thoughts, opinions, and judgements. So for example, even if others strongly wished for an individual to eat meat, if the individual does not care at all about these outside opinions, the influence of these outside opinions is expected to be lessened and possibly even nullified (i.e. stronger effect than simply additive, hence multiplicative effect), making the overall subjective norm score much lower in comparison with an individual who cares more about the thoughts and opinions of others. Thus, the product (rather than sum) of the scores for these two statements provided the overall subjective norm score.

PBC was measured with three scalar questions, which were prefaced with the statement, "In regards to your current meat consumption habits…". These questions included, "I am confident I could change my habits if I wanted to", "Whether I change my habits is entirely up to me", and "Changing my habits is not something that is under my control". Each statement had the anchors 'strongly disagree' to 'strongly agree'. Internal consistency was poor (α = .59) as expected; however, composite reliability for the three item scale was 0.79 (reliability above 0.7 is considered acceptable).

MAQ. The fourth section contained 16 statements (consistent with Graca et al., 2015a) intended to measure meat attachment. Participants were asked, "On a scale from 1 to 7, please rate your agreement with the following group of statements." with the anchors 'strongly disagree', 'strongly agree', and middle anchor 'neither agree nor disagree'. Four statements comprised the hedonism subscale: "A good steak is without comparison", "To eat meat is one of the good pleasures in life", "I love meals with meat", and "I'm a big fan of meat". The affinity subscale was also comprised of four statements, which were all reverse coded for analyses: "I feel bad when I think of eating meat", "To eat meat is disrespectful towards life and the environment", "Meat reminds me of diseases", and "By eating meat I'm reminded of the death and suffering of animals". The entitlement subscale was comprised of three statements: "According to our position in the food chain, we have a right to eat meat", "To eat meat is an unquestionable right of every person", and "Eating meat is a natural and indisputable practice". Lastly, the dependence subscale was comprised of five statements: "Meat is irreplaceable in my diet", "I would feel fine with a meatless diet", "If I couldn't eat meat I would feel weak", "If I was forced to stop eating meat I would feel sad", and "I can't picture myself not eating meat regularly". The statement, "I would feel fine with a meatless diet" was reverse coded for analyses. Each group of statements was averaged to create a subscale score, while all statements were averaged to create a global scale score. The four subscales, hedonism ($\alpha = .91$), affinity ($\alpha = .87$), entitlement ($\alpha = .81$) and dependence ($\alpha = .84$) in addition to the global scale ($\alpha = .93$) all had high internal consistency. Composite reliability (CR) and average variance extracted (AVE) were both calculated (CR = 0.92; AVE = 0.73) and both were above accepted thresholds (0.7 and 0.5, respectively; Hair et al., 2010). Correlations between MAQ subscales and global scale show strong associations (Table 7.1), indicating strong internal consistency and reliability.

Table 7.1 Correlations of MAQ global and subscales

Scale	M	SD	1	2	3	4	5
1. Hedonism	5.1	1.4	-				
2. Affinity	5.2	1.5	.56*	-			
3. Entitlement	4.8	1.4	.67*	.47*	-		
4. Dependence	4.2	1.4	.77*	.44*	.65*	-	
5. Global	4.8	1.2	.90*	.74*	.80*	.88*	-

Note. M = mean, SD = standard deviation.
* $p < .05$

Structural Measures. The fifth part of the questionnaire contained six statements regarding potential 'top-down' actions that governments could take in order to promote meat reduction, which were organized into two sets of three questions each based upon framing of the policy (i.e. either for public health or environmental concerns). The policy measures proposed to participants in order to address environmental concerns included, 'To reduce food-related environmental impact, an 'environmental tax' should be placed on meat and meat products to make them more expensive.', 'To reduce food-related environmental impact, an 'environmental subsidy' should be applied to plant-based foods to make them more affordable.', and 'Public institutions like universities should reduce the environmental impact of their food catering by providing more meatless meals.'. The policy measures proposed to participants in order to address public health concerns included, 'To promote improved public health, a 'health tax' should be placed on meat and meat products to make them more expensive.', 'To promote improved public health, a 'health subsidy' should be applied to plant-based foods to make them more affordable.', 'Public institutions like universities should promote public health by providing more meatless meals among their food options'. The public health and environmental frames were included in order to determine if support for the proposed measures differed depending upon the justification provided (i.e. framing of the policy). Respondents were asked to rate these different proposed measures on scales from 1 to 7 with the respective anchors of 'strongly disagree' to 'strongly agree'. The scores between the two frames (i.e. health and environment) were also averaged for each policy proposal (i.e. subsidies, procurement, taxes) in order to determine overall agreement with each measure without focusing on differences in framing.

Demographics. The sixth and final section of the questionnaire contained demographic questions in order to elucidate any trends in meat consumption within the population.

7.2.2 Questionnaire testing and distribution

A random representative sample of adults was acquired from ResearchNow (online marketing organization, now renamed Dynata; www.Dynata.com), who distributed the supplied questionnaire link to panellists throughout New Zealand during the months of March and April, 2017. Participants were offered monetary compensation for completing the questionnaire and were told that they could remove themselves from the study at any

time without any disadvantage to themselves. Comparisons between consumer groups was a main research goal, and since only a small percentage of the New Zealand population follows a strict vegetarian diet, a large enough sample to capture most of the variation within this group was desired. Therefore, a target of around 800 respondents was sought in order to provide large enough samples of each consumer group, including vegetarians (e.g. if 5% were vegetarian this would result in a sample of 40 for that group), so that statistically sound comparisons could be made.

Once a target sample size was chosen, ResearchNow was provided a link to the Qualtrics-designed questionnaire, which was then given to their affiliated survey panels, which include Peanut Labs, Valued Opinions, and e-Rewards (ResearchNow, 2017). ResearchNow is a key provider of online representative samples for those wishing to conduct survey research in New Zealand. The company ensures quality data by assigning each panel member an ID number upon panel enrolment and through tracking their actions, including profile updates, survey screeners, and past survey participation in order to determine the consistency and reliability of the panel member (ResearchNow, 2017). In addition, established privacy policies like identifying panel members by unique IDs when supplying data ensures that respondents' confidentiality is protected. ResearchNow asserts that they are fully committed to bringing clients quality data that accurately represent consumer opinions.

Profile data was utilised in order to access consumers from different regions of New Zealand, ensuring that respondents in multiple cities across the nation were given the opportunity to take part in the questionnaire. Once panellists completed the questionnaire, ResearchNow would grant virtual currency, which could then be used towards redeeming items such as gift cards, vouchers, and/or physical merchandise. Once the target of 800 respondents was passed, the questionnaire link was removed from the panel websites shortly afterwards, preventing further access by panellists, and the researcher was contacted by ResearchNow. After reviewing the obtained data on Qualtrics, the researcher approved of the sample (n = 843) and ResearchNow was compensated the amount agreed upon in a prior quote.

7.2.3 Statistical analysis

Before any analysis took place, data from the questionnaire was checked for missing and/or incomplete values. The questionnaire design required participants to answer each question before advancing and as expected, no missing values were detected in the data. Data was analysed using SPSS 23 (SPSS Inc., Chicago, IL) and descriptive statistics were used to report means and standard deviations.

Perceived environmental friendliness. For the first research question, means and 95% confidence intervals were used in order to determine the presence (or lack thereof) of statistically significant differences in perceived environmental friendliness between sustainable food behaviours (e.g. buying foods with less packaging, eating seasonally, eating less meat, etc.). In addition to determining if there were significant differences, the researcher also wished to understand *how large* these differences were and therefore, effect size was also calculated. Effect size, as described by Kelly and Preacher (2012), is a quantitative measure of magnitude of some phenomenon that is used in order to address a specified question. The researcher aimed to understand the magnitude of the difference between the means of the highest and lowest rated sustainable behaviours, so effect size based upon those means was calculated (i.e. Cohen's *d*; Lakens, 2013). Calculating Cohen's *d* involves taking the mean difference between groups and dividing that value by their pooled standard deviations, as represented by the following equation:

$$d = \frac{(M_2 - M_1)}{SD_{pooled}}$$

$$SD_{pooled} = \sqrt{\frac{(SD_1^2 + SD_2^2)}{2}}$$

Once calculated, 'rules of thumb' established in the literature can be utilised in order to ascertain the magnitude of the Cohen's *d* value. As described by Sawilowsky (2009), the 'rules of thumb' are as follows: d ~ 0.01 = very small; d ~ 0.2 = small; d ~ 0.5 = medium; d ~ 0.8 = large; d ~ 1.2 = very large; d ~ 2.0 = huge. Thus, a small Cohen's *d* means that the magnitude of difference between the two items being compared is small, while a larger Cohen's d means the magnitude of difference between the items is large.

Motivations for Reduction. The motivations that drive individuals to reduce their meat consumption may differ depending on the consumer group of the individual (e.g. heavy meat eater, meat reducer, vegetarian, etc.), and so the aim was to compare between these groups in order to see if differences in motivations exist and if so, determine their statistical significance. Normally, a single t-test could be used in order to determine significant differences between two groups, but since there are more than two consumer groups wanting to be compared, multiple t-tests would have to be performed. This sounds reasonable, however, each t-test contributes 5% of error (if seeking 95% confidence) to the type 1 error rate (i.e. probability of false positive), which can become quite large when this error is compounded over multiple calculations. In order to avoid this, an analysis of variance (ANOVA) can be performed instead, which is specifically designed to compare the means of three or more groups simultaneously, thereby bypassing the multiple t-test conundrum and maintaining the standard type 1 error rate (Kim, 2014). Therefore, one-way ANOVA and Tukey HSD post hoc calculations were performed in order to determine if statistically significant differences in motivations between consumer groups were present for each potential meat reduction motivation. The ANOVA determined whether there were significant differences in mean motivation scores between consumer groups, while the Tukey HSD post hoc test would provide directionality and additional details on exactly how consumer groups differ (or not) for each motivation. To ensure an adequate sample size, G*Power software was used to perform an a-priori power analysis for the proposed ANOVA analysis. To achieve a power of 90% with a Cohen's f effect size of 0.18 (small to medium effect) and type 1 error rate of 0.05, the required sample size was 396 which is well below the realised sample size of 841.

MAQ Validity and Model Fit. Before further research questions involving the TPB and MAQ were explored, the researchers wished to confirm the conceptual structure of the MAQ, since this was only the second study to the authors' knowledge to utilise the construct. When using a psychometric tool (e.g. MAQ) to measure a theoretical construct (e.g. meat attachment) the validity of the tool must first be supported with evidence before results can be stated confidently. Validity can be shown in many ways, but can be broken down into three main phases as described by Flake and colleagues (2017): substantive, structural, and external. Substantive validity is the relevance and representativeness of the tool's content with findings already established in the literature, and is therefore of primary importance when first designing a measurement tool.

Structural validity, also known as internal or construct validity, is the tool's ability to accurately and consistently measure the theoretical construct under investigation. Finally, external validity is the comparison of the tool's output with trends, variables, and/or factors that are expected or known to be related to the construct trying to be measured. By combining these different forms of validity, the researcher is able to take a more holistic perspective of the tool to ensure that it is indeed a valid instrument for measuring a real-world phenomenon.

During its creation, the MAQ's substantive validity was established by Graca et al. (2015a) through the use of relevant literature on meat consumption attitudes and behaviours, but its structural and external validity may not hold when studying New Zealand consumers, since the MAQ's validity has only been tested among Portuguese consumers. Therefore, it would prove useful to establish the MAQ's structural and external validity in the context of New Zealand to ensure that the proposed theoretical model for meat attachment is still a valid and reliable measurement tool for measuring meat attachment among New Zealand consumers.

The structural validity of the MAQ can be determined with structural equation modelling techniques. One such technique often utilised for validating theoretical models is confirmatory factor analysis (CFA), which verifies structural relationships of observed variables and the latent factors they form (Harrington, 2009; pg. 5). To clarify, observed variables are the items directly measured by the researcher (e.g. questions, scales, etc.), while latent factors are psychological constructs that comprise the theoretical model. For instance, when utilizing the MAQ the observed variables are the questions asked by the researcher and the latent factors are the psychological constructs meant to be measured by those questions. The MAQ has four first order latent factors: hedonism, entitlement, affinity, and dependence, which combine to create the second order latent factor of meat attachment. There is no direct way to measure these constructs (Harrington, 2009; pg. 5), and so the first order latent factors are measured through questions on the MAQ, that when grouped can represent each construct. For example, there are four questions of the MAQ that are designed to measure hedonism, and CFA allows the researcher to determine if these four questions actually load onto the same latent factor (i.e. hedonism) as expected. Loadings are basically correlations between observed variables (i.e. questions) and latent factors, so questions that theoretically measure a factor are expected to load strongly on that particular factor. This is known as convergent validity, which

when combined with discriminant validity form the model's overall structural validity (Koeske, 1994).

Discriminant validity, as it sounds, is when observed variables of different latent factors are distinct (Bagozzi et al., 1991). So using the above example once again, each of the four questions that measure hedonism should not load strongly onto another latent factor (e.g. entitlement), because these latent factors are outlined in the theoretical model of meat attachment as being independent constructs. Of course, small amounts of covariance are expected because constructs may not be completely discrete in people's minds, questions used to measure the constructs will inevitably have measurement error, etc.; however, this covariance should not be strong if these latent factors are indeed independent as outlined by theory.

Once all observed variables (i.e. questions) are verified to load onto their particular first order latent factors (e.g. hedonism, entitlement, etc.), the loadings of those first order factors on the second order factor of meat attachment can be determined. So in summary, the global scale of meat attachment is measured by four latent factor subscales, which are comprised of the sixteen questions in the MAQ. Through calculated factor loadings, CFA allows the researcher to determine the strength of the relationships between the questions, first order latent factors, and second order latent factor. The strength (or lack thereof) of these relationships can help the researcher better understand the construct validity of the MAQ, which ultimately determines whether the proposed theoretical structure can be trusted to accurately measure the real world phenomenon of meat attachment. Therefore, in order to establish construct validity, CFA calculations were performed using IBM SPSS AMOS 23 with guidance from the AMOS 22 user's guide by Arbuckle (2013; pg. 137-144).

In addition to determining the structural validity of the MAQ, utilizing variables that lie outside the theoretical structure of meat attachment, but are expected to be strongly interlinked with the construct, can provide external validity. Two primary methods by which external validity can be determined is through the use of predictive correlations with variables that should be associated with the construct and by comparing across groups where differences in meat-related attitudes and behaviours have been noted previously in the literature (Flake et al., 2017). Standard Pearson correlation coefficients can be calculated in order to determine the strength of linear relationship between MAQ

scores and additional variables that are expected to be associated with the construct of meat attachment (e.g. respondents' current meat intake, attitudes towards meat, etc.). Correlation coefficients can range in values from -1 to 1 (i.e. perfect negative correlation to perfect positive correlation), with values closer to zero being weaker and values closer to 1 (both positive and negative) being stronger. MAQ scores are expected to have a weak to moderate (~ .3 to .5) positive correlation with positive attitudes towards and increased frequency of meat consumption, but a strong to perfect correlation (~ .7 to 1) is not expected as many other factors besides meat attachment can influence an individual's attitudes and meat eating habits (e.g. tradition, social norms, availability of meat alternatives, etc.). In other words, due to the complex nature of studying human psychology and behaviour, perfect correlations are not necessary or even expected; rather, correlations that show a consistent pattern across multiple associated variables is sufficient and desirable.

In addition to correlating MAQ scores with associated variables, determining significant differences in average scores between consumer groups can also add to external validity if findings concur with patterns established in past research (Flake et al., 2017). For instance, past literature has found women to be less attached to meat and more willing to reduce its consumption when compared to men; so, if MAQ scores are found to be lower on average for women when compared to men, it is a good indication that the MAQ is indeed measuring the real-world phenomenon of meat attachment, thereby increasing its generalizability. An independent samples t-test was performed in order to determine if there was a significant difference between male and female meat attachment scores.

TPB and MAQ predictive ability. Establishing the theoretical structure of the MAQ is great for building upon theory, but the primary interest of study 1 is to understand how it can be used as a tool to predict willingness and intentions to reduce meat consumption, as well as agreement with proposed structural measures (i.e. dependent variables). Alongside the TPB which is an already established model within behavioural research, the verification of the MAQ in the New Zealand context would provide further support to its use as a reliable measurement tool. If the TPB and MAQ are found to accurately predict the dependent variables described above, their extended use as 'measuring sticks' to track changes in individuals over time or in response to an intervention becomes more promising.

Linear regression provides a method by which this predictive ability can be measured, through modelling the relationship between two variables (e.g. MAQ and willingness to reduce). Although a simple linear regression would provide the relationship between an entire theoretical construct and a dependent variable, understanding how each component of the construct interacts with the variable would be even more insightful. Multiple regression allows for the modelling of multiple predictors for a dependent variable, which is useful for models like the TPB and MAQ that are comprised of multiple components. However, the aim is to understand the predictive power of the entire constructs *and* the individual components that form them. Therefore, hierarchical multiple regressions were utilised due to their ability to nest multiple concepts into an overall construct when testing relationships with a specified dependent variable (Petrocelli, 2003). This technique not only provided the predictive power of the entire TPB and MAQ models, but also simultaneously measured the predictive strength of their individual components. To provide an example for clarity, the relationship between the global scale of the MAQ (i.e. meat attachment) and its component subscales (i.e. hedonism, entitlement, etc.) can be modelled simultaneously with willingness to reduce meat in the diet. This would inform the researcher of the MAQ's predictive ability for willingness, as well as elucidate the relative predictive strengths of the MAQ subscales for this particular variable. So hypothetically, the strongest subscale predictor for willingness to reduce could be hedonism, while the strongest predictor for agreement with proposed structural measures could be entitlement. Through the use of hierarchical multiple regression, these intricacies can be measured while also maintaining the measurement of the overarching theoretical models. An a-priori power analysis was performed for the regressions to ensure that an adequate sample size would be attained for these analyses. With a power of 90%, Cohen's f^2 effect size of 0.02 (small effect), and type 1 error rate of 0.05 the required sample size was 776 which is below the realised sample size of 841.

7.3 Results

The results section will outline the findings of Study 1, which are based upon the statistical analyses outlined in the previous section (7.2.3). Sample characteristics are provided first, so the reader can better understand the sample and how closely it matches New Zealand's national demographics. Next, descriptive statistics for all dependent

variables are provided to the reader for easy referencing, followed by four separate sections, each dedicated to one of the four main research questions. The first section will compare consumer opinions on the environmental benefits of different food behaviours, including reducing meat intake; the second will compare consumer motivations for (potentially) reducing meat intake; the third will establish the validity of the MAQ; and the fourth will focus on the ability of the TPB and MAQ components to predict willingness and intentions to reduce meat consumption as well as agreement with proposed structural measures (e.g. meat tax).

7.3.1 Sample characteristics

A sample of 843 responses were received, with two participants being removed due to the age criteria for the questionnaire (both were under the age of 18 and therefore, not adults), leaving 841 valid responses. The sample had a 49:50 male to female gender ratio (close to expected 49:51) (stats.govt.nz, accessed 2017), had a median age in the range of 36 to 40 (expected 38) (stats.govt.nz, accessed 2017), and matched expected ethnicity profiles except for underrepresenting Maori (6.5%) and Pacific Islanders (3.2%) (expected 15% and 7%, respectively; see Table 7.2 below, on next page; stats.govt.nz, accessed 2017). The sample was biased towards individuals with higher education (41% compared to expected 20% with a Bachelor's degree or higher qualification; stats.govt.nz, accessed 2017).

Table 7.2 Demographic information

Sex	n	%
Female	424	50.4
Male	413	49.1
Gender diverse	4	0.5
Age		
18-30	269	32.0
31-40	203	24.1
41-50	169	20.1
51-60	141	16.8
61-70	59	7.0
Ethnicity		
European	599	71.2
Maori	55	6.5
Pacific Islander	27	3.2
Asian	101	12
Mixed	45	5.4
Other	12	1.4
Undisclosed	2	0.2
Income		
Less than $20,000	125	14.9
$20,000 to $49,999	213	25.3
$50,000 to $99,999	306	36.4
$100,000 to $199,999	163	19.4
$200,000 or more	34	4
Education/Qualifications		
No qualification	57	6.8
Completed NCEA levels 1-3 or equivalent	259	30.9
Apprenticeship or trade certification	176	21
Bachelor's degree or higher	346	41.3
Undisclosed	3	0.4
Area of Residence	n	%
Inner city	195	23.2
Outer city	358	42.6
Town	199	23.7
Rural	89	10.6

7.3.2 Descriptive statistics for dependent variables

Summarised below in Table 7.3 are the descriptive statistics for all dependent variables.

Table 7.3 Descriptive statistics for dependent variables

	All Groups n = 841		Abstainer n = 25			Reducers n = 312			Standard n = 504		
	M	SD	M	SD	Min Max	M	SD	Min Max	M	SD	Min Max
Behaviours											
Less packaging	5.6	1.4	6.0	1.4	2, 7	5.7	1.3	1, 7	5.5	1.4	1, 7
Eating seasonal	5.4	1.4	5.5	1.6	2, 7	5.7	1.3	1, 7	5.2	1.4	1, 7
Eating local	5.2	1.4	5.6	1.4	2, 7	5.5	1.1	1, 7	4.9	1.4	1, 7
Avoiding air transported	4.6	1.5	5.8	1.2	3, 7	5.0	1.4	1, 7	4.4	1.5	1, 7
Eating organic	4.5	1.7	5.2	1.6	2, 7	4.9	1.6	1, 7	4.2	1.6	1, 7
Eating less meat	4.2	1.6	6.3	1.5	2, 7	5.0	1.5	1, 7	3.6	1.4	1, 7
Motivations											
High cost of meat	4.9	1.5	2.2	1.7	1, 7	5.0	1.4	1, 7	4.9	1.5	1, 7
Health benefits	4.8	1.6	3.9	2.4	1, 7	5.4	1.3	1, 7	4.5	1.5	1, 7
Taste preferences	4.4	1.6	4.2	2.7	1, 7	4.4	1.6	1, 7	4.4	1.5	1, 7
Animal welfare	4.3	1.8	5.9	1.6	1, 7	4.6	1.7	1, 7	3.9	1.7	1, 7
Weight control	4.2	1.7	2.7	2.0	1, 7	4.7	1.7	1, 7	4.0	1.6	1, 7
Environmental concerns	4.1	1.6	4.4	2.4	1, 7	4.7	1.5	1, 7	3.7	1.5	1, 7

Table 7.3 (continued) Descriptive statistics for dependent variables

Meat Reduction Variables	All Groups n = 841		Abstainer n = 25			Reducers n = 312			Standard n = 504		
	M	SD	M	SD	Min Max	M	SD	Min Max	M	SD	Min Max
Willingness to reduce	3.0	1.5	-	-	-	-	-	-	3.0	1.5	1, 7
Intentions to reduce	2.4	1.4	-	-	-	-	-	-	2.4	1.4	1, 7
Agreement with measures	3.9	1.3	5.8	1.0	3, 7	4.4	1.2	1, 7	3.5	1.1	1, 6

Note. M = mean, SD = standard deviation, and Min/Max = minimum and maximum. Variables measured on seven point scale. Some variable names are shortened in order to fit the table. Please refer to the main sections these dependent variables are discussed for more detail. "Agreement with measures" is an average based on the six scalar questions for gauging agreement with proposed structural measures: three focused on public health and three focused on environmental sustainability. Abstainers and reducers have no values for willingness and intentions to reduce because they have already reduced their meat consumption. Minimum and maximum values for the "All Groups" category follow that seen for "Reducers", with all values ranging from one to seven.

7.3.3 Perceived environmental friendliness of sustainable food behaviours (RQ1)

Concerning the six sustainable food behaviours, consumers believed that buying foods with less packaging had the greatest positive impact on the environment, followed by eating seasonally, buying local, avoiding air-transported foods, buying organic, and eating less meat (see Figure 7.1). The environmental benefit of consuming less meat was rated significantly lower by respondents in comparison to all other sustainable food behaviours. Cohen's *d* was calculated using results from a two sample t-test and effect size between buying foods with less packaging and eating less meat was large ($d = .77$) ($d \sim .80$ is considered large; Sawilowsky, 2009). Descriptive statistics for the sustainable food behaviours are presented in Table 7.3 (in sub-section 7.3.2).

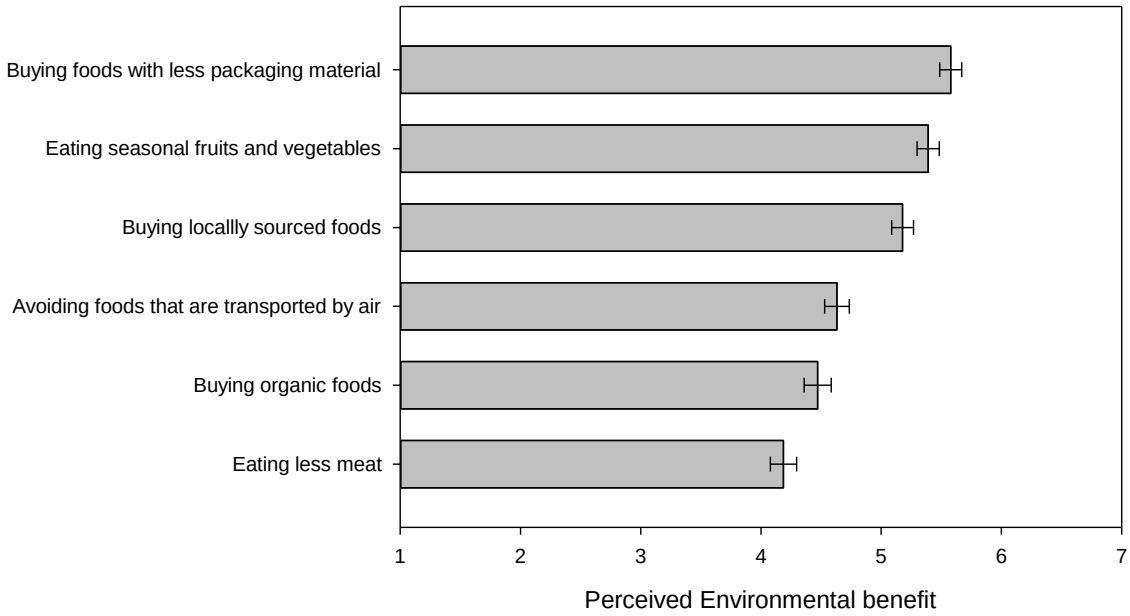

Figure 7.1 Means and 95% CI for perceived environmental benefit of six sustainable food behaviours

7.3.4 Motivations for reduction (RQ2)

Across the entire sample, high cost of meat was the strongest motivational factor in reducing meat consumption (M = 4.9), followed by health benefits (M = 4.8), taste preferences (M = 4.4), animal welfare concerns (M = 4.3), weight control (M = 4.2) and more environmentally friendly (M = 4.1). These values were more influenced by standard consumers (n = 504), less so by reducers (312), and hardly at all by abstainers (n = 25) due to sheer numbers in each consumer group.

Strength of motivations to reduce were statistically different between consumer groups (abstainer, reducer, standard), except for taste preferences based upon one-way ANOVA results: high cost of meat [$F(2, 838) = 44.9$, $p < .001$], health benefits [$F(2, 838) = 47$, $p < .001$], taste preferences [$F(2, 838) = 0.2$, $p = .809$], animal welfare concerns [$F(2, 838) = 29$, $p < .001$], weight control [$F(2, 838) = 28.6$, $p < .001$], more environmentally friendly [$F(2, 838) = 50.3$, $p < .001$]. Health benefits, environmental concerns and taste preferences all failed Levene's Test of homogenous variances, so the more robust Welch statistic was used in place of the ANOVA values for those three motivations. In addition to the ANOVA, a Tukey HSD post hoc test was performed in order to compare all consumer groups with one another for each of the six motivations.

To supplement the post hoc test, a bar graph was created in order to aid visualization of the differences between groups (Figure 7.2).

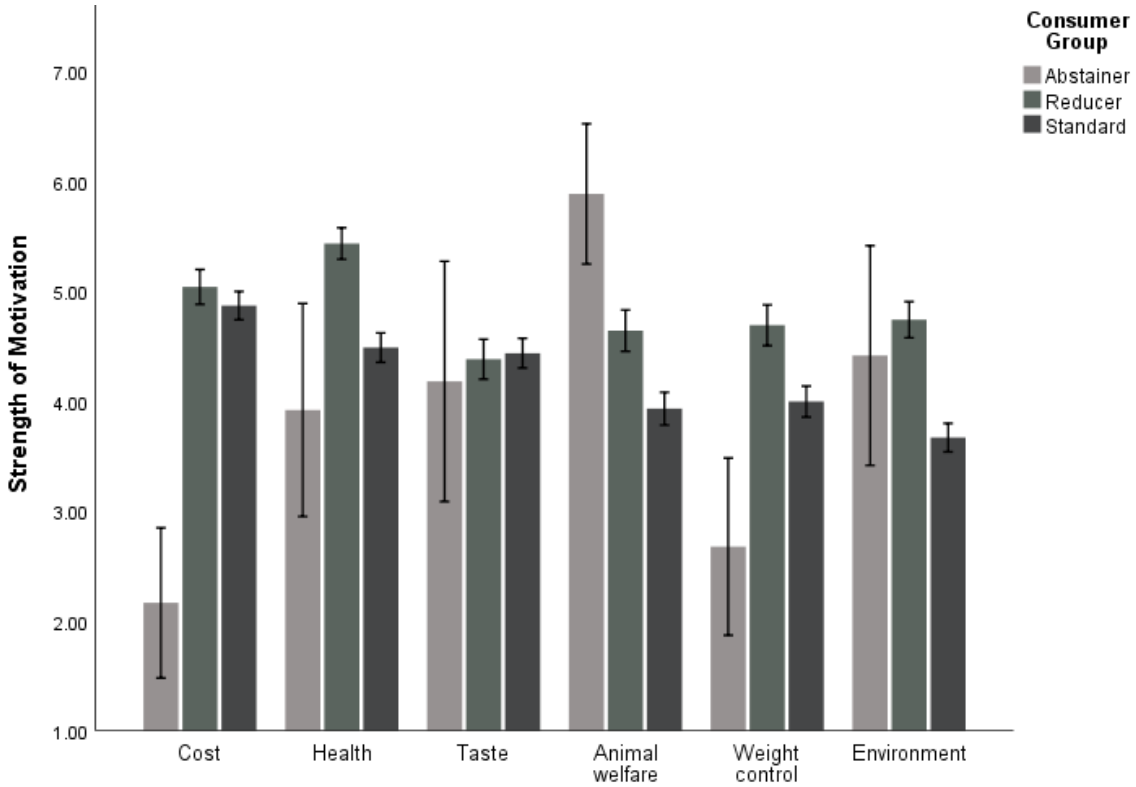

Figure 7.2 Strength of motivations to reduce meat intake (means with 95% confidence intervals) by consumer category (abstainer, reducer, standard). Cost = high cost of meat, Health = health benefits, Taste = taste preferences, Animal welfare = animal welfare concerns, Weight control = weight control, Environment = more environmentally friendly.

Based upon the post hoc analysis, all non-overlapping confidence intervals between consumer groups indicated a significant difference, while overlapping confidence intervals were found to be non-significant (see Figure 7.2), except for the difference between abstainers and standard consumers on the 'more environmentally friendly' motivation which was found to be significant despite overlap ($p = 0.043$). As shown, ranking of motivations to reduce meat consumption differ depending upon consumer group. For reducers, the 'health benefits' motivation is statistically higher when compared to standard consumers, even outweighing the motivation of 'high cost of meat'. The motivations 'more environmentally friendly', 'animal welfare concerns', and 'weight control' are also statistically higher among reducers when compared to standard consumers. Among abstainers, the 'weight control' and 'high cost of meat' motivations are statistically lower, while the 'animal welfare concerns' motivation is statistically higher when compared to reducers and standard consumers. In addition, the 'more

environmentally friendly' motivation is statistically higher among abstainers when compared to standard consumers, but not reducers.

7.3.5 MAQ validity

Pearson correlation was used in order to determine associations between the MAQ subscales and global scale with the TPB components and meat intake level. All variables had skewness values (between -.89 to .84) and kurtosis values (between -.55 to .74) that fell within acceptable ranges for normality (± 2; George and Mallery, 2016) As seen in Table 7.4, the MAQ subscales and global scale were moderately and strongly correlated with attitudes and meat intake level, while correlations were much weaker with subjective norm and PBC. This was expected, as Graca and colleagues (2015a) found attitudes to be heavily correlated with the MAQ subscales and global scale, with weaker correlations for subjective norm and PBC observed. Subjective norm and PBC showed statistically significant positive correlations with each subscale, except for affinity and dependence, respectively. All four subscales and the global scale showed positive correlations with actual meat intake.

Table 7.4 Pearson correlations for MAQ with TPB components and meat intake

Scale	Attitudes	Subjective norm	PBC	Meat intake
1. Hedonism	.76**	.13**	.09*	.53**
2. Affinity	.54**	-.15**	.29**	.30**
3. Entitlement	.61**	.12**	.09**	.39**
4. Dependence	.66**	.19**	-.08*	.48**
5. Global	.77**	.09**	.11**	.51**

* p < .05; ** p < .01

A t-test was performed in order to determine if meat-attachment scores differed depending on gender of the respondent. A statistically significant difference in scores was found between men (M = 5.06, SD = 1.08) and women (M = 4.54, SD = 1.23); t (835) = 6.50, p = .000. This adds further support to the validity of the MAQ, as studies in the past have shown women to be less attached to meat when compared with their male counterparts (Rothgerber, 2013; Ruby et al., 2016; Schosler et al., 2015).

MAQ confirmatory factor analysis. A confirmatory factor analysis (CFA) was performed with maximum likelihood estimation method using AMOS 23 (Arbuckle,

2013) in order to assess model fit for the MAQ theoretical structure (See Figure 7.3 below).

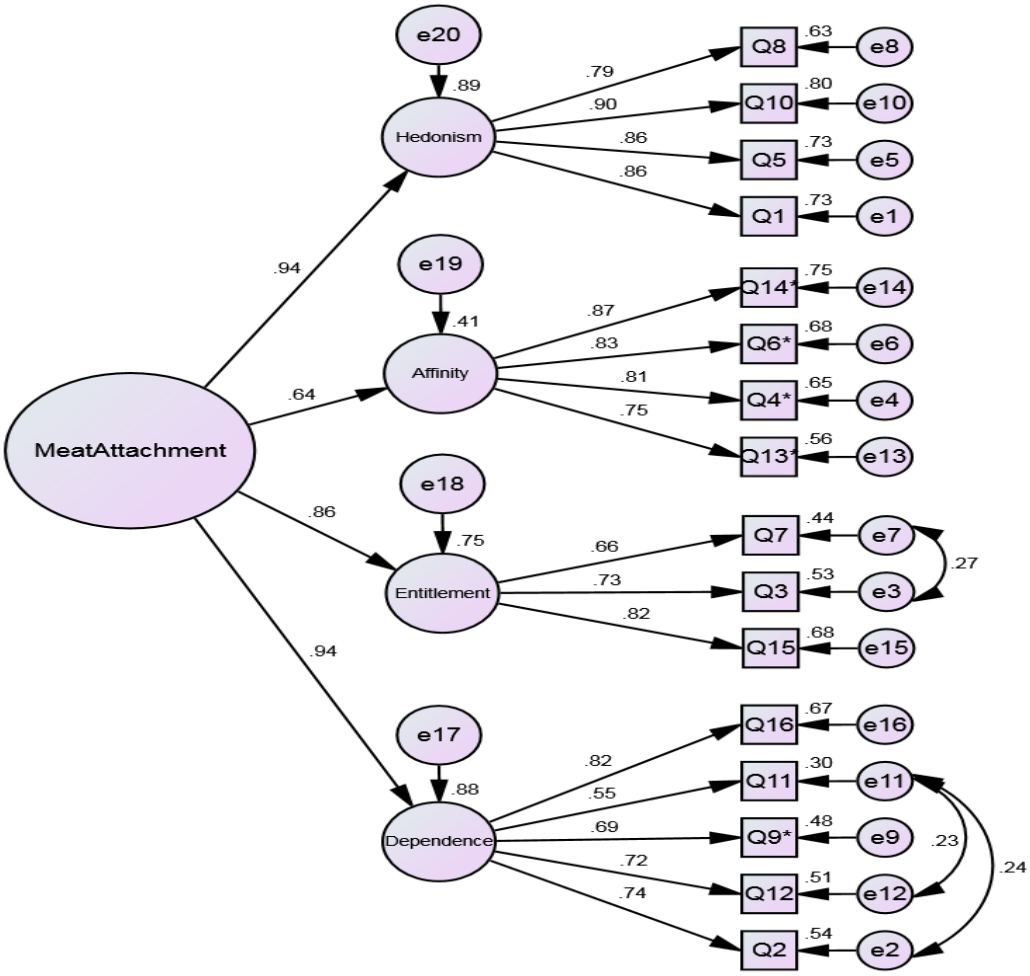

Figure 7.3 Confirmatory factor analysis of the Meat Attachment Questionnaire theoretical structure. Model included four latent factors (subscales) and one second order dimension (global scale; i.e. average score of combined subscales). Standardized coefficients for both factor loadings (e.g. .94 for Hedonism and MeatAttachment) and R^2 values (e.g. .88 for Dependence) are presented.
* Reverse coded items (e.g. Q14*)

The chi-square value ($x^2/df = 5.92$) was higher than the accepted standard ($x^2/df \leq 3$); however, chi-square is known to be overly sensitive to larger sample sizes ($N \geq 200$) (Schermelleh-Engel et al., 2003; Vandenberg, 2006). Therefore, alternative fit indices were utilised as additional indicators for adequate model fit (CFI = .95, TLI = .93, RMSEA = .08 [.07, .08]) and all indices performed within accepted standards (CFI ≥ .90, TLI ≥ .90, RMSEA ≤ .08; Hoe, 2008). Hence, the theoretical model for meat attachment

was accepted as having good fit. Correlated error variances were included between the questions 'According to our position in the food chain, we have a right to eat meat' (Q3) and 'To eat meat is an unquestionable right of every person' (Q7) due to their similar wording and ideas (i.e. right to eat meat). Correlated error variances were also included between the questions 'Meat is irreplaceable in my diet' (Q2), 'If I couldn't eat meat I would feel weak' (Q11), and 'If I was forced to stop eating meat I would feel sad' (Q12) due to their relatedness on the subject of meat no longer being present in the diet (hypothetically). Including these correlated error variances based on theoretical backing allowed for improved model fit (Brown, 2014), and fit indices, factor loadings and R^2 coefficients all had similar values and/or followed similar patterns as those seen in previous work by Graca and colleagues (2015a).

7.3.6 TPB and MAQ prediction of willingness, intentions, and support of structural measures (RQ3 and RQ4)

TPB and MAQ predictive power. In order to determine the predictive ability of the TPB components and MAQ subscales and global scale, eleven hierarchical regressions were performed: five for willingness to reduce, five for intention to reduce, and one for agreement with proposed structural measures (Table 7.5 below). Step 1 involved inputting the TPB components (i.e. attitudes, subjective norms, PBC), while Step 2 included either one of four MAQ subscales (e.g. Hedonism) or the global scale. As shown in Tables 7.5 through 7.7, attitudes consistently held the highest beta weights in comparison to subjective norm and PBC, which were statistically non-significant in all of the regression analyses except for subjective norm with the dependence subscale for predicting intentions. For each regression analysis, significance values were adjusted using the sequentially rejective Bonferroni test, which controls for both type I and type II errors, proposed by Holm (1979).

Willingness and Intentions to Reduce. Skewness (between .41 and .80) and kurtosis (between -.49 and -.72) values for both willingness and intentions were within acceptable ranges for normality (± 2; George and Mallery, 2016). Multicollinearity was not observed to be an issue within the regression analyses, as variance inflation factor (VIF) values ranged from 1.06 to 2.00 with tolerance values between .50 to .94 (VIF < 10 and tolerance > .10 considered acceptable; O'Brien, 2007). Additional explained

variance offered by the MAQ subscales and global scale were all significant, ranging from 4% (Affinity) to 13% (Global) for willingness to reduce and 2% (Hedonism and Entitlement) to 7% (Global) for intentions to reduce. Explained variance overlap occurred between attitudes and the MAQ, as beta weights for attitudes dropped drastically when MAQ subscales or global scale were included in the regression analyses. Dependence shared almost complete explanatory variance with the global scale in regards to willingness to reduce meat consumption, and explained variance offered by the global scale for intentions to reduce was almost half that of the explained variance for willingness to reduce.

Table 7.5 Hierarchical regressions: TPB and MAQ predicting willingness and intentions to reduce

Predictor Variables	*Reg 1*	*Reg 2*	*Reg 3*	*Reg 4*	*Reg 5*	*Reg 6*
Willingness						
TPB components						
Attitudes	-.50***	-.27***	-.36***	-.37***	-.26***	-.15*
Subjective norm	.07	.07	.05	.08	.11	.08
PBC	.05	.06	.10	.07	.00	.07
MAQ scales						
Hedonism		-.35***				
Affinity			-.26***			
Entitlement				-.25***		
Dependence					-.41***	
Global						-.51***
R^2	.24	.30	.28	.28	.36	.37
R^2 change	.24***	.07***	.04***	.05***	.12***	.13***
Intentions						
TPB components						
Attitudes	-.42***	-.28***	-.26***	-.34***	-.26***	-.16*
Subjective norm	.06	.06	.03	.06	.08*	.07
PBC	-.03	-.02	.03	-.02	-.06	-.01
MAQ scales						
Hedonism		-.21***				
Affinity			-.31***			
Entitlement				-.15**		
Dependence					-.27***	
Global						-.38***
R^2	.19	.21	.24	.20	.24	.26
R^2 change	.19***	.02***	.06***	.02**	.05***	.07***

Note. Reg = regression. Standardized beta weights are displayed for each regression. The first regression only included the TPB components, with each successive regression adding a MAQ subscale or global scale (i.e. all subscales combined).

* p < .05; ** p < .01, *** p < .001.

Structural Measures. Based upon paired sample t-tests, agreement with plant-based food subsidies (Health frame (HF): M = 4.94, SD = 1.83; Environmental frame (EF): M = 4.89, SD = 1.78) was significantly higher than agreement with plant-based friendly public food procurement practices (HF: M = 3.95, SD = 1.70; EF: M = 3.92, SD = 1.66) for both the health: t (840) = 14.14, p <.001; and environmental: t (840) = 15, p < .001; frames. In addition, agreement with plant-based friendly public food procurement practices was higher than agreement with taxes on meat and meat products (HF: M =

2.64, SD = 1.72; EF: M = 2.85, SD = 1.79) for both the health: t (840) = 23, p <.001 and environmental: t (840) = 19.51, p < .001 frames. The disparities in agreement level between proposed measures were therefore all significant; however, the reason for implementation (i.e. for improved public health or environmental sustainability) only had a statistically significant impact on respondents' agreeableness to the meat tax proposal, in which the environmental frame had an increased agreement level when compared to the public health frame; t (840) = 5.34, p <.001.

A hierarchical regression was performed in order to determine the predictive ability of the TPB components and MAQ global scale. Positive attitudes towards meat along with high attachment were found to have strong negative relationships with overall agreement towards proposed measures (score averaged across all six proposals) (Table 7.6). Skewness (-.15, SE = .08) and kurtosis (.14, SE = .17) values of summed proposal scores were both within the acceptable range of ± 2 for normality. Multicollinearity was not observed to be an issue within the regression analyses, as variance inflation factor (VIF) values ranged from 1.05 to 2.54 with tolerance values between .39 and .95.

Table 7.6 Hierarchical regressions: TPB and MAQ predicting agreement with proposed measures

Predictor Variables	*Reg 1*	*Reg 2*	*Reg 3*	*Reg 4*	*Reg 5*	*Reg 6*
TPB components						
Attitudes	-.49***	-.32***	-.23***	-.36***	-.32***	-.10*
Subjective norm	.17***	.17***	.07*	.18***	.18***	.15***
PBC	-.04	-.05	.04	-.04	-.09**	-.05
MAQ scales						
Hedonism		-.22***				
Affinity			-.49***			
Entitlement				-.21***		
Dependence					-.25***	
Global						-.50***
R^2	.26	.28	.41	.29	.29	.36
R^2 change	.26***	.02***	.15***	.03***	.03***	.10***

Note. Reg = regression. Standardized beta weights are displayed for each regression. The first regression only included the TPB components, with each successive regression adding a MAQ subscale or global scale (i.e. all subscales combined).
* p < .05; ** p < .01, *** p < .001

7.4 Discussion

This section will be dedicated to discussing the results for the proposed research questions. To begin, a summary of the results, organised by research question, is provided to refresh the reader's mind of the main findings and to act as a quick and easy reference. The general discussion follows, outlining not only how the findings compare and add to past studies on meat consumption/reduction, but also what these findings may mean for the promotion of meat reduction at both the individual and societal level. The end of the general discussion then leads into the final two sections of the chapter, which will discuss the limitations of Study 1 and future research prospects (7.5) and the final conclusions for the study (7.6).

7.4.1 Summary of results

RQ and Hypothesis 1 - Perceived environmental friendliness. The environmental benefit of consuming less meat was rated significantly lower by respondents in comparison to all other sustainable food behaviours and the difference between the highest rated (i.e. buying foods with less packaging material) and the lowest (i.e. eating less meat) was large based on effect size calculations. Eating less meat was rated by consumers as less environmentally friendly when compared to the other five sustainable behaviours, even though reduced meat consumption would likely have much greater and more widespread environmental benefits compared to many if not all of the other listed behaviours. Thus, hypothesis 1 was confirmed.

RQ and Hypothesis 2 - Motivations for Reduction. Strength of motivations for meat reduction differ depending on individuals' meat eating habits. Standard consumers, reducers, and abstainers all seem to have differing priorities when it comes to the reduction of meat. High cost of meats and health concerns seem to be primary motivations for standard consumers and reducers, with raised ethical motivations such as environmental friendliness and animal welfare concerns buttressing the more primary motivations among reducers. For abstainers, more ethical motivations such as animal welfare concerns and environmental friendliness are primary, while motivations such as health concerns and taste preferences play more minor roles. Based on these results, hypothesis 2 was confirmed.

RQs and Hypotheses 3 and 4 - TPB and MAQ predicting willingness, intentions and proposed measures. The construct of meat attachment, as outlined by the MAQ, was validated based upon a confirmatory factor analysis which establishes its usefulness along with the TPB components in studying meat-related opinions and habits. Based upon hierarchical multiple regressions, attitudes and meat attachment (all subscales and global scale) were significant predictors of willingness and intentions to reduce meat intake, while subjective norm and PBC had no consistent significant predictive ability. Attitudes and meat attachment were also the most significant predictors for agreement with proposed structural measures aimed at reducing meat consumption; however, subjective norm was also a significant predictor. For the proposed structural measures, consumers agreed most with the plant-based food subsidy proposal, followed by the proposal for more plant-based foods in public institutions, and finally the proposal for a tax on meat and meat products. The framing used for the proposals seemed to have little impact on consumer agreement, except for the proposal of meat taxes, where agreement levels were significantly higher given the environmental frame versus the public health frame. Therefore, hypotheses 3 and 4 were confirmed.

7.4.2 General discussion

In New Zealand, consumer awareness of the relative environmental impacts of meat consumption in comparison to other sustainable food behaviours seems to be low, matching a trend seen in other nations (Campbell-Arvai, 2015; Lea and Worsley, 2008; Macdiarmid et al., 2016; Siegrist et al., 2015). This is important to note, as a non-existent or tentative connection between meat eating and its associated environmental impacts could potentially explain why environmental concern was the weakest motivation for consumers when considering to reduce their meat intake. Increasing knowledge concerning meat consumption's link to environmental impacts could be an important component in intervention efforts that strive to motivate dietary change among consumers, as meat reduction among vegetarians has been shown to be influenced by multiple motivations, with some acting as catalysts for dietary change and others as supplementary support for continued meat reduction (Fox and Ward, 2008). Awareness raising of associated environmental issues could therefore complement other motivations that are already more salient among consumers, such as health and cost concerns, in order

to further promote reduced meat consumption. However, knowledge alone does not necessarily result in direct action (i.e. value-action gap; Kollmuss and Agyeman, 2002) and more studies are needed in order to understand the limitations of awareness raising on prevalent habits like high levels of meat consumption.

High costs of meat and health benefits associated with its reduced intake were the most prevalent motivations for consumers, which could be a result of heightened awareness regarding these issues, or simply a tendency for individuals to think about and prioritize personal wellbeing and safety above other concerns that are less salient and tangible in their daily lives (e.g. animal welfare, environmental impacts). This is relevant for businesses that market meat substitutes, as relaying health benefits and ensuring competitive prices will likely be important selling points for transitioning standard consumers from meat to alternative plant-based products. However, as also observed by de Boer et al. (2017), motivations to reduce shift between consumer groups, with health benefits, animal welfare and environmental concerns being stronger motivations for reducers when compared with standard meat eaters. Also, animal welfare concerns was the most prominent reason given by abstainers for reducing their meat consumption (followed by environmental concerns), further supporting the link between ethical beliefs and stronger commitments to meat reduction (Hoffman et al., 2013; Penny et al., 2015). Therefore, awareness raising campaigns that focus on multiple motivations for reduction rather than one alone (utilizing animal welfare and environmental arguments alongside more salient health benefits) could have the potential to convince standard consumers to reduce, while also making the commitment for those who have already reduced even stronger.

One interesting point to note, is that although standard consumers rated high costs of meat as being their strongest motivation to reduce intake, the idea of meat taxes had the least support of all structural measures. Thus, price manipulations are likely to be an effective policy tool to help curb meat intake; however, the implementation of such a proposal would likely be met with strong resistance from consumers (in addition to the meat industry), as expressed by multiple authors (Edjabou and Smed, 2013; Nordgren, 2012; Sall and Gren, 2015). Therefore, it will likely be necessary to first address these positive attitudes and attachments that consumers hold towards meat in order to gain enough support for these structural measures that have the potential for more widespread dietary change. Thinking ahead, the effectiveness of such policies will have to be assessed if

implemented, as retailers and consumers find ways to subvert certain measures. For example, in response to newly implemented food taxes, retailers can offer price adjustments or discounts to keep sales of the product high and/or consumers can shift to cheaper stores to purchase the same or similar products at lower prices, ultimately weakening the policy's impact on consumption rates (Jensen and Smed, 2013). Thus, even if a proposed measure such as a meat tax is implemented, it will likely take complementary efforts in awareness raising campaigns and/or framing strategies that increase other motivations in addition to cost (e.g. environmental sustainability benefits).

Attitudes and meat attachment were observed to be a strong predictors of willingness, intention, and agreement with proposed structural measures, but subjective norm and PBC offered no clear predictive power (except for subjective norm for agreement with proposed measures). This slightly contradicts Povey et al. (2001), who found all TPB components to be significant predictors of intentions to reduce meat consumption, although subjective norm and PBC were much weaker predictors than attitudes. These findings fall in line with more recent results from Graca et al. (2015a), who found attitudes and meat attachment to be the strongest predictors in regards to willingness and intentions to reduce meat intake, with subjective norm and PBC being only minor and inconsistent predictors. Implications based on these findings are relevant to any strategy aiming to change current consumption patterns, whether to benefit the environment or public health. In order to increase willingness and intentions to reduce meat intake, as well as promote acceptance of more meat reduction structural measures, positive attitudes and attachment that consumers hold towards meat will likely need to be addressed.

This is only the second study (to the author's knowledge) to utilize the MAQ, adding further to its validity and reliability as a tool for measuring meat attachment and predicting intentions and willingness to reduce meat consumption. Also, being used in a New Zealand context gives the MAQ some cross-cultural validity, as attachment patterns and their associations with current meat eating behaviours and considerations for future reduction seem to be consistent across the two nations presently studied (Portugal and New Zealand). Expanded use of the questionnaire in other nations will be vital in order to determine whether the MAQ has universal qualities, or if major inconsistencies in theory or measurement of meat attachment exist between cultures. Findings from this study give further support to the use of the MAQ as a measurement and predictive tool in regards to meat attachment and willingness to reduce meat intake. This tool could prove

useful for not only further scientific enquiry into meat attachment and behaviour change studies, but also for organizations wishing to understand and measure changes in meat attachment across populations and/or over time.

7.5 Limitations and future directions

The limitations of Study 1 are mainly a result of the methods utilised, as well as the scope with which the research questions were addressed. For example, participants' meat intake levels were self-reported and therefore, not free from personal biases or fallacies. In addition, Maori and Pacific Islanders were underrepresented, and it is unknown how this impacts the results. Also, fish and seafood were included under the category 'meat' due to associated environmental impacts (e.g. ocean pollution, overfishing, etc.), but more recent studies have found that those who consume land-raised meats (e.g. beef, chicken, pork, etc.) differ in food orientations in comparison to those who consume fish (Graca et al., 2019). In addition, frequency of meat consumption was measured, but amounts of meat consumed each meal was not. Future studies could utilize food journals (either written or photographed) in order to gain a more fine-tuned portrayal of both frequencies and quantities of meat consumed. Also, due to the nature of being a cross-sectional study, no patterns or changes in meat consumption attitudes or attachments can be seen over time. Future studies are likely to benefit from utilizing the TPB and MAQ in a time series fashion, in order to elucidate changes that occur from controlled interventions, media campaigns, or any other event that might have significant impacts on consumers' meat-related thoughts and behaviours. Also, this study mainly focused on understanding the MAQ as a practical tool for measuring meat attachment and predicting willingness and intentions to reduce, along with agreement towards proposed structural measures. Future studies wishing to add further to the theoretical understanding of meat attachment are likely to benefit from taking a more psychological approach, possibly by comparing or combining other established theoretical models (e.g. 4Ns of meat consumption; Piazza et al., 2015) with the MAQ. Lastly, the current study only looked at consumers in New Zealand. Although findings from the MAQ seem to be quite similar between the two nations currently studied (Portugal and New Zealand), evidence from other nations is needed in order to determine whether the theoretical structure of meat attachment has universal properties that can be applicable across regions and cultures.

7.6 Study 1 conclusions

In summary, awareness of meat's environmental impacts is low among New Zealand consumers, matching trends seen in other western nations. Educating consumers about these issues has the potential to act as a complementary motivation along with already more prevalent and salient motivations such as cost and health considerations. Attitudes and meat attachment are strong predictors of willingness and intentions to reduce personal meat intake, as well as agreement with structural measures aiming to reduce meat consumption at a broader scale. These findings have implications on interventions trying to promote meat reduction at the personal and/or societal level, which will likely need to address the positive attitudes and attachment towards meat that consumers hold, if significant dietary shifts are to occur. The utilization of the MAQ as a measurement and predictive tool seems promising, and its implementation in experimental studies and within organizations to gauge the effectiveness of meat reduction interventions is a logical next step.

7.7 Implications for Study 2

The focus of Study 2 is to determine whether an informational intervention will have short and/or long-term impacts on consumer's meat-related attitudes and behaviours. Study 1 found that there is much room for improvement in terms of consumers' awareness of meat's environmental impacts, and that consumer attitudes and meat attachment do predict individual willingness and intentions to reduce personal meat intake. Thus, Study 2 can build upon the findings of Study 1 by determining how changes in consumer attitudes, meat attachment, and actual meat eating behaviour occur and differ as a result of being exposed to (different) informational interventions. The application of Study 1's findings and how Study 2 intends to expand upon them will be outlined and explored further in the following chapter (Chapter 8).

CHAPTER 8

STUDY 2 – POTENTIAL IMPACTS OF FILM ON PROMOTING MEAT REDUCTION

8.1 Chapter outline

This Chapter will build upon Chapter 7 (Study 1) by first outlining the insights of Study 1, explaining how Study 2 may expand upon them, and describing the specific research questions aimed to be answered in Study 2 (8.2). Then, the proposed study design and methods for Study 2 will be reviewed (8.3), followed by a results section where answers to the proposed research questions are provided (8.4). Afterwards, a discussion section is presented, which gives a short summary of the study results along with what they mean for future studies and for promoting societal meat reduction through information provision measures (8.5). Finally, limitations and future directions are outlined (8.6) in addition to the final conclusions for Study 2 (8.7).

8.2 Overview of Study 1 and research questions for Study 2

Although Study 1 offered an overview of New Zealand consumers in terms of awareness of meat's environmental impacts, motivations to reduce intake, attitudes towards meat, meat attachment, and actual meat intake, it did not provide any insights on how meat-related variables may be *changed* through information provision measures and whether such changes would maintain themselves over time. With the establishment that more longitudinal studies are needed on how more thorough information provision methods can shape meat intake and how the use of film may be an effective medium of choice to communicate such information (see Chapter 6), a study that aims to fill this gap by measuring changes in meat-related variables over time in response to informational films seems sensible. Thus, with university students being recognised as a favourable target group for such an intervention (for reasons outlined in Chapter 6), and with three primary motivations identified as being more prevalent among reducers and abstainers (of meat) in Study 1 (health, environmental concern, and animal welfare), the following research questions (RQ) are proposed for Study 2:

RQ1 Does the viewing of a meat-related film have any significant immediate and/or long-term impacts on willingness and/or intentions to reduce meat in the diet, reduction frequencies, and/or meat intake frequencies?

RQ2 Does the viewing of a film dealing with a meat-related topic (e.g. human health, environmental sustainability, or animal welfare) have any immediate and/or long-term impacts on motivations to reduce meat consumption? If films do have an impact, which topics are influential and on which motivations? Do these impacts increase, persist, or diminish over time (e.g. 1 month after film viewing)?

RQ3 Does the viewing of a meat-related film have any immediate and/or long-term impacts on TPB variables, meat attachment (and/or its sub-components) or agreement with proposed meat-reduction structural measures (e.g. meat tax) either within or between different film groups? Also, which films cause which impacts, and do these impacts increase, persist, or diminish over time?

RQ4 Additionally, do post-viewing factors (e.g. acquiring additional information, discussing with others, etc.) offer any additional explanatory support for the findings of previous research questions?

8.3 Methods

The following section will outline the methods used for Study 2, starting with the power analysis that was conducted prior to the study in order to get an estimated required sample size based upon the intended study design and proposed statistical analyses (comparing questionnaire answers over three points in time between four groups: health, environmental, animal welfare, and control, 8.3.1). Next, the structural design and execution of Study 2 are outlined (e.g. format of film viewing sessions, provision of questionnaires, formation of groups, etc., 8.3.2), followed by the content and design of the four questionnaires given to participants (screener, pre, post, and 1 month follow-up, 8.3.3). The section concludes with an overview of the statistical analyses utilised in order to answer the proposed research questions (8.3.4).

8.3.1 Power analysis

A power analysis was performed before advertising for the study in order to get an estimate for a sample size that would provide adequate power for the questions wanting to be answered. Using G*power software for a repeated measures ANOVA with a desired power and effect size of 0.9 and 0.2, the corresponding total sample size for three measurements (pre, post, and one month follow-up questionnaires) with four groups (health, environment, animal welfare, and control) was calculated to be 76. With this figure in mind, the researchers aimed to recruit around 80 total participants for the study.

8.3.2 Study design and execution

The study sought to recruit around 80 New Zealand university students between the ages of 18 and 30, who would be broken into groups to watch one of four films: What the Health (run time: 1 hour and 37 minutes), Cowspiracy: The Sustainability Secret (run time: 1 hour and 31 minutes), Earthlings (run time: 1 hour and 48 minutes), and Jim & Andy: The Great Beyond (run time: 1 hour and 34 minutes) (Synopses for each can be found in Appendix F). The first three titles would act as meat-related films focused on the health, environmental, and animal welfare impacts of meat consumption, respectively; while the last title would act as an unrelated control film (on the topic of Jim Carrey and method acting). The three meat-related films were selected based upon their popularity and quality as rated by a popular film-rating website (www.imdb.com), along with their focus on the particular topic sought by the researcher. The three meat-related films are all persuasive in nature, and although the health (What the Health) and environmental (Cowspiracy) films use more statistics by nature of their arguments (e.g. rates of heart disease, deforestation, etc.) in comparison to the animal welfare (Earthlings) film, they are all quite similar in their messaging that stopping the consumption of animal products is a primary solution to the issues discussed. The www.imdb.com ratings for the films, as of January 2018, were as follows: What the Health (7.7 out of 10 with ~ 22,000 reviews), Cowspiracy: The Sustainability Secret (8.3 out of 10 with ~ 17,000 reviews), and Earthlings (8.7 out of 10 with ~ 16,000 reviews). The films were all viewed and timed by the researcher before being incorporated into the study to ensure that each film spent the majority of screen time (i.e. > 90%) focused on the main topic associated with each film (e.g. the topic of 'health' for What the Health).

A screener questionnaire was made using Qualtrics software, in order to make sure that participants were young (18 to 30 years of age), currently enrolled in a university in New Zealand, and currently consuming meat (i.e. not vegetarian or vegan). Vegetarians and/or vegans were not included in the study due to the researchers wishing to study the impacts of the films on meat consumers. In addition, the screener also asked whether participants had viewed films related to any meat-related topics in the past (12 names of meat-related documentary films provided, including What the Health, Cowspiracy: The Sustainability Secret, and Earthlings). This information was used to prioritise the selection of participants who had not seen a film, while still keeping those who had as 'potential participants' in case sample quotas were not met.

Advertisements for the study were designed and distributed across the university in the form of small posters and online bulletins, to attract participants in February[4] of 2018. The advertisements included a brief outline of the study, the screener questionnaire link, the expected incentive for participation (food provided at film screenings and $20 at completion of study) and the main researcher's name and contact information. Once enough qualified participants were gathered through the screener questionnaire, they were randomly partitioned by gender into one of four film groups (since gender has been found to be a key variable in meat consumption, as mentioned previously in section 3.6; see Sanchez-Sabate and Sabate, 2019). Afterwards, the four film groups were compared in order to assure that no significant differences were present between the groups in terms of important variables (e.g. attachment and meat intake) that might influence meat-related factors (e.g. response to films, changes in attitudes or behaviours, etc.). Three scheduled screening sessions would be held for each film in order to offer flexibility to participants. The three dates were chosen by participants, who marked their available dates and times using Doodle polls hosted for each group on doodle.com. Once dates were chosen, emails were sent out to participants notifying them of the dates and times of the film, as well as informing them that there would be snacks and refreshments provided.

The film sessions were held in rented rooms within the School of Geography at the University of Otago. Participants were given tablets which were pre-loaded with the questionnaires in order to make the sessions hassle-free and streamlined. A short passage was given before the start of the pre-questionnaire introducing the topic of the study,

[4] February was chosen as it is the beginning of the semester at the University of Otago

along with the information sheet for participants that outlined the study (Appendix G), confidentiality and security of data, ethics information, and contact details of researchers involved. After reading the information sheet, participants were asked to give consent before being directed to the questionnaire. Once all participants completed the pre-questionnaire, they were asked to help themselves to the snacks and refreshments provided and to make themselves comfortable. Before showing the film, the main researcher read a scripted introduction, which included a welcome, a brief reminder of the study's purpose, and request to pay attention for the entirety of the film. Once the film finished, participants were given tablets again in order to answer the post-questionnaire. Before leaving, participants were reminded that they would receive a follow-up questionnaire in their emails exactly one month from the date they watched the film.

One month after each screening date, participants were sent a follow-up questionnaire to their emails. Once completed, participants were sent instructions for collecting their $20 gift as a thank you for their time and participation in the study. Once all follow-up responses were received and gifts distributed, the main researcher sent a de-briefing email to all participant emails in order to elucidate the full purpose and design of the study.

8.3.3 Questionnaires

Screener Questionnaire. After a short introduction, participants were asked for their names and contact details (i.e. email and phone number); then, questions about age, gender, study at a university in New Zealand, current meat consumption, and the MAQ followed. Participants were screened out if they stated that they were younger than 18 or older than 30, were not currently studying at a university in New Zealand, or were not consuming meat. Out of 197 responses, six were rejected due to violation of age requirements, one was rejected because of the university attendance requirement, and two were rejected based on the requirement of meat consumption, leaving 188 valid participants.

Next, participants were asked, "Have you ever watched any films specifically about meat consumption's impacts on personal health, animal welfare, or environmental sustainability?" with the answer options 'Yes', 'Maybe', and 'No'; while also being asked, "More specifically, do you remember having seen any of the films listed below? (Select all that apply). If not, please select an appropriate 'other' option (if applicable)

and put the name of the film you have seen in the corresponding text box. If you cannot remember the name of the film, simply leave the text box blank. If you have never seen any film related to these topics, simply select 'Have not seen'" with the answer options consisting of twelve popular meat-related documentaries, three open answer text boxes (other film on meat and: personal health, environmental sustainability, animal welfare) and a final 'Have not seen' option. Participants who answered that they had seen a meat-related documentary before were screened out of the study (55 in total, leaving 133 valid participants), as the researchers wished to focus on individuals who were relatively unfamiliar with documentaries on meat-related topics. Those who were screened out of the study were thanked for their time and participation, while those who qualified were sent a follow-up email in order to confirm their status as a qualified candidate for the study. If more detail is desired, the full screener questionnaire is provided in Appendix H.

Formation of groups. Not all 133 eligible participants were included in the study for various reasons (e.g. some never responded to follow-up emails, some could not attend any of the scheduled film session dates, etc.). Thus, the 85 participants who were eligible based upon the screener questionnaire, who responded to follow-up emails, and who were available on the scheduled film session dates were randomly assorted into four groups: health, environment, animal welfare, and control. The random assortment was done separately for men and women, in order to balance gender between the groups, since it has been linked to meat-related variables and trends in past studies. Chi-square analyses were used to determine if there were any significant differences between groups in relation to gender ($p = .91$), meat intake ($p = .78$), and consumer category ($p = .74$) and as shown, no significant differences between the four groups was detected for any of these variables. In addition, a one-way ANOVA was run in order to ensure that meat attachment levels between the groups were not significantly different. The ANOVA failed the assumption of homogeneous variances with a significant Levene's test result ($p = .01$), so the robust Welch statistic was utilised; however, no significant differences in meat attachment were observed between groups ($p = .14$).

Pre, post, and follow-up questionnaire design. Three questionnaires in total were designed and used for the study (after the screener): pre, post, and follow-up; which were all mostly comprised of questions originally designed and utilised in Study 1. Each film group received pre, post, and specifically tailored follow-up questionnaires which had wording and questions to match the film they viewed. Prior to distribution, the

preliminary questionnaires were reviewed by two fellow researchers, and once changes to questions were complete (mostly grammatical and word choice adjustments), they were deemed ready for distribution. In efforts to reduce redundancy, the sub-sections below outline the questions and answer options that were provided across all questionnaires. The sections that were utilised across all questionnaires share many of the same themes and/or were derived from those seen in the questionnaire used in Study 1; therefore, if more detail is desired for the questions, answers, themes, etc. of the sub-sections provided below, please refer back to the methods section of Study 1 (7.2) or to Appendix I, J, or K.

Current meat consumption habits with willingness, intentions and motivations to reduce. In all questionnaires, each section of questions dealing with meat consumption was prefaced with the following statement, "Please keep in mind when answering questions, that the word "meat" refers to red and white meats (e.g. beef, lamb, pork, chicken, turkey, fish, seafood etc.) that are either unprocessed (e.g. chicken breast, steak, fish filet) or processed (e.g. sausage, salami, meat mince, chicken nuggets, crab cakes)". Meat intake frequency and reduction were measured with two questions and answers were used in order to break the respondents into two consumer groups (standard and reducers), not three (as in Study 1), since abstainers were removed from the study through the screener questionnaire. Willingness, intention and motivations to reduce were all measured on seven point scales. Answer options for motivations included 'health benefits', 'more environmentally friendly', 'animal welfare concerns', 'high cost of meat', 'taste preferences', and 'weight control'.

TPB components. This section of the questionnaires consisted of scalar questions designed to understand each TPB component: attitudes, subjective norms, and PBC. Attitudes were measured with five sematic differential scales, subjective norm was determined with two scalar questions, and PBC was measured with three scalar questions. The scores on these two scales for subjective norm were expected to have a multiplicative rather than additive effect (See sub-section 7.2.1 for more detail), and were therefore multiplied together to produce a subjective norm score, as consistent with Graca et al. (2015a).

MAQ. This section of the questionnaires contained 16 statements (consistent with Graca et al., 2015a) intended to measure meat attachment. The hedonism and affinity subscales were both comprised of four statements, the entitlement subscale was

comprised of three statements, and the dependence subscale was comprised of five statements. Each group of statements was averaged to create a subscale score, while all 16 statements were averaged to create a global scale score (i.e. meat attachment).

Structural Measures. This section of the questionnaires contained three statements regarding potential 'top-down' structural measures that governments could take in order to promote meat reduction, which were framed differently to each group, dependent upon the film viewed. The group that watched the health-focused film received the public health frame; the group that watched the environmental film received the environmental frame; and the group that watched the animal welfare film received the animal welfare frame. As an example, the policy measures proposed to the health group in order to address public health concerns included, "To promote improved public health, a 'health tax' should be placed on meat and meat products to make them more expensive." (i.e. tax agreement), "To promote improved public health, a 'health subsidy' should be applied to plant-based foods to make them more affordable." (i.e. subsidy agreement), and "Public institutions like universities should promote public health by providing more meatless meals among their food options" (i.e. procurement agreement). These same questions were asked of the environmental and animal welfare groups, but with the appropriate framing used for each (e.g. 'environmental tax', to promote animal welfare, etc.). The control group received the same questions, but with no framing included (e.g. A subsidy should be applied to plant-based foods to make them more affordable).

Pre-Questionnaire. A short passage was given before the start of the pre-questionnaire introducing the topic of the study, along with the information sheet for participants that outlined the study (Appendix G), confidentiality and security of data, ethics information, and contact details of researchers involved. After reading the information sheet, participants were asked to give consent before being directed to the questionnaire. Participants were asked about their meat consumption, willingness and intentions to reduce, motivations to reduce, TPB components, MAQ, and agreement with proposed structural measures. In addition, participants were asked for their age and gender, as well as, "What is your ethnic background? (Select all that apply)" with the answer options: 'European / Pākehā', 'Māori', 'Pacific Islander', and 'Other (please specify)'; "Please describe your current level of education" with the answer options: 'Bachelor's student', 'Master's student', 'PhD (Doctor of Philosophy) student', and

'Other (Please specify)'; and "Which of the following best describes your living situation?" with the answer options: 'Living with parents', 'Living alone', 'Flatting / Multi-person household', 'Living with partner', 'Living with partner plus other adults (e.g. flatmates, children aged over 16)', and 'Other (Please specify). The full pre-questionnaire can be viewed in Appendix I.

Post-Questionnaire. Immediately after viewing the film, participants were asked again about their meat consumption, willingness and intentions to reduce, motivations to reduce, TPB components, MAQ, and agreement with proposed structural measures. These questions mirrored those provided in the pre-questionnaire in order to maintain consistency for participants and to make comparative analyses easier (see Appendix J).

One month follow-up questionnaire. For each follow-up questionnaire, participants were provided with the title and synopsis of the film they viewed one month prior in order to help them remember and prepare for the questions in the questionnaire. Participants were first asked questions related to actions or feelings they may have taken/had after watching their respective films. The first question asked, "Since the film, have you acquired any additional information on the topic discussed in the film or other topics related to meat consumption? (e.g. from films, websites, articles, blogs, knowledgeable friends)." With the answer options: 'Yes, I searched for additional information', 'Yes, but it was only by chance', and 'No'. This was followed by three questions "Have you discussed the film or any of its messages with others?"; "Has the film had any lasting impact on your attitudes, beliefs, or thoughts on eating meat?"; and "Has the film had any lasting impact on your meat eating habits?" with each having 'Yes' and 'No' answer options. The sections that followed matched the pre and post questionnaires by asking about participants' meat consumption, potential reductions since film viewing, willingness and intentions to reduce, motivations to reduce, TPB components, MAQ, and agreement with proposed structural measures. If more detail is desired, the full follow-up questionnaire can be viewed in Appendix K.

8.3.4 Statistical analysis

Before any analysis took place, data from the questionnaire was checked for missing and/or incomplete values. The questionnaire design required participants to answer each

question before advancing and as expected, no missing values were detected in the data. Data was analysed using SPSS 23 (SPSS Inc., Chicago, IL) and descriptive statistics were used to report means and standard deviations.

Summary of variables and relationships to be tested. The researcher wished to explore multiple variables and relationships, including how motivations, TPB components, meat attachment, and agreement with proposed structural measures may change over time in response to exposure to a meat-related film, while also identifying any differences between film framings (i.e. health, environmental, animal welfare). In order to address the first question, the three experimental groups were partitioned into one group (name 'experimental) and the single control group was kept on its own as a baseline for comparison. The author labelled these two distinct groups under the category 'film vs control' in order to mitigate confusion. In order to answer the second question, the four groups (i.e. health, environmental, animal welfare, control) were all kept as distinct groups in order to test the more nuanced differences between them (e.g. maybe the health group has a greater drop in attitudes compared to the other three groups, or possibly the animal welfare group will experience greater declines in meat intake in comparison to the environmental group, etc.). By partitioning the groups into 'film vs control' as well as their already-formed 'groups', the researcher would be able to answer both whether meat-related films seem to have an impact (in general) in comparison to a control and/or whether meat-related films on specific subjects result in more or less of an impact on the dependent variables listed above.

In addition, the researcher also intended to measure whether any changes (both within and between groups) more directly related to meat intake occurred. Thus, changes in willingness and intentions to reduce, reduction rates (i.e. percentage of group that has reduced), and meat intake frequencies were compared over three points in time. These comparisons were made between all groups (both experimental and control) in order to ascertain whether specific framings had more or less of an impact on specific variables (e.g. health information led to the greatest reduction rates, environmental information led to the greatest increases in willingness and intentions to reduce, etc.).

Changes to motivations, TPB components, meat attachment, and agreement with proposed policies. The researchers wished to analyse potential changes in motivations, TPB components, meat attachment, and agreement with structural measures

across three points in time. Analysis of variance (ANOVA) could be performed in order to compare differences in mean scores within each group over the three measurements (pre, post, and follow-up); however, ANOVA assumes that the variables are unrelated when making comparisons. This assumption is violated, because these variables are in fact related, due to scores coming from the same individuals, just measured at different moments in time. Therefore, a repeated measures ANOVA was utilised, due to its ability to account for multiple comparisons made over time for related variables (Girden, 1992). Analysis of Covariance (ANCOVA) could be used in conjunction with the ANOVA, but instead of measuring changes over time within groups, a repeated measures ANCOVA allows for comparing changes over time between groups while controlling for each group's initial scores on a given variable. Thus, repeated measures ANOVAs and ANCOVAs were performed in order to determine whether any significant changes both within and between groups occurred over time. In addition to analysing by individual groups, the same procedures were also performed based upon film vs control (either experimental or control), where the experimental group is comprised of all three non-control groups while the control group remains constant. This was performed in order to gain a greater understanding of the differences between being exposed to any meat-related film (i.e. experimental group) or not (i.e. control group), rather than differences between meat-related films based upon framings.

The ANOVA and ANCOVA results would identify whether significant changes occurred within and between groups, but the researchers wished for more detail in regards to which changes occurred, the magnitude of these changes, which groups these changes occurred within/between, and how these changes differed over time. Therefore, post-hoc paired sample t-tests were performed for all variables with significant changes as identified by the ANOVA analyses in order to determine which groups showed significant changes and whether these changes were observed in the post-questionnaire and/or follow-up questionnaire when compared with initial pre-questionnaire values. Post-hoc pairwise comparisons were also performed for all variables with significant differences between groups as identified by the ANCOVA analyses in order to identify the groups that differed and the relative magnitude of these differences.

ANOVA and ANCOVA assumptions. There are many assumptions that must first be met in order for the results of repeated measures ANOVAs and ANCOVAs to be trusted for accuracy. First, the independent variable(s) should be comprised of at least two

independent categorical groups (e.g. health, environmental, animal welfare, and control groups). Second, residual distributions of the dependent variables' (e.g. attitudes, meat attachment, agreement with structural measures), for each level of the independent variable, should be relatively normal; however, as long as this assumption is not severely violated, type I error rates should remain reliable (Nimon, 2012). Third, for ANOVA analyses, the variances of the differences between the groups must be relatively equal (i.e. assumption of sphericity; Quene and Van den Bergh, 2004). Mauchley's test can be interpreted in order to determine if the sphericity assumption holds. If the p value from the test is less than .05, sphericity cannot be assumed, whereas if the p value is greater than .05, sphericity can be assumed. Epsilon (ε) is a measurement of departure from sphericity, with a value of 1 equating to no departure and values further from 1 showing greater departure. If Mauchley's test fails, three corrections can be used, listed from least to most conservative: Huynh-Feldt, Greenhouse-Geisser, and lower-bound (Nimon, 2012). It has been advised that when epsilon is >.75, the less conservative Huynh-Feldt correction should be used, and when it epsilon is <.75 the more conservative Greenhouse-Geisser correction should be applied; while the lower-bound correction is almost always too conservative and is therefore rarely used (Girden, 1992).
Repeated measures ANCOVAs must meet the same assumptions as those listed above for repeated measures ANOVAs; however, Levene's Test should be used instead of Mauchley's Test because ANCOVA requires the assumption of equal variances between groups rather than the assumption of equal variances in *differences* between groups (i.e. sphericity). Furthermore, there are additional assumptions for repeated measures ANCOVAs that must also be met. For each independent variable(s) (e.g. film vs control or group), the dependent variables (e.g. attitude post-questionnaire score) and covariates (e.g. attitude pre-questionnaire score) must have linear relationships and the regression slopes for these relationships must be homogenous (Nimon, 2012). In addition, there must be verification that the independent variables and covariates have no relationship with one another (i.e. assumption of homoscedasticity).

Due to the sheer number of ANOVAs and ANCOVAs being performed and the resulting increased probability of finding significant differences by chance, multiple comparisons must be accounted for and adjustments to p-values must be made in order to maintain acceptable type I error rates. The Bonferroni adjustment is a popular option, but can sometimes be too conservative (i.e. reduction in power), especially for studies with

smaller sample sizes and/or when a large number of hypotheses are being tested (Chen et al., 2017; Benjamini and Hochberg, 1995). In addition, the Bonferroni adjustment assumes that the hypothesis tests are statistically independent; however in many studies, there are relationships between the dependent variables. This assumption of independence therefore can create an over-correction, resulting in a severe reduction in power, making the traditional Bonferroni correction too conservative in studies that utilize multiple related dependent variables (Benjamini and Hochberg, 1995). Thus, since this study involved multiple variables that were indeed interrelated (i.e. attitudes towards meat, meat attachments variables, etc.), for each ANOVA/ANCOVA analysis, significance values were adjusted using the false discovery rate (FDR) procedure, as developed by Benjamini and Hochberg (1995). Adjustments were made across all major groups of variables being tested (i.e. Motivations, TPB, Meat Attachment, and Agreement with policy measures) for each test performed (i.e. ANOVA, ANCOVA, post-hoc paired t-tests, and post-hoc pairwise comparisons). For example, the TPB, meat attachment, and policy agreement variables were each analysed by group (i.e. health, environmental, animal welfare, and control) and film vs control (i.e. experimental or control) for both within and between subjects effects (i.e. ANOVA and ANCOVA). This amounted to twelve dependent variables with two sets of categorical independent variables (i.e. group and film vs control) being included for each analysis. Therefore, the value used for 'number of hypothesis tests' when performing the correction for this set of dependent variables was twenty-four (twelve tests for both categorical variables) for each respective analysis (i.e. ANOVA and ANCOVA). The correction was also applied to both sets of paired t-test comparisons (pre to post and pre to follow-up) and the pairwise comparisons that followed the ANOVA and ANCOVA analyses.

For the post-hoc t-tests and pairwise comparisons, the only assumptions are that the observations are independent, that the dependent variable is approximately normally distributed, and that there are no outliers present. Tests for equal variances could be performed as well, but since sample sizes are nearly equal in all cases, a test for this assumption is not needed for these analyses.

Summary of Assumption Analyses. Before any analyses occurred, the dependent variables (and their residuals) were checked in order to ascertain whether the assumptions of normality, sphericity, equality of variances, linearity, and homoscedasticity were met as prerequisites for the proposed ANOVA/ANCOVA analyses outlined above. Normality

and outliers were also checked among the dependent variables before conducting post-hoc t-tests.

Dependent variables (and residuals) were found to have 'reasonably normal' distributions separated by both group and film vs control for each of the three time points (although ANOVAs are robust to non-normality; see Blanca et al., 2017). Most histogram plots showed no concerning deviations from normality, and although some significant Shapiro-Wilk results occurred for some of the dependent variables at some of the time points (mostly seen in the subjective norm, PBC, Tax, Subsidy, and Procurement variables), the ANOVA's robustness to non-normality led the authors to feel comfortable in continuing with the analyses.

For each repeated measure ANOVA, Mauchly's Test was performed in order to determine sphericity, with significant test results (i.e. violation of sphericity) being followed by corrections based on epsilon values. The less conservative Huynh-Feldt correction was utilised for epsilon values greater than 0.75, while the more conservative Greenhouse-Geisser correction was not used for any epsilon values as none fell below the 0.75 cut-off.

For the ANCOVA, Levene's Test was non-significant (in both the post and follow-up questionnaires) for all six motivations and all three TPB components, so the assumption of equal variances was accepted for these variables. However, Levene's Test was significant at various time points for all the meat attachment subscales (except affinity) and global scale. These violations were noted, but due to ANCOVA's robustness to unequal variances when sample sizes between groups are equal or near equal (Barrett, 2011), and the primary implication of this violation being non-identification of a significant difference in means, the authors felt the analyses could continue.

Scatterplots of the covariate (i.e. pre-questionnaire) and the dependent variables (i.e. post and follow-up questionnaires) were created for each 'level' of the independent variables (i.e. 'experimental' and 'control' for the 'film vs control' variable and 'health group', 'environmental group', 'animal welfare group', and 'control' for the 'group' variable) in order to assess the assumption of linearity, as explained by Johnson (2016). All relationships at all levels of the independent variables were linear in shape, and so the assumption of linearity was accepted. Homogeneity of regression slopes was also tested, and non-significant relationships ($p > .05$) between the covariate (pre-questionnaire) and

the independent variables were observed for (almost) all dependent variables (including both post and follow-up scores for each). Variables that violated the assumption were the motivations: health and animal welfare; the TPB variable: subjective norm; all MAQ variables: hedonism, affinity, entitlement, dependence, and meat attachment (global scale) when analysing follow-up scores by group; in addition to the structural measures: tax and subsidy agreement when analysing post-questionnaire scores by group. Also, the assumption was violated for the MAQ variable: dependence; as well as the structural measures: tax, subsidy, and overall policy agreement (i.e. tax, subsidy, and procurement agreement combined) when analysing post-questionnaire scores by film vs control. Therefore, homogeneity was assumed for all dependent variables, excluding the exceptions outlined above (Johnson, 2016). Variables that violated assumptions (see Table 8.1 on following page) were noted within their respective analyses and the authors were more cautious when interpreting ANCOVA outputs for these cases. However, this violation results in an increased likelihood of incorrectly accepting the null hypothesis (increase in Type II error) and so therefore, non-significant results should be treated with more caution, while significant results are expected to remain relatively unaffected.

Table 8.1 Summary table of variables that violated homogeneity of regression slopes assumption

		Motivation	TPB	MAQ	Structural Measures
Post Questionnaire	*Group*	N/A	N/A	N/A	1. Tax 2. Subsidy
	Film vs Control	N/A	N/A	1. Dependence	1. Tax 2. Subsidy 3. Overall agreement
Follow-up Questionnaire	*Group*	1. Health 2. Animal welfare	1. Subjective norm	1. Hedonism 2. Affinity 3. Entitlement 4. Dependence 5. Global scale	N/A
	Film vs Control	N/A	N/A	N/A	N/A

Note. These violations increase type II error rates, and so non-significant results for these variables should be interpreted with caution.

In order to verify the final assumption for the ANCOVA analyses, scatterplots of the dependent variables and the standardized residuals of the covariate were observed in order to test the assumption of homoscedasticity, and with no significant deviations in variance observed in any of the variables at any of the time points, the assumption was accepted for all analyses.

Lastly, before any post-hoc t-tests were conducted, normality and outliers for the dependent variables were checked in order to test assumptions. No severe deviations from normality nor any outliers were observed, and so the researchers felt that the t-tests would provide accurate results.

Paired t-tests, Chi-square test, and Wilcoxon Signed-Ranks tests to determine changes in willingness and intentions to reduce, reduction rates, and meat intake frequencies. The researchers wished to understand how willingness and intentions to reduce, reduction rates, and actual meat intake frequencies changed over the course of the study. Willingness and intentions were measured on Likert scales, and so paired t-tests

would be appropriate to determine whether significant changes occurred within groups over time. Reduction rates, on the contrary, were measured on a binary scale (Yes / No) and so a Chi-square test would be appropriate when determining whether watching a meat-related film, in comparison to a control film, would result in a higher frequency of reductions in meat intake. Actual meat intake was measured on an ordinal scale with only five discrete answer options, and for this reason the researchers felt that the more robust Wilcoxon Signed-Ranks Test would be a more appropriate test, in comparison to paired t-tests, to determine whether changes in meat intake occurred within each group from time of pre-questionnaire to time of follow-up.

Before any t-tests were performed on willingness and intentions, the variables were checked for normal distributions and outliers. For willingness to reduce meat consumption, normality could not be assumed as the kurtosis value (4.12) for the health group exceeded that of the range to assume normality of ± 2 suggested by George and Mallery (2016). However, t-tests have been shown to be robust to deviations from normality (Edgell and Noon, 1984; Lumley et al., 2002; Sawilowsky and Blair, 1992) and small sample sizes (De Winter, 2013); and when using likert scale data, Meek and colleagues (2007) found that paired sample t-tests outperformed Wilcoxon tests, even when assumptions were violated. Therefore, the researchers determined that continuing with the paired sample t-tests analyses was preferred, due to their ability to offer more statistical power, despite the violation of normality in the health group. Boxplots were created for each dependent variable, but no outliers were detected and so the outlier assumption was accepted.

Only two assumptions need to be met for chi-square analyses, which include having at least two independent variables and a dependent variable that is measured at an ordinal or nominal scale. In the proposed analysis for changes in reduction rates, there are at least two independent variables (i.e. groups) and the dependent variables are measured on a nominal scale (i.e. have or have not reduced) and so the assumptions for the analysis are met.

The Wilcoxon Signed-Ranks test makes four assumptions. First, the two samples being compared must be dependent or related, which is true for the proposed analyses as the pre and follow-up answers of the same question are being compared within groups. Second, the samples being compared must be independently taken, and since they were taken at

two distinct time points, this assumption is also met. Third, the dependent variable should theoretically be continuous in nature, and since frequency of meat intake can be placed on a theoretically continuous spectrum of never consuming meat to consuming meat at every moment possible, this assumption was also accepted. Lastly, both samples being compared must be measured at the ordinal scale, and since meat intake was measured on an ordinal scale with five answer options in both questionnaires, this final assumption is met.

8.4 Results

The following section will outline the findings of Study 2, starting with an overview of the characteristics of both the sample (8.4.1) and the separate film groups (8.4.2). Next, findings for whether and how the different meat-related films (i.e. experimental films in comparison to the control film) influenced willingness and intentions to reduce, reduction frequencies (from pre-questionnaire to one month follow-up), and meat intake frequencies over time will be outlined (8.4.3). With the knowledge of whether meat-related films had an impact in comparison to the control film on meat eating intentions and behaviours, more detailed information can be gathered, such as if (and which) motivations shifted over time (and in which groups), along with whether the same was observed for TPB variables, meat attachment global scale and subscales, and agreement with structural measures (sections 8.4.4 and 8.4.5, respectively). The final section will review factors related to film viewing (e.g. searched for more information, discussed film with others, etc.), measured at time of follow-up, in order to determine if individuals acted differently in reaction to the film after viewing (8.4.6).

8.4.1 Sample characteristics

This section will review the characteristics of the final sample of 85 university students. Sociodemographic details for the sample are summarised below in Table 8.2.

Table 8.2 Meat Intake and sociodemographic characteristics

	n	%
Meat Intake		
Several times a day	11	12.9
Daily	31	36.5
Several times a week	38	44.7
Rarely	5	5.9
Consumer category		
Standard	52	61.2
Reducer	33	38.8
Sex		
Female	65	76.5
Male	20	23.5
Ethnicity		
European	42	49.4
Maori	2	2.4
Asian	29	34.1
Mixed	6	7.1
Other	6	7.1
Education/Qualifications		
Bachelor's Student	71	83.5
Master's Student	8	9.4
PhD (Doctor of Philosophy) student	5	5.9
Other	1	1.2
Housing		
Flatting / Multi-person household	70	82.4
Living with partner plus other adults	5	42.6
Living with partner	2	2.4
Living with parents	3	23.7
Living alone	2	2.4
Living in Hall of Residence	3	3.5

The majority of students identified themselves as standard consumers who have not reduced their meat consumption. The sample was biased towards women and those of European and Asian ethnicity; while being mostly comprised of students pursuing Bachelor's degrees and living in multi-person households.

8.4.2 Group characteristics

This section will outline the group characteristics of the four groups that were formed from the sample of 85 students (see Table 8.3 below). The majority of participants stated that they consumed meat several times a week or daily, and the ratio of standard consumers to reducers was similar across groups; in addition, the ratio of females to males across groups was comparable. In terms of ethnicity, the health and environmental groups were similar to one another, while the animal welfare group was most analogous to the control group. The level of education and format of housing was similar across all groups. ANOVAs were run between the groups based on the various sociodemographic variables (e.g. meat consumption, ethnicity, gender, etc.) and no significant differences were found.

Table 8.3 Group characteristics

	Health group	Environmental group	Animal welfare group	Control
Meat Intake				
Several times a day	3	2	2	4
Daily	9	5	9	8
Several times a week	9	12	8	9
Rarely	1	2	2	0
Consumer category				
Standard	13	12	12	15
Reducer	9	9	9	6
Sex				
Female	17	15	17	16
Male	5	6	4	5
Ethnicity				
European	10	14	9	9
Maori	0	0	1	1
Asian	7	6	9	7
Mixed	1	0	2	3
Other	4	1	0	1
Education/Qualifications				
Bachelor's Student	19	17	18	17
Master's Student	1	1	2	3
PhD (Doctor of Philosophy) student	1	3	0	1
Other	1	0	1	0
Housing				
Flatting / Multi-person household	17	19	18	16
Living with partner plus other adults	1	0	2	2
Living with partner	1	0	0	1
Living with parents	1	1	1	0
Living alone	1	1	0	0
Living in Hall of Residence	1	0	0	2

Note. Number of participants is provided for each variable separated by group

8.4.3 Willingness, intentions, reduction, and meat intake

This section will outline: how willingness and intentions to reduce changed between the pre and post-questionnaires within each group, how reduction frequencies differed (if at

all) between groups at time of follow-up, and if changes in meat intake frequencies differed between groups from the time of the pre-questionnaire to the follow-up.

Willingness and intentions to reduce. The results from paired sample t-tests in order to determine how willingness and intentions to reduce meat intake changed over time within groups is provided in Table 8.4 below. As shown, willingness and intentions to reduce increased significantly in all experimental groups, except for willingness in the environmental group which still increased, but was non-significant (likely due to ceiling effect caused by higher pre-questionnaire score). In contrast, the control group saw no significant increases in either willingness or intentions to reduce meat intake.

Table 8.4 Paired sample t-tests for willingness and intentions to reduce meat intake

	Pre M	Post M	*T*	df
Willingness				
Health group	4.06	5.70	4.85*	21
Env. group	4.94	5.51	2.23	20
Animal group	4.30	5.59	3.54*	20
Control	5.02	4.60	-3.05*	20
Intentions				
Health group	3.28	5.60	5.57*	21
Env. group	3.87	5.15	4.56*	20
Animal group	3.42	5.50	5.91*	20
Control	3.84	3.58	-.81	20

Note. Significance values are adjusted for multiple comparisons through the FDR procedure. 'Pre M' = pre-questionnaire mean, 'Post M' = post-questionnaire mean, 'Env.' = environmental, and 'Animal' = animal welfare. * $p < .05$

Reduction over time. A Chi-square test was performed in order to compare whether watching a meat-related film, in comparison to a control film, resulted in a greater frequency of reported reductions in meat intake in the follow-up questionnaire. A significant interaction was observed (X^2 (1) = 11.21, $p < .001$), and so the viewing of a meat-related film was determined to have significant impact on the choice to reduce meat in the diet over time. This effect was similar across experimental groups, with the animal welfare group having the largest number of reducers, followed by the health and environmental groups, respectively (see Table 8.5). As an aside, two participants, one in the health group and one in the animal welfare group stated that they are no longer consuming meat since watching their respective films.

Table 8.5 Group frequencies of meat reduction at time of follow-up

	No reduction	Reduced	Total
Health group	10	12	22
Environmental group	10	11	21
Animal welfare group	8	13	21
Control	18	3	21

Note. 'No reduction' are individuals who have not reduced their meat consumption since the viewing of their film, while 'Reduced' are those who have reduced since the viewing of their film.

Changes in Meat Intake. Due to meat intake level being measured on an ordinal scale with only five discrete answer options, as well as concerns over assumptions of normality and equal variances being violated, the more robust Wilcoxon Signed-Ranks Test was performed in order to determine if any significant changes occurred in meat intake levels within each group. The test for the health group ($Z = -1.9$, $p = .040$), the environmental group ($Z = -2.6$, $p = .009$), and the animal welfare group ($Z = -1.8$, $p = 0.047$) indicated that meat intake levels at time of the follow-up were significantly lower than meat intake levels at time of the pre-questionnaire. The test for the control group ($Z = -1.3$, $p = .172$) showed no significant shifts in meat intake levels from the time of the pre-questionnaire to the follow-up. One-tailed significance tests were used, due to the authors only being interested in whether watching a film decreased meat intake levels. For the focus of this study, increases in meat intake levels would be viewed the same as no change in meat intake levels (no reduction in meat intake); therefore, one-tailed significance tests were deemed appropriate (Ruxton and Neuhauser, 2010).

Summary. From the results outlined above, it is clear that the viewing of a meat-related film significantly increased willingness and intentions to reduce meat intake, promoted higher reduction frequencies in comparison to a control film, and resulted in reduced self-reported meat intake frequencies within groups over time (except in the control group). With this established, hypothesis 1 was confirmed and the sections below (8.4.4 to 8.4.6) will provide more detail on which (if any) measured variables (e.g. motivations, TPB variables, meat attachment, etc.) changed over time and if these changes differed between groups.

8.4.4 Motivations

This section will focus on the motivations to reduce meat intake measured throughout the study (from pre-questionnaire to follow-up), starting with the descriptive statistics for the motivations. Next, the ANOVA and ANCOVA results will be reviewed in order to determine if (and how) changes in motivations differed within groups over time or between groups, respectively. Finally, the post-hoc tests that were conducted will be discussed (paired t-tests for the ANOVA and pairwise comparisons for the ANCOVA) in order to gather more detail on which groups differed.

Descriptive Statistics. Descriptive statistics for the six motivations, separated by groups and questionnaire, are provided in Table 8.6 below.

Table 8.6 Means and standard deviations of motivation scores by group

	M (SD)		
	Pre	Post	Follow-up
Health Benefits			
Health group	5.2 (1.1)	6.1 (0.7)	5.2 (1.6)
Environmental group	5.3 (1.1)	4.7 (1.3)	4.9 (1.2)
Animal welfare group	4.6 (1.4)	5.4 (1.3)	5.1 (1.0)
Control	5.5 (0.9)	5.4 (1.4)	5.4 (1.3)
Environmental Benefit			
Health group	4.7 (1.5)	5.5 (1.3)	5.1 (1.5)
Environmental group	5.0 (1.0)	5.9 (0.9)	5.5 (1.1)
Animal welfare group	4.8 (1.4)	5.7 (1.3)	5.4 (1.4)
Control	5.4 (1.3)	5.4 (1.3)	5.1 (1.5)
Animal Welfare			
Health group	4.5 (1.2)	5.1 (1.4)	4.7 (1.7)
Environmental group	4.9 (1.5)	5.1 (1.5)	4.8 (1.6)
Animal welfare group	5.0 (1.1)	6.2 (0.9)	5.8 (0.9)
Control	4.9 (1.4)	4.9 (1.5)	4.8 (1.8)
High Cost			
Health group	4.6 (1.4)	4.6 (1.5)	4.6 (1.9)
Environmental group	4.9 (1.6)	4.2 (1.5)	4.4 (1.5)
Animal welfare group	4.9 (1.4)	4.5 (1.7)	4.8 (1.5)
Control	5.1 (1.2)	5.1 (1.3)	5.4 (1.2)
Taste Preferences			
Health group	4.5 (1.5)	4.3 (1.5)	4.6 (1.8)
Environmental group	4.2 (1.3)	3.7 (1.1)	3.9 (1.2)
Animal welfare group	3.4 (1.9)	3.1 (1.5)	3.8 (1.4)
Control	4.4 (1.8)	4.4 (1.6)	4.2 (1.8)
Weight Control			
Health group	4.0 (1.9)	4.5 (1.7)	4.0 (2.0)
Environmental group	3.9 (1.7)	3.5 (1.7)	4.0 (1.7)
Animal welfare group	3.9 (1.9)	3.9 (1.7)	3.7 (1.7)
Control	3.9 (1.8)	3.7 (1.9)	4.6 (1.5)

Note. M = Mean, SD = Standard Deviation. Pre, post, and follow-up refer to questionnaire from which scores were taken. Motivations were measured on seven-point scales.

ANOVA and ANCOVA analyses. A one-way ANOVA was performed in order to determine if there were any significant differences in motivations to reduce meat consumption between the four groups at the time of the pre-questionnaire. Non-significant results were obtained for all six motivations ($P > .05$), and so it was determined, that for each motivation, no significant differences were present between

groups before film viewing. As shown in Table 8.7 below, significant differences in strength of motivation both within and between groups over time were only detected for two motivations: health and animal welfare. The health motivation showed the largest changes within groups over time (F = 4.11), while animal welfare showed the greatest changes between groups over time (F = 6.45).

Table 8.7 Repeated measures ANOVA and ANCOVA for motivations to reduce meat intake

	Within-group effects			*Between-group effects*		
	df	M^2	F	df	M^2	F
Health [#]	6	3.07	4.11*	3	6.66	3.90*
Environmental Benefit	5.90[+]	1.26	1.95	3	4.04	3.04
Animal welfare [#]	6	1.65	3.23*	3	9.84	6.45*
High Cost	5.67[+]	0.96	0.99	3	5.70	1.95
Taste Preferences	6	1.09	0.79	3	3.21	1.38
Weight control	6	2.82	2.46	3	2.76	1.25

Note. All p values were adjusted for multiple comparisons with the FDR procedure before determining significance.
[#] Homogeneity of regression slopes violated (ANCOVA)
[+] Huynh-Feldt corrected
* $p < .05$

Paired t-tests. Paired sample t-tests were performed for all groups for both the 'health benefits' and 'animal welfare' motivations in order to determine which groups showed significant changes in the relative strengths of these motivations and whether these changes were observed in the post-questionnaire and/or follow-up questionnaire when compared with initial pre-questionnaire values (see Table 8.8 below). The health group showed significant increases in the health and animal welfare motivations immediately after watching their film; however, these increases were no longer significant at the one month follow-up. The animal welfare group showed increases in the health motivation, but none of these increases were significant; while significant increases were observed for the animal welfare motivation for both the post and follow-up questionnaires. Interestingly, the environmental group showed a significant decrease in the health motivation in the post questionnaire, although this significance was not present in the follow-up. The control group showed no significant increases or decreases in either motivation.

Table 8.8 Paired t-test results for changes in motivations over time

	Post			*Follow-up*		
	Δ Mean	SD	t	Δ Mean	SD	t
Health benefits						
Health group	0.87	1.15	3.5*	- 0.02	1.14	- 0.09
Env. group	-0.62	0.76	- 3.8*	- 0.40	0.89	- 2.0
Animal group	0.76	1.79	1.94	0.45	1.63	1.26
Control	- 0.10	0.23	- 0.46	- 0.13	1.23	- 0.50
Animal welfare						
Health group	0.66	1.04	2.99*	0.23	1.42	0.77
Env. group	0.18	0.82	1.01	- 0.13	0.97	- 0.61
Animal group	1.20	1.06	5.19*	0.81	1.21	3.08*
Control	0.07	0.66	0.50	- 0.09	0.82	- 0.48

Note. 'Health benefits' and 'Animal welfare' are motivations for meat reduction. All p values were adjusted for multiple comparisons with the FDR procedure before determining significance. 'Env.' = environmental and 'Animal' = animal welfare. * $p < .05$

ANCOVA pairwise comparisons. The repeated measures ANCOVA determined that significant differences were present over the three measured time points between the groups for both the health and animal welfare motivations. However, the authors wished to determine which groups differed and the magnitude of these differences and thus, post-hoc pairwise comparisons were performed alongside the ANCOVA analysis (see Table 8.9 below).

The pairwise comparisons allowed for examining the changes in motivations between groups over time while controlling for pre-questionnaire scores. The environmental group had significantly lower values for the health motivation in comparison to both the health and animal welfare groups over time; however, the health group did not have significantly higher health motivation scores in comparison to the control and the animal welfare group (See Table 8.9 below). In contrast, the animal welfare group had significantly higher values for the animal welfare motivation in comparison to all other groups, with the largest differences being seen when compared to the control, the environmental group, and the health group, respectively.

Table 8.9 ANCOVA pairwise comparisons for motivations

	Comparison Group	M Difference	Std. Error
Health benefits			
Health group	Env.	0.90*	0.28
	Animal	0.12	0.29
	Control	0.40	0.28
Env. group	Animal	- 0.78*	0.29
	Control	- 0.50	0.29
Animal group	Control	0.28	0.30
Animal Welfare			
Health group	Env.	0.33	0.27
	Animal	- 0.67*	0.27
	Control	0.38	0.27
Env. group	Animal	- 1.01*	0.27
	Control	0.04	0.27
Animal group	Control	1.05*	0.27

Note. 'Health benefits' and 'Animal welfare' are motivations for meat reduction. Mean differences are derived from estimated marginal means which have been adjusted based upon pre-questionnaire scores. All p values were adjusted for multiple comparisons with the FDR procedure before determining significance. 'Env.' = environmental and 'Animal' = animal welfare. * $p < .05$

8.4.5 TPB, meat attachment, and agreement with structural measures

The following section will review whether (and how) changes to measured variables (i.e. TPB, meat attachment, and agreement with proposed structural measures) occurred over time both within and/or between groups. Descriptive statistics for the variables will first be given, followed by the findings of the repeated measures ANOVAs and ANCOVAs in order to determine changes in these variables both within and between groups, respectively. Lastly, the post-hoc tests that were conducted will be discussed (paired t-tests for the repeated measures ANOVA and pairwise comparisons for the repeated measures ANCOVA) in order to gather more detail on how groups differed on each variable.

Descriptive Statistics. Descriptive statistics for the TPB components (i.e. attitudes, subjective norm, perceived behavioural control), meat attachment and

subscales, and agreement with proposed policy measures are provided in Table 8.10. The environmental group's pre-questionnaire scores were lower than other groups for variables like attitudes, meat attachment, and the meat attachment subscales, and this was simply due to there being a higher number of individuals within that group that scored lower on those variables. Given that the differences between groups were within one standard deviation of one another, such differences were determined to likely have no major implications on proposed analyses; however, caution must still be taken when interpreting results for the environmental group due to these differences.

Table 8.10 Means and standard deviations between experimental and control groups for TPB components, meat attachment and subscales, and agreement with policies

	M (SD)		
	Pre	Post	Follow-up
Attitudes			
Experimental	4.8 (1.0)	3.5 (1.1)	4.0 (1.1)
Health group	4.9 (1.1)	3.5 (1.2)	4.1 (1.4)
Environmental group	4.4 (1.0)	3.5 (1.1)	3.9 (0.9)
Animal welfare group	4.9 (0.8)	3.6 (1.2)	4.0 (0.9)
Control	4.8 (0.8)	4.7 (0.9)	4.9 (1.0)
Subjective Norm			
Experimental	17.9 (12.1)	16.2 (11.6)	16.9 (13.1)
Health group	17.5 (11.8)	16.4 (11.5)	16.4 (15.5)
Environmental group	14.1 (12.3)	12.1 (10.8)	15.2 (12.0)
Animal welfare group	22.1 (11.6)	20.0 (11.7)	19.2 (11.5)
Control	21.5 (9.6)	21.8 (9.8)	20.0 (13.0)
PBC			
Experimental	5.6 (0.9)	5.9 (0.8)	5.7 (1.0)
Health group	5.6 (0.7)	6.0 (0.5)	5.9 (0.7)
Environmental group	5.9 (0.9)	5.8 (0.8)	5.8 (1.1)
Animal welfare group	5.4 (1.0)	5.9 (0.9)	5.4 (1.1)
Control	5.8 (0.8)	5.9 (0.9)	5.5 (0.9)
Hedonism			
Experimental	4.9 (1.3)	4.0 (1.2)	4.2 (1.3)
Health group	4.8 (1.7)	4.0 (1.4)	4.2 (1.7)
Environmental group	4.5 (1.0)	3.9 (1.1)	4.1 (1.0)
Animal welfare group	5.3 (0.9)	4.1 (1.2)	4.1 (1.1)
Control	4.7 (0.9)	4.7 (1.0)	4.7 (0.9)
Affinity			
Experimental	5.2 (1.2)	3.7 (1.3)	4.3 (1.3)
Health group	5.3 (1.2)	3.8 (1.4)	4.5 (1.4)
Environmental group	5.0 (1.4)	4.0 (1.2)	4.5 (1.2)
Animal welfare group	5.2 (1.0)	3.3 (1.0)	3.9 (1.3)
Control	5.0 (1.0)	4.9 (1.1)	4.9 (1.2)
Entitlement			
Experimental	4.5 (1.4)	3.4 (1.4)	3.6 (1.6)
Health group	4.7 (1.6)	3.9 (1.9)	4.0 (2.1)
Environmental group	4.2 (1.4)	3.0 (1.1)	3.6 (1.3)
Animal welfare group	4.5 (1.2)	3.4 (1.1)	3.4 (1.2)
Control	4.1 (1.1)	4.0 (1.1)	3.9 (1.1)

Table 8.10 (continued) Means and standard deviations between experimental and control groups for TPB components, meat attachment and subscales, and agreement with policies

	M (SD)		
	Pre	Post	Follow-up
Dependence			
Experimental	3.9 (1.3)	3.0 (1.1)	3.2 (1.2)
Health group	4.3 (1.6)	3.1 (1.2)	3.2 (1.3)
Environmental group	3.3 (1.3)	2.8 (1.1)	2.9 (1.2)
Animal welfare group	4.2 (0.9)	3.0 (1.1)	3.4 (1.1)
Control	4.1 (0.9)	3.9 (1.0)	4.0 (1.0)
Meat Attachment			
Experimental	4.6 (1.1)	3.5 (1.0)	3.8 (1.1)
Health group	4.7 (1.4)	3.6 (1.3)	3.9 (1.4)
Environmental group	4.2 (0.9)	3.4 (0.8)	3.7 (0.9)
Animal welfare group	4.8 (0.7)	3.4 (0.7)	3.7 (0.9)
Control	4.5 (0.7)	4.3 (0.7)	4.4 (0.7)
Tax Agreement			
Experimental	2.9 (1.5)	4.6 (1.6)	3.9 (1.8)
Health group	2.5 (1.7)	4.2 (1.7)	3.5 (1.8)
Environmental group	2.9 (1.1)	4.7 (1.1)	3.7 (1.6)
Animal welfare group	3.4 (1.4)	4.7 (2.1)	4.6 (1.8)
Control	3.1 (1.7)	3.3 (1.9)	3.2 (1.8)
Subsidy Agreement			
Experimental	5.3 (1.2)	6.1 (0.9)	6.0 (1.2)
Health group	5.4 (1.5)	6.2 (0.9)	6.0 (1.7)
Environmental group	5.3 (0.9)	5.9 (1.0)	6.0 (0.9)
Animal welfare group	5.1 (1.0)	6.0 (0.8)	5.9 (0.9)
Control	4.9 (1.7)	5.3 (1.8)	5.6 (1.4)
Procurement Agreement			
Experimental	4.6 (1.3)	5.7 (1.3)	5.3 (1.4)
Health group	4.2 (1.4)	5.6 (1.4)	4.8 (1.7)
Environmental group	4.6 (1.1)	5.8 (1.2)	5.4 (1.3)
Animal welfare group	5.0 (1.2)	5.9 (1.1)	5.6 (1.2)
Control	5.1 (1.2)	5.5 (1.1)	5.3 (1.1)
Policy Agreement			
Experimental	4.3 (0.9)	5.5 (0.9)	5.1 (1.2)
Health group	4.0 (1.1)	5.4 (0.8)	4.8 (1.5)
Environmental group	4.3 (0.6)	5.5 (0.9)	5.1 (0.9)
Animal welfare group	4.5 (0.8)	5.5 (1.0)	5.4 (1.1)
Control	4.3 (1.0)	4.7 (1.1)	4.7 (1.0)

Note. The experimental group is comprised of the health, environmental, and animal welfare groups. ‘Pre’, ‘Post’, and ‘Follow-up’ refer to the questionnaire from which the value was taken.

Repeated Measures ANOVAs and ANCOVAs. Repeated Measures ANOVAs and ANCOVAs were performed in order to test for significant changes in dependent variables over time, both within and between the experimental and control groups. As shown in Table 8.11 below, ANOVA results showed significant differences within groups over time for attitudes, PBC, meat attachment and its subscales, tax agreement, and overall policy agreement. In addition, ANCOVA results identified significant differences between groups over time for these same variables, excluding PBC; therefore, significant changes occurred to the variables identified above not only within groups, but also between groups (excluding PBC) over time. No significant changes either within or between groups over time were seen for subjective norm, subsidy agreement, or procurement agreement; however, the assumption of homogenous regression slopes was violated between the groups for subjective norm, and so a non-significant result must be interpreted with caution. Two sociodemographic variables, gender and ethnicity, were included as covariates in preliminary ANOVAs and ANCOVAs in order to test whether they significantly impacted any observed changes for the dependent variables; however, no significant effects were observed and therefore, gender and ethnicity were not included in final analyses.

Table 8.11 Repeated measures ANOVA and ANCOVA for dependent variables by film vs control (experimental or control) and group from pre-questionnaire to follow-up

	Within-group effects			*Between-group effects*		
	df	M^2	F	df	M^2	F
Attitudes						
Film vs Control	1.79^{+}	5.79	11.57^{*}	1	30.12	29.73^{*}
Group	5.49^{+}	2.16	4.42^{*}	3	10.43	10.19^{*}
Subjective Norm						
Film vs Control	2	27.50	0.47	1	133.91	0.99
Group $^{\#}$	6	29.23	0.49	3	52.48	0.38
PBC						
Film vs Control	2	0.62	2.40	1	2.00	3.32
Group	6	0.68	2.74^{*}	3	1.66	2.87
Hedonism						
Film vs Control	2	3.76	7.78^{*}	1	15.66	14.23^{*}
Group $^{\#}$	6	1.55	4.12^{*}	3	6.66	6.20^{*}

Table 8.11 (continued) Repeated measures ANOVA and ANCOVA for dependent variables by film vs control (experimental or control) and group from pre-questionnaire to follow-up

	Within-group effects			*Between-group effects*		
	Df	M [2]	F	Df	M [2]	F
Affinity						
Film vs Control	2	6.83	15.07*	1	31.29	23.63*
Group #	6	3.25	7.58*	3	15.20	12.90*
Entitlement						
Film vs Control	2	3.35	7.10*	1	14.77	10.62*
Group #	6	1.67	3.60*	3	6.39	4.66*
Dependence						
Film vs Control #	2	2.54	5.69*	1	18.65	17.02*
Group #	6	1.50	3.46*	3	6.77	6.14*
Meat Attachment						
Film vs Control	2	3.66	14.36*	1	19.10	25.87*
Group #	6	1.64	6.67*	3	7.56	10.62*
Tax Agreement						
Film vs Control #	2	8.13	8.90*	1	40.06	15.97*
Group #	6	3.43	3.78*	3	13.53	5.27*
Subsidy Agreement						
Film vs Control	2	0.83	1.34	1	3.41	2.79
Group	6	0.44	0.69	3	1.15	0.92
Procurement Agreement						
Film vs Control	2	2.23	2.51	1	3.65	2.47
Group	6	1.15	1.28	3	1.46	0.97
Policy Agreement						
Film vs Control	2	2.84	7.53*	1	11.97	11.35*
Group	6	1.15	3.04*	3	4.03	3.73*

Note. Within-group effects measured whether changes occurred within each categorical variable (control or experimental for 'Film vs Control' and groups 1, 2, 3, and 4 for 'group') over time. Between-subjects effects measured whether significant differences were present between these same categorical variables over time. All significance values were adjusted with the FDR procedure.
Homogeneity of regression slopes violated (ANCOVA assumption)
+ Huynh-Feldt corrected
* $p < .05$

While controlling for pre-questionnaire values, Film vs Control had significant changes (* = $p < .05$) both within and between groups over time (respectively) for the variables:

- Attitudes – F (1.79, 83) = 11.57* and F (1, 84) =29.73*
- Hedonism – F (2, 83) = 7.78* and F (1, 84) = 14.23*
- Affinity – F (2, 83) = 15.07* and F (1, 84) = 23.63*
- Entitlement – F (2,83) = 7.10* and F (1, 84) = 10.62*
- Dependence – F (2, 83) = 5.69* and F (1, 84) = 17.02*
- Meat Attachment – F (2,83) = 14.36* and F (1, 84) = 25.87*
- Tax Agreement – F (1, 84) = 8.90* and F (2, 83) = 15.97*
- Policy Agreement – F (1, 84) = 7.53* and F (2, 83) = 11.35*.

Groups showed significant changes (* = $p < .05$) both within and between groups over time (respectively) for these same variables while controlling for pre-questionnaire values, with the addition of PBC (but only within group effects):

- Attitudes – F (5.49, 79) = 4.42* and F (3, 82) = 10.19*
- PBC – F (6, 79) = 2.74*
- Hedonism – F (6, 79) = 4.12* and F (3, 82) = 6.20*
- Affinity – F (6, 79) = 7.58* and F (3, 82) = 12.90*
- Entitlement – F (6,79) = 3.60* and F (3, 82) = 4.66*
- Dependence – F (6, 79) = 3.46* and F (3, 82) =6.14*
- Meat Attachment – F (6,79) = 6.67* and F (3, 82) = 10.62*
- Tax Agreement – F (6, 79) = 3.78* and F (3, 82) = 5.27*
- Policy Agreement – F (6, 79) = 3.04* and F (3, 82) = 3,73*.

Paired t-tests. The repeated measures ANOVA determined that significant differences were present for some variables within groups over time; however, the authors wished to determine the nature and magnitude of these differences for both the post and follow-up questionnaires in comparison to the pre-questionnaire; therefore, post-hoc paired t-tests were performed.

Attitudes. As shown in Table 8.12 below, attitudes were significantly lower in the post and follow-up questionnaires for all experimental groups (compared to the pre-questionnaire), but no significant differences were seen over time in the control group. The most significant declines were seen within the health and animal welfare groups, with

animal welfare maintaining the greatest decline in attitudes in the follow-up questionnaire. It is important to note that the smaller declines in attitudes seen within the environmental group could be due to a floor effect, as the pre-questionnaire mean value for attitudes in this group was already lower than the values seen for other groups (see Table 8.10). In addition, although attitudes became more positive between the post-questionnaire and follow-up, the follow-up values among the three experimental groups were still significantly lower when compared to their pre-questionnaire counterparts. Therefore, the health, environmental, and animal welfare groups were all determined to have significantly lower attitudes (i.e. viewed meat less favourably) as a result of watching their respective films, and this effect was still present one month after viewing.

PBC. Interestingly, PBC showed significant increases for both the health and animal welfare groups, but only the health group maintained this significant increase in the follow-up questionnaire. Once again, a non-significant result for the environmental group might be due to a ceiling effect, as the pre-questionnaire mean value for the PBC variable was already greater for the environmental group when compared to the other experimental groups (see Table 8.10). Therefore, only the health and animal welfare groups were determined to have significantly higher PBC (i.e. higher feelings of control over whether they consume meat) as a result of watching their respective films, with this effect still being seen in the health group one month after viewing.

Meat attachment. Meat attachment, along with all four of its subscales, were significantly lower in the post and follow-up questionnaires for all experimental groups, except the dependence subscale for the environmental group in the follow-up. It is important to note once again that this non-significant result in the follow-up for the environmental group might be due to a floor effect, as the pre-questionnaire mean value for the dependence variable was lower for the environmental group when compared to the other groups (see Table 8.10). In addition, although meat attachment and its subscales became more positive between the post-questionnaire and follow-up, the follow-up values were still significantly lower when compared to their pre-questionnaire counterparts. Therefore, the health, environmental, and animal welfare groups were determined to have significantly lower meat attachment after viewing their respective films, and despite a weakening of this effect over time, meat attachment was still significantly lower even one month after viewing.

Among the experimental groups, the animal welfare group showed the greatest declines (both in terms of frequency and magnitude) in mean scores for meat attachment and all four subscales both in the post and follow-up questionnaires, excluding follow-up mean values for dependence. However, these greater declines in raw means did not necessarily result in the highest t-values due to the increased standard deviations seen for the animal welfare group across all meat attachment variables. Thus, the impacts of the film in the animal welfare group were more inconsistent among individuals when compared with the impacts of the other films on individuals in other groups. Based upon t-values, the animal welfare group had the greatest declines in hedonism, affinity, and overall meat attachment in the post-questionnaire, with the environmental group having the greatest declines among these variables in the follow-up. The environmental group initially had the greatest declines in entitlement in the post-questionnaire, but the animal welfare group maintained the lowest values in the follow-up. The health group showed the greatest decline in dependence in the post-questionnaire and this trend remained in the follow-up as well. When assessing changes in overall meat attachment, the health group saw the greatest declines, followed by the animal welfare and environmental groups, respectively. Interestingly, the control group saw significant declines in dependence and meat attachment in the post-questionnaire but not in the follow-up, although the raw magnitude of these declines (i.e. change in means) was much smaller than those seen in the experimental groups.

Agreement with structural measures. Significant increases in tax and overall policy agreement were seen for all experimental groups in both post and follow-up questionnaires (see Table 8.12 below). All groups, including the control, moved from being centred on agreement with such measures, to being favourable of such measures (although less so for the control; see Table 8.10). The environmental group showed the greatest increases in tax and overall policy agreement in both the post and follow-up questionnaires, except for tax agreement in the follow-up, in which the animal welfare group showed the greatest increase in agreement. Although much smaller in magnitude, it is worth noting that the control group also had a significant increase in overall policy agreement.

Table 8.12 Paired t-test results for changes in dependent variables over time

	Post			*Follow-up*		
	Δ Mean	SD	t	Δ Mean	SD	t
Attitudes						
Health group	- 1.36	1.29	- 5.0*	- 0.77	0.80	- 4.5*
Env. group	- 0.91	0.95	- 4.4*	- 0.55	0.78	- 3.3*
Animal group	- 1.36	1.32	- 4.7*	- 0.88	0.73	- 5.6*
Control	- 0.12	0.40	- 1.4	0.10	0.66	0.7
PBC						
Health group	0.42	0.57	3.4*	0.33	0.59	2.6*
Env. group	- 0.09	0.49	- 0.8	- 0.06	0.64	- 0.4
Animal group	0.45	0.77	2.7*	- 0.01	1.02	- 0.1
Control	0.03	0.55	0.25	- 0.30	0.76	- 1.8
Hedonism						
Health group	- 0.81	1.09	-3.5*	- 0.59	0.64	- 4.3*
Env. group	- 0.63	0.80	- 3.6*	- 0.40	0.48	- 3.8*
Animal group	- 1.17	1.27	- 4.2*	-1.14	1.35	- 3.9*
Control	- 0.05	0.35	- 0.67	-0.03	0.56	-0.3
Affinity						
Health group	- 1.46	1.06	- 6.4*	- 0.78	0.81	- 4.5*
Env. group	- 0.94	1.09	- 4.0*	- 0.52	0.65	- 3.6*
Animal group	- 1.90	1.04	- 8.4*	- 1.33	1.43	- 4.3*
Control	- 0.12	0.39	- 1.5	- 0.11	0.59	- 0.8
Entitlement						
Health group	- 0.75	1.22	- 2.9*	- 0.71	0.93	- 3.6*
Env. group	- 1.16	0.97	- 5.5*	- 0.59	0.94	- 2.9*
Animal group	- 1.16	1.16	- 4.6*	- 1.14	1.33	- 3.9*
Control	- 0.13	0.33	- 1.8	- 0.18	0.55	- 1.5
Dependence						
Health group	- 1.19	1.03	- 5.4*	- 1.04	0.91	- 5.4*
Env. group	- 0.52	0.90	- 2.7*	- 0.40	1.11	- 1.7
Animal group	- 1.20	1.04	- 5.3*	- 0.79	1.42	- 2.6*
Control	- 0.24	0.41	- 2.7*	- 0.10	0.51	- 0.9

Table 8.12 (continued) Paired t-test results for changes in dependent variables over time

	Post			*Follow-up*		
	Δ Mean	SD	t	Δ Mean	SD	t
Meat Attachment						
Health group	- 1.08	0.84	- 6.0*	- 0.80	0.46	- 8.3*
Env. group	- 0.78	0.76	- 4.7*	- 0.47	0.60	- 3.6*
Animal group	- 1.36	0.92	- 6.8*	- 1.08	1.22	- 4.1*
Control	- 0.14	0.18	- 3.7*	- 0.10	0.34	- 1.4
Tax Agreement						
Health group	1.71	1.71	4.7*	0.97	1.63	2.8*
Env. group	1.86	1.42	6.0*	0.84	1.31	2.9*
Animal group	1.33	1.60	3.8*	1.20	1.20	4.6*
Control	0.22	0.61	1.7	0.10	0.99	0.5
Policy Agreement						
Health group	1.31	0.99	6.2*	0.71	1.16	2.9*
Env. group	1.22	0.81	6.9*	0.80	0.78	4.8*
Animal group	1.01	0.96	4.8*	0.88	0.90	4.5*
Control	0.34	0.51	3.1*	0.37	0.78	2.1

Note. All p values were corrected for multiple comparisons with the FDR procedure. 'Env.' = environmental and 'Animal' = animal welfare. * $p < .05$

ANCOVA pairwise comparisons. The repeated measures ANCOVA determined that significant differences were present over the three measured time points between the experimental groups and control group for specific dependent variables, while controlling for pre-questionnaire values. However, which groups differed and the magnitude of these differences still needed to be determined and thus, post-hoc pairwise comparisons were performed alongside the ANCOVA analysis (See Table 8.13 below).

Attitudes. Across the post and follow-up questionnaires, attitude scores for all experimental groups were significantly lower than scores for the control group (See Table 8.13 below). The animal welfare group had the most significant declines, followed by the health and environmental groups, respectively; however, it is important to keep in mind that these means are adjusted based upon pre-questionnaire scores, and so the smaller decline seen in the environmental group could be due to their lower pre-questionnaire attitude score (floor effect).

Meat attachment. Meat attachment and all its subscales showed significant declines in all experimental groups in comparison to the control group, except the health group for the entitlement variable. In comparison to the control group, the animal welfare group showed the greatest declines in hedonism, affinity, entitlement, and meat attachment; with the health group having the strongest decline in dependence. For overall

meat attachment, the animal welfare group showed the greatest declines, followed by the health and the environmental groups, respectively.

Agreement with structural measures. Significant increases in agreement were observed for the proposal of a meat tax and overall policies (all three policy proposals combined) in comparison to the control group. Increases in agreement were quite similar across the experimental groups, with the animal welfare and environmental groups showing similarly large increases in tax and overall policy agreement.

Table 8.13 ANCOVA pairwise comparisons between experimental groups and control group

	M Difference	Std. Error
Attitudes		
Health group	- 1.02*	0.22
Env. group	- 0.84*	0.22
Animal group	- 1.07*	0.22
Hedonism		
Health group	- 0.64*	0.22
Env. group	- 0.52*	0.23
Animal group	- 0.97*	0.23
Affinity		
Health group	- 0.93*	0.24
Env. group	- 0.62*	0.24
Animal group	- 1.44*	0.24
Entitlement		
Health group	- 0.46	0.26
Env. group	- 0.71*	0.26
Animal group	- 0.91*	0.26
Dependence		
Health group	- 0.89*	0.23
Env. group	- 0.60*	0.24
Animal group	- 0.80*	0.23
Meat Attachment		
Health group	- 0.74*	0.18
Env. group	- 0.59*	0.19
Animal group	- 1.01*	0.19
Tax Agreement		
Health group	1.04*	0.35
Env. group	1.14*	0.35
Animal group	1.20*	0.35
Policy Agreement		
Health group	0.57*	0.23
Env. group	0.64*	0.23
Animal group	0.64*	0.23

Note. Mean differences are derived from estimated marginal means which have been adjusted based upon pre-questionnaire scores. Significance values are corrected for multiple comparisons using the FDR procedure. ‘Env.’ = environmental and ‘Animal’ = animal welfare. * $p < .05$

8.4.6 Follow-up film factors

This section will briefly review the answers provided by participants on additional questions included in the follow-up questionnaire, which were designed and included in an effort to acquire more context and detail on the actions and/or reactions of individuals to their respective films.

Descriptive statistics. Descriptive statistics are provided for factors measured at time of follow-up in order to (roughly) determine if any differences were present between groups in terms of meat-related actions, thoughts, and/or feelings after watching a film. The health group had the largest number of participants who searched for additional information after watching the film, followed by the animal welfare, environmental, and control groups respectively. A majority in all groups (and all participants in the animal welfare group) discussed content of their respective films with others after the viewing and thus, sharing with others seemed to be a common theme, regardless of film content. The animal welfare group had the greatest number of participants who stated that they felt the film had a lasting impact on their meat-related attitudes and behaviours, followed by the health, environmental, and the control groups respectively.

Table 8.14 Descriptive statistics for factors measured at time of follow-up

	Health group	Env. group	Animal group	Control
Additional Information				
Yes, Searched	12	5	6	0
Yes, by chance	4	3	9	5
No	6	13	6	16
Discussed with Others				
Yes	20	18	21	18
No	2	3	0	3
Lasting Impact-Attitudes				
Yes	13	10	17	2
No	9	11	4	19
Lasting Impact-Habits				
Yes	11	9	13	2
No	11	12	8	19

Note. ‘Env.’ = environmental and ‘Animal’ = animal welfare.

8.5 Discussion

This section will be dedicated to discussing the results for Study 2's research questions. To begin, a summary of the results, organised by research question, is provided to refresh the reader's mind of the main findings and to act as a quick and easy reference (8.5.1). The general discussion follows (8.5.2), outlining not only how the findings compare and add to past studies on how informative interventions can influence meat consumption/reduction, but also what these findings may mean for the promotion of meat reduction throughout society. After the general discussion, the final two sections of the chapter are presented, which will include the limitations and future directions of Study 2 (8.6) followed by the final conclusions (8.7).

8.5.1 Summary of results

RQ1, RQ2, and RQ3. A summary of the results for the first three research questions (and their respective hypotheses) is provided below (Table 8.15).

Table 8.15 Results summary for research questions 1, 2, and 3

	Health	Env.	Animal	Control
RQ1 – Intake				
Willingness to reduce	**+**	**+**	**+**	**-**
Intentions to reduce	**+**	**+**	**+**	**-**
Reduction frequency	**+**	**+**	**+**	**-**
Reduced meat intake	**+**	**+**	**+**	**-**
RQ2 – Motivations				
Health benefits	**+/-**	**-**	**-**	**-**
Environmental benefits	**-**	**-**	**-**	**-**
Animal welfare	**+/-**	**-**	**+/+**	**-**
High cost	**-**	**-**	**-**	**-**
Taste preferences	**-**	**-**	**-**	**-**
Weight control	**-**	**-**	**-**	**-**
RQ3 – Meat-related variables				
Reduced positive attitudes	**+/+**	**+/+**	**+/+**	**-**
Reduced subjective norm	**-**	**-**	**-**	**-**
Increased perceived behavioural control	**+**	**-**	**+/-**	**-**
Reduced meat attachment	**+/+**	**+/+**	**+/+**	**-**
Increased agreement structural measures	**+/+**	**+/+**	**+/+**	**+/-**

Note. **+/+** = positive for post and follow up; **+/-** = positive for post and null for follow up

RQ4 – Post-viewing actions and reactions. The majority of participants in the health and animal welfare groups received additional information, whether on purpose or by chance, after watching their film; while a minority of participants did so in the environmental and control groups. As measured at time of follow-up, a majority of participants in all groups discussed film content with others, and so further sharing and engagement occurred regardless of film content. Participants' self-evaluated impacts of the film on their meat-related attitudes and habits were highest in the animal welfare group, followed by the health, environmental, and control groups respectively.

8.5.2 General discussion

Meat intake and associated variables. Exposure to health, environmental, and animal welfare information all significantly increased willingness and intentions to reduce, in addition to reduction frequencies among individuals; while significantly decreasing meat intake frequencies. These findings align with results from Cordts et al. (2014), who found that intentions to reduce meat intake increased after having read an article about meat's impacts on either personal health, climate change, or animal welfare. In addition, past health studies found that the provision of health information reduced intentions to eat meat (Bertolotti et al., 2016), as well as self-reported processed meat intake frequencies (Carfora et al., 2017). However, these findings do slightly contradict Loy et al. (2016) who found that information alone (in the form of text messages) increased intentions to reduce meat, but did not lead to actual meat reductions among study participants. They did find however, that when mental contrasting with implementation intentions (MCII) was provided alongside informative messages, increased intentions did result in reduced meat intake (i.e. closing the intention-behaviour gap). The differences in results between the present study and the one conducted by Loy and colleagues could be a result of differences in the presentation of information (documentary film vs text messages) and/or the measurement tool utilised (questionnaire vs dietary diaries). Nonetheless, it would seem that the provisioning of information in regards to meat-related impacts can be effective at promoting meat reduction, depending upon what types (and in what forms) information is presented and what techniques might be used alongside them (e.g. MCII) to further promote behaviour change. As more studies

are conducted on the topic, a clearer picture should materialise in regards to what types (and in what forms) of information are effective at promoting sustained meat reduction.

Among different types of information, this study found that environmentally-focused information, specifically, seems to promote the greatest declines in meat intake frequency, which is somewhat surprising considering its weaker effects on attitudes and meat attachment in comparison to the health and animal welfare frames (although possibly influenced by ceiling or floor effects within the environmental group). This finding contrasts that of de Boer et al. (2013) and de Groeve and Bleys (2017), who found the provision of information in regards to animal agriculture's contribution to climate change to be ineffective at promoting reduced-meat diets or support for "less meat initiatives", respectively. This could be due to a difference in the scope of the environmental framing of the information however (e.g. specifically focusing on one environmental issue like climate change vs multiple issues like climate change, pollution, land degradation, etc.). Furthermore, the finding from the current study aligns with more recent research which found that consumer willingness (Stea and Pickering, 2018) and intentions (Graham and Abrahamse, 2017) to reduce meat intake could be increased through the use of environmental frames. Making definitive conclusions for why the environmental frame was so effective at reducing meat intake frequency, in comparison to the health and animal welfare frames, resides outside the scope of this study; however, one could postulate a few potential factors. First, as established in previous studies, awareness of meat's environmental impacts is still quite low (Campbell-Arvai, 2015; Lea and Worsley, 2008; Lentz et al., 2018; Macdiarmid et al., 2016; Siegrist et al., 2015), meaning that consumers who are exposed to environmental information could find the presented concepts, ideas, and facts to be relatively novel in comparison to receiving information about subjects that are potentially more salient (i.e. health and animal welfare concerns). This novelty factor may create new, and possibly stronger motivations to reduce meat intake, by either creating a new motivational factor (i.e. environmental sustainability concerns), and/or expanding upon other motivational factors that are already more salient and established in the individual's mind (i.e. health or animal welfare concerns). Also, as mentioned previously, environmental issues and animal agriculture's contribution to them have garnered more media attention over the past few years, and so this may be a factor as well.

In addition, 'environmental impact' is a broad term that encompasses multiple issues, which may evoke varying thoughts in a consumer's mind, from ideas that are more self-interested (e.g. preservation of natural systems that sustain human health and life) to those that are more moral/ethical (e.g. preservation of ecosystems and biodiversity for their own sake). Through its potential to create concerns over both personal preservation and ethical treatment of other entities, concerns about environmental impacts may potentially offer a more robust reasoning for reducing meat consumption, in comparison to other motivations that are likely more unidimensional, such as health (primarily self-preservation) or animal welfare (primarily ethical) concerns.

Motivations. In regards to motivations, it would seem that exposure to health and environmental information show no significant lasting changes, at least in the present study. However, results support that exposure to animal welfare information can significantly increase the motivation to reduce meat consumption and this increase can persist over time. This result is not surprising, considering the findings of past studies, which found that consumers who hold ethical beliefs have stronger and more long-lasting commitments to meat reduction (Hoffman et al., 2013; Penny et al., 2015). However, it is important to note that animal welfare information will likely have greater variation in terms of individual impact in comparison to other forms of information that showed less variation in impact across individuals, as standard deviations were higher on average in the animal welfare group when compared to the health and environmental groups. Thus, animal welfare information will likely be quite effective for those particularly sensitive to such information, while health and environmental information will likely have more consistent impacts across individuals. Still, these findings are promising for the utilization of animal welfare information to promote reduced meat consumption among consumers, as those who are receptive do maintain higher motivational levels and those who are not receptive can be targeted with other forms of information.

It is important to note that these findings may not be cross-culturally applicable as animal welfare concerns have traditionally been a central tenet of vegetarian movements in 'western' cultures (Preece, 2009). As stated previously by Ruby and Heine (2012), as a result of cultural differences, it is likely that 'western' consumers are more preoccupied with the mental states of animals they consume when compared to 'non-western' consumers. Therefore, animal welfare information, as presented in this study, may have minimal to no impact on audiences from nations whose cultures differ greatly from those

found in many 'western' nations. Of course, interest in vegetarianism and plant-based diets have become more popular, especially within western nations and specifically over the last decade, and animal welfare concerns could have shifted into a position that is more prominent than once was within 'non-western' cultures due to rising globalization and cultural exchange. Studies on 'non-western' populations would need to be conducted in order to ascertain the validity of such an assertion however, as motivations for plant-based eating have mainly been focused within 'western' nations.

TPB. Exposure to health, environmental, and animal welfare information were all found to significantly reduce positive attitudes towards meat. Past studies have found attitudes to be a strong predictor of willingness and intentions to reduce meat intake (Berndsen and Van der Pligt, 2004; Graca et al., 2015a; Povey et al., 2001; Saba and Di Natale, 1998), as well as agreement with potential meat-reduction structural measures like taxes (Lentz et al., 2018) and thus, reductions in positive attitudes were expected to be accompanied by increases in these associated variables. Findings support this assertion, as a decline in positive attitudes was mirrored by increased intentions and willingness to reduce, in addition to greater agreement with policy measures; thus, it can be concluded that the provision of health, environmental, and/or animal welfare information can be effective tools for not only shifting more personal variables, such as attitudes, attachment, or intentions to reduce meat intake but also variables that are more societal in scope, such as agreement and support for potential meat-reduction structural measures (e.g. taxes).

Exposure to animal welfare information seemed to result in the greatest reductions in attitudes towards meat, followed by health and environmental information, respectively. Significant shifts in meat-related attitudes in response to exposure to environmentally-framed information slightly contradict findings of a previous study also conducted within New Zealand (Graham and Abrahamse, 2017), which found that the provision of information in regards to meat's climate change impacts did not significantly alter consumer attitudes, but did however increase concerns and decrease intentions to consume meat. These disparities in attitude change may simply be a result of sample size and composition differences between studies (i.e. smaller student sample in current study), greater media coverage in recent years regarding environmental issues like climate change and animal agriculture's contribution to such issues, or could also be linked to differences in the scope of environmental information provided within each study, as discussed previously. For instance, Graham and Abrahamse (2017) provided information

solely on meat's contribution to climate change, while the current study provided information that encompassed multiple environmental issues linked to meat consumption (e.g. pollution, inefficient conversion of crops, habitat and species loss, etc.).

The issue of scope could be an important factor, as two previous studies (Graham and Abrahamse, 2017; Vainio et al., 2018), found responses to climate-related messages and associated framings to be heavily influenced by prior beliefs, and that consumers who were sceptical of meat's climate change impacts were less influenced by the information in comparison to those who found climate change impacts to be believable. Prior beliefs can result in confirmation bias (i.e. seeking information that confirms held beliefs while ignoring contradictory information), which has been found in past food studies (White et al., 2003), including those focused on meat consumption (de Boer et al., 2013). Hence, an information provision method that takes a more holistic approach towards meat's environmental impacts, as was the case for the current study, may be a more effective method for overcoming issues of confirmation bias than methods that focus on only one impact, such as climate change. After all, it is relatively easy for an individual to be sceptical and/or ignore information about a habit's (e.g. meat intake) contribution to a single issue (e.g. climate change), while doing the same for a habit's contribution to a wide array of issues (e.g. climate change, soil and water degradation, deforestation, habitat loss, biodiversity loss, pollution, etc.) is much more difficult. For example, if an individual doubted climate change and/or meat's contribution to it, a climate change information framing could easily be dismissed by the individual and any potential shifts in motivation as a result of concern for the environment would likely go unrealised. However, if this same individual was presented with a more holistic environmental framing; multiple issues, in addition to climate change, would have to be dismissed by the individual in order to expect no potential change in motivation. This is an unlikely scenario (unless the individual carries little to no concern for the environment in general) due to the increased difficulty of refuting and/or ignoring information on meat's contribution to a wide array of issues, that when taken together, form an interconnected and cohesive 'picture' of its overall environmental impact. Thus, it would seem that there may be an advantage in taking a more holistic approach in the form of offering information on multiple environmental issues rather than just one, if changes in meat-related attitudes are desired; of course, future studies would have to compare provision methods in order to determine if such an advantage actually exists. Also, this advantage is

discussed in regards to environmental issues, but the same may be true for other issues as well, like those that are health or animal welfare related.

Exposure to health information was the only treatment that retained any long-term impacts on PBC and thus it seems that exposure to information on the health benefits of reducing meat in the diet gives consumers a feeling of having more personal control over whether they consume meat or not. This is likely due to consumer beliefs about the necessity of meat in the diet and that without its consumption, one would lack essential nutrients and/or minerals, resulting in possible health complications (i.e. the 'necessary' dimension of the 4Ns model by Piazza et al., 2015[5]). If consumers believe they require meat in order to maintain proper health, it would only stand to reason that a vast majority of individuals would view potential meat reduction, at least partially, as unsafe and impractical. After all, not many would take the risk of changing a behaviour if they believed such a change could potentially result in negative outcomes for personal health and/or wellbeing, even if other strong motivations existed for reducing meat consumption (e.g. environmental or animal welfare concerns). However, as a result of being presented with information that not only contradicts the idea of meat's dietary necessity, but also links its overconsumption to potential health risks, it only follows that consumers might question the true necessity of meat intake, as well as its positive contributions to health and wellbeing. Further study could offer more insight into how and to what extent health information impacts PBC in regards to meat intake, but initial findings from this study support the use of health information in order to increase feelings of control and choice among individuals who perceive meat as a requirement in the diet.

Exposure to information had no significant short-term or long-term effects on subjective norms. Thus, it seems that information on the health, environmental, or animal welfare impacts of meat consumption do not significantly affect individual thoughts or feelings on how others perceive the act of consuming meat. This could be the result of a potentially tenuous connection between personal knowledge and social relationships. For example, even if an individual created new thoughts and/or beliefs about performing a behaviour in response to receiving new information, this change would likely have limited to no effect on how others perceive the behaviour and the value an individual grants to these

[5] The 4Ns model by Piazza and colleagues frames the societal justifications for meat consumption. These 4Ns include meat intake being perceived as being: natural, necessary, nice, and normal. The 4Ns model is discussed further in Chapter 9.

perceptions (at least in the short-term). Thus, any changes in the social desirability of consuming meat are more likely to result from long-term shifts at the societal rather than individual scale, like some of the trends seen over the last decade (e.g. increased rates of vegetarianism among consumers, shifts towards more plant-based food products by food companies, media recognition of plant-based diets' environmental benefits).

MAQ. Exposure to health, environmental, or animal welfare information were all found to significantly reduce meat attachment; however, the animal welfare frame seemed to be most effective at reducing attachment, hedonism, affinity, and entitlement towards meat. Thus, the acquisition of animal welfare information can be quite effective at reducing individual feelings of meat attachment, pleasure in consuming meat, positive attributes given to meat, and the notion of having a right to consume meat; however, it is important to note that disparities in variable scores were more prevalent between those exposed to animal welfare information (i.e. higher standard deviations) in comparison to the other two experimental groups, meaning that exposure to animal welfare information will likely have more variation in terms of impacts while health and environmental information will have more consistent impacts across individuals. This is supported further by findings from Dowsett et al. (2018), who found that female participants exposed to animal welfare information (i.e. information that encouraged participants to make a connection between meat and an animal's life) showed lower levels of meat attachment when compared to a control, while male participants showed higher levels of meat attachment. This contradictory response between females and males shows that the provision of animal welfare information can be divisive in terms of lowering meat attachment. Although no differences were found between females and males in the current study, likely due to a relatively small sample size and female bias, animal welfare information still showed the greatest variations in terms of individual response, and thus the inconsistent impact of such information across individuals must be kept in mind. The provisioning of animal welfare information to promote lower meat attachment will likely have strong impacts, but on smaller numbers of individuals (likely females), meaning that the inclusion of other topics (e.g. health and environmental impacts) is likely advisable if the goal is to promote meat-reduction at the societal scale.

In contrast to other meat attachment subscales, dependence was most influenced by exposure to health information, and this finding aligns well with the significant impacts of health information on PBC. The reductions in dependence are likely due to a similar

mechanism as described for the reductions in PBC: individuals questioning their perception of meat as a necessity in the human diet, in response to the contradictory information received about its health implications. Interestingly, exposure to health information does not seem to significantly reduce feelings of entitlement towards consuming meat, contrasting effects seen for exposure to environmental or animal welfare information. This seems logical, as the idea of 'rights' is heavily based upon its use and applicability in relation to other entities (e.g. other humans, animals, and the environment). Since the impacts of personal choice in relation to health would primarily result in consequences for the self (e.g. living with a disease, decreased life expectancy, etc.), while the impacts of personal choice in relation to the environment or animals would primarily result in consequences for other living beings (humans included); it seems logical that information on the latter would more readily call into question one's right to consume meat. Thus, animal welfare information may be the most effective at reducing consumer meat attachment (particularly in females), while health information would likely be best utilised with consumers who feel that the consumption of meat is a dietary requirement.

Agreement with structural measures. Exposure to health, environmental, or animal welfare information were all found to significantly increase agreement with proposed meat-reduction structural measures over time, but exposure to environmental information, specifically, showed slightly higher increases in agreement when analysing within-group effects, followed respectively by animal welfare and health information. This slightly higher increase in agreement could be due to the sheer scale of environmental matters in comparison to other issues (e.g. health concerns), and/or perhaps the view that environmental degradation is a result of collective actions and should therefore be addressed through collective (e.g. policy) rather than personal (e.g. consumer choice) solutions. These reasons are purely speculative however, and analyses of between-group effects revealed similar impacts across information framings, matching results found by Whitley et al. (2017).

Therefore, despite slight differences between information framings, the only definitive conclusion is that exposure to health, environmental, or animal welfare information can all significantly increase agreement with meat-reduction structural measures, and these increases can persist over time (at least one month after exposure). In addition, results from the control group seem to support that policy agreement can increase even without

exposure to a targeted informational framework. Whether these effects were a result of exposure to the policy statements alone, exposure to another section of the questionnaire, or a combination cannot be determined; however, there is a possibility that simply exposing individuals to proposed structural measures increases agreement with the potential implementation of those measures (i.e. mere-exposure effect; Zajonc, 2001). If this is the case, individuals, governments, and/or organizations who seek to promote meat reduction through policy might gain public support through simply increasing the policy's exposure. Even if a hypothetical measure has not been fully designed, the priming of the public in advance may increase support for the measure and create a policy window whereby progressive policies can be effectively introduced and implemented (Rose et al., 2017). This information could be of use to any entity wishing to promote and/or implement long-term meat reduction policies, as garnering public support will likely be necessary, especially considering the expected resistance from both meat industries and consumers (Edjabou and Smed, 2013; Nordgren, 2012; Sall and Gren, 2015). Of course, exposure to information and to potential policies are not one-sided and do not operate in a vacuum, as consumers will likely be bombarded with pro-meat messaging and anti-tax propaganda through mass marketing, lobbying by the meat industry, etc.

8.6 Limitations and future directions

The limitations of this research are mainly a result of the sample acquired, methods utilised, and the specific scope of the research questions. The sample consisted of relatively young undergraduate students, a group that has been shown to not only have higher awareness in regards to environmental issues, but also potentially face more social pressure to behave in more environmentally-friendly ways, like reducing meat intake (Šedová et al., 2016). In addition, although gender was balanced between groups, the overall sample was female biased. Women, on average, eat less meat than men (Kiefer et al., 2005), are more risk-averse (Byrnes et al., 1999), and express higher concerns in response to information on meat's health, environmental, or animal welfare impacts (Cordts et al., 2014). Meat reduction has been found to be more prevalent among those who are younger, educated, and female; so findings from this study are likely to be a 'best case scenario' for the effectiveness of information provision. Thus, future studies would benefit from studying alternative consumer groups (e.g. male, older, less educated, non-

western, etc.) in order to determine how responses to different informational framings differ between consumer groups. Also, due to the nature of the study, the sample size was relatively small (85 total participants), and thus broad generalizations from results of the study must be made with caution.

Furthermore, effects of information framings and delivery method (i.e. film) were only studied in isolation, and so no interactive effects could be observed. A past study found no significant differences in intentions to reduce meat intake when a combination of health and climate change frames were combined (Vainio et al., 2018); however, future studies could utilize different combinations of framings (i.e. health and financial benefits, environmental and animal welfare benefits, etc.) and/or methods of delivery (e.g. films, written information, discussions with peers) to study their effectiveness either compared to one another, or to their use in isolation. Also, the method of delivery (i.e. film) was chosen due to its accessibility and ease of use among individuals, communities, organizations, and/or governments; but there is no guarantee that films (or at least, the appropriate type or quality of films) will be voluntarily viewed and/or absorbed by the target audience (Hirsch and Nisbet, 2007). Participants in this study were recruited knowing they would be watching a film, and so were likely prepared to listen and give full attention to the material presented; while the public's use of film will likely show more variation, due to sheer differences of interest in and/or attention paid to the material. In addition, only impacts in relation to the information contained within the films were studied, while supplementary factors that normally pair with a film's release such as marketing, social media discussions, and/or activist events were not explored.

Lastly, this study is limited by its scope. Although it provides information on how information framings differ in regards to their effects on meat-related variables and self-reported intake frequencies, no further information is provided on how individuals incorporated (or not) the information into their daily lives. For example, meat intake frequencies were lower, but it is unknown whether individuals reduced intake across all meat categories or whether reductions mainly occurred for a particular type of meat (e.g. red meat). Also, changes in other behaviours, such as shopping habits (e.g. buying different meats, shopping at different stores); consumption of other foods (e.g. eggs, milk, processed products); or other dietary factors (e.g. dining out less, dieting), were not measured. Qualitative research that takes a more intimate look at how meat-related

information is applied (or not) by consumers, and how this impacts people's daily lives is therefore another potentially fruitful avenue of study.

8.7 Study 2 conclusions

In summary, exposure to a film about meat's impacts on personal health, environmental sustainability, or animal welfare all increased intentions to curb meat intake, while reducing self-reported meat intake frequencies and positive attitudes and attachment towards meat. Environmental information showed the greatest declines in self-reported meat intake frequencies, meaning it may be particularly useful in campaigns aiming for direct behavioural change. Animal welfare information was most effective at reducing positive attitudes and attachment towards meat, while also raising (and sustaining) the motivation to reduce meat intake. Health information was particularly effective at reducing perceived dependence on meat, while simultaneously increasing perceived control in regards to its consumption. All three informational frames were found to increase agreement with proposed meat-reduction structural measures, and a potential mere-exposure effect may apply when promoting such policies to the public. These findings have implications for interventions trying to promote meat reduction, as information provision through the use of film would seem to be an effective method for promoting dietary shifts both at the personal and/or societal level. The utilization of different framings, methods, and techniques used in order to promote meat reduction for environmental and/or public health benefits holds promise, and further study into the nuanced effectiveness of these strategies will aid current and future researchers and practitioners.

8.8 Study 1 and Study 2 Implications

Study 1 found that consumers have relatively low awareness in regards to meat's environmental impacts, that those who reduce or abstain from meat are more driven by ethical concerns (e.g. environmental or animal welfare concerns) over those that are more personal (e.g. cost or health concerns), and that attitudes and meat attachment can accurately predict willingness and intentions to reduce meat consumption as well as agreement with proposed meat-reduction structural measures (e.g. environmental meat

tax). Study 2 has built upon those findings by determining how different informational framings (i.e. health, environmental, animal welfare) impact consumers' willingness and intentions to reduce meat intake, actual meat eating frequency, attitudes and attachment towards meat, and agreement with proposed structural measures. When presented in the format of a film, all informational framings resulted in significant impacts on the meat-related variables, with slight nuances between framings.

What do the results from both studies mean for shifting individuals (and eventually, society) to more plant-based diets? A logical first step would be to address the gap in consumer understanding between the choice to consume meat and its associated environmental impacts (identified in Study 1). By better informing consumers of meat's impacts through media (e.g. film, social media, commercials, etc.), their positive attitudes and attachment towards meat, which seem to be likely drivers behind consumption (as identified in Study 1), will shift or at least be more susceptible to shifts towards more plant-based eating. One avenue for raising awareness and/or encouraging attitude and behavioural change among consumers towards more plant-based diets is the utilization of film (as found in Study 2). Encouraging citizens/consumers to watch such films (through paid advertisements on television promoting such films, public endorsements of films by governments, etc.) could lead to individual (small-scale) shifts in personal attitudes and attachment towards meat, in addition to more support for more structural measures like an environmental tax on meat. As more and more individuals begin to shift their attitudes and behaviours, larger shifts in society are likely to be observed as consumer demand, social acceptance, and the collective conscience of society begin to increasingly support plant-based diets over time. Eventually, consumers and food suppliers (whether public or private) will likely enter a cyclical feedback loop whereby consumers increase demands for plant-based foods and suppliers respond by fulfilling those demands with greater supply. As supply of plant-based options increases and becomes more normalized, it will in turn likely increase demands from consumers even further as social and economic barriers weaken (i.e. increased social acceptance as new norm forms and products become more prevalent and cheaper due to increased availability and competition between companies to capture newly created market).

Based on the results from Study 1 and Study 2 and their potential for shifting society towards more plant-based diets at the societal level, Chapter 9 to follow will discuss what such a shift might look like and the evidence that such a shift is already underway.

CHAPTER 9

THESIS DISCUSSION AND CONCLUSIONS

9.1 Chapter outline

This Chapter will bring together findings from both Study 1 and Study 2, discuss their contributions to the understanding of current societal meat intake, and explore their implications for individuals, organisations, and/or governments who wish to promote meat reduction for environmental and/or public health reasons. Lack of public awareness in regards to meat's environmental impacts and how it is a missed opportunity for shifting meat eating behaviours is discussed (9.2), followed by the impacts of different informational framings on common meat reduction motivations (9.3) and the utility of the Theory of Planned Behaviour (TPB) and Meat-Attachment Questionnaire (MAQ) in studying meat intake and its reduction over time (9.4). Next, consumer agreement with proposed meat-reduction structural measures (9.5), consumer meat intake (9.6), how these two variables change in response to different information provision framings, and what this might mean for encouraging meat reduction at the individual and/or societal scale are outlined. Finally, an overview of how society might be shifted towards more plant-based diets, using a recently developed theoretical model for the justifications of meat consumption, is discussed (9.7), ending with the final conclusions of the thesis (9.8).

9.2 Awareness of environmental impacts and influence of information provision

Awareness of meat's environmental impacts seems to be low among New Zealand consumers, matching a trend that has also been found in other nations (Austgulen et al., 2018; Campbell-Arvai, 2015; Lea and Worsley, 2008; Macdiarmid et al., 2016; Siegrist et al., 2015). This is important, as a lack of understanding and awareness in regards to meat's environmental impacts may result in unconcerned consumers, who might otherwise contemplate and/or change their meat eating behaviours, if informed. Based upon findings from the current work, exposure to information on meat's environmental impacts can decrease positive attitudes and attachment towards meat, while also

increasing agreement with proposed meat-reduction policies, intentions to reduce meat intake, and self-reported intake frequencies. Therefore, it would seem that a lack of awareness in regards to meat's environmental impacts among consumers is a missed opportunity to promote not only personal meat intake reductions, but also potential meat-reduction policies that could encourage reduced-meat diets at a larger scale.

Of course, even if the environmental impacts of meat can be fully communicated to and understood by consumers, actual behaviour change outcomes will likely depend upon the characteristics of the target audience. As an example, consumers with low environmental concern and/or high attachment may be quite resistant to messages aimed at changing their meat intake, while consumers with high environmental concern and/or moderate to low levels of attachment are likely to be more amenable. Furthermore, within this consumer spectrum, there may be an 'optimal range' of individuals to target, based upon level of environmental concern and/or meat attachment, where information provision initiatives are most worthwhile in terms of absolute impact (i.e. greatest shifts in attitudes, attachment, meat intake, etc.). Consumers who are too low on this spectrum (i.e. low environmental concern and/or high meat attachment) may completely ignore or be strongly resistant to meat reduction messages, while those who are too high (i.e. high environmental concern and/or low meat attachment) may have already understood and/or incorporated similar messages into their outlooks on meat, making the provision of further information to these individuals a case of diminishing returns (i.e. ever smaller shifts in meat-related attitudes, attachment and intake reductions). Thus, identifying consumers who are concerned *enough* about the environment and/or who are not too particularly attached to meat, and directing environmental messages towards them will likely be the most effective awareness raising strategy in terms of promoting the greatest shifts in meat intake reductions. Future information provision studies that include large, diverse samples that encompass the full consumer spectrum will likely be able to provide more specific details on this 'optimal range' and how important its consideration may be for practitioners seeking to promote meat reductions for environmental benefit.

It is clear that providing information on meat's environmental impacts can promote reductions in consumer meat intake frequency; however, it remains unclear why the provisioning of environmental information, specifically, seems to be more effective than that of health or animal welfare information. Acquisition of information has been found to not always translate into behaviour change (i.e. value-action gap; Kollmuss and

Agyeman, 2002), however there may be exceptions if the information is novel to the target audience (e.g. Lee and Kotler, 2012; Mckenzie-Mohr, 2013). Thus, since consumer awareness in regards to meat's environmental impacts is low, it is plausible that receiving information on the topic is relatively novel for a majority of individuals. This novelty may result in more original contemplations and/or evaluations of current eating behaviours when compared to other topics that are possibly more salient among consumers, such as meat's health and animal welfare implications.

Novelty could be a key explanatory factor, but other aspects, such as the potential for simultaneous arousal of both self-preservation and ethical concerns within individuals, may also explain why exposure to environmental information has a particularly strong influence on meat intake. Provision of health or animal welfare information are likely more unidimensional in terms of fears and motivations raised when compared to environmental information (e.g. primarily those of self-preservation in response to health risks and primarily those of ethical quandaries in response to animal welfare concerns). After all, not only can awareness of environmental issues bring about questions of ethics and morality in terms of human-made damages (e.g. Is it acceptable to damage ecosystems?) but also potentially raise concerns about self-preservation (e.g. Can humans live long and healthy lives in a degraded environment?). Thus, in addition to novelty, the variety and severity of potential impacts (e.g. harm to self, other humans, and non-human entities) along with their associated costs and benefits, could explain environmental information's particular effectiveness in promoting reduced meat intake.

9.3 Meat reduction motivations and information provision impacts

Understanding motivations is complex, no matter the behaviour being studied. For meat reduction, strength of motivations are heavily dependent upon current meat eating habits. Consumers that have not reduced their meat intake (i.e. standard consumers), those that have reduced (i.e. reducers), and those who no longer consume meat (i.e. abstainers) differ in priorities in regards to why they have (or may consider) reducing their meat intake. More personal concerns (i.e. high cost of meat and health benefits) seem to be primary motivators for standard consumers and reducers when contemplating (or enacting) meat reduction. In contrast, more ethical or prosocial concerns (i.e. animal

welfare and environmental sustainability) were found as primary initial drivers for meat reduction among abstainers, while also being stronger motivators among reducers, in comparison to standard consumers. Thus, incentivising initial interest in plant-based diets among meat eating consumers may best be done through messages that address potential financial savings and health benefits, followed by more thorough discussions about the environmental and/or animal welfare impacts in order to shift initial interest into long-term motivation to reduce meat intake.

Overall, the degree of meat reduction seems to correlate with increases in more ethically-driven motivations and concurrent decreases in more personally-driven ones. This finding was not necessarily surprising, as those who abstain from meat for primarily ethical concerns have already been found to have stronger convictions (i.e. consume fewer animal products and abstain for longer periods of time) than those who abstain due to more personal concerns, like health benefits (Hoffman et al., 2013; Jabs et al., 1998; Radnitz et al., 2015). Thus, it was expected that exposure to ethical arguments would create stronger and more long-lasting increases in motivations to reduce meat intake, and this was in fact found in the current work. Among consumers who were exposed to information in regards to meat's impacts on personal health, the environment, or animal welfare, only those exposed to the animal welfare frame sustained an increase in its accompanying motivation to reduce meat consumption, although variations between individuals was greater. Thus, even though ethically-focused messaging may be more inconsistent in terms of generating motivations to reduce meat intake across individuals, findings support that overall, motivations driven primarily by ethical concerns (e.g. concern for another animal's wellbeing), once established and/or raised, are more likely to be retained over time in comparison to motivations that are driven by more personal concerns.

Some might argue that the motivation to reduce meat intake for environmental benefit would also be driven mainly by ethical concerns, and that exposure to environmental information should therefore have similar long-term effects on motivations when compared to exposure to animal welfare information, but this may not necessarily be the case. Environmental issues are holistic by nature, and such issues are likely quite abstract to the average person due to a number of reasons: many issues have no single discrete source (e.g. pollution run-off), the full impacts of some issues do not manifest themselves unless observed in the aggregate (e.g. greenhouse gas emissions and global climate

change), and there is difficultly in making clear and direct connections between human actions, their associated environmental effects, and the resulting ethical implications. Thus, although environmental issues are holistic in nature (as previously mentioned), individuals may focus on such issues in a discrete, rather than holistic way. This contrasts with issues of animal welfare, which due to their simplicity and clarity, are likely easier for consumers to understand fully while also taking less mental effort in terms of making connections between current behaviour and its associated impact(s).

For instance, when consuming meat, it is quite easy for one to realize that it has come from an animal that has been killed, and so the ethical implications of that action (i.e. is it right or wrong if an animal had to die for this piece of meat?) are much simpler and easier to make in comparison to something more complicated and abstract, like a piece of meat's contribution to climate change which would require much more complex thought and mental effort from an individual (e.g. inefficient conversions of crops to feed animals, methane emissions as a result of animals' digestion process, deforestation to grow feed crops and for raising animals, resulting climatic impacts from various emission sources, etc.). Thus, due to the sometimes abstract and complex nature of environmental issues, along with people's frequent detachment from their ethical implications (e.g. loss of biodiversity in other parts of the world), it would not be surprising if people's concerns in regards to the environment were mainly a result of more personal concerns (e.g. pollution negatively impacting human health, land degradation not sustaining human food supplies, etc.). Thus, motivations for reducing meat intake based upon environmental concerns may not sustain themselves in the long-term, at least to the same degree as motivations based upon animal welfare concerns, due to these environmental concerns potentially arising from more personally-driven fears (e.g. human safety and wellbeing). There is also a possibility that environmentally-driven motivations are not fully or partially within either the 'ethical' or 'personal' motivational categories and are instead situated in a completely distinct 'prosocial' category, which is defined as a desire to perform voluntary actions in order to benefit something apart from oneself (Rosenfeld and Burrow, 2017). If environmental motivations are distinct in this way, it may offer a potential explanation for why increases in motivation to reduce meat intake in response to environmental information exposure were not sustained as observed with exposure to ethically-focused animal welfare information.

Due to the finding that sustained increases in reduction motivation can result from exposure to animal welfare information, one might conclude that promoting motivations which are primarily driven by ethical considerations is the optimal strategy when aiming for long-term reductions in meat intake among consumers, but it is important to note that motivations can have impacts beyond just dietary change. For example, based upon findings from Rothgerber (2014), motivations for choosing a dietary pattern (e.g. personal or ethical) can potentially have greater impacts on self-perceptions (i.e. comparing oneself to others) than actual dietary behaviour (i.e. level of meat consumption). Put simply, even if two individuals abstain equally from eating meat, their reasons for doing so may contribute more to their perceived 'sameness' than their shared behaviour. So an individual who abstains from eating meat primarily out of concerns for personal health may perceive themselves as being more similar to another individual who only reduces their consumption (but also out of concerns for personal health), in comparison to a fellow abstainer who does not consume meat for alternate reasons (e.g. ethical concerns like animal welfare).

Not only can motivations behind the decision to reduce meat intake influence self-perceptions, but they can also impact how an individual is perceived by others. For instance, MacInnis and Hodson (2017) found that meat eaters view ethically-motivated vegetarians more negatively than health-motivated vegetarians. These negative feelings towards ethically-motivated vegetarians may result in perceptions of greater division between consumer groups (Rosenfeld and Burrow, 2017), and may even foment stronger and/or more frequent conflicts between these groups (LeRette, 2014). Taking this information into account, it would be interesting for future studies to investigate how different information provision framings and/or techniques impact not only self-perceptions, but also the perceptions that non-reducers hold towards reducers and abstainers. For instance, could the provision of environmental information impact these social variables differently in comparison to animal welfare information, and if so, how? It would be interesting to determine whether exposure to different information framings result in different or similar evaluations of oneself and/or others, and how the framings and motivations might interact with one another (e.g. does providing environmental information to a non-reducer make them see reducers motivated by animal welfare concerns more favourably, and does providing animal welfare information result in similar evaluations of reducers motivated by environmental concerns?). Finding such

similarities or differences between message framings and/or perceptions would provide more insight into the wider social implications of knowledge acquisition (e.g. how the messages and individuals who carry them are viewed) and by extension, possibly even offer indications of how messages are (or could be) transferred between individuals and whether they are met with acceptance and/or resistance in wider society.

For example, maybe animal welfare messages and those who reduce meat intake for animal welfare concerns are largely ignored and/or encounter resistance by the majority of non-reducers, while health and/or environmental messages (and those who reduce for those concerns) are met with greater openness and acceptance. This information could be vital, because although exposure to ethical arguments seems to be particularly effective in sustaining long-term motivations to reduce among individuals, initiatives that seek to spread meat reduction motivations across larger populations will have to consider the social challenges involved when seeking to spread such messages to those who currently consume meat regularly (e,g, establishing initial contact and starting conversations about issues without angering or annoying the target audience). Therefore, further study on the societal perceptions of different meat reduction motivations (and feelings towards those who hold them) may offer insight into whether messages will be supressed or openly spread between individuals and organizations, and by extension, whether these messages will permeate throughout society (i.e. across communities, regions, countries, etc.). The spread and acceptance of such messages throughout society will help to foster opportunities to not only make plant-based diets more normalised, but also potentially create greater public support for more structural measures, like the increased procurement of plant-based foods in public institutions like schools and hospitals. If such endeavours are successful, the barrier of meat consumption being the ‘norm’ will likely weaken, making it easier for individuals to follow a more plant-based diet as a result of diminishing stigma and increased availability of plant-based options throughout society.

In addition to societal factors, more attention should also be paid to the potential interactive effects of motivational messages. Although current findings offer greater understanding of information’s influence on singular meat-reduction motivations, the potential interactions of these motivations in real-world scenarios is still not fully understood. For example, Lazzarini et al. (2016) found that Swiss consumers associated the healthiness and environmental friendliness of foods, even when this association was unwarranted. Therefore, assuming that similar associations are also held by consumers in

other nations, could the combination of messages on meat reduction's potential health and environmental benefits be more readily accepted by and/or easily incorporated into consumers' decision making processes, as it would already align with pre-established beliefs? Furthermore, if these topics are already linked in consumers' minds, could providing information on only one aspect of meat reduction (e.g. potential health benefits) also inadvertently increase consumer perceptions of its positive impact on additional, seemingly unrelated topics (e.g. environmental benefits)? Finding answers to these questions could be extremely important for meat reduction efforts, as the combination of complementary message framings could result in even greater motivations to reduce meat consumption among individuals. Of course, caution must be taken when making generalizations, as pre-established associations will likely differ depending on consumer demographics and cultural background, but further study on these potential associations and how they may aid or hinder meat reduction efforts will likely be beneficial for practitioners and/or policy makers.

It is important for future studies to continue focusing on meat reduction motivations, the differences in perception across consumer groups and throughout society for each motivation, and the potential interactive effects these motivations may have in real-world scenarios; however, it is also vital to not adopt too narrow of a focus, as addressing other factors such as barriers (e.g. necessity of meat for nutrition, difficulty in breaking traditions and/or habits, etc.) can also be important when promoting meat reduction amongst consumers (Rosenfeld and Burrow, 2017). For instance, individuals may be deterred from ideas of meat reduction due more to the presence of perceived barriers rather than a lack of motivations, which has been found to be a more common theme among men (Cordts et al., 2014). Thus, women may more easily translate isolated motivational messages into actual meat reduction outcomes, while men may struggle to do the same. Thus, initial prioritization of messages that aim to reduce perceived barriers, followed by messages that are more motivationally-focused, may be the most effective strategy for male audiences, while female audiences may already be more receptive to direct motivational messages. In addition to gender, ideology also seems to predict not only meat reduction likelihood and behaviour, but also support for vegetarians and vegetarianism. MacInnis and Hodson (2017) found that those with more conservative ideologies (i.e. high value towards tradition, authority, etc.) are more likely to have negative views towards vegetarians, and as a result are less likely to support meat

reduction behaviours amongst individuals. Thus, psychological associations such as perceived power and masculinity and pre-existing political ideologies likely influence the formation of perceived meat-reduction barriers and the resistance one has towards other individuals who reduce meat (and their associated philosophies). Hence, shifting commonly held stigmas associated with the choice to eat a plant-based diet, as found by Markowski and Roxburgh (2019), is advised when addressing populations where pre-established barriers to meat reduction are known to be prevalent.

9.4 Theory of Planned Behaviour (TPB) and Meat-Attachment Questionnaire (MAQ) predictive ability and alterations based on information provision

Similar to findings of past studies, willingness and intentions to reduce meat intake (Berndsen and Van der Pligt, 2004; Graca et al., 2015a; Povey et al., 2001; Saba and Di Natale, 1998), as well as agreement with proposed meat-reduction policies (Lentz et al., 2018), were best predicted by the attitudes component of the TPB model, with the subjective norm and perceived behavioural control (PBC) components showing much weaker and inconsistent predictive power. PBC was found to increase among individuals after exposure to information on the potential health benefits of reducing meat intake, an effect that would be expected to contribute positively towards potential meat reduction based upon past study findings, where PBC was determined to be a positive predictor of actual behavioural change (Terry and O'Leary, 1995); however, this same study found that PBC was not a significant predictor of behavioural intentions, and this finding matched with results from the present research, where PBC did not predict willingness and intentions to reduce meat intake nor agreement with potential meat-reduction policies. This makes drawing conclusive insights about PBC's resulting effects on meat reduction difficult, in addition to the lack of information on its potential interactive effects with actual behavioural control. Thus, further study on this topic could prove quite useful in enhancing current understanding of what role PBC plays in terms of influencing perceived barriers and primary motivations among consumers when contemplating meat reduction, as well as how different levels of PBC may impact outlooks and/or habits in regards to meat-related behaviours.

For example, consumers who have higher PBC may be more interested in listening to new information about meat's impacts on a given topic and as a result, also be more likely to integrate that new information into their general perceptions of meat simply due to the perceived 'applicability' of the information. To be more specific, those with lower PBC may view meat consumption as more of a necessary behaviour, and that information regarding its impacts is irrelevant due to meat eating being a requirement; while those with higher PBC likely view meat consumption as a preference-based choice, making information regarding its impacts both pertinent and applicable. In addition, an individuals' PBC may be able to predict how likely newly acquired motivations (possibly in response to acquisition of new information) are to be translated into committed, long-term meat reduction efforts. In comparison to consumers with low levels of PBC, high PBC individuals might be more willing to put time and effort into researching information, making adjustments, and overcoming challenges when trying to perform behaviours that better align with meat reduction motivations, due to a greater sense of command and control over the decision to eat meat or not. Even if the strength of motivation(s) to reduce are equivalent in individuals with lower PBC, resulting behaviour changes may not manifest simply due to feelings that hamper the likelihood of implementing and sustaining those changes (e.g. feelings of helplessness, impossibility, pointlessness, etc.). Based upon past and current research, it would seem that PBC does *not* have a particularly strong influence over willingness or intentions to reduce meat intake, but it may contribute in minor ways or in specific instances in translating willingness or intentions into actual behaviour change, as discussed above. Therefore, further inquiries should focus on alternative methods than those used in the current work, in order to expand upon the information known so far about PBC, and its resulting influence on meat eating behaviour. For example, future studies could measure individuals' PBC and track their eating habits over prolonged periods of time to determine if PBC is a significant mediator in translating changed attitudes and intentions to reduce meat intake in response to an intervention.

In contrast to PBC's unclear significance, the construct of meat attachment, in addition to attitudes, was shown to be a strong indicator of willingness and intentions to reduce, as well as agreement with meat-reduction policies. In addition, it has now been verified as a valid measurement within three nations: Portugal (Graca et al., 2015a), New Zealand (Lentz et al., 2018), and Australia (Dowsett et al., 2018). Thus, not only does the

construct of attachment seem to be an important predictor of meat-related variables, but it may also have cross-cultural validity and applicability. The three nations studied thus far are all 'western' (i.e. predominately European culture) however, and so similar studies on meat attachment within 'non-western' nations would offer further detail in regards to the extent of meat attachment's cross-cultural validity and predictive power.

Exposure to information about meat's impacts on personal health, environmental sustainability, or animal welfare can all decrease positive attitudes and attachment towards meat, although with slight differences in magnitude and varying effects depending on the meat attachment subscale under investigation. This is important, as positive attitudes towards meat have been identified as strong predictors of consumers' lack of willingness and intentions to reduce personal meat intake, in addition to being linked to cross-cultural prejudice directed towards abstainers (Earle and Hodson, 2017). Thus, it would stand to reason that all three information framings could not only make consumers more open to the idea of reducing meat intake, but also potentially reduce their negative beliefs and prejudice held towards those that have already reduced or removed meat from their diet. So not only can information provision promote individual behaviour change, but also potentially create a more positive, encouraging, and less discriminatory social environment for reducers and abstainers through changing the perceptions and opinions that non-reducers hold towards those with different dietary habits.

Exposure to animal welfare information has the greatest potential for reducing attitudes, attachment, and hedonism, affinity, and entitlement towards meat; while exposure to health information is most effective at reducing feelings of dependence towards meat. So it would seem that animal welfare information is the most effective at changing consumers' general perceptions of meat (e.g. is it a behaviour that is 'positive', 'wanted', 'likeable', 'good', and 'right') while health information is specifically effective at altering consumer perceptions of meat's necessity as a dietary component. These findings could be quite useful for anyone who wishes to promote reduced meat consumption, as the provision of animal welfare information can promote the strongest shifts in general outlook towards meat, while the provision of health information could address some common barriers that consumers hold when contemplating meat reduction. Not only does this provide some guidelines as far as what information should be prioritized when communicating with a specific target audience (i.e. primarily animal welfare information for those with low motivations to reduce or primarily health information for those with

perceived barriers), but it also identifies a potential synergy between the information framings that may be wasted if they are only utilised in isolation. Simultaneously reducing barriers while promoting motivations is likely to have an even greater long-term impact on individual meat consumption behaviours, and thus, a combination of framings is strongly advised, where possible, when designing and implementing meat reduction interventions. Of course, funding and time do not always allow for approaches that incorporate and address multiple influential factors (e.g. motivations and barriers), and in these cases it is advised that practitioners prioritize their meat-reduction messages based upon the target audience, using the guidelines discussed above.

Nonetheless, it would seem film is an effective medium by which to communicate meat's impacts on human health, the environment, and/or animals. Based upon the findings from Study 2, it would appear that communicating these messages through film not only shifted attitudes and attachment (as found in previous information provision studies), but actually resulted in lower self-reported meat intake, which has not been found in prior studies. Thus, a key component to providing messages that result in behaviour change among individuals could be the format in which such information is provided. For example, prior studies utilised methods such as short passages or newspaper articles in order to inform consumers about meat's impacts (e.g. Cordts et al., 2014; Graham and Abrahamse, 2017; Whitley et al., 2017). Short (and arguably less engaging) forms of information provision may fail to encourage behaviour changes in individuals while more lengthy, thorough, and arguably engaging formats like film may result in not only attitude changes, but also shifts in meat eating behaviour. I believe this is a vital point for future research on potential meat reduction involving information provision and is also extremely useful for practitioners who seek to promote meat reduction amongst the public.

9.5 Agreement with proposed structural measures

Framings used for proposed meat-reduction policies (e.g. for either improved public health or environmental sustainability) would seem to have little impact on consumer agreement, except in the case of proposed meat taxes, where agreement levels are significantly higher when paired with an environmental justification rather than a public health justification. Exposure to information on the health, environmental, or animal

welfare impacts of meat all increase agreement with proposed meat-reduction policies; but environmental information, specifically, seems to be the most impactful. This could be a result of consumer perceptions of what is 'appropriate', in the sense that environmental issues are caused by society and should therefore be solved with solutions that are more societal in scale (e.g. taxes across individuals and businesses). In addition, consumers may believe that environmental issues are too large in scale to address through individual action alone, and that societal-level solutions are not only advisable, but necessary if significant improvements are to be made. Also, debates on whether carbon taxes (e.g. for transportation) should primarily be the responsibility of producers or eventual consumers have resulted in suggested 'emission added' approaches, where producers and consumers share responsibility and therefore each pay a share in terms of the accumulative carbon burden (Bastianoni et al., 2004). It would be interesting to see how awareness raising among consumers would change, if at all, their ideas of how proposed environmental taxes on meat could best be distributed in terms of shared responsibility to pay between producers and consumers.

Many of the above statements are only conjectures for why environmental information was so effective at increasing policy agreement, and future studies can help determine whether such claims are valid by focusing on methods that would provide more detailed information on the interactions between consumers and potential meat reduction policies. This could possibly be in the form of consumer interviews and/or focus groups, which would focus on understanding how consumer knowledge and awareness in regards to specific meat-related impacts affects thoughts, agreement, and support for various proposed policy measures. This may offer insight into whether the presence or absence of information, and how the type of information and/or depth of understanding, dictates an individuals' thoughts on the nature of meat-related issues and how they may best be addressed (e.g. personal choice to alter consumption, companies' responsibility to implement mitigation practices, government's role in creating policy to influence consumers and/or industry, etc.). In addition, more detail can be gathered on conflicting narratives that consumers may be exposed to from state and non-state actors (e.g. meat industry, governmental dietary panels, etc.; Godfray et al., 2018) and how these conflicting messages might be handled by consumers, both in terms of shaping their general opinions towards meat and their agreement with proposed reduction policies.

In addition to research on the interactions between consumer knowledge and meat-reduction policy agreement, further study on the potential presence of a mere-exposure effect (i.e. increased agreement with policy simply through exposure) may offer beneficial insights as well. If such an effect exists, its confirmation may be useful for practitioners or policy makers wishing to promote specific meat-reduction measures, especially if public support is needed and/or the would-be-affected consumers are initially resistant to even considering such proposals. By simply raising awareness about the potential policies, through various forms of media (e.g. television news, online advertisements, social media posts, etc.), consumers may begin to be more accepting of the proposals, or at least be more open to learning about them. This would likely not work in isolation as a long-term solution, but simple promotion could be an important first step in not only making consumers more aware of proposed meat-reduction policies, but also in increasing agreeableness and openness towards the future implementation of such policies.

9.6 Meat intake

Exposure to information on meat's health, environmental, or animal welfare impacts all increased not only willingness and intentions to reduce, but also self-reported meat intake frequencies. Not only does this provide validity to the effectiveness of information provision (both broadly and for these specific framings) in increasing the likelihood of meat reduction among individuals, but it also offers evidence that information provision can influence meat-eating behaviour.

It would seem that exposure to information on meat's environmental impacts, specifically, is quite effective at promoting meat reduction among individuals even though it produces some of the weakest reductions in variables that have been shown to predict meat intake (i.e. attitudes and attachment). This could be due to a variety of reasons, one being the potential influence of environmental information on alternative meat-related variables that have not received proper attention in the studies to date, meaning that additional variables of importance related to environmental concerns specifically, may not be properly captured by the methods utilised within this thesis (e.g. TPB and MAQ). This is just one potential explanation for the somewhat conflicting findings of environmental information's impacts on meat-related variables and actual

meat intake, and future studies that seek to build upon these initial findings would likely benefit from more detailed investigations into changes in consumer attitudes and meat attachment post-exposure. Such investigations could provide more specifics about the mechanisms and pathways by which environmental information impacts meat-eating behaviours, which could support an expansion of meat attachment to include additional variables of importance that better capture the impacts of exposure to environmental information

9.7 Overview: Shifting society towards more plant-based diets

If the ultimate goal is to promote improved environmental sustainability and public health by means of reducing or possibly even reversing current and projected global meat consumption trends, it is important to recognize and address the major justifications for meat consumption that are held by people, both individually and as a collective. Piazza and colleagues (2015) summarize the four main justifications (i.e. 4Ns) for meat consumption:

1. It is natural (fundamental part of being 'human')
2. It is necessary (required for health and wellbeing)
3. It is nice (positive feelings associated with consumption)
4. It is normal (majority of people eat meat)

Since these four justifications encompass the majority of reasons provided by consumers when explaining and/or defending their meat intake (Piazza et al., 2015), it would stand to reason that prioritising these four key justifications is currently the most advisable strategy for promoting meat reduction. Findings from the current work can offer insight for how some of these justifications may best be addressed; but depending upon the justification being targeted, other methods and strategies besides information provision may be more effective. For example, evidence suggests that information provision can be an effective method by which to counteract some of these justifications (e.g. meat eating is natural, necessary, and nice); however, alternative methods (e.g. increased availability, variety, and affordability of meat substitutes; shifts in government support and endorsements from animal-based to plant-based industry; etc.) are likely to be more appropriate for addressing justifications that are not as readily influenced by information provision (e.g. meat eating is normal). Also, information provision is not uni-directional,

as consumers will likely receive more frequent and powerful messaging encouraging them to consume meat (e.g. advertisements directly from meat industry, fast food restaurants, or supermarkets). Thus, using evidence from the current work in addition to the findings of other sources (to be discussed below), avenues for promoting meat reduction will be explored using the framework provided by the 4Ns of meat consumption as outlined above.

9.7.1 Natural and necessary

Based upon findings from the current work, when consumers are exposed to information on the negative health impacts of consuming meat and the positive health impacts of consuming plant-based foods, their feelings of dependence towards meat drop significantly. Decreased dependence addresses the 'necessary' justification by shifting the act of meat consumption more from a realm of necessity (to maintain health and wellbeing) to one of choice in the consumer's mind. Thus, the implementation of awareness raising efforts in regards to the health benefits of plant-based diets is expected to be one of the most effective methods in addressing the 'necessary' meat consumption justification. These awareness raising efforts could be conducted by private individuals and/or organizations, especially those who are already considered to be trusted authorities on such matters (e.g. nutritionists, doctors, hospitals, etc.), but endorsements of plant-based diets by other authorities, such as government organizations and representatives (e.g. New Zealand Ministry of Health, United States Surgeon General, etc.), will likely be needed as well if quicker and more sustained shifts towards plant-based diets are to be achieved. Health Canada, a government organization that works to promote greater public health within Canada, has released a new food guide that promotes eating less animal products (including meat) in comparison to past food guides, and even encourages frequent substitution of plant-based alternatives (e.g. beans, lentils, tofu) for meat (Health Canada, 2018). Other government health organizations should follow the example set by Health Canada and promote more plant-based foods through direct endorsement and/or the release of revised food guides.

Spreading messages and raising awareness of the health benefits associated with going more plant-based may also help counter the justification of meat eating being 'natural' for humans. Naturalness and healthiness of foods are associated with one another in the minds of western consumers (Rozin et al., 2012a); thus, raising awareness on the

detrimental health impacts of meat eating may cause the idea of its 'naturalness' to be called into question by individuals who may ask themselves and/or others, "If humans *were* designed to eat meat, shouldn't one expect its increased consumption to result in positive health outcomes, instead of the negative ones that have been observed?". These questions, and the resulting doubts that would likely follow, are therefore expected to weaken the 'natural' justification of meat consumption, without the need for detailed information and/or complex debates on human physiology and whether our bodies are scientifically 'designed' to eat meat. Once again, targeted health messages such as these are likely to have an even greater impact if endorsed and/or propagated by those who are seen as trusted authorities on matters of public health, such as the New Zealand Ministry of Health.

9.7.2 Nice

Consumer attitudes towards meat were found to decline significantly after exposure to information on meat's health, environmental, or animal welfare impacts. Thus, awareness raising efforts that focus on any of these impacts will likely be an effective method by which to address the justification that eating meat is 'nice'. However, animal welfare information seemed to show the greatest declines in attitudes towards meat, and so it would be expected to also have the most significant impact on this justification. It would seem that many consumers care for the wellbeing of animals, and this is even true for the animals that are consumed (i.e. meat paradox), but mental defence mechanisms are often utilised in order to resolve this dissonance so that individuals can maintain their meat eating behaviour (e.g. moral disengagement; Bastian and Loughnan, 2017). Being confronted with the fact that animals must be killed to produce meat does seem to overcome (at least to some degree for most individuals) these defence mechanisms and can result in relatively long-term attitude changes.

The exact reasoning for why animal welfare information is so effective at changing attitudes in comparison to the health and environmental information framings is not fully clear, but it may be due to increases in primarily ethical concerns as a result of exposure, rather than increases in more personal concerns (e.g. as a result of receiving health-related information). Ethical concerns may result in stronger negative feelings in comparison to personal concerns, due to differences in perceived value of the thing being 'sacrificed' or 'lost' and/or the way in which these losses are positioned, and this may result in

individuals being less willing to make trade-offs for performing the behaviour (e.g. killing an animal vs enjoying the taste of meat). To clarify, individuals may place more value on an animal's life than they do their own health (i.e. the things potentially being 'lost' by eating meat), and so the thought of an animal being killed is seen as a greater loss in their minds in comparison to the prospect of personal harm, like developing a disease linked to meat consumption.

In addition to this value judgement, individuals may also deem an action that negatively impacts other entities (e.g. other humans, animals, etc.) as more undesirable than an equivalent action that negatively impacts the self. Thus, ethical arguments presented to an individual may make an action seem more undesirable than if personal arguments were presented, due to the repercussions of those actions being held in a place of greater importance in the individual's mind. In addition to these inherent differences between ethical and personal concerns, the ethical implications of eating animals are likely quite simple and straightforward (i.e. *certain* impact) for consumers in terms of connecting the behaviour to said impacts (i.e. eating meat requires killing animals), at least in comparison to other implications that are also potentially 'ethical' in nature, but are more complex and convoluted (i.e. *potential* impact of meat's environmental burden). Therefore, either solely or in combination, the greater importance given to animal welfare considerations and the ease with which impacts to animals can be linked to meat eating behaviour may explain why animal welfare has such a profound impact on attitudes towards meat, and by extension the justification of meat eating being 'nice'.

Explanation of potential mechanisms aside, it is clear that providing information to consumers on the animal welfare impacts of eating meat will help to address the 'nice' justification, but it is important to note that it will likely be necessary to maintain and/or repeat exposure over time. As stated above, consumers have mental defences in order to maintain current behaviours, and so even if messages have high potential to spark meat-reduction motivations, consumers may consciously or subconsciously ignore or forget information in order to avoid cognitive dissonance. However, repeated messaging is an advisable strategy in counteracting or subverting these mental defences, because it is likely the case that over time, these mental defences will either become ineffectual or tiring for the individual. Thus, if significant reductions in the 'nice' justification are sought among the majority of consumers, repeated and/or consistent messaging about meat's animal welfare impacts is strongly advised.

9.7.3 Normal

Currently, meat consumption is the societal norm, as most people eat meat and view the behaviour as 'normal' or 'standard' dietary practice. Unfortunately, although information provision and awareness raising efforts are effective tools for changing individual meat-related opinions and behaviours, these strategies may not be as well-suited for uprooting meat eating's place as a societal norm, at least within a relatively short time frame. Due to logistical constraints (e.g. limited time, money, access to audience, etc.), information provision endeavours are usually limited in scope (i.e. targeted to small groups) and are focused on changing individual thoughts and behaviours. Societal norms however, are based upon the aggregation of *all* individual thoughts and behaviours, and so methods that are tailored for much smaller groups (i.e. information provision) can only be expected to make very small, incremental contributions towards shifting such large-scale factors over time. In addition, people's perceptions of what is 'normal' will largely be based upon the frequency with which other individuals perform the behaviour, and so even if information provision is effective at changing one's personal thoughts and feelings towards meat, the reality of meat consumption being a societal norm still remains. Thus, in contrast to the three justifications discussed previously ('natural', 'necessary', and 'nice'), information provision will likely be a much longer and more difficult process in terms of counteracting the 'normal' justification.

Luckily, although global rates of meat consumption are increasing and are projected to continue doing so over the next decade, plant-based diets have become more popular over time, especially over the past decade. Plant-based diets have become increasingly endorsed and practiced by celebrities (Sunday, Naturespath, 2018) and top-tier athletes (Berger, CNBC News, 2018), as well as among average consumers, as global interest (i.e. total searches including the keyword 'veganism') has increased by about 500% over the past decade, with the greatest interest occurring in Israel, Australia, Canada, New Zealand, and Austria, respectively (Google Trends, 2018). In addition to increased interest, the adoption of plant-based diets has also increased greatly over the past decade, especially in western nations. For example, a third of New Zealand consumers have already reduced their meat consumption (1 News, 2019) and Beef and Lamb (2018a) has already recognized that the future of meat will be providing lower quantities of "higher

quality" meats to consumers in the coming decades as plant-based diets and products continue to rise in popularity.

A U.K. survey conducted in 2013 found that one in every four consumers had reduced their meat consumption over the previous year and that 17% of young consumers indicated that they abstained from meat (Vegetarian Society, 2013); and a 2017 report on food trends noted that adoption of low-meat diets have increased by 18% in Germany from 2014 to 2017, while those who identify as vegan has increased from 1% to 6% in the U.S. within the same timeframe (GlobalData, 2017). Another report, based in Portugal, found that the number of vegetarians has increased fourfold from 2007 to 2017, from about 30,000 to 120,000 (The Portugal News, 2017). These trends would seem to be present in New Zealand as well, as a 2016 poll found that Kiwis who state that all, or almost all, of their food is vegetarian has grown by 27% since 2011 (Roy Morgan Research, 2016); and in the current work, a large proportion (40%) of consumers indicated that they have either already reduced or are in the process of reducing their meat intake (337 reducers and abstainers among sample of 841 consumers). Even in non-western countries and/or communities, interest and adoption of plant-based diets is growing, as seen in China where the vegan market is expected to grow by at least 17% between 2015 and 2020 (Inside Retail Asia, 2016); and at New Zealand's largest Marae (Turangawaewae), more plant-based foods are being provisioned in order to promote health improvements within Maori communities (Tahana, 1 News, 2018).

Unsurprisingly, coupled with increased consumer interest and adoption, sales of plant-based foods have increased and are projected to continue, while sales of *some* animal-based foods are in decline. In New Zealand, sales of alternative plant-based 'meat' products have "soared" over the past five years, while purchase of meat products have declined overall (Caldwell, 2019). Based upon research conducted by Nielsen in the U.S., sales of plant-based foods increased by about 8% from 2016 to 2017, with plant-based 'meat' alternatives seeing 6% growth and dairy alternatives seeing 20% growth (Simon, 2017). Research by Mintel found similar trends, with almost half (49%) of all Americans drinking non-dairy milk(s) and sales of traditional dairy dropping by 7% in 2015 (Mintel Press Office, 2016). Egg sales are declining as well, as one of the largest egg producers in the U.S., Cal-Maine, reported its first annual loss in over a decade, with the CEO Dolph Baker citing increased popularity of plant-based egg alternatives as the primary driver for reduced demand (Sheetz, CNBC News, 2017). Increased sales of plant-based alternatives

are not just limited to the U.S., as the global plant-based 'meat' market is expected to grow by 7.7% per year (compounded annually), with the Asia-Pacific region having the highest expected increases (9.4%), based upon projections from 2018 to 2025 (Prasannan, Allied Market Research, 2018), in addition to the global plant-based 'cheese' market, which is expected to have a similar yearly growth (7.6% per year, compounded annually) from 2016 to 2024, based upon a report by Variant Market Research (Bharat Book Bureau, 2017).

So with increased interest and consumer demand for plant-based foods, and specifically direct alternatives to traditionally animal-based products, markets have responded as expected, through companies adjusting their focus and efforts in order to produce more plant-based products in order to meet or 'get ahead' of shifting consumer demand. Thus, companies that focus on producing plant-based alternatives have recently received hefty investments, with some even coming from companies that have traditionally sold animal-based products. In New Zealand, popular brands like Fry's and Linda McCartney along with newer brands like Bean Supreme, Gardein, and many others have released plant-based products that were not available to consumers 5 or 10 years ago and more of these products are being released over time by established and new companies in the New Zealand market. Other products like the Beyond Burger, which is popular in both New Zealand and other countries a direct plant-based alternative that is claimed to mimic the qualities of a traditional animal-based burger, has received investments from wealthy individuals including Bill Gates (Microsoft founder), Leonardo DeCaprio (actor), and Biz Stone and Evan Williams (Twitter co-founders), as well as Tyson Foods (U.S. meat company) and is now sold in over 5,000 stores across the U.S. (Fox, 2017, Forbes). Beyond Meat, the company behind the Beyond Burger, recently had its initial public offering (i.e. stock market launch) and was the highest performing public offering by a U.S. company since the year 2000 (Murphy, 2019, www.marketwatch.com). Sweet Earth, a plant-based company that offers direct meat substitutes, was acquired by multi-billion dollar, multinational manufacturer and marketer of food products, Nestlé (Nestlé, 2017), and similarly Kite Hill, a plant-based company specializing in direct cheese substitutes, has secured an investment of $18 million from General Mills (BusinessWire, 2016). Small plant-based brands being bought by large corporations are not isolated incidences, as this trend is being seen across the food industry in multiple nations:

- Maple Leaf Foods (Canadian meat company): Acquired plant-based companies Field Roast (Maple Leaf Foods, 2018) and Lightlife Foods (Maple Leaf Foods, 2017)
- Dean Foods (American dairy company): Majority shareholder of plant-based company Good Karma (PR Newswire, 2018)
- Otsuka (Japanese pharmaceutical company): Acquired plant-based 'dairy' company Daiya (Otsuka Pharmaceutical, 2017)
- Danone (American multinational dairy company): Acquired plant-based 'dairy' company WhiteWave (Globe News Wire, 2017)
- Beef and Lamb New Zealand: Recently recognised that alternative proteins are likely to become a major competitor, and that alternative proteins could reach mass manufacturing within the next five years (Beef and Lamb New Zealand, 2018a; www.beeflambnz.com)

These increased investments in plant-based products from large corporations are not restricted only to foods that ultimately end up on store shelves, but also those that are served at restaurant tables and through fast food drive-throughs. Multiple fast food chains have made efforts to offer more plant-based options, with McDonald's offering a vegan burger as a permanent menu option in Finland and Sweden (Pechter, 2017, Business Insider), Taco Bell outlining how to order entirely plant-based meals off their menu (TacoBell, 2018), and White Castle initially offering the plant-based Impossible Burger at 140 of their 377 locations across the U.S, which due to market shares being 250% higher in those 140 locations, has now been expanded to all locations nationwide (Fortune, 2018). Burger King has since followed suit, recently announcing that they will also be bringing their Impossible Whopper (made using Impossible Burger) to their stores across the U.S. in 2019 (Wiener-Bronner, 2019, CNN business).

In addition to large fast-food companies shifting towards more plant-based offerings, smaller start-up plant-based fast food chains, like Veggie Grill, Next Level Burger, and Lord of the Fries are finding success and expanding to more locations (Peters, 2018, FastCompany; Shaw, 2018, NZ Herald). Plant-based options are even beginning to penetrate more niche markets such as food catering, as evidenced by Air New Zealand's move to offer the Impossible Burger as part of their Business Premier menu on their flights from Los Angeles to Auckland (Air NZ, 2018).

Thus, it is clear, that in response to increased consumer interest and demand for plant-based foods, major investments in plant-based food brands and products has begun to rise substantially and these investments have resulted in innovations such as direct plant-based alternatives that mimic animal-based products (e.g. burgers, milks, cheeses, etc.) in addition to wider and more varied offerings of plant-based foods, both in stores and when dining out. Each of these factors promote one another in a cyclical feedback loop:

1. Increased demand encourages further investment in plant-based products and companies
2. More investment results in better and wider offerings of plant-based products for consumers
3. Tastier, cheaper, and more available plant-based products encourage greater consumer interest and adoption
4. Greater interest and adoption pushes for an even greater demand for plant-based products

This feedback loop is already occurring within multiple nations and across the food industry, and this is already beginning to address (or challenge) the justification of meat consumption as being 'normal'. As interest in and adoption of plant-based diets rises, and plant-based companies and products become more ubiquitous, perceptions of eating plant-based products will become standard practice for many consumers, and as a result, the 'normal' justification for meat consumption would be expected to continue weakening over time, especially if information provision initiatives promoting meat reduction occur concurrently. Also, as more plant-based products are released that can be effectively substituted for traditionally animal-based foods, consumers will find the adoption of plant-based diets as more convenient (i.e. habits do not need to change as foods are simply swapped), which has been shown to be a major factor in adopting reduced-meat diets (Schenk et al., 2018).

9.7.4 Potential state involvement

The majority of the discussions above are focused on consumer and/or market-based topics related to promoting more plant-based diets, but what about the potentiality of more state involvement? Although government backing would likely contribute significantly to meat reduction efforts (e.g. as seen for tobacco in the past), controversy

exists over what matters governments should or should not be involved in, and this is true in New Zealand as well. In many nations, food choice is likely regarded as a personal matter and that institutions such as government should have limited to no influence and/or control over such decisions and thus, the potential implementation of policies that promote meat reduction will likely depend upon the nation under question and how the policy will be viewed by government officials, citizens, and the local food industries. Many modernized nations believe in free and open trade, with minimal interference from state bodies, and this is true in New Zealand as they have had a "hands off" approach to farmers and agricultural products since the 1980s. In addition, policies that negatively impact imports and/or exports of meat will likely be viewed as a direct threat to international agreements on free trade between nations. In fact, the World Trade Organization (WTO) prohibits the use of tariffs on meat products, no matter the justification behind it, except in the case of food safety (e.g. preventing transmission of livestock diseases) (Godfray et al., 2018). For example, Samoa was forced to reverse a ban on fatty-meat imports that were introduced as an anti-obesity campaign in order to be allowed to join the WTO (Snowdon and Thow, 2013). Therefore, trade regulations and international agreements will make meat reduction policies much more difficult to implement in nations that wish to retain trading partnerships with other nations. Of course, nations who strongly wish to reduce their meat consumption may find ways to circumvent such regulations (e.g. implementation of regulations or permits to accommodate the 'true cost' of meat without violating trade agreements), but specifics for how this may be accomplished is beyond the scope of this work.

Despite potential conflicts, some governments have already begun taking steps towards promoting more plant-based diets within their nations. Health Canada has released new dietary guidelines promoting more plant-based food consumption in order to promote greater public health within Canada, as discussed above, and China's government has released new dietary guidelines that aim to cut the country's meat consumption by 50% in order to promote both public health and environmental sustainability. China's new dietary guidelines and endorsements of plant-based foods have also been accompanied by advertisements with famous celebrities aimed at encouraging Chinese citizens to eat less meat in order to promote environmental sustainability (e.g. in order to meet GHG emission targets set under the Kyoto protocol; Milman and Leavenworth, 2016; The Guardian). As suggested by Bogueva at al. (2017), other nations should follow China's

example, and make strong commitments to reducing meat consumption within their nations through government supported marketing campaigns and policy measures.

Direct education will likely be necessary if stronger policy changes are sought (e.g. taxes, bans, etc.), as was the case for tobacco (Royal College of Physicians, 2012), and 'nudging' interventions in public institutions (e.g. adjusting of menus or portion sizes of meat to promote more plant-based foods in schools, hospitals, government offices, etc.) may offer simple, but effective methods by which to shift consumers more towards plant-based options without the need for more direct interventions (as summarized by Godfray et al., 2018). Of course, some may view such 'nudging' measures as being 'under-handed' and potentially unethical, but individuals in society are already being constantly nudged into making relatively poor decisions (e.g. encouraged to eat unhealthy fast food through product placements and/or indirect advertisements) and others see its potential implementation as more favourable in comparison to traditional market-based solutions. I believe that such measures could be effective, but may not yet be necessary, as governments that wish to promote effective and long-term meat reduction can follow the example set by China, and introduce 'softer' policies (e.g. changes in dietary guidelines, endorsement of plant-based diets, education campaigns, etc.), that will work in conjunction with the plant-based trends that are already occurring among consumers and the market, as discussed above. Based upon results from the current work, it is clear that education and awareness raising can significantly impact meat-related perceptions and behaviours, and such measures can be implemented alongside 'softer' policies in order to further encourage and hasten the movements towards plant-based foods that have already been observed in multiple nations and worldwide.

Within New Zealand, many changes that promote plant-based eating have occurred in recent years, but many barriers still remain. A third of consumers have already reduced their meat consumption, and plant-based products that can more easily be substituted for traditionally meat based products and meals are becoming more prevalent from multiple companies. Also, Beef and Lamb NZ has already predicted that the future of meat is going to be comprised of more plant-based alternatives and that meat sales overall are expected to decline in the coming years. Despite these advancements, consumers are still consuming high amounts of meats, which have serious implications for not only their own health, but also New Zealand's environment and the health of ecosystems around the

globe. The government is unlikely to take any direct actions beyond small changes in recommendations to encourage reduced-meat consumption, as it would likely be seen as an overreach by the government, especially considering the history between the government and farmers over recent decades.

Therefore, more significant action from consumers and the government would be ideal, but being realistic, more significant changes are likely to come from where they have already been seen: consumers and the industry. With greater awareness raising among consumers and other incentives like the increased availability of more plant-based alternatives, the trends that have already been seen like declining meat consumption and increased uptake of plant-based products is likely to continue. This process can be encouraged and would likely occur at a faster pace by changing consumer attitudes, intentions, and meat intake behaviours even further. By exposing consumers to information in an engaging format (like in Study 2), they are more likely to eat less meat and even promote structural measures that would support plant-based eating even further. Given that the New Zealand government is unlikely to implement any major policies like a meat tax without support and pressure from the majority of citizens, awareness raising efforts by private individuals and non-profit organizations within New Zealand are likely still a key component to the growing movement. And in the meantime, industries will continue to shift and cater to these changing consumers as demands for plant-based products rise and animal-based ones decline. Once awareness of impacts tied to consumption reaches a critical level amongst consumers, it is likely that addressing impacts of meat and dairy exports will also become a contentious issue, and over time, alternatives that are less environmentally damaging will likely be found for them as well, just as plant-based alternatives are now becoming a norm for consumers. Overall, consumers and industry are already shifting towards more plant-based foods, and it is further investment into consumer awareness by individuals, scientists, and non-profit organizations that will likely continue to make plant-based eating more prevalent. Once enough support is reached, governments will likely have greater ability to begin dismantling and replacing their environmentally damaging industries with those that promote health in both people and the environment, without the risk for as much public or industry-led backlash.

9.8 Conclusions

The main objectives of the current work were to better understand meat consumption and how its reduction may be promoted for improved environmental and/or public health in the New Zealand context. Main findings from the two studies that comprise the bulk of the thesis are as follows:

1. Awareness of meat's environmental impacts in relation to other food behaviours is low among consumers.
2. Strength of reduction motivations differ depending upon current meat consumption behaviour, but generally, strength of ethically-driven motivations correlate with lower rates of meat consumption, while more personally-driven motivations correlate with higher rates.
3. Attitudes and meat attachment were found to be the strongest predictors of willingness and intentions to reduce meat consumption.
4. Expanding upon initial findings, the psychological model of meat attachment was verified as a valid tool of measurement within New Zealand.
5. Exposure to information on meat's animal welfare impacts significantly increased motivations to reduce meat intake out of concern for animal welfare, and this effect persisted over time.
6. Exposure to information on meat's health, environmental, or animal welfare impacts resulted in significant reductions in attitudes and attachment towards meat, with the animal welfare group showing the strongest persistent declines over time for attitudes and meat attachment (global scale and subscales, except for dependence subscale).
7. Exposure to information on meat's health impacts, specifically, showed significant increases in PBC and significant declines in dependence, and these effects persisted over time.
8. Exposure to information on meat's health, environmental, or animal welfare impacts resulted in significantly higher levels of agreement with proposed meat reduction policies, with the environmental group showing the greatest increases.
9. Exposure to information on meat's health, environmental, or animal welfare impacts resulted in significant increases in willingness and intentions to reduce, paired with increased rates of meat reduction within groups, while self-reported

meat intake frequencies declined significantly, with the environmental group showing the greatest declines in intake.

Thus, based upon these findings combined with those from the literature, some main conclusions can be drawn. Awareness of meat's environmental impacts is low within New Zealand (as found in other Western nations), and educating consumers on this topic can influence not only their personal meat-related attitudes, but also encourage reductions in meat consumption (in fact, environmental framing resulted in *the largest* reductions), as well as increased agreement with proposed meat-reduction policies. Therefore, addressing the lack of awareness in regards to meat's environmental impacts would seem to be a missed opportunity for encouraging more sustainable and healthy plant-based diets. Also, the benefits of information provision, if implemented, would likely be twofold, as it encourages personal changes while simultaneously promoting greater endorsement of more structural actions, such as environmental taxes on meat or increased subsidies for plant-based foods. Currently, the New Zealand government and many others are not taking significant steps to promote meat reduction, and this is likely due to the economic importance of meat production for the country. However, as New Zealand consumers become more aware of the impacts of consuming meat on their health and the health of the local environment, there is much greater chance that these consumers will also call into question the negative environmental impacts of producing such products for export. If consumers become aware of these impacts, they can begin to place pressure on the government and/or the meat industry to encourage and invest in other exports that will not be as damaging to the country's environment.

In addition, meat consumption would seem to be mostly predicated upon people's attitudes and attachment towards meat, rather than beliefs surrounding its social desirability and/or perceived necessity in the diet, which makes shifting the behaviour a matter of primarily addressing the positive attitudes that people hold towards the behaviour. In fact, this is what was found, as information provision on meat's negative impacts effectively reduced attitudes and attachment, resulting in meat intake reductions regardless of information framing (i.e. focusing on either health, environmental, or animal welfare impacts). In more rare cases, where an individual mainly consumes meat due to its perceived necessity in the diet, the provision of information on the health benefits of adopting a plant-based diet will be helpful in encouraging the individual to reduce their meat intake, as such information will increase feelings of control (i.e. feeling capable of

making a choice of what to consume) while simultaneously decreasing feelings of dependence on meat.

So, despite the potentiality of an awareness-action gap as discussed thoroughly in the behaviour change literature, it would seem that in the case of meat consumption and its various negative impacts, information provision can result in actual behaviour change, in the form of reduced (self-reported) meat intake frequencies. Therefore, individuals, organizations, and/or governments who wish to promote environmental sustainability and/or improved public health, and who are not already making efforts to raise awareness in regards to meat's impacts, are missing an opportunity to shift consumer perceptions and behaviours in a way that could have significant benefits for both the natural environment and human health. Of course, information provision does not have to be utilised in isolation, as efforts that strive to make plant-based diets more normalized (e.g. through changing national dietary guidelines, encouraging innovation in plant-based products that directly substitute for those that are traditionally animal-based, etc.) will also be beneficial in encouraging the consumption of more plant-based foods. It will be interesting to see whether new 'reduced-meat' guidelines, as put forward by Health Canada (discussed previously), will lead to reduced meat consumption within Canada and to what degree consumers follow the new guidelines.

Thus, by educating consumers, not only can we have a more informed citizenry, but we can also encourage reduced-meat diets through shifting individual attitudes and consumption behaviours, while increasing support for policy measures that if implemented, can help to promote wider societal-level behavioural shifts to diets that are more sustainable and healthy. Information provision can increase consumer awareness with regards to meat's impacts and encourage individual agency (i.e. choice to consume less meat), and it can also encourage changes to how society currently produces, distributes, and consumes food (i.e. structure) both in the short-term (immediate and lasting increases in meat-reduction policy agreement) and long-term (slow, gradual changes in consumer attitudes and behaviours over time, which can ultimately change both the structure and function of food systems based on shifting demand). Thus, as suggested by Springmann et al. (2018) and Willett et al. (2019), promoting plant-based diets, along with other measures (e.g. reductions in food waste and loss, implementation of more efficient and environmentally-friendly farming technologies and/or techniques, etc.), can act as an integral piece in shifting our currently unsustainable food systems to

ones that can not only provide nutritious foods that promote healthy living, but also stay within suggested planetary boundaries (e.g. GHGs, biomass appropriation, and reactive nitrogen; Pelletier and Tyedmers, 2010). Therefore, although providing information to consumers in regards to the various impacts of meat consumption may seem small and insignificant, when scaled to the societal level, it does offer an opportunity to revolutionise the food system through changes in consumer demand, support for more structural changes (i.e. environmental taxes on high impact foods like meat), and/or support for other environmentally-friendly food system changes that can work in conjunction with meat reduction (e.g. reduced food waste, adoption of composting systems for food waste, etc.).

If a societal shift toward more plant-based diets can be achieved, food-related GHG emissions and resulting climate change impacts would decline, the vast area of land currently occupied by livestock and their feed crops would be freed, food-related land and water pollution and/or degradation would be reduced, habitat and biodiversity loss would be slowed, greater food security would be encouraged (especially amongst the poorest individuals), public health outcomes would be improved, other societal costs linked to disease and illness would be diminished, and the number of animals that are harmed and/or killed for food would be reduced. Put simply, by encouraging plant-based eating, one promotes a society that uses resources more efficiently, is more environmentally-friendly, is healthier, and is more compassionate towards other species while promoting a world that is cleaner, safer, less polluted, and more capable of sustaining natural ecosystems and biodiversity. Therefore, any individual, whether they be a teacher, lawyer, politician, environmentalist, social worker, doctor, or general consumer can find strong reasons to encourage more plant-based eating. In acknowledging this, it is my hope that individuals and/or organizations (whether private or state), will encourage more sustainable and healthy forms of eating through the promotion of plant-based diets, and that efforts to improve both the environment and human society will extend beyond just food and encompass multiple behaviours that are currently practised. Our abilities as humans provide the capacity to grow and improve society, but if not utilised in a thoughtful and considerate way, these same gifts can also contribute to the destruction of the natural environment as well as undermine the health and wellbeing of both individuals and society itself. Thus, we must remain vigilant, taking a critical and rational view of how we currently live in order to identify areas of improvement and implement changes

that make life better for humans, for other species, and the world. Shifting society away from meat-heavy diets to those that are more plant-based is such an area of improvement, and it is my hope that we can implement the changes necessary to encourage such a shift.

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Appendix A

Maori Consultation

NGĀI TAHU RESEARCH CONSULTATION COMMITTEE
TE KOMITI RAKAHAU KI KĀI TAHU

Tuesday, 06 December 2016.

Dr Sean Connelly, Department of
Geography, DUNEDIN.

Tēnā Koe Dr Sean Connelly,

Food-related Opinions and Behaviours of Young New Zealand Adults

The Ngāi Tahu Research Consultation Committee (the committee) met on Tuesday, 06 December 2016 to discuss your research proposition.

By way of introduction, this response from The Committee is provided as part of the Memorandum of Understanding between Te Rūnanga o Ngāi Tahu and the University. In the statement of principles of the memorandum it states ″Ngāi Tahu acknowledges that the consultation process outline in this policy provides no power of veto by Ngāi Tahu to research undertaken at the University of Otago″. As such, this response is not ″approval″ or ″mandate″ for the research, rather it is a mandated response from a Ngāi Tahu appointed committee. This process is part of a number of requirements for researchers to undertake and does not cover other issues relating to ethics, including methodology they are separate requirements with other committees, for example the Human Ethics Committee, etc.

Within the context of the Policy for Research Consultation with Māori, the Committee base consultation on that defined by Justice McGechan:

"Consultation does not mean negotiation or agreement. It means: setting out a proposal not fully decided upon; adequately informing a party about relevant information upon which the proposal is based; listening to what the others have to say with an open mind (in that there is room to be persuaded against the proposal); undertaking that task in a genuine and not cosmetic manner.

Reaching a decision that may or may not alter the original proposal."

The Ngāi Tahu Research Consultation Committee has membership from:

Te Rūnanga o Ōtākou Incorporated
Kāti Huirapa Rūnaka ki Puketeraki
Te Rūnanga o Moeraki

The Committee considers the research to be of importance to Māori health.

As this study involves human participants, the Committee strongly encourage that ethnicity data be collected as part of the research project as a right to express their self-identity. That is the questions on self-identified ethnicity and descent, these questions are contained in the latest census.

The Committee suggests dissemination of the research findings to Māori health organisations and to Donna Matahaere-Atariki who is a board member of the Health and Disability NonGovernment Organisations and a member on Oranga Tamariki - Protecting our children initiative within Ngai Tahi..

We wish you every success in your research and the committee also requests a copy of the research findings.

This letter of suggestion, recommendation and advice is current for an 18 month period from Tuesday, 06 December 2016 to 6 June 2018.

Nāhaku noa, nā

P.P. NTRCC

Mark Brunton
Kaiwhakahaere Rangahau Māori
Research Manager Māori
Research Division Te Whare
Wānanga o Otāgo Ph: +64 3
479 8738
Email: mark.brunton@otago.ac.nz Web:
www.otago.ac.nz

The Ngāi Tahu Research Consultation Committee has membership from:
Te Rūnanga o Ōtākou Incorporated
Kāti Huirapa Rūnaka ki Puketeraki
Te Rūnanga o Moeraki

Appendix B

University of Otago Ethics Approval

D17/064

Dr S Connelly
Department of Geography
Division of Humanities

7 April 2017

Dear Dr Connelly,

I am writing to confirm for you the status of your proposal entitled "**Food-related Opinions and Behaviours of Young New Zealand Adults**", which was originally received on March 9, 2017. The Human Ethics Committee's reference number for this proposal is **D17/064**.

The above application was Category B and had therefore been considered within the
Department or School. The outcome was subsequently reviewed by the University of Otago Human Ethics Committee. The outcome of that consideration was that the proposal was approved.

Approval is for up to three years from the date of HOD approval. If this project has not been completed within three years of this date, re-approval must be requested. If the nature, consent, location, procedures or personnel of your approved application change, please advise me in writing.

Yours sincerely,

Mr Gary Witte
Manager, Academic Committees
Tel: 479 8256
Email: gary.witte@otago.ac.nz

Appendix C

Study 1 Publication

Title: Gauging Attitudes and Behaviours: Meat Consumption and Potential Reduction

Authors: Garrett Lentz, Sean Connelly, Miranda Mirosa, Tim Jowett

Year: 2018

Journal: Appetite

Web link: https://www.sciencedirect.com/science/article/pii/S0195666317313934

Abstract:

The present study focused on adding to the understanding of meat consumption and potential drivers for its reduction in New Zealand. Using the Theory of Planned Behaviour (TPB) and the recently developed Meat-Attachment Questionnaire (MAQ), this study investigated New Zealand consumers' attitudes, motivations and behaviours in regards to meat consumption. Results derive from a questionnaire sent across New Zealand in March 2017, in which 841 responses were obtained from representative consumer panels. Consumer awareness of the severity of meat's environmental impacts was found to be quite low in comparison to other sustainable food behaviours. Motivations for reduction seem to shift across consumer groups, with different considerations rising and falling in importance depending on current meat consumption habits. Among the TPB components, only attitudes were found to accurately and consistently predict willingness and intentions to reduce personal meat intake, while both attitudes and subjective norms predicted agreement with proposed structural measures that would promote meat reduction and/or plant-based food consumption. In addition, the MAQ was found to provide explanatory power above and beyond that of the TPB components alone and this research supports its use as a tool to further understand meat consumption and potential motivations for reduction. The authors believe these results could be useful for governments or organizations wishing to implement meat reduction strategies, as well as providing a stepping stone for further research inquiry into motivations behind meat consumption and its potential reduction.

Appendix D

Study 1 Participant Information Sheet

[Reference Number: *D17/064*]
[*02/03/2017*]

Food-related Opinions and Behaviours of New Zealand Adults: Baseline Survey

INFORMATION SHEET FOR PARTICIPANTS

Thank you for showing an interest in this project. Please read this information sheet carefully before deciding whether or not to participate. If you decide to participate we thank you. If you decide not to take part there will be no disadvantage to you and we thank you for considering our request.

What is the Aim of the Project?

The overall aim of this project is to provide a general understanding of food-related beliefs, opinions and behaviours among New Zealand adults.

This research is being conducted as a partial fulfilment for a Doctor of Philosophy.

What Types of Participants are being sought?

I am seeking participants that are interested in sharing their food-related opinions and behaviours. Participants will include a wide array of adults across New Zealand. Recruitment will occur through emails distributed by ResearchNow.

What will Participants be asked to do?

Should you agree to take part in this project, you will be asked to answer a questionnaire on food-related beliefs and behaviours. The survey is expected to take 10 to 15 minutes.

What Data or Information will be collected and what use will be made of it?

The data collected during this survey will include thoughts and opinions on food-related subjects, as well as some personal information such as age, gender, and ethnicity in order to identify potential trends among New Zealand adults.

All collected data will be stored on password protected devices that are only accessible by the researcher. Any personal information will be deleted at the conclusion of the study, but

the derived data will be kept as a reference for the research, possibly indefinitely. All precautions will be taken to ensure that any sensitive data sent over email or submitted through the survey will be protected and secure.

The results of the project may be published and will be available in the University of Otago Library (Dunedin, New Zealand) but every attempt will be made to preserve individual anonymity.

Can Participants change their mind and withdraw from the project?

You may withdraw your participation in the project at any time and without any disadvantage to yourself.

What if Participants have any Questions?

If you have any questions about our project, either now or in the future, please feel free to contact either:

Garrett Lentz

Department of Geography

University Telephone Number: 03 479 8777

Email Address: garrett.lentz@postgrad.otago.ac.nz

and

Dr. *Sean Connelly*

Department of Geography

University telephone number: 03 479 8771

Email Address: sean.connelly@otago.ac.nz

and

Dr *Miranda Mirosa*

Department of Food Science

University telephone number: 03 479 7953 Ext. 7953

Email Address: miranda.mirosa@otago.ac.nz

This study has been approved by the Department stated above. However, if you have any concerns about the ethical conduct of the research you may contact the University of Otago Human Ethics Committee through the Human Ethics Committee Administrator (ph 03 479-8256). Any issues you raise will be treated in confidence and investigated and you will be informed of the outcome.

Appendix E

Study 1 Questionnaire

The purpose of this survey is to gauge attitudes, beliefs and behaviours in regards to food consumption among citizens in New Zealand. Results from this survey will be used in a PhD thesis at the University of Otago.

All participants will remain anonymous as sociodemographic details will be gathered, but none that could be used to identify you. Participation in this survey is voluntary.

This survey is meant to assess your personal thoughts and opinions, so there are no right or wrong answers to any of the survey questions. There are no screen-out questions, so please be completely honest when answering questions, as your responses will not at all affect your ability to complete the survey.

Please confirm that:

I have read the Information Sheet concerning this project and understand what it is about. All my questions have been answered to my satisfaction. I understand that I am free to request further information at any stage. I know that:

1. My participation in the project is entirely voluntary.

2. I am free to withdraw from the project at any time without any disadvantage.

3. No personal identifying information will be requested, however raw data on which the results of the project depend will be retained in secure storage for at least five years.

4. During this study I will be asked about my opinions, beliefs and behaviours in regards to what foods I buy and consume. If any of the questions make me feel uncomfortable, I understand that I can stop the survey and withdraw my answers at any time.

5. The results of the project may be published and will be available in the University of Otago Library (Dunedin, New Zealand) but every attempt will be made to preserve my anonymity.

By clicking next, I agree to take part in this project.

Thank you for your participation, Garrett Lentz, PhD student; Sean Connelly, Supervisor; Miranda Mirosa, Supervisor

[page break]

This first question will be asking about opinions on different food purchasing and dietary habits.

[Q1] On a scale from 1 to 7, rate how you believe each of the following eating behaviours affects the environmental friendliness of your diet.

	Very small environmental benefit					Very large environmental benefit	
	1	2	3	4	5	6	7
Buying locally sourced foods							
Buying foods with less packaging material							
Avoiding foods that are transported by air							
Eating less meat							
Buying organic foods							
Eating seasonal fruits and vegetables							

[page break]

The remainder of the survey will be asking about your meat consumption. Please keep in mind when answering questions, that the word “meat” refers to red and white meats (e.g. beef, lamb, pork, chicken, turkey, fish, seafood etc.) that are either unprocessed (e.g. chicken breast, steak, fish filet) or processed (e.g. sausage, salami, meat mince, chicken nuggets, crab cakes).

[page break]

[Q2] On average, how often do you consume meat or products that include meat?

- ○ Several times a day
- ○ Daily
- ○ Several times a week
- ○ Rarely
- ○ Never

Skip To: Q7 If Q2 = Never

[page break]

[Q3] Have you already or are you currently making any efforts to reduce your personal meat consumption?

- ○ Yes
- ○ No

Skip To: Q7 If Q3 = Yes

[page break]

[Q4] On a scale from 1 to 7, how willing would you be to consider reducing your meat consumption sometime in the near future?

	Not at all willing					Extremely willing
	1	2	3	4	5	6	7

[Q5] Specifically, in the next six months do you intend to reduce your meat consumption?

	Do not intend at all					Fully intend
	1	2	3	4	5	6	7

[page break]

[Q6] How important, if at all, would each of the following factors be in strengthening your consideration to reduce your overall meat intake?

	Not at all important			Moderately Important			Extremely Important
	1	2	3	4	5	6	7
Health benefits							
More environmentally friendly							
Animal welfare concerns							
High cost of meat							
Taste preferences							
Weight Control							
Other: (Please Specify)							

[page break]

Display Q7:

If Q2 = Never

Or Q3 = Yes

[Q7] Please think back to when you first decided to reduce your meat consumption. How important were each of the following factors in influencing your initial decision to lower your overall meat intake?

	Not at all important			Moderately important			Extremely important
	1	2	3	4	5	6	7
Health benefits							
More environmentally friendly							
Animal welfare concerns							
Price of meat							
Taste preferences							
Weight Control							
Other: (Please specify)							

[page break]

On the scales provided, please choose what most closely aligns with your thoughts and attitudes towards the act of consuming meat. NOTE: Scores closer to 1 mean you agree more with the attribute on the left and scores closer to 7 mean you agree more with the attribute on the right.

[Q8] Bad/Good:

	Bad					Good	
	1	2	3	4	5	6	7

[Q9] Unpleasant/Pleasant:

	Unpleasant					Pleasant	
	1	2	3	4	5	6	7

[Q10] Against/For:

	Against					For	
	1	2	3	4	5	6	7

[Q11] Unfavorable/Favorable:

	Unfavorable					Favorable	
	1	2	3	4	5	6	7

[Q12] Negative/Positive:

	Negative					Positive	
	1	2	3	4	5	6	7

[page break]

On a scale from 1 to 7, please rate your agreement with the following statements.

[Q13] People who are important to me think that I should eat meat.

	Strongly disagree			Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7

[Q14] In regards to people who are important to you, how much do they influence your actions to either consume or not consume meat?

Not at all	A little	A moderate amount	A lot	Don't Know
1	2	3	4	5	6	7

[page break]

Please rate your agreement with the following statements in regards to your current meat consumption habits

[Q15] I am confident that I could change my habits **if** I wanted to

Strongly disagree	Neither agree nor disagree	Strongly agree
1	2	3	4	5	6	7

[Q16] Whether I change my habits is entirely up to me

Strongly disagree	Neither agree nor disagree	Strongly agree
1	2	3	4	5	6	7

[Q17] Changing my habits is **not** something that is under my control

Strongly disagree	Neither agree nor disagree	Strongly agree
1	2	3	4	5	6	7

[page break]

[Q18] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7
To eat meat is one of the good pleasures in life							
Meat is irreplaceable in my diet							
According to our position in the food chain, we have a right to eat meat							
I feel bad when I think of eating meat							
I love meals with meat							
To eat meat is disrespectful towards life and the environment							

[Q19] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7
To eat meat is an unquestionable right of every person							
A good steak is without comparison							
I would feel fine with a meatless diet							
I'm a big fan of meat							
If I couldn't eat meat I would feel weak							
If I was forced to stop eating meat I would feel sad							

[Q20] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7
Meat reminds me of diseases							
By eating meat I'm reminded of the death and suffering of animals							
Eating meat is a natural and indisputable practice							
I can't picture myself not eating meat regularly							

[page break]

Now we would like your opinions on meat-focused policies that address **environmental** concerns.

Please rate your agreement with the following statements.

[Q21] To reduce food-related environmental impact, an 'environmental tax' should be placed on meat and meat products to make them more expensive.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7

[Q22] To reduce food-related environmental impact, an 'environmental subsidy' should be applied to plant-based foods to make them more affordable.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7

[Q23] Public institutions like universities should reduce the environmental impact of their food catering by providing more meatless meals.

Strongly disagree		Neither agree nor disagree				Strongly agree
1	2	3	4	5	6	7

[page break]

Now we would like your opinions on meat-focused policies that address **public health** concerns.

Please rate your agreement with the following statements.

[Q24] To promote improved public health, a 'health tax' should be placed on meat and meat products to make them more expensive.

Strongly disagree		Neither agree nor disagree				Strongly agree
1	2	3	4	5	6	7

[Q25] To promote improved public health, a 'health subsidy' should be applied to plant-based foods to make them more affordable.

Strongly disagree		Neither agree nor disagree				Strongly agree
1	2	3	4	5	6	7

[Q26] Public institutions like universities should promote public health by providing more meatless meals among their food options

Strongly disagree		Neither agree nor disagree				Strongly agree
1	2	3	4	5	6	7

[page break]

Almost done. These last few questions will be used to gain sociodemographic information.

[Q27] What is your age?

- ○ Under 18
- ○ 18 - 24
- ○ 25 - 30
- ○ 31 - 35
- ○ 36 - 40
- ○ 41 - 50
- ○ 51 - 60
- ○ 61 - 70
- ○ 71 or over

[Q28] What is your gender?

- ○ Male
- ○ Female
- ○ Gender diverse

[Q29] What is your ethnic background? (Select all that apply)

- ☐ European / Pākehā
- ☐ Māori
- ☐ Pacific Islander
- ☐ Other (Please specify):

[Q30] What is the highest level of education you have attained?

- ○ No qualification
- ○ Completed NCEA levels 1 - 3 or equivalent
- ○ Apprenticeship, trade certification, or similar
- ○ Bachelor's Degree
- ○ Master's Degree
- ○ PhD (Doctor of Philosophy)
- ○ Other (Please specify):

[Q31] How many adults live in your household?

- ○ 1
- ○ 2
- ○ 3
- ○ 4
- ○ 5 or more

[Q32] How many children aged under 18 years live in your household?

- ○ None
- ○ 1
- ○ 2
- ○ 3
- ○ 4
- ○ 5 or more

[Q33] Which of the following best describes your household?

- ○ One-person household
- ○ Couple only
- ○ Couple with children aged under 16 years
- ○ Couple only plus other adults (e.g. children aged over 16)
- ○ Couple with children under 16 years, plus other adults
- ○ One-parent with children under 16 years, plus other adults
- ○ One-parent with children under 16 years
- ○ Multiple families with children under 16 years
- ○ Multi-person adult household / Flatting
- ○ Other: (Please specify)

[Q34] What is your average yearly personal or household income? (If flatting or sharing a home with other non-family members, please put only personal income)

- ○ Less than $20,000
- ○ $20,000 to $34,999
- ○ $35,000 to $49,999
- ○ $50,000 to $74,999
- ○ $75,000 to $99,999
- ○ $100,000 to $149,999
- ○ $150,000 to $199,999
- ○ $200,000 or more

[Q35] Which of the following best describes the area you live in?

- ○ Inner city
- ○ Outer city
- ○ Town
- ○ Rural

[page break]

You have reached the end of the survey. Thank you for your participation! Please click the next (>>) button to be redirected to the survey site to collect your compensation.

Appendix F

Study 2 Film Synopses

The following film synopses were copied in May, 2019 from the popular movie-review site (www.imdb.com), which is also where ratings for the films were taken.

What the Health:

"An intrepid filmmaker on a journey of discovery as he uncovers possibly the largest health secret of our time and the collusion between industry, government, pharmaceutical and health organizations keeping this information from us."

Cowspiracy: the Sustainability Secret:

"Follow the shocking, yet humorous, journey of an aspiring environmentalist, as he daringly seeks to find the real solution to the most pressing environmental issues and true path to sustainability."

Earthlings:

"Using hidden cameras and never-before-seen footage, Earthlings chronicles the day-to-day practices of the largest industries in the world, all of which rely entirely on animals for profit."

Jim & Andy: The Great Beyond:

"A behind-the-scenes look at how Jim Carrey adopted the persona of idiosyncratic comedian Andy Kaufman on the set of Man on the Moon (1999)."

Appendix G

Study 2 Participant Information Sheet

[Reference Number: *D17/064*]
[*02/03/2017*]

Food-related Opinions and Behaviours of Young New Zealand Adults: Educational Film Workshop

INFORMATION SHEET FOR PARTICIPANTS

Thank you for showing an interest in this project. Please read this information sheet carefully before deciding whether or not to participate. If you decide to participate we thank you. If you decide not to take part there will be no disadvantage to you and we thank you for considering our request.

What is the Aim of the Project?

The aim of the project is to understand how documentaries might influence food-related opinions and behaviours of university students. Specifically, it aims to understand how varying types of information independently and/or mutually influence an individual's opinions, behaviours, and outlook on food and food-related issues and policies.

This research is being conducted as a partial fulfilment for a Doctor of Philosophy.

What Types of Participants are being sought?

I am seeking participants that are university students, are between the ages of 18 and 30, and are open to sharing their food-related opinions and behaviours.

What will Participants be asked to do?

Should you agree to take part in this project, you will be asked to take a pre-survey, watch a film (snacks and refreshments provided), and take two post-surveys, one immediately after watching the film and one 1 month follow-up. The expected total time spent on answering surveys and watching the film is about 2 to 2.5 hours total, depending on how fast or slow each survey is taken. Films are intended to be shown in the Geography department at the University of Otago and participants will be notified if changes to location occur. To recognize the actual or reasonable costs involved with participating in this project, all participants will be reimbursed $20 after the full completion of the study.

It is the study participant's responsibility to correctly account for tax on the receipt of any cash payment.

What Data or Information will be collected and what use will be made of it?

The data collected during this survey will include thoughts and opinions on food-related subjects, as well as some personal information such as age, gender, and ethnicity in order to identify potential demographic patterns.

All collected data will be stored on password protected devices that are only accessible by the researcher. Any personal information will be deleted at the conclusion of the study, but the derived data will be kept as a reference for the research, possibly indefinitely. All precautions will be taken to ensure that any sensitive data sent over email or submitted through the survey will be protected and secure.

The results of the project may be published and will be available in the University of Otago Library (Dunedin, New Zealand) but every attempt will be made to preserve individual anonymity.

Can Participants change their mind and withdraw from the project?

You may withdraw your participation in the project at any time and without any disadvantage to yourself.

What if Participants have any Questions?

If you have any questions about our project, either now or in the future, please feel free to contact either:

Garrett Lentz

Department of Geography

University Telephone Number: 03 479 8777

Email Address: garrett.lentz@postgrad.otago.ac.nz

and

Dr. *Sean Connelly*

Department of Geography

University telephone number: 03 479 8771

Email Address: sean.connelly@otago.ac.nz

and

Dr *Miranda Mirosa*

Department of Food Science

University telephone number: 03 479 7953 Ext. 7953

Email Address: miranda.mirosa@otago.ac.nz

This study has been approved by the Department stated above. However, if you have any concerns about the ethical conduct of the research you may contact the University of Otago Human Ethics Committee through the Human Ethics Committee Administrator (ph 03 479-8256). Any issues you raise will be treated in confidence and investigated and you will be informed of the outcome.

Appendix H

Study 2 Screener Questionnaire

Welcome to the eligibility questionnaire for the film workshop study. Completing the survey should only take a few minutes and we encourage you to be as honest and thoughtful as possible with your answers.

[Q1] To start, please provide your contact details so that the researcher can contact you with further information about the study. This information will only be used as a means to communicate information related to the study.

- ○ First name _____
- ○ Last name _____
- ○ Email address _____
- ○ Phone number _____

[page break]

These first questions are meant to provide some background information about yourself.

[Q2] What is your age?

Answer:_____

Skip To: END If Q2 <= 17
Skip To: END If Q2 >= 31

[Q3] What is your gender?

- ○ Male
- ○ Female
- ○ Gender Diverse

[Q4] Are you currently studying at a university in New Zealand?

- ○ Yes, I am an undergraduate
- ○ Yes, I am a postgraduate
- ○ No

Skip To: END If Q4 = No

[Q5] Are you an international student? If so, please write in your home country.

- ○ Yes _____
- ○ No

[Q6] What is your field of study at the university? (e.g. physics, psychology, food science, geography, etc.)

Answer:_____

[page break]

The remainder of the survey will be asking about your meat consumption. Please keep in mind when answering questions, that the word “meat” refers to red and white meats, including seafood (e.g. beef, lamb, pork, chicken, turkey, fish, shrimp, etc.) that are either unprocessed (e.g. chicken breast, steak, fish filet) or processed (e.g. sausage, salami, meat mince, chicken nuggets, crab cakes).

[Q7] On average, how often do you consume meat or products that include meat?

- ○ Several times a day
- ○ Once a day
- ○ Several times a week
- ○ A few times a month
- ○ Less than a few times a month
- ○ Never

Skip To: END If Q7 = Never

[Q8] Have you already or are you currently making any efforts to reduce your personal meat consumption?

- ○ Yes
- ○ No

[page break]

[Q9] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7
To eat meat is one of the good pleasures in life							
Meat is irreplaceable in my diet							
According to our position in the food chain, we have a right to eat meat							
I feel bad when I think of eating meat							
I love meals with meat							
To eat meat is disrespectful towards life and the environment							

[Q10] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7
To eat meat is an unquestionable right of every person							
A good steak is without comparison							
I would feel fine with a meatless diet							
I'm a big fan of meat							
If I couldn't eat meat I would feel weak							
If I was forced to stop eating meat I would feel sad							

[Q11] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7
Meat reminds me of diseases							
By eating meat I'm reminded of the death and suffering of animals							
Eating meat is a natural and indisputable practice							
I can't picture myself not eating meat regularly							

[page break]

The last two questions of the survey will be asking about films you may have seen.

[Q12] Have you ever watched any films specifically about meat consumption's impacts on personal health, animal welfare, or environmental sustainability?

- ○ Yes
- ○ Maybe
- ○ No

[Q13] More specifically, do you remember having seen any of the films listed below? (Select all that apply). If not, please select an appropriate 'other' option (if applicable) and put the name of the film you have seen in the corresponding text box. If you cannot remember the name of the film, simply leave the text box blank. If you have never seen any film related to these topics, simply select 'Have not seen'.

[Continued on next page]

- ○ What the Health
- ○ Forks Over Knives
- ○ PlantPure Nation
- ○ PLANEAT
- ○ Eating You Alive
- ○ Cowspiracy: The Sustainability Secret
- ○ Earthlings
- ○ Vegucated
- ○ Live and Let Live
- ○ Peaceable Kingdom
- ○ Speciesism: The Movie
- ○ Meet Your Meat
- ○ Other film on meat and personal health:
- ○ Other film on meat and environmental sustainability:
- ○ Other film on meat and animal welfare:
- ○ Have not seen

[page break]

You have reached the end of the eligibility questionnaire. When clicking the "Next" button, if you are redirected to an "error" display, there is no need to worry as your answers have been successfully recorded. The researcher will be in contact with you soon and thank you again for your time and participation!

Appendix I

Study 2 Pre-questionnaire(s)

Before beginning the survey, please verify that you have read over the Information Sheet. By clicking 'Next', you agree to take part in the study and begin the survey.

[page break]

[Q1] Before beginning the survey, please put in your first and last name:

First name _____

Last name _____

[page break]

The first section of the survey will be asking about your meat consumption. Please keep in mind when answering questions, that the word “meat” refers to red and white meats (e.g. beef, lamb, pork, chicken, turkey, fish, seafood etc.) that are either unprocessed (e.g. chicken breast, steak, fish filet) or processed (e.g. sausage, salami, meat mince, chicken nuggets, crab cakes).

[page break]

[Q2] On average, how often do you consume meat or products that include meat?

- ○ Several times a day
- ○ Daily
- ○ Several times a week
- ○ Rarely
- ○ Never

[page break]

[Q3] On a scale from 1 to 7, how willing would you be to consider reducing your meat consumption sometime in the near future?

	Not at all willing					Extremely willing
	1	2	3	4	5	6	7

[Q4] Specifically, in the next six months do you intend to reduce your meat consumption?

	Do not intend at all					Fully intend
	1	2	3	4	5	6	7

[page break]

[Q5] How important, if at all, would each of the following factors be in strengthening your consideration to reduce your overall meat intake?

	Not at all important		Moderately Important		Extremely Important	
	1	2	3	4	5	6	7
Health benefits							
More environmentally friendly							
Animal welfare concerns							
High cost of meat							
Taste preferences							
Weight Control							
Other: (Please Specify)							

[page break]

On the scales provided, please choose what most closely aligns with your thoughts and attitudes towards the act of consuming meat. NOTE: Scores closer to 1 mean you agree more with the attribute on the left and scores closer to 7 mean you agree more with the attribute on the right.

[Q6] Bad/Good:

	Bad					Good
1	2	3	4	5	6	7

[Q7] Unpleasant/Pleasant:

	Unpleasant					Pleasant
1	2	3	4	5	6	7

[Q8] Against/For:

	Against					For
1	2	3	4	5	6	7

[Q9] Unfavorable/Favorable:

	Unfavorable					Favorable
1	2	3	4	5	6	7

[Q10] Negative/Positive:

	Negative					Positive
1	2	3	4	5	6	7

[page break]

Please use the scales provided to answer the following statements.

[Q11] People who are important to me think that I should eat meat.

Not at all	Somewhat	Mostly	Absolutely

1 2 3 4 5 6 7

[Q12] In regards to people who are important to you, how much do they influence your actions to consume meat?

Not at all	A little	A moderate amount	A lot	Don't Know

1 2 3 4 5 6 7

[page break]

Please rate your agreement with the following statements in regards to your current meat consumption habits

[Q13] I am confident that I could change my habits **if** I wanted to

Strongly disagree	Neither agree nor disagree	Strongly agree

1 2 3 4 5 6 7

[Q14] Whether I change my habits is entirely up to me

Strongly disagree	Neither agree nor disagree	Strongly agree

1 2 3 4 5 6 7

[Q15] Changing my habits is **not** something that is under my control

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7

[page break]

[Q16] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7
To eat meat is one of the good pleasures in life							
Meat is irreplaceable in my diet							
According to our position in the food chain, we have a right to eat meat							
I feel bad when I think of eating meat							
I love meals with meat							
To eat meat is disrespectful towards life and the environment							

[Q17] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7
To eat meat is an unquestionable right of every person							
A good steak is without comparison							
I would feel fine with a meatless diet							
I'm a big fan of meat							
If I couldn't eat meat I would feel weak							
If I was forced to stop eating meat I would feel sad							

[Q18] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7
Meat reminds me of diseases							
By eating meat I'm reminded of the death and suffering of animals							
Eating meat is a natural and indisputable practice							
I can't picture myself not eating meat regularly							

[page break]

Now we would like your opinions on some hypothetical policies that could affect public meat consumption if proposed and implemented. Please rate your agreement with the following statements.

[Q19] An additional tax should be placed on meats and products that include meat in order to promote public health

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7

[Q20] Subsidies should be provided to plant-based foods and products in order to promote public health

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7

[Q21] Public institutions like universities and hospitals should provide more meatless meals through their food catering in order to promote public health

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7

[page break]

Almost done. These last few questions will be used to gain sociodemographic information.

[Q22] What is your age?

Answer:_____

[Q23] What is your gender?

- ○ Male
- ○ Female
- ○ Gender Diverse

[Q24] What is your ethnic background? (Select all that apply)

- ☐ European / Pākehā
- ☐ Māori
- ☐ Pacific Islander
- ☐ Other (Please specify): _____

[Q25] Please describe your current level of education.

- ○ Bachelor's student
- ○ Master's student
- ○ PhD (Doctor of Philosophy) student
- ○ Other (Please specify): _____

[Q26] Which of the following best describes your living situation?

- ○ Living with parents
- ○ Living alone
- ○ Flatting / Multi-person household
- ○ Living with partner
- ○ Living with partner plus other adults (e.g. flatmates, children aged over 16)
- ○ Other: (Please specify) _____

[page break]

You have reached the end of the survey. Thank you for your participation!

Appendix J

Study 2 Post-questionnaire(s)

[Q1] Before beginning the survey, please put in your first and last name:

First name _____

Last name _____

[page break]

This first section of the survey will be asking questions about the film you just watched.

[Q2] In general, what did you think about the film?

I did not like it at all			No strong opinion			I liked it a lot
1	2	3	4	5	6	7

[Q3] How much of the information in the film was new to you?

None at all		A little		A lot		All of it
1	2	3	4	5	6	7

[Q4] How believable was the information presented in the film?

Not at all believable		A little believable		Very believable		Completely believable
1	2	3	4	5	6	7

[Q5] Thinking about the messages in the film, do they provide strong reasons for reducing your meat consumption?

Not at all						Absolutely
1	2	3	4	5	6	7

[Q6] Do you think you will share the messages in the film (or the film itself) with others?

Not at all	Possibly	Probably	Absolutely			
1	2	3	4	5	6	7

[page break]

[Q7] On a scale from 1 to 7, how willing would you be to consider reducing your meat consumption sometime in the near future?

Not at all willing					Extremely willing	
1	2	3	4	5	6	7

[Q8] Specifically, in the next six months do you intend to reduce your meat consumption?

Do not intend at all					Fully intend	
1	2	3	4	5	6	7

[page break]

[Q9] How important, if at all, would each of the following factors be in strengthening your consideration to reduce your overall meat intake?

	Not at all important			Moderately Important			Extremely Important
	1	2	3	4	5	6	7
Health benefits							
More environmentally friendly							
Animal welfare concerns							
High cost of meat							
Taste preferences							
Weight Control							
Other: (Please Specify)							

[page break]

On the scales provided, please choose what most closely aligns with your thoughts and attitudes towards the act of consuming meat. NOTE: Scores closer to 1 mean you agree more with the attribute on the left and scores closer to 7 mean you agree more with the attribute on the right.

[Q10] Bad/Good:

	Bad					Good	
	1	2	3	4	5	6	7

[Q11] Unpleasant/Pleasant:

	Unpleasant					Pleasant	
	1	2	3	4	5	6	7

[Q12] Against/For:

	Against					For
	1	2	3	4	5	6	7

[Q13] Unfavorable/Favorable:

	Unfavorable					Favorable
	1	2	3	4	5	6	7

[Q14] Negative/Positive:

	Negative					Positive
	1	2	3	4	5	6	7

[page break]

Please use the scales provided to answer the following statements.

[Q15] People who are important to me think that I should eat meat.

	Not at all	Somewhat			Mostly		Absolutely
	1	2	3	4	5	6	7

[Q16] In regards to people who are important to you, how much do they influence your actions to consume meat?

	Not at all	A little		A moderate amount		A lot	Don't Know
	1	2	3	4	5	6	7

[page break]

Please rate your agreement with the following statements in regards to your current meat consumption habits

[Q17] I am confident that I could change my habits **if** I wanted to

Strongly disagree		Neither agree nor disagree				Strongly agree
1	2	3	4	5	6	7

[Q18] Whether I change my habits is entirely up to me

Strongly disagree		Neither agree nor disagree				Strongly agree
1	2	3	4	5	6	7

[Q19] Changing my habits is **not** something that is under my control

Strongly disagree		Neither agree nor disagree				Strongly agree
1	2	3	4	5	6	7

[page break]

[Q20] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7
To eat meat is one of the good pleasures in life							
Meat is irreplaceable in my diet							
According to our position in the food chain, we have a right to eat meat							
I feel bad when I think of eating meat							
I love meals with meat							
To eat meat is disrespectful towards life and the environment							

[Q21] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7
To eat meat is an unquestionable right of every person							
A good steak is without comparison							
I would feel fine with a meatless diet							
I'm a big fan of meat							
If I couldn't eat meat I would feel weak							
If I was forced to stop eating meat I would feel sad							

[Q22] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7
Meat reminds me of diseases							
By eating meat I'm reminded of the death and suffering of animals							
Eating meat is a natural and indisputable practice							
I can't picture myself not eating meat regularly							

[page break]

Now we would like your opinions on some hypothetical policies that could affect public meat consumption if proposed and implemented. Please rate your agreement with the following statements.

[Q23] An additional tax should be placed on meats and food products that include meat in order to promote public health.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7

[Q24] Subsidies should be provided to plant-based foods and products in order to promote public health.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7

[Q25] Public institutions like universities and hospitals should provide more meatless meals through their food catering in order to promote public health.

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7

[page break]

You have reached the end of the survey. Thank you for your participation!

Appendix K

Study 2 One Month Follow-up Questionnaire(s)

[Q1] Before beginning the survey, please put in your first and last name:

First name _____

Last name _____

[page break]

The first section of the survey will be asking about the film titled "[Film Title]" that you watched about a month ago for the study. To refresh your memory, the film discussed [short film synopsis].

[Q2] How much have you thought about the messages in the film and/or the film itself since you watched it?

	Not at all		A little		A good amount		A lot
	1	2	3	4	5	6	7

[page break]

[Q3] Since the film, have you acquired any additional information on the topic discussed in the film or other topics related to meat consumption? (e.g. from films, websites, articles, blogs, knowledgeable friends).

- ○ Yes, I searched for additional information
- ○ Yes, but it was only by chance
- ○ No

Skip To: Q5 If Q3 = No

[page break]

[Q4] Which of the following topics, in regards to meat consumption, have you acquired more information about since watching the film? (Select all that apply)

- ☐ Health-related information
- ☐ Environmental sustainability information
- ☐ Animal welfare information
- ☐ Other (please specify): _____

[page break]

[Q5] Have you discussed the film or any of its messages with others? If so, can you please give a brief description or example of the discussions you have had? If not, can you please state why.

- ○ Yes _____
- ○ No _____

[page break]

[Q6] Please keep in mind when answering the following questions, that the word “meat” refers to red and white meats (e.g. beef, lamb, pork, chicken, turkey, fish, seafood etc.) that are either unprocessed (e.g. chicken breast, steak, fish filet) or processed (e.g. sausage, salami, meat mince, chicken nuggets, crab cakes).

[Q7] Has the film had any lasting impact on your attitudes, beliefs, or thoughts on eating meat? If so, please briefly describe how your views have changed.

- ○ Yes _____
- ○ No

[page break]

[Q8] Has the film had any lasting impact on your meat eating habits? If so, can you please give a brief description of how your meat eating habits have changed? If not, can you please explain why your habits have not changed? (e.g. taste preferences, lack of resources, not enough motivation, etc.)

- ○ Yes _____
- ○ No _____

[page break]

[Q9] On average, how often do you consume meat or products that include meat?

- ○ Several times a day
- ○ Once a day
- ○ Several times a week
- ○ A few times a month
- ○ Less than a few times a month
- ○ Never

Skip To: Q13 If Q9 = Never

[page break]

[Q10] Since watching the film, have you reduced your meat consumption?

- ○ Yes
- ○ No

Skip To: Q13 If Q10 = Yes

[page break]

[Q11] On a scale from 1 to 7, how willing would you be to consider reducing your meat consumption sometime in the near future?

	Not at all willing					Extremely willing
	1	2	3	4	5	6	7

[Q12] Specifically, in the next six months do you intend to reduce your meat consumption?

	Do not intend at all					Fully intend
	1	2	3	4	5	6	7

[page break]

[Q13] How important, if at all, would each of the following factors be in strengthening your consideration to reduce your overall meat intake?

	Not at all important		Moderately Important		Extremely Important
	1	2	3	4	5	6	7
Health benefits							
More environmentally friendly							
Animal welfare concerns							
High cost of meat							
Taste preferences							
Weight Control							
Other: (Please Specify)							

[page break]

On the scales provided, please choose what most closely aligns with your thoughts and attitudes towards the act of consuming meat. NOTE: Scores closer to 1 mean you agree more with the attribute on the left and scores closer to 7 mean you agree more with the attribute on the right.

[Q14] Bad/Good:

	Bad					Good
	1	2	3	4	5	6	7

[Q15] Unpleasant/Pleasant:

	Unpleasant					Pleasant
	1	2	3	4	5	6	7

[Q16] Against/For:

	Against					For
	1	2	3	4	5	6	7

[Q17] Unfavorable/Favorable:

	Unfavorable					Favorable
	1	2	3	4	5	6	7

[Q18] Negative/Positive:

	Negative					Positive
	1	2	3	4	5	6	7

[page break]

Please use the scales provided to answer the following statements.

[Q19] People who are important to me think that I should eat meat.

Not at all	Somewhat	Mostly	Absolutely

1	2	3	4	5	6	7

[Q20] In regards to people who are important to you, how much do they influence your actions to consume meat?

Not at all	A little	A moderate amount	A lot	Don't Know

1	2	3	4	5	6	7

[page break]

Please rate your agreement with the following statements in regards to your current meat consumption habits

[Q21] I am confident that I could change my habits **if** I wanted to

Strongly disagree	Neither agree nor disagree	Strongly agree

1	2	3	4	5	6	7

[Q22] Whether I change my habits is entirely up to me

Strongly disagree	Neither agree nor disagree	Strongly agree

1	2	3	4	5	6	7

[Q23] Changing my habits is **not** something that is under my control

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7

[page break]

[Q24] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7
To eat meat is one of the good pleasures in life							
Meat is irreplaceable in my diet							
According to our position in the food chain, we have a right to eat meat							
I feel bad when I think of eating meat							
I love meals with meat							
To eat meat is disrespectful towards life and the environment							

[Q25] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7
To eat meat is an unquestionable right of every person							
A good steak is without comparison							
I would feel fine with a meatless diet							
I'm a big fan of meat							
If I couldn't eat meat I would feel weak							
If I was forced to stop eating meat I would feel sad							

[Q26] On a scale from 1 to 7, please rate your agreement with the following group of statements.

	Strongly disagree		Neither agree nor disagree				Strongly agree
	1	2	3	4	5	6	7
Meat reminds me of diseases							
By eating meat I'm reminded of the death and suffering of animals							
Eating meat is a natural and indisputable practice							
I can't picture myself not eating meat regularly							

[page break]

Now we would like your opinions on some hypothetical policies that could affect public meat consumption if proposed and implemented. Please rate your agreement with the following statements.

[Q27] An additional tax should be placed on meats and food products that include meat in order to promote public health.

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7

[Q28] Subsidies should be provided to plant-based foods and products in order to promote public health.

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7

[Q29] Public institutions like universities and hospitals should provide more meatless meals through their food catering in order to promote public health.

	Strongly disagree		Neither agree nor disagree			Strongly agree
	1	2	3	4	5	6	7

[page break]

You have reached the end of the survey. Thank you for your participation!